GM Powertrain Electronics Integration and Software

GMPT GMLAN Diagnostic Test Mode Configuration Specification

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1. Introduction

This document specifies the configuration of GMLAN diagnostic test modes for GMPT controllers using the C core software.

1.1 Applicability

The intended audience for this document includes the algorithm, software, service, assembly verification, and calibration communities.

1.2 Assumptions

It is assumed that the standard data types in [2] shall be used for the scaling / encoding of all signals specified within this document.

2. Diagnostic Services

The Legislated diagnostic services in Table 1shall be implemented according to [3]. Additional details (e.g., the set of sub-functions supported, the set of parameter values supported, specific conditions for negative response return codes, etc.) are specified in the following sections.

Service Identifier	Diagnostic Service	Application Software	Boot Software
$01	RequestCurrentPowertrainDiagnosticData	X	
$02	RequestPowertrainFreezeFrameData	X	
$03	RequestEmission-RelatedDiagnosticTroubleCodes-Confirmed	X	
$04	Clear/ResetDiagnosticInformation	X	
$06	RequestOnBoardMonitoringTestResults	X	
$07	RequestEmission-RelatedDiagnosticTroubleCodes-Pending	X	
$08	RequestControlOfOnBoardSystem		
$09	RequestVehicleInformation	X	
$0A	Request Emission-Related Diagnostic Trouble Codes With Permanent Status	X	

Table 1: Legislated Diagnostic Services

The Enhanced diagnostic services in Table 2: Diagnostic Services
shall be implemented according to [1]. Additional details (e.g., the set of sub-functions supported, the set of parameter values supported, specific conditions for negative response return codes, etc.) are specified in the following sections.

Service Identifier	Diagnostic Service	Application Software	Boot Software
$10	InitiateDiagnosticOperation	X	
$12	ReadFailureRecordData	X	
$1A	ReadDataByIdentifier	X	X
$20	ReturnToNormalOperation	X	X
$22	ReadDataByParameterIdentifier	X	
$23	ReadMemoryByAddress	X	
$27	SecurityAccess	X	X
$28	DisableNormalCommunication	X	X
$2C	DynamicallyDefineMessage	X	
$2D	DefinePIDByAddress	X	
$34	RequestDownload	X	X
$36	TransferData	X	X
$3B	WriteDataByIdentifier	X	
$3E	TesterPresent	X	X
$7F	NegativeResponse	X	X
$A2	ReportProgrammedState	X	X
$A5	ProgrammingMode	X	X
$A9	ReadDiagnosticInformation	X	
$AA	ReadDataByPacketIdentifier	X	
$AE	DeviceControl	X	

Table 2: Diagnostic Services

2.1 OVIR / BFPR - NegativeResponse ($7F) Service

The values supported for the returnCode parameter of this diagnostic service are specified in Table 3: Descriptions of Negative Response Return Codes
.

If a request is made for a diagnostic service which is not supported or unknown, a negative response return code of ServiceNotSupported ($11) shall be sent.

Return Code	Description
$11	ServiceNotSupported
$12	SubFunctionNotSupported-InvalidFormat
$22	ConditionsNotCorrectOrRequestSequenceError
$31	RequestOutOfRange
$35	InvalidKey
$36	ExceededNumberOfAttempts
$37	RequiredTimeDelayNotExpired
$78	RequestCorrectlyReceived-ResponsePending
$81	SchedulerFull
$83	VoltageOutOfRangeFault (High / Low)
$85	GeneralProgrammingFailure
$89	DeviceTypeError
$99	ReadyForDownload-DTCStored
$E3	DeviceControlLimitsExceeded

Table 3: Descriptions of Negative Response Return Codes

2.2 ODAR - RequestCurrentPowertrainDiagnosticData ($01) Service

The values supported for the Parameter Identifiers (PIDs) reported by this legislated diagnostic service are specified in Section 3.1, Legislated PIDs. The individual PID specifications, scaling information, and Generic Scan Tool display formats are found in [3]. Not all PIDs are applicable or supported by all systems.

PID $00 is a bit-encoded PID that indicates, for each module, which PIDs that module supports. PID $00 must be supported by all modules that respond to a Mode $01 request as defined in this document, because diagnostic tools that conform to SAE J1978 use the presence of a response by the vehicle to this request to determine which protocol is supported for OBD II communications.

The maximum time after a valid request for a supported PID is received and a response is transmitted by the ECU is 50 milliseconds.

There are no conditions for a negative response return code for this diagnostic service. No response should be generated when a request is made for a PID that is not supported by the ECU. The length of PID response data is limited to 255 bytes.

Additional details can be found in the Implementation Specification for SAE J1979 Test Modes [11].

2.3 FFRR - RequestPowertrainFreezeFrameData ($02) Service

The current direction within GMPT is to implement Service $02 in the PCM/ECM only. Service $02 should not be supported by a TCM.

The values supported for the Freeze Frame Parameter Identifiers (PIDs) reported by this legislated diagnostic service are specified in the Implementation Specification for SAE J1979 Test Modes [11]. The individual PID specifications, scaling information, and Generic Scan Tool display formats are found in [3]. Not all PIDs are applicable or supported by all systems.

Frame $00 is the only defined Legislative Freeze Frame that must be supported by all ECUs that respond to a Service $02 request as defined in [3].

Frame $00 / PID $00 is a bit-encoded PID that indicates, for each ECU, which Frame $00 PIDs are supported. Frame $00 / PID $00 must be supported by all ECUs that respond to a Service $02 request as defined in [3]. The maximum time after a valid request for a supported Frame $00 PID is received and a response is transmitted by the ECU is 50 milliseconds.

There are no conditions for a negative response return code for this legislated diagnostic service. No response should be generated when a request is made for a Freeze Frame or PID that is not supported by the module.

Additional details can be found in the Implementation Specification for SAE J1979 Test Modes [11].

2.4 DFIR - RequestEmission-RelatedDiagnosticTroubleCodes ($03) Service

The purpose of this mode is to obtain the confirmed emission related trouble codes (DTCs) from all emission related controllers.

There are no conditions for a negative response return code for this legislated diagnostic service.

Additional details can be found in the Implementation Specification for SAE J1979 Test Modes [11].

2.5 DRER - ClearDiagnosticInformation ($04) Service

The purpose of this mode is to request to clear or reset all diagnostic information. This request will initialize both emission and non-emission related powertrain diagnostics which must include all engine, transmission, electronic throttle control, et al. diagnostics. The list of variables and corresponding initialization values is extensive. Along with clearing DTCs, freeze frame and failure records Fault ID's must be cleared, diagnostic timers and counters initialized, Mode 06 test results reset, etc. Some variable initialization is conditional on whether a DTC is stored. A typical application will likely be required to initialize hundreds of variables. "Initialization" in general does not mean “clear”.

Other manufacturer specific “clearing/resetting” actions may also occur in response to this request message. All ECUs shall respond to this request message with ignition ON regardless of whether or not the engine is running. If the ignition key is not in the RUN position (as determined by each ECU), the ECU shall send a negative response message with response code $22 - conditionsNotCorrect.

The maximum time after a positive response to complete clearing of all DTC information is 2.0 seconds.

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 4: Negative Response Return Codes for ClearDiagnosticInformation ($04) Service
.

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect. • Request message sent with the physically addressed CAN identifier.
$22	ConditionsNotCorrectOrRequestSequenceError • Ignition key not in RUN position.

Table 4: Negative Response Return Codes for ClearDiagnosticInformation ($04) Service

Additional details can be found in the Implementation Specification for SAE J1979 Test Modes [11].

2.6 ODAR - RequestOnBoardMonitoringTestResults ($06) Service

The current direction within GMPT is to implement Service $06 in the PCM/ECM only. Service $06 should not be supported by a TCM. The purpose of this service is to request the on-board diagnostic tests supported by the ECU by On-Board Diagnostic Monitor ID (OBDMID).

Positive responses are required for all ranges up to the largest supported OBDMID corresponding to an actual test.

Responses to a Service $06 will include all supported on-board diagnostic Test Identifiers (TIDs) along with their associated Unit and Scaling ID (UASID)

The individual OBDMID, TID, and UASID specifications are found in [10]. Not all OBDMIDs and TIDs are applicable or supported by all systems.

There are no conditions for a negative response return code for this legislated diagnostic service.

Additional details can be found in the Implementation Specification for SAE J1979 Test Modes [11].

2.7 DFIR - RequestPendingEmission-RelatedDiagnosticTroubleCodes ($07) Service

The purpose of this mode is to obtain the emission related DTCs that have failed either this key-on or during the previous trip. This provides an indication of the DTCs that have failed this ignition cycle or on the prior trip in which they had run.

There are no conditions for a negative response return code for this legislated diagnostic service.

Additional details can be found in the Implementation Specification for SAE J1979 Test Modes [11].

2.8 ODAR - RequestVehicleInformation ($09) Service

The purpose of this mode is to enable the off-board test device to request vehicle specific information such as Vehicle Identification Number (VIN), Calibration IDs, Calibration Verification Numbers, and Rate Based OBD Monitoring Numbers.

The individual InfoType specifications and Generic Scan Tool display formats are found in [3]. Not all InfoTypes are applicable or supported by all systems. The InfoTypes reported by this service are specified in the Implementation Specification for SAE J1979 Test Modes [11].

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 5.

Return Code	Conditions
$78	RequestCorrectlyReceived-ResponsePending • Waiting to calculate Calibration Verification Numbers.

Table 5: Negative Response Return Codes for RequestVehicleInformation ($09) Service

Additional details can be found in the Implementation Specification for SAE J1979 Test Modes [11].

2.9 DFIR – Request Emission-Related Diagnostic Trouble Codes With Permanent Status ($0A) Service

The purpose of this service is to enable the external test equipment to obtain all DTCs with "permanentDTC" status. The confirmed and/or pending DTCs have been previously cleared by a successful execution of a $04 Clear/reset emission-related diagnostic information service but permanent DTCs remain in the non-volatile memory of the server until the appropriate monitors for each DTC have successfully passed.

Service $0A is required for all emissions-related DTCs. The intended use of this data is to prevent vehicles from passing an in-use inspection simply by disconnecting the battery or clearing DTCs with a scan tool prior to the inspection. The presence of permanent DTCs at an inspection without the MIL illuminated is an indication that a proper repair was not verified by the on-board monitoring system.

Permanent DTCs shall be stored in non-volatile memory. These DTCs cannot be cleared by any test equipment (e.g. on-board tester, off-board tester). The OBD system shall clear these DTCs itself by completing and passing the on-board monitor and also may require some CARB-defined driving cycle. This prevents clearing DTCs simply by disconnecting the battery.

A confirmed DTC shall be stored as a permanent DTC no later than the end of the ignition cycle and subsequently at all times that the confirmed DTC is commanding the Malfunction Indicator on (e.g., for currently failing systems but not during the 40 warm-up cycle self-healing process).

There are no conditions for a negative response return code for this legislated diagnostic service.

Additional details can be found in the Implementation Specification for SAE J1979 Test Modes [11].

2.10 DRER - InitiateDiagnosticOperation ($10) Service

The sub-functions supported for this diagnostic service are specified in Table 6.

Level	Sub-Function
$02	disableAllDTCs
$03	enableDTCsDuringDeviceCntrl

Table 6: Sub-Functions for InitiateDiagnosticOperation ($10) Service

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 7: Negative Response Return Codes for InitiateDiagnosticOperation ($10) Service
.

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect. • Sub-function not supported.
$22	ConditionsNotCorrectOrRequestSequenceError • A request message is received with sub-function enableDTCsDuringDeviceCntrl ($03) while the DeviceControl ($AE) diagnostic service is active. • A request message is received with sub-function enableDTCsDuringDeviceCntrl ($03) and DTC's have been disabled by another diagnostic service.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 7: Negative Response Return Codes for InitiateDiagnosticOperation ($10) Service

2.11 FFRR - ReadFailureRecordData ($12) Service

The sub-functions supported for this diagnostic service are specified in Table 8: Sub-Functions for ReadFailureRecordData ($12) Service
.

Level	Sub-Function
$01	readFailureRecordIdentifiers
$02	readFailureRecordParameters

Table 8: Sub-Functions for ReadFailureRecordData ($12) Service

A positive response to sub-function readFailureRecordIdentifiers ($01) of this diagnostic service shall always have a value of PID ($00) for the failureRecordDataStructureIdentifier parameter.

The content of the failure records supported for this diagnostic service are specified in Section 7.1 Corporate Standard Freeze Frame
The conditions for each of the negative response return codes for this diagnostic service are documented in Table 9: Negative Response Return Codes for ReadFailureRecordData ($12) Service
.

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect. • Sub-function not supported.
$31	RequestOutOfRange • Failure record identifier requested is not valid (i.e., does not match a stored identifier in the failure records).
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 9: Negative Response Return Codes for ReadFailureRecordData ($12) Service

2.12 ODAR / BFPR - ReadDataByIdentifier ($1A) Service

The values supported for the dataIdentifier parameter of this diagnostic service are specified in Section 4, Data Identifiers (DIDs) (only DIDs $B0, $C1, and $CC are supported in boot).

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 10: Negative Response Return Codes for ReadDataByIdentifier ($1A) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.
$22	ConditionsNotCorrectOrRequestSequenceError • Not supported.
$31	RequestOutOfRange • Requested DID not supported.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 10: Negative Response Return Codes for ReadDataByIdentifier ($1A) Service

2.13 OVIR / BFPR - ReturnToNormalMode ($20) Service

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 11: Negative Response Return Codes for ReturnToNormalMode ($20) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.

Table 11: Negative Response Return Codes for ReturnToNormalMode ($20) Service

2.14 ODAR - ReadDataByParameterIdentifier ($22) Service

The values supported for the parameterIdentifier parameter of this diagnostic service are specified in Section 3, Parameter Identifiers (PIDs).

The Powertrain implementation of this service provides a USDT positive response to a request for one PID, provided that the request message is properly formatted and the requested PID is supported in the ECU. The length of PID response data is limited to 255 bytes.

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 12: Negative Response Return Codes for ReadDataByParameterIdentifier ($22) Service
.

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.
$22	ConditionsNotCorrectOrRequestSequenceError • Not supported.
$31	RequestOutOfRange • Requested PID is not supported (and request message was physically addressed).
$78	RequestCorrectlyReceived-ResponsePending • Sent when positive response may take longer than 100 ms to generate (request for a secondary processor PID).

Table 12: Negative Response Return Codes for ReadDataByParameterIdentifier ($22) Service

2.15 ODAR - ReadMemoryByAddress ($23) Service

The size of the memoryAddress parameter of this diagnostic service shall be 4 bytes in length. The maximum value of the memorySize parameter of this diagnostic service shall be 16 bytes.

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 13: Negative Response Return Codes for ReadMemoryByAddress ($23) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.
$31	RequestOutOfRange • Memory address requested is invalid, restricted, or secure while the ECU is locked. • More than 16 bytes requested.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 13: Negative Response Return Codes for ReadMemoryByAddress ($23) Service

2.16 OSAR / BFPR - SecurityAccess ($27) Service

The sub-functions supported for this diagnostic service are specified in Table 14: Sub-Functions for SecurityAccess ($27) Service
.

Level	Sub-Function
$01	SPSrequestSeed
$02	SPSsendKey
$03	DevCtrlrequestSeed
$04	DevCtrlsendKey

Table 14: Sub-Functions for SecurityAccess ($27) Service

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 15: Negative Response Return Codes for SecurityAccess ($27) Service
.

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect. • Sub-function not supported.
$22	ConditionsNotCorrectOrRequestSequenceError • A request message is received with sub-function sendKey ($02 or $04) prior to receiving a request message with sub-function requestSeed ($01 or $03, respectively). • A request message is received with sub-function DevCtrlrequestSeed ($03) while the DeviceControl ($AE) diagnostic service is active.
$35	InvalidKey • Value of the key is not valid.
$36	ExceededNumberOfAttempts • Too many attempts with invalid key values.
$37	RequiredTimeDelayNotExpired • Delay timer is active when a request is received.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 15: Negative Response Return Codes for SecurityAccess ($27) Service

2.17 OVIR / BFPR - DisableNormalComm ($28) Service

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 16: Negative Response Return Codes for DisableNormalComm ($28) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.
$22	ConditionsNotCorrectOrRequestSequenceError • Not supported.

Table 16: Negative Response Return Codes for DisableNormalComm ($28) Service

2.18 ODAR - DynamicallyDefineMessage ($2C) Service

The values supported for the DPIDIdentifier parameter of this diagnostic service are $EF to $FE, for a total of 16 dynamic DPIDs.

The PID numbers supported for the PIDData parameter of this diagnostic service are specified in Section 3, Parameter Identifiers (PIDs).

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 17: Negative Response Return Codes for DynamicallyDefineMessage ($2C) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length too short (must be at least 4 bytes). • Request message data length not an even number (each PID is a 2 byte value). • The total number of data bytes requested to be reported in a UUDT message exceeds 7 data bytes.
$31	RequestOutOfRange • Requested PID is not supported. • The dynamic DPID number is not supported.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 17: Negative Response Return Codes for DynamicallyDefineMessage ($2C) Service

2.19 ODAR - DefinePIDByAddress ($2D) Service

The values supported for the parameterIdentifier parameter of this diagnostic service are $FE00 to $FE3F, for a total of 64 main processor dynamic PIDs, and $FF00 to $FF3F, for a total of 64 secondary processor dynamic PIDs (if present).

The size of the memoryAddress parameter of this diagnostic service shall be 4 bytes in length.

The value of the memorySize parameter of this diagnostic service shall be 1, 2, or 4 bytes for main processor dynamic PIDs and shall be 2 bytes for secondary processor dynamic PIDs (if present).

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 18: Negative Response Return Codes for DefinePIDByAddress ($2D) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.
$31	RequestOutOfRange • Memory address requested is invalid, restricted, or secure while the ECU is locked. • The size of a main processor dynamic PID is not 1, 2, or 4 bytes or the size of a secondary processor dynamic PID is not 2 bytes. • The dynamic PID number is not supported.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 18: Negative Response Return Codes for DefinePIDByAddress ($2D) Service

2.20 OFPR / BFPR - RequestDownload ($34) Service

Since encryption and compression techniques are not supported with the TransferData ($36) diagnostic service, the values of the dataFormatIdentifier and unCompressedMemorySize parameters of this diagnostic service shall be ignored.

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 19: Negative Response Return Codes for RequestDownload ($34) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.
$22	ConditionsNotCorrectOrRequestSequenceError • The DisableNormalComm ($28) diagnostic service is not active when a request message is received for this diagnostic service. • The ProgrammingMode ($A5) diagnostic service is not active when a request message is received for this diagnostic service. • The ECU is secure (i.e., has not been unlocked via the SecurityAccess ($27) diagnostic service) when a request message is received for this diagnostic service. • The ECU is in the process of receiving a download, via the TransferData ($36) diagnostic service, when a request message is received for this diagnostic service.
$78	RequestCorrectlyReceived-ResponsePending • Transmitted prior to transitioning to boot (positive response is transmitted from boot).
$99	ReadyForDownload-DTCStored • Not supported.

Table 19: Negative Response Return Codes for RequestDownload ($34) Service

2.21 BFPR / OFPR - TransferData ($36) Service

The sub-functions supported for this diagnostic service are specified in Table 20: Sub-Functions TransferData ($36) Service

Level	Sub-Function
$00	download
$80	downloadAndExecuteOrExecute

Table 20: Sub-Functions TransferData ($36) Service

The size of the startingAddress parameter of this diagnostic service shall be 4 bytes in length.

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 21: Negative Response Return Codes for TransferData ($36) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Not supported.
$22	ConditionsNotCorrectOrRequestSequenceError • RequestDownload ($34) service has not been received prior to the TransferData ($36) service. • The order of software and/or calibration parts downloaded is invalid
$31	RequestOutOfRange • Not supported.
$78	RequestCorrectlyReceived-ResponsePending • Transmitted after each TransferData ($36) message (positive response is transmitted after the data is processed).
$83	VoltageOutOfRangeFault (High / Low) • Not supported.
$85	GeneralProgrammingFailure • Write or erase error during programming.
$89	DeviceTypeError • Not supported.

Table 21: Negative Response Return Codes for TransferData ($36) Service

2.22 ODAR - WriteDataByIdentifier ($3B) Service

The values supported for the dataIdentifier parameter of this diagnostic service are specified in Section 4, Data Identifiers (DIDs).

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 22: Negative Response Return Codes for WriteDataByIdentifier ($3B) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect (correct number of data bytes not provided with the requested DID).
$22	ConditionsNotCorrectOrRequestSequenceError • Not supported.
$31	RequestOutOfRange • Requested DID not supported or read-only.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 22: Negative Response Return Codes for WriteDataByIdentifier ($3B) Service

2.23 OVIR / BFPR - TesterPresent ($3E) Service

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 23: Negative Response Return Codes for TesterPresent ($3E) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.

Table 23: Negative Response Return Codes for TesterPresent ($3E) Service

2.24 BFPR / OFPR - ReportProgrammedState ($A2) Service

The values supported for the programmedState parameter of this diagnostic service are fully programmed ($00), op s/w present, cal data missing ($02), and op s/w present, default or no start cal present ($03).

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 24: Negative Response Return Codes for ReportProgrammedState ($A2) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 24: Negative Response Return Codes for ReportProgrammedState ($A2) Service

2.25 OFPR / BFPR - ProgrammingMode ($A5) Service

The sub-functions supported for this diagnostic service are specified in Table 25: Sub-Functions for ProgrammingMode ($A5) Service

Level	Sub-Function
$01	requestProgrammingMode
$03	enableProgrammingMode

Table 25: Sub-Functions for ProgrammingMode ($A5) Service

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 26: Negative Response Return Codes for ProgrammingMode ($A5) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect. • Sub-function not supported.
$22	ConditionsNotCorrectOrRequestSequenceError • A request message is received with sub-function enableProgrammingMode ($03) prior to receiving a request message with sub-function requestProgrammingMode ($01). • DisableNormalComm ($28) service has not been received prior to the ProgrammingMode ($A5) service. • Programming mode is already active.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 26: Negative Response Return Codes for ProgrammingMode ($A5) Service

2.26 DFIR - ReadDiagnosticInformation ($A9) Service

The sub-functions supported for this diagnostic service are specified in Table 27: Sub-Functions for ReadDiagnosticInformation ($A9) Service

Level	Sub-Function
$80	readStatusOfDTCByNumber
$81	readStatusOfDTCByStatusMask
$82	sendOnChangeDTCCount

Table 27: Sub-Functions for ReadDiagnosticInformation ($A9) Service

The values supported for the DTCMaskRecord parameter of sub-function readStatusOfDTCByNumber ($80) of this diagnostic service are specified in Section 6, Diagnostic Trouble Codes (DTCs).

The DTC count for sub-function sendOnChangeDTCCount ($82) of this diagnostic service is updated at an interval of 1.0 second.

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 28: Negative Response Return Codes for ReadDiagnosticInformation ($A9) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect. • Sub-function not supported.
$31	RequestOutOfRange • DTC and fault type requested with sub-function readStatusOfDTCByNumber ($80) is not valid.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.

Table 28: Negative Response Return Codes for ReadDiagnosticInformation ($A9) Service

2.27 ODAR - ReadDataByPacketIdentifier ($AA) Service

The sub-functions supported for this diagnostic service are specified in Table 29: Sub-Functions for ReadDataByPacketIdentifier ($AA) Service

Level	Sub-Function
$00	stopSending
$01	sendOneResponse
$02	scheduleAtSlowRate
$03	scheduleAtMediumRate
$04	scheduleAtFastRate

Table 29: Sub-Functions for ReadDataByPacketIdentifier ($AA) Service

The values supported for each element of the request_DPIDs parameter of this diagnostic service are $EF to $FE, for a total of 16 dynamic DPIDs. Static DPIDs are not supported.

The ECU shall support up to 5 DPID numbers in a request for sub-function sendOneResponse ($01) of this diagnostic service.

The ECU shall have a Periodic DPID Scheduler (PDS) with up to 16 DPIDs. The ECU's actual scheduling rates for the PDS are specified in Table 30: Actual Scheduling Periods for ReadDataByPacketIdentifier ($AA) Service

Sub-Function	Actual Scheduling Period
scheduleAtSlowRate ($02)	1000.0 ms
scheduleAtMediumRate ($03)	200.0 ms
scheduleAtFastRate ($04)	25.0 ms

Table 30: Actual Scheduling Periods for ReadDataByPacketIdentifier ($AA) Service

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 31: Negative Response Return Codes for ReadDataByPacketIdentifier ($AA) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect. • Sub-function not supported. • Number of DPIDs requested in a one shot is greater than 5. • Request contains more than 16 DPIDs to be scheduled.
$31	RequestOutOfRange • DPID requested is not supported or has not yet been defined.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.
$81	SchedulerFull • Request would result in more than 16 DPIDs being scheduled.

Table 31: Negative Response Return Codes for ReadDataByPacketIdentifier ($AA) Service

2.28 ODCR - DeviceControl ($AE) Service

The values supported for the CPIDNumber parameter of this diagnostic service are specified in Section 5, Control Packet Identifiers (CPIDs).

The conditions for each of the negative response return codes for this diagnostic service are documented in Table 32: Negative Response Return Codes for DeviceControl ($AE) Service

Return Code	Conditions
$12	SubFunctionNotSupported-InvalidFormat • Request message data length incorrect.
$31	RequestOutOfRange • CPID requested is not supported.
$78	RequestCorrectlyReceived-ResponsePending • Not supported.
$E3 $xx $yy	DeviceControlLimitsExceeded • Device control exited due to internal protection algorithms. A device control limits exceeded return code ($xx $yy) is appended to the negative response to indicate which limit was exceeded. The device control limits exceeded return codes supported for each device control are specified in Section 5, Control Packet Identifiers (CPIDs).

Table 32: Negative Response Return Codes for DeviceControl ($AE) Service

3. Parameter Identifiers (PIDs)

PIDs are divided into several ranges that determine their usage and the ownership of their requirements, as shown in Table 33: PID Ranges

Definitions for legislated PIDs, corporate PIDs, and development PIDs are provided in the following sections. Definitions for controller specific PIDs must be provided in the associated controller specific diagnostic CTS.

Range	Usage
$0000 to $3FFF	Powertrain PIDs
$0000 to $00FF	Legislated PIDs (see Section 3.1)
$0100 to $0FFF	Reserved
$1000 to $2FFF	Powertrain Corporate Standard PIDs (see Section 3.2)
$3000 to $3FFF	Powertrain Controller Specific PIDs
$4000 to $7FFF	Chassis PIDs (Reserved)
$8000 to $BFFF	Body PIDs (Reserved)
$C000 to $FFFF	Development PIDs
$C000 to $FBFF	Reserved
$FC00 to $FCFF	Main Processor Development PIDs (see Section 3.3)
$FD00 to $FDFF	Secondary Processor Development PIDs (see Section 3.3)
$FE00 to $FEFF	Secondary Processor Dynamic PIDs
$FF00 to $FFFF	Main Processor Dynamic PIDs

Table 33: PID Ranges

3.1 Legislated PIDs

The legislated PIDs below are also specified within [3]. In the event of a conflict between this document and [3], the later shall take precedence. Refer to [11] (Implementation Specification for SAE J1979 Test Modes) for the GM interpretation of the PIDs required to be supported.

3.1.1 ODAR – PID $0000 – Parameter Identifiers Supported - $01 to $20

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0000	Parameter Identifiers Supported - $01 to $20	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PID $01 Supported	0	7	1	BLN	N/A	$1=True; $0=False
PID $02 Supported	0	6	1	BLN	N/A	$1=True; $0=False
PID $03 Supported	0	5	1	BLN	N/A	$1=True; $0=False
PID $04 Supported	0	4	1	BLN	N/A	$1=True; $0=False
PID $05 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PID $06 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PID $07 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PID $08 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PID $09 Supported	1	7	1	BLN	N/A	$1=True; $0=False
PID $0A Supported	1	6	1	BLN	N/A	$1=True; $0=False
PID $0B Supported	1	5	1	BLN	N/A	$1=True; $0=False
PID $0C Supported	1	4	1	BLN	N/A	$1=True; $0=False
PID $0D Supported	1	3	1	BLN	N/A	$1=True; $0=False
PID $0E Supported	1	2	1	BLN	N/A	$1=True; $0=False
PID $0F Supported	1	1	1	BLN	N/A	$1=True; $0=False
PID $10 Supported	1	0	1	BLN	N/A	$1=True; $0=False
PID $11 Supported	2	7	1	BLN	N/A	$1=True; $0=False
PID $12 Supported	2	6	1	BLN	N/A	$1=True; $0=False
PID $13 Supported	2	5	1	BLN	N/A	$1=True; $0=False
PID $14 Supported	2	4	1	BLN	N/A	$1=True; $0=False
PID $15 Supported	2	3	1	BLN	N/A	$1=True; $0=False
PID $16 Supported	2	2	1	BLN	N/A	$1=True; $0=False
PID $17 Supported	2	1	1	BLN	N/A	$1=True; $0=False
PID $18 Supported	2	0	1	BLN	N/A	$1=True; $0=False
PID $19 Supported	3	7	1	BLN	N/A	$1=True; $0=False
PID $1A Supported	3	6	1	BLN	N/A	$1=True; $0=False
PID $1B Supported	3	5	1	BLN	N/A	$1=True; $0=False
PID $1C Supported	3	4	1	BLN	N/A	$1=True; $0=False
PID $1D Supported	3	3	1	BLN	N/A	$1=True; $0=False
PID $1E Supported	3	2	1	BLN	N/A	$1=True; $0=False
PID $1F Supported	3	1	1	BLN	N/A	$1=True; $0=False
PID $20 Supported	3	0	1	BLN	N/A	$1=True; $0=False

This PID indicates which legislated PIDs, within the range of $01 to $20, are supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.2 DFIR – PID $0001 – Emission Related Monitor Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0001	Emission Related Monitor Status	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Malfunction Indicator Lamp Status	0	7	1	ENM	N/A	$0=Off; $1=On
Number of Emission Related DTCs	0	6	7	UNM	0 - 127	E = N * 1
Reserved	1	7	1			
Comprehensive Component Monitoring Status	1	6	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Fuel System Monitoring Status	1	5	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Misfire Monitoring Status	1	4	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Compression ignition monitoring supported	1	3	1	ENM	N/A	$0 = Spark ignition monitors supported $1 = Compression ignition monitors supported
Comprehensive Component Monitoring Supported	1	2	1	BLN	N/A	$1=True; $0=False
Fuel System Monitoring Supported	1	1	1	BLN	N/A	$1=True; $0=False
Misfire Monitoring Supported	1	0	1	BLN	N/A	$1=True; $0=False
	The following descriptions for Bytes 2 and 3 are to be used for spark ignition vehicles only.					
Exhaust Gas Recirculation and/or Variable Valve Timing System Monitoring Supported	2	7	1	BLN	N/A	$1=True; $0=False
Oxygen Sensor Heater Monitoring Supported	2	6	1	BLN	N/A	$1=True; $0=False
Oxygen Sensor Monitoring Supported	2	5	1	BLN	N/A	$1=True; $0=False
~~Air Conditioning System Refrigerant Monitoring Supported~~	2	4	1	~~BLN~~	~~N/A~~	~~$1=True; $0=False~~
Secondary Air System Monitoring Supported	2	3	1	BLN	N/A	$1=True; $0=False
Evaporative System Monitoring Supported	2	2	1	BLN	N/A	$1=True; $0=False
Heated Catalyst Monitoring Supported	2	1	1	BLN	N/A	$1=True; $0=False
Catalyst Monitoring Supported	2	0	1	BLN	N/A	$1=True; $0=False
Exhaust Gas Recirculation and/or Variable Valve Timing System Monitoring Status	3	7	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Oxygen Sensor Heater Monitoring Status	3	6	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Oxygen Sensor Monitoring Status	3	5	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
~~Air Conditioning System Refrigerant Monitoring Status~~	3	4	1	~~ENM~~	~~N/A~~	~~$0=Monitor Complete;~~ ~~$1=Monitor Not Complete~~
Secondary Air System Monitoring Status	3	3	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Evaporative System Monitoring Status	3	2	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Heated Catalyst Monitoring Status	3	1	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Catalyst Monitoring Status	3	0	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
	The following descriptions for Bytes 2 and 3 are to be used for compression ignition vehicles only.					

EGR and/or VVT system monitoring supported	2	7	1	BLN	N/A	$1=True; $0=False
PM filter monitoring supported	2	6	1	BLN	N/A	$1=True; $0=False
Exhaust gas sensor monitoring supported	2	5	1	BLN	N/A	$1=True; $0=False
Reserved	2	4	1			
Boost pressure system monitoring supported	2	3	1	BLN	N/A	$1=True; $0=False
Reserved	2	2	1			
NOx/SCR aftertreatment monitoring supported	2	1	1	BLN	N/A	$1=True; $0=False
NMHC catalyst monitoring supported	2	0	1	BLN	N/A	$1=True; $0=False
EGR and/or VVT system monitoring Status	3	7	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
PM filter monitoring status	3	6	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Exhaust gas sensor monitoring status	3	5	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Reserved	3	4	1			
Boost pressure system monitoring status	3	3	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Reserved	3	2	1			
NOx/SCR aftertreatment monitoring status	3	1	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
NMHC catalyst monitoring status	3	0	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete

This PID contains the following information associated with the status of the legislated emission related monitors for spark and compression ignition vehicles.

The status of the malfunction indicator lamp (MIL). Note that the MIL status shall indicate “Off” during the key on, engine off bulb check unless the MIL has also been commanded “On” for a detected malfunction.

The number of emission related DTCs stored in the ECU.

The set of continuous monitors for spark ignition vehicles (comprehensive component, fuel system, and misfire) and non-continuous monitors (exhaust gas recirculation system, oxygen sensor heater, oxygen sensor, air conditioning system refrigerant, secondary air system, evaporative system, heated catalyst, catalyst) supported and the status of each monitor since DTCs were cleared.

For compression ignition vehicles, the set of non-continuous monitors (NMHC catalyst, NOx/SCR aftertreatment, Bosst pressure system, Exhaust gas sensor, Particulate Matter filter, EGR/VVT system) supported and the status of each monitor since DTCs were cleared

Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5. The displayed order of bytes 0-3 between this document and SAE J1979 are in opposite order.

3.1.3 FFRR – PID $0002 – Freeze Frame Diagnostic Trouble Code

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0002	Freeze Frame Diagnostic Trouble Code	2	UNM	0 - 65535	E = N * 1

This PID contains the diagnostic trouble code (DTC) that caused the freeze frame to be stored. The mapping of values to DTCs is specified by [4]. A value of zero indicates that a freeze frame has not been stored. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.4 FULR – PID $0003 – Fuel System Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0003	Fuel System Status	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Bank 1 Closed Loop - Alternate Control	0	4	1	BLN	N/A	$1=True; $0=False
Bank 1 Open Loop - System Fault	0	3	1	BLN	N/A	$1=True; $0=False
Bank 1 Open Loop - Driving Conditions	0	2	1	BLN	N/A	$1=True; $0=False
Bank 1 Closed Loop	0	1	1	BLN	N/A	$1=True; $0=False
Bank 1 Open Loop - Minimum Criteria for Closed Loop Not Met	0	0	1	BLN	N/A	$1=True; $0=False
Reserved	1	7	1			
Reserved	1	6	1			
Reserved	1	5	1			
Bank 2 Closed Loop - Alternate Control	1	4	1	BLN	N/A	$1=True; $0=False
Bank 2 Open Loop - System Fault	1	3	1	BLN	N/A	$1=True; $0=False
Bank 2 Open Loop - Driving Conditions	1	2	1	BLN	N/A	$1=True; $0=False
Bank 2 Closed Loop	1	1	1	BLN	N/A	$1=True; $0=False
Bank 2 Open Loop - Minimum Criteria for Closed Loop Not Met	1	0	1	BLN	N/A	$1=True; $0=False

This PID contains the status of each bank of the closed loop fuel control system. For each fuel control bank, exactly one signal within this PID shall have a value of True. The value of the PID for each fuel control bank should equal 0 when the engine is not running. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.5 ETQR/MAFR – PID $0004 – Calculated Engine Load

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0004	Calculated Engine Load	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the calculated engine load, mass airflow as a percentage of the maximum mass airflow, where the maximum mass airflow is determined at wide open throttle (WOT) and the current operating conditions (ambient temperature, ambient pressure, and engine speed). In a Diesel engine, the engine load is calculated using the current engine torqueas a percentage of the maximum available engine torque calculated at standard ambient pressure and temperature function of engine speed. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.6 ECTR – PID $0005 – Engine Coolant Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0005	Engine Coolant Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the undefaulted engine coolant temperature measured by a sensor. If the data used to compute engine coolant temperature is obtained from a serial data signal, then, if the signal is not received, the value of PID $0005 shall be set to $00 = -40 °C. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.7 FADR – PID $0006 – Short Term Fuel Trim Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0006	Short Term Fuel Trim Bank 1	1	UNM	-100 to +99.21875 %	E = N * 100 / 128 - 100

This PID contains the short term closed loop fuel correction for bank 1. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.8 FADR – PID $0007 – Long Term Fuel Trim Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0007	Long Term Fuel Trim Bank 1	1	UNM	-100 to +99.21875 %	E = N * 100 / 128 - 100

This PID contains the long term fuel correction for bank 1. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.9 FADR – PID $0008 – Short Term Fuel Trim Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0008	Short Term Fuel Trim Bank 2	1	UNM	-100 to +99.21875 %	E = N * 100 / 128 - 100

This PID contains the short term closed loop fuel correction for bank 2. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.10 FADR – PID $0009 – Long Term Fuel Trim Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0009	Long Term Fuel Trim Bank 2	1	UNM	-100 to +99.21875 %	E = N * 100 / 128 - 100

This PID contains the long term fuel correction for bank 2. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.11 FRPR/FCBR – PID $000A – Fuel Pressure Relative to Atmosphere

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$000A	Fuel Pressure Relative to Atmosphere	1	UNM	0 to 765 kPa	E = N * 3

For Port Fuel Injected (PFI) systems, this PID displays undefaulted fuel rail pressure at the engine. For high pressure fuel systems, this PID indicates the undefaulted low side fuel pressure before the high pressure pump. In all cases, this pressure reading is referenced to atmosphere (gauge pressure). If the data used to compute fuel pressure is obtained from a serial data signal, then, if the signal is not received, the value of PID $000A shall be set to $00 = 0 kPaG. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.12 MAPR – PID $000B – Intake Manifold Absolute Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$000B	Intake Manifold Absolute Pressure	1	UNM	0 - 255 kPa	E = N * 1

This PID contains the unfiltered engine intake manifold absolute pressure (MAP). Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.13 EPSR – PID $000C – Engine Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$000C	Engine Speed	2	UNM	0 - 16383.75 rpm	E = N * 0.25

This PID contains the undefaulted engine speed determined from the low resolution (LORES) period of the crank position sensor. If the data used to compute engine speed is obtained from a serial data signal, then, if the signal is not received, the value of PID $000C shall be set to $FF = 16383.75 rpm. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.14 VSPR – PID $000D – Vehicle Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$000D	Vehicle Speed	1	UNM	0 - 255 kph	E = N * 1

This PID contains the undefaulted vehicle speed calculated from the transmission output speed. If the data used to compute vehicle speed is obtained from a serial data signal, then, if the signal is not received, the value of PID $000D shall be set to $FF = 255. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.15 SPKR – PID $000E – Spark Advance

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$000E	Spark Advance	1	UNM	-64 - 63.5 °	E = N * 0.5 - 64

This PID contains the final spark advance in terms of a crankshaft angle. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.16 EITR – PID $000F – Intake Air Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$000F	Intake Air Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the undefaulted intake air temperature measured by a sensor that is typically integrated with the MAF sensor and located near the throttle body. For multiple IAT systems, PID $000F shall represent IAT 1. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.17 MAFR – PID $0010 – MAF Sensor Mass Airflow

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0010	MAF Sensor Mass Airflow	2	UNM	0 to 655.35 g/s	E = N * 0.01

This PID contains the mass airflow measured by a MAF sensor. On two-sensor applications, PID $0010 will represent MAF sensor A. MAF sensor A is associated with engine bank 1. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.18 TPSR – PID $0011 – Absolute Throttle Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0011	Absolute Throttle Position	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the absolute throttle position (no correction based upon learned minimum position) from throttle position sensor 1. If the data used to compute absolute throttle position is obtained from a serial data signal, then, if the signal is not received, the value of PID $0011 shall be set to $00 = 0 %. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.19 AIRR – PID $0012 – Commanded Secondary Air Injection Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0012	Commanded Secondary Air Injection Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
AIR Pump Commanded on for Diagnostics	0	3	1	BLN	N/A	$1=True; $0=False
Air Diverted to Atmosphere / Pump Off	0	2	1	BLN	N/A	$1=True; $0=False
Air Diverted Downstream of First Catalyst	0	1	1	BLN	N/A	$1=True; $0=False
Air Diverted Upstream of First Catalyst	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains the commanded state of the secondary air injection (SAI) system. When the SAI system is active, the signal Air Diverted Upstream of First Catalyst shall be set to True, and the other signals within this PID shall be set to False. When the SAI system is not active, the signal Air Diverted to Atmosphere / Pump Off shall be set to True, and the other signals within this PID shall be set to False. Note that the signal Air Diverted Downstream of First Catalyst is always set to False. The signal AIR Pump Commanded on for Diagnostics is not used by GMPT AIR system monitoring at this time because the AIR monitor does not include an intrusive part to the test. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.20 OXYR – PID $0013 – Location of Oxygen Sensors

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0013	Location of Oxygen Sensors	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Bank 2 Sensor 4 Present	0	7	1	BLN	N/A	$1=True; $0=False
Bank 2 Sensor 3 Present	0	6	1	BLN	N/A	$1=True; $0=False
Bank 2 Sensor 2 Present	0	5	1	BLN	N/A	$1=True; $0=False
Bank 2 Sensor 1 Present	0	4	1	BLN	N/A	$1=True; $0=False
Bank 1 Sensor 4 Present	0	3	1	BLN	N/A	$1=True; $0=False
Bank 1 Sensor 3 Present	0	2	1	BLN	N/A	$1=True; $0=False
Bank 1 Sensor 2 Present	0	1	1	BLN	N/A	$1=True; $0=False
Bank 1 Sensor 1 Present	0	0	1	BLN	N/A	$1=True; $0=False

This PID indicates the locations of the set of oxygen sensors that are present on the vehicle. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.21 OXYR – PID $0014 – Oxygen Sensor Voltage Bank 1 Sensor 1 / Short Term Fuel Trim Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0014	Oxygen Sensor Voltage Bank 1 Sensor 1 / Short Term Fuel Trim Bank 1	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Oxygen Sensor Voltage	0	7	8	UNM	0 - 1.275 V	E = N * 0.005
Short Term Fuel Trim	1	7	8	UNM	-100 to +99.21875 %	E = N * 100 / 128 - 100

This PID contains the oxygen sensor voltage for bank 1 sensor 1 and the short term fuel correction based upon feedback from this sensor. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.22 OXYR – PID $0015 – Oxygen Sensor Voltage Bank 1 Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0015	Oxygen Sensor Voltage Bank 1 Sensor 2	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Oxygen Sensor Voltage	0	7	8	UNM	0 - 1.275 V	E = N * 0.005
Short Term Fuel Trim	1	7	8	UNM	-100 to +99.21875 %	E = N * 100 / 128 - 100

This PID contains the oxygen sensor voltage for bank 1 sensor 2 (the short term fuel trim signal shall always contain its maximum representable value since short term fuel correction is not based upon feedback from this sensor). Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.23 OXYR – PID $0018 – Oxygen Sensor Voltage Bank 2 Sensor 1 / Short Term Fuel Trim Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0018	Oxygen Sensor Voltage Bank 2 Sensor 1 / Short Term Fuel Trim Bank 2	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Oxygen Sensor Voltage	0	7	8	UNM	0 - 1.275 V	E = N * 0.005
Short Term Fuel Trim	1	7	8	UNM	-100 to +99.21875 %	E = N * 100 / 128 - 100

This PID contains the oxygen sensor voltage for bank 2 sensor 1 and the short term fuel correction based upon feedback from this sensor. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.24 OXYR – PID $0019 – Oxygen Sensor Voltage Bank 2 Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0019	Oxygen Sensor Voltage Bank 2 Sensor 2	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Oxygen Sensor Voltage	0	7	8	UNM	0 - 1.275 V	E = N * 0.005
Short Term Fuel Trim	1	7	8	UNM	-100 to +99.21875 %	E = N * 100 / 128 - 100

This PID contains the oxygen sensor voltage for bank 2 sensor 2 (the short term fuel trim signal shall always contain its maximum representable value since short term fuel correction is not based upon feedback from this sensor). Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.25 DFIR – PID $001C – On Board Diagnostic System Type

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$001C	On Board Diagnostic System Type	1	ENM	N/A	$01=OBD II (CARB); $02=OBD (Federal EPA); $03=OBD and OBD II; $04=OBD I; $05=Not OBD Compliant; $06=EOBD; $07=EOBD and OBD II; $08=EOBD and OBD; $09=EOBD, OBD, and OBD II; $0A=JOBD; $0B=JOBD and OBD II; $0C=JOBD and EOBD; $0D=JOBD, EOBD, and OBD II; $0E= OBD, EOBD and KOBD; $0F= OBD, OBD II, EOBD and KOBD; $10=ISO/SAE reserved;Heavy Duty Vehicles (EURO EEC) C (gas engines) $11=Engine Manufacturer Diagnostics (EMD); $12=Engine Manufacturer Diagnostics Enhanced (EMD+); $13=Heavy Duty On-Board Diagnostics (Child/Partial); $14=Heavy Duty On-Board Diagnostics; $15=World Wide Harmonized OBD; $16=ISO/SAE reserved; $17=Heavy Duty Euro OBD Stage I without NOx control; $18=Heavy Duty Euro OBD Stage I with NOx control; $19=Heavy Duty Euro OBD Stage II without NOx control; $1A=Heavy Duty Euro OBD Stage II with NOx control; $1B=ISO/SAE reserved; $1C=Brazil OBD Phase 1; $1D=Brazil OBD Phase 2; $1E=Korean OBD; $1F=India OBD I; $20=India OBD II; $21=Heavy Duty Euro OBD Stage VI; $22=OBD, OBD II and HD OBD; $23 - $FA =ISO/SAE reserved

This PID contains the on board diagnostic (OBD) system type required by worldwide OBD regulations. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.26 PTOR – PID $001E – Power Take Off Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$001E	Power Take Off Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Power Take Off Active	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains an indication that power take off (PTO) is active. If the data used to compute the power take off status is obtained from a serial data signal, then, if the signal is not received, the value of PID $001E shall be set to 0000 0000 (binary) = False for bit 0. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.27 PSAR – PID $001F – Engine Run Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$001F	Engine Run Time	2	UNM	0 - 65535 s	E = N * 1

For non-hybrid vehicles, this PID contains the time that the engine has been running. The value of this PID shall freeze if an engine stall occurs and shall be reset to zero at power-up (Run/Crank transition from low to high).

For hybrid vehicles, the value of this PID shall increment after the ignition switch is turned to the on position and the engine is running, or, if the vehicle can be started in electric-only mode, Engine Run Time shall increment after the ignition switch is turned to the on position and the vehicle starts to move. It shall continue to increment even if the engine is turned off by the vehicle control system. Engine Run Time shall be reset to zero during every control module power-up and when entering the key-on, engine off position.

Engine Run Time is limited to 65,535 seconds and shall not wrap around to zero. If the data used to compute engine run time is obtained from a serial data signal, then, if the signal is not received, the value of PID $001F shall be set to $FF = 65535 s. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.28 ODAR – PID $0020 – Parameter Identifiers Supported - $21 to $40

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0020	Parameter Identifiers Supported - $21 to $40	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PID $21 Supported	0	7	1	BLN	N/A	$1=True; $0=False
PID $22 Supported	0	6	1	BLN	N/A	$1=True; $0=False
PID $23 Supported	0	5	1	BLN	N/A	$1=True; $0=False
PID $24 Supported	0	4	1	BLN	N/A	$1=True; $0=False
PID $25 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PID $26 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PID $27 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PID $28 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PID $29 Supported	1	7	1	BLN	N/A	$1=True; $0=False
PID $2A Supported	1	6	1	BLN	N/A	$1=True; $0=False
PID $2B Supported	1	5	1	BLN	N/A	$1=True; $0=False
PID $2C Supported	1	4	1	BLN	N/A	$1=True; $0=False
PID $2D Supported	1	3	1	BLN	N/A	$1=True; $0=False
PID $2E Supported	1	2	1	BLN	N/A	$1=True; $0=False
PID $2F Supported	1	1	1	BLN	N/A	$1=True; $0=False
PID $30 Supported	1	0	1	BLN	N/A	$1=True; $0=False
PID $31 Supported	2	7	1	BLN	N/A	$1=True; $0=False
PID $32 Supported	2	6	1	BLN	N/A	$1=True; $0=False
PID $33 Supported	2	5	1	BLN	N/A	$1=True; $0=False
PID $34 Supported	2	4	1	BLN	N/A	$1=True; $0=False
PID $35 Supported	2	3	1	BLN	N/A	$1=True; $0=False
PID $36 Supported	2	2	1	BLN	N/A	$1=True; $0=False
PID $37 Supported	2	1	1	BLN	N/A	$1=True; $0=False
PID $38 Supported	2	0	1	BLN	N/A	$1=True; $0=False
PID $39 Supported	3	7	1	BLN	N/A	$1=True; $0=False
PID $3A Supported	3	6	1	BLN	N/A	$1=True; $0=False
PID $3B Supported	3	5	1	BLN	N/A	$1=True; $0=False
PID $3C Supported	3	4	1	BLN	N/A	$1=True; $0=False
PID $3D Supported	3	3	1	BLN	N/A	$1=True; $0=False
PID $3E Supported	3	2	1	BLN	N/A	$1=True; $0=False
PID $3F Supported	3	1	1	BLN	N/A	$1=True; $0=False
PID $40 Supported	3	0	1	BLN	N/A	$1=True; $0=False

This PID indicates which legislated PIDs, within the range of $21 to $40, are supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.29 VSPR – PID $0021 – Distance With Malfunction Indicator Lamp On

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0021	Distance With Malfunction Indicator Lamp On	2	UNM	0 - 65535 km	E = N * 1

This PID contains the accumulated distance driven with the malfunction indicator lamp (MIL) on. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.30 FULR – PID $0022 – Fuel Rail Pressure Relative to Manifold Vacuum

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0022	Fuel Rail Pressure Relative to Manifold Vacuum	2	UNM	0 to 5178 kPa	E = N * 5178 / 65535

This PID displays an 'undefaulted' fuel rail pressure relative to manifold vacuum. This PID is intended for gaseous fuel injection alternative fuel applications (CNG/LPG Mono-Fuel and Bi-Fuel) where it is common for the fuel pressure regulator to be 'biased' to manifold vacuum. In all cases, the pressure reading provided by the PID is to represent fuel rail pressure relative to manifold vacuum. In cases where the physical pressure sensor measurement is 'gauge' or 'absolute' in nature, the undefaulted sensor output will be 'adjusted' to an equivalent 'relative to manifold vacuum' value. For Bi-Fuel applications, this PID should display data for the alternative fuel source, regardless of the fuel system currently active. When the data used to compute this PID is obtained from a serial data signal, if the signal is not received, the value of PID $0022 shall be set to $0000 = 0.0 kPa. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.31 FHPR – PID $0023 – Fuel Rail Pressure Relative to Atmosphere

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0023	Fuel Rail Pressure Relative to Atmosphere	2	UNM	0 to 655350 kPa	E = N * 10

This PID displays undefaulted fuel rail pressure at the engine when the reading is referenced to atmosphere (gauge pressure). This PID is used for the high side pressure for fuel systems using a high pressure fuel pump. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.32 EGRR – PID $002C – Exhaust Gas Recirculation Desired Pintle Lift

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$002C	Exhaust Gas Recirculation Desired Pintle Lift	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the exhaust gas recirculation (EGR) desired pintle lift as a percentage of maximum pintle lift. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.33 EGRR – PID $002D – Exhaust Gas Recirculation Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$002D	Exhaust Gas Recirculation Error	1	UNM	-100 to 99.21875 %	E = N * 100 / 128 - 100

This PID contains the exhaust gas recirculation (EGR) error, the difference between the actual and desired EGR pintle lift, as a percentage of the desired EGR pintle lift. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.34 EVPR – PID $002E – Canister Purge Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$002E	Canister Purge Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the commanded duty cycle for the canister purge output. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.35 FLVR – PID $002F – Total Fuel Level Percentage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$002F	Total Fuel Level Percentage	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the undefaulted, total fuel level as a percentage of the rated (advertised) capacity of the fuel tank(s). If the data used to compute fuel level is obtained from a serial data signal, then, if the signal is not received, the value of PID $002F shall be set to $00=0 %. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.36 DRER – PID $0030 – Warm-Up Cycles Since Code Clear

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0030	Warm-Up Cycles Since Code Clear	1	UNM	0 - 255 counts	E = N * 1

This PID contains the number of OBD defined warm-up cycles since the last time codes were cleared with a scan tool or, when applicable, with a battery disconnect. Once this counter reaches its maximum value, it will stop counting and will remain at its maximum value until codes are cleared. For Bi-Fuel capable vehicles (i.e. CNG/LPG alternative fuel system), this PID represents the fuel level for the active fuel system. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.37 VSPR – PID $0031 – Distance Since Code Clear

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0031	Distance Since Code Clear	2	UNM	0 - 65535 km	E = N * 1

This PID contains the accumulated distance driven since the last code clear or memory reset. Note that this distance is not reset by a natural code clear (i.e., 40 warm up cycles without a failure being reported). Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.38 EVPR – PID $0032 – Fuel Tank Vapor Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0032	Fuel Tank Vapor Pressure	2	SNM	-8192 to +8191.75 Pa	E = N * 0.25

This PID contains the undefaulted fuel tank vapor pressure measured by a sensor. If the data used to compute fuel tank vapor pressure is obtained from a serial data signal, then if the signal is not received, the value of PID $0032 shall be set to $8000 = -8192 Pa. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.39 AAPR – PID $0033 – Barometric Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0033	Barometric Pressure	1	UNM	0 - 255 kPa	E = N * 1

This PID contains the ambient air (barometric) pressure. Ambient air pressure is determined directly from a barometric pressure sensor in applications where this sensor is available and is estimated (as a function of MAP and other parameters) in all other applications. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.40 OXYR – PID $0034 – Wide Range Oxygen Sensor Bank 1 Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0034	Wide Range Oxygen Sensor Bank 1 Sensor 1	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bit)	Data Type	Range	Conversion
Oxygen Sensor Lambda	0	16	16	UNM	0 to 1.999 λ	E = N * 2 / 65535
Oxygen Sensor Current	2	16	16	UNM	-128 to 127.59 mA	E = N * 128 / 32768 - 128

PID $34 shall be used for linear or wide-ratio oxygen sensors when equivalence ratio and current are displayed.

If PID $4F is not supported for this ECU, or if PID $4F is supported and includes $00 for either Equivalence Ratio or Maximum Oxygen Sensor Current, the external test equipment shall use the scaling values included in this table for those values. If PID $4F is supported for this ECU, the external test equipment shall calculate scaling and range for these PIDs as explained in the PID $4F definition. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.41 OXYR – PID $0035 – Wide Range Oxygen Sensor Bank 1 Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0035	Wide Range Oxygen Sensor Bank 1 Sensor 2	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bit)	Data Type	Range	Conversion
Oxygen Sensor Lambda	0	16	16	UNM	0 to 1.999 λ	E = N * 2 / 65535
Oxygen Sensor Current	2	16	16	UNM	-128 to 127.59 mA	E = N * 128 / 32768 - 128

PID $35 shall be used for linear or wide-ratio oxygen sensors when equivalence ratio and current are displayed.

If PID $4F is not supported for this ECU, or if PID $4F is supported and includes $00 for either Equivalence Ratio or Maximum Oxygen Sensor Current, the external test equipment shall use the scaling values included in this table for those values. If PID $4F is supported for this ECU, the external test equipment shall calculate scaling and range for these PIDs as explained in the PID $4F definition. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.42 OXYR – PID $0038 – Bank 2 – Sensor 1 (wide range O2S)

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0038	Bank 2 – Sensor 1 (wide range O2S)	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bit)	Data Type	Range	Conversion
Oxygen Sensor Lambda	0	16	16	UNM	0 to 1.999 λ	E = N * 2 / 65535
Oxygen Sensor Current	2	16	16	UNM	-128 to 127.59 mA	E = N * 128 / 32768 - 128

PID $38 shall be used for linear or wide-ratio oxygen sensors when equivalence ratio and current are displayed.

If PID $4F is not supported for this ECU, or if PID $4F is supported and includes $00 for either Equivalence Ratio or Maximum Oxygen Sensor Current, the external test equipment shall use the scaling values included in this table for those values. If PID $4F is supported for this ECU, the external test equipment shall calculate scaling and range for these PIDs as explained in the PID $4F definition. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.43 CATR – PID $003C – Catalyst Temperature Bank 1 Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$003C	Catalyst Temperature Bank 1 Sensor 1	2	UNM	-40 - 6513.5 °C	E = N * 0.1 - 40

This PID contains the estimated catalyst temperature for bank 1 sensor 1 (as a function of engine speed, engine airflow, and fuel ethanol percentage) for use in the oxygen sensor and catalyst monitor diagnostic algorithms. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.44 CATR – PID $003D – Catalyst Temperature Bank 2 Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$003D	Catalyst Temperature Bank 2 Sensor 1	2	UNM	-40 - 6513.5 °C	E = N * 0.1 - 40

This PID contains the estimated catalyst temperature for bank 2 sensor 1 (as a function of engine speed, engine airflow, and fuel ethanol percentage) for use in the oxygen sensor and catalyst monitor diagnostic algorithms. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.45 ODAR – PID $0040 – Parameter Identifiers Supported - $41 to $60

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0040	Parameter Identifiers Supported - $41 to $60	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PID $41 Supported	0	7	1	BLN	N/A	$1=True; $0=False
PID $42 Supported	0	6	1	BLN	N/A	$1=True; $0=False
PID $43 Supported	0	5	1	BLN	N/A	$1=True; $0=False
PID $44 Supported	0	4	1	BLN	N/A	$1=True; $0=False
PID $45 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PID $46 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PID $47 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PID $48 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PID $49 Supported	1	7	1	BLN	N/A	$1=True; $0=False
PID $4A Supported	1	6	1	BLN	N/A	$1=True; $0=False
PID $4B Supported	1	5	1	BLN	N/A	$1=True; $0=False
PID $4C Supported	1	4	1	BLN	N/A	$1=True; $0=False
PID $4D Supported	1	3	1	BLN	N/A	$1=True; $0=False
PID $4E Supported	1	2	1	BLN	N/A	$1=True; $0=False
PID $4F Supported	1	1	1	BLN	N/A	$1=True; $0=False
PID $50 Supported	1	0	1	BLN	N/A	$1=True; $0=False
PID $51 Supported	2	7	1	BLN	N/A	$1=True; $0=False
PID $52 Supported	2	6	1	BLN	N/A	$1=True; $0=False
PID $53 Supported	2	5	1	BLN	N/A	$1=True; $0=False
PID $54 Supported	2	4	1	BLN	N/A	$1=True; $0=False
PID $55 Supported	2	3	1	BLN	N/A	$1=True; $0=False
PID $56 Supported	2	2	1	BLN	N/A	$1=True; $0=False
PID $57 Supported	2	1	1	BLN	N/A	$1=True; $0=False
PID $58 Supported	2	0	1	BLN	N/A	$1=True; $0=False
PID $59 Supported	3	7	1	BLN	N/A	$1=True; $0=False
PID $5A Supported	3	6	1	BLN	N/A	$1=True; $0=False
PID $5B Supported	3	5	1	BLN	N/A	$1=True; $0=False
PID $5C Supported	3	4	1	BLN	N/A	$1=True; $0=False
PID $5D Supported	3	3	1	BLN	N/A	$1=True; $0=False
PID $5E Supported	3	2	1	BLN	N/A	$1=True; $0=False
PID $5F Supported	3	1	1	BLN	N/A	$1=True; $0=False
PID $60 Supported	3	0	1	BLN	N/A	$1=True; $0=False

This PID indicates which legislated PIDs, within the range of $41 to $60, are supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.46 DFIR – PID $0041 – Emission Related Monitor Status This Driving Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0041	Emission Related Monitor Status This Driving Cycle	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	8			
Reserved	1	7	1			
Comprehensive Component Monitoring Status This Cycle	1	6	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Fuel System Monitoring Status This Cycle	1	5	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Misfire Monitoring Status This Cycle	1	4	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Compression ignition monitoring supported	1	3	1	ENM	N/A	$0 = Spark ignition monitors supported $1 = Compression ignition monitors supported
Comprehensive Component Monitoring Enabled For This Cycle	1	2	1	BLN	N/A	$1=True; $0=False
Fuel System Monitoring Enabled For This Cycle	1	1	1	BLN	N/A	$1=True; $0=False
Misfire Monitoring Enabled For This Cycle	1	0	1	BLN	N/A	$1=True; $0=False
	The following descriptions for Bytes 2 and 3 are to be used for spark ignition vehicles only.					
Exhaust Gas Recirculation and/or Variable Valve Timing System Monitoring Enabled For This Cycle	2	7	1	BLN	N/A	$1=True; $0=False
Oxygen Sensor Heater Monitoring Enabled For This Cycle	2	6	1	BLN	N/A	$1=True; $0=False
Oxygen Sensor Monitoring Enabled For This Cycle	2	5	1	BLN	N/A	$1=True; $0=False
~~Air Conditioning System Refrigerant Monitoring Enabled For This Cycle~~	2	4	1	~~BLN~~	~~N/A~~	~~$1=True; $0=False~~
Secondary Air System Monitoring Enabled For This Cycle	2	3	1	BLN	N/A	$1=True; $0=False
Evaporative System Monitoring Enabled For This Cycle	2	2	1	BLN	N/A	$1=True; $0=False
Heated Catalyst Monitoring Enabled For This Cycle	2	1	1	BLN	N/A	$1=True; $0=False
Catalyst Monitoring Enabled For This Cycle	2	0	1	BLN	N/A	$1=True; $0=False
Exhaust Gas Recirculation and/or Variable Valve Timing System Monitoring Status This Cycle	3	7	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Oxygen Sensor Heater Monitoring Status This Cycle	3	6	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Oxygen Sensor Monitoring Status This Cycle	3	5	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
~~Air Conditioning System Refrigerant Monitoring Status This Cycle~~	3	4	1	~~ENM~~	~~N/A~~	~~$0=Monitor Complete; $1=Monitor Not Complete~~
Secondary Air System Monitoring Status This Cycle	3	3	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Evaporative System Monitoring Status This Cycle	3	2	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete

Heated Catalyst Monitoring Status This Cycle	3	1	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Catalyst Monitoring Status This Cycle	3	0	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
	The following descriptions for Bytes 2 and 3 are to be used for compression ignition vehicles only.					
EGR and/or VVT system Monitoring Enabled For This Cycle	2	7	1	BLN	N/A	$1=True; $0=False
PM filter Monitoring Enabled For This Cycle	2	6	1	BLN	N/A	$1=True; $0=False
Exhaust gas sensor Monitoring Enabled For This Cycle	2	5	1	BLN	N/A	$1=True; $0=False
Reserved	2	4	1			
Boost pressure system Monitoring Enabled For This Cycle	2	3	1	BLN	N/A	$1=True; $0=False
Reserved	2	2	1			
NOx/SCR aftertreatment Monitoring Enabled For This Cycle	2	1	1	BLN	N/A	$1=True; $0=False
NMHC catalyst Monitoring Enabled For This Cycle	2	0	1	BLN	N/A	$1=True; $0=False
EGR and/or VVT system Monitoring Status This Cycle	3	7	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
PM filter Monitoring Status This Cycle	3	6	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Exhaust gas Monitoring Status This Cycle	3	5	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Reserved	3	4	1			
Boost pressure system Monitoring Status This Cycle	3	3	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
Reserved	3	2	1			
NOx/SCR aftertreatment Monitoring Status This Cycle	3	1	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete
NMHC catalyst Monitoring Status This Cycle	3	0	1	ENM	N/A	$0=Monitor Complete; $1=Monitor Not Complete

This PID contains the enable and completion status during the current driving / monitoring cycle of each continuous legislated emission related monitor (comprehensive component, fuel system, and misfire) and non-continuous legislated emission related monitor (exhaust gas recirculation system, oxygen sensor heater, oxygen sensor, air conditioning system refrigerant, secondary air system, evaporative system, heated catalyst, catalyst).

For compression ignition vehicles, the set of non-continuous monitors are (NMHC catalyst, NOx/SCR aftertreatment, Bosst pressure system, Exhaust gas sensor, Particulate Matter filter, EGR/VVT system).

Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.
The displayed order of bytes 0-3 between this document and SAE J1979 are in opposite order.

3.1.47 LVTR – PID $0042 – Control Module Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0042	Control Module Voltage	2	UNM	0 - 65.535 V	E = N * 0.001

This PID contains the voltage of the power input to the control module. Since battery voltage is not directly measured by powertrain controllers, switched battery voltage, available from the Run/Crank input to the controller, shall be used to populate this PID. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.48 MAFR – PID $0043 – Absolute Engine Load

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0043	Absolute Engine Load	2	UNM	0 - 25700 %	E = N * 100 / 255

This PID contains the absolute engine load, air mass per cylinder as a percentage of the maximum air mass per cylinder, where the maximum air mass per cylinder is determined at wide open throttle (WOT), standard temperature and pressure (STP), and 100% volumetric efficiency. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.49 FULR – PID $0044 – Equivalence Ratio

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0044	Equivalence Ratio	2	UNM	0 - 1.999969482422	E = N / 32768

This PID contains the commanded equivalence ratio. The equivalence ratio should be reported as 1.0 when PID $0003 indicates that the vehicle is closed loop. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.50 TPSR – PID $0045 – Relative Throttle Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0045	Relative Throttle Position	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the relative throttle position (corrected based upon its learned minimum position) from throttle position sensor 1. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.51 OATR – PID $0046 – Ambient Air Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0046	Ambient Air Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID shall display the estimated ambient air temperature. For applications supporting an OBD-compliant outside air temperature (OAT) sensor, this PID shall display the undefaulted temperature measured by that sensor. For applications without an OBD-compliant OAT,the variable used for the estimated ambient air temperature shall be the variable that estimates the ambient air temperature by using minimum intake air temperature sampled at a calibrated number of counts. If the data used to compute the outside air temperature is obtained from a serial data signal, then, if the signal is not received, the value of PID $0046 shall be set to $00=-40°C s. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.52 TPSR – PID $0047 – Absolute Throttle Position B

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0047	Absolute Throttle Position B	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the absolute throttle position (no correction based upon learned minimum position) from throttle position sensor 2. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.53 APSR – PID $0049 – Accelerator Pedal Position D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0049	Accelerator Pedal Position D	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the absolute accelerator pedal position (no correction or normalization based upon learned minimum or maximum positions) from accelerator pedal position sensor 1. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.54 APSR – PID $004A – Accelerator Pedal Position E

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$004A	Accelerator Pedal Position E	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the absolute accelerator pedal position (no correction or normalization based upon learned minimum or maximum positions) from accelerator pedal position sensor 2. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.55 TPSR – PID $004C – Commanded Throttle Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$004C	Commanded Throttle Position	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the commanded throttle position used to control the throttle motor, which is determined from the arbitration of the various requests for the authority to control the throttle, including remedial actions. A value of 0% indicates that closed throttle is commanded, and a value of 100% indicates that wide open throttle (WOT) is commanded. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.56 OXYR/ENGR – PID $004F – External Test Equipment Configuration Information 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$004F	External Test Equipment Configuration Information 1	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bit)	Data Type	Range	Conversion
Maximum Value for Equivalence Ratio	0	7	8	UNM	0 to 255	E = N * 1
Maximum Value for Oxygen Sensor Voltage	1	7	8	UNM	0 to 255 V	E = N * 1
Maximum Value for Oxygen Sensor Current	2	7	8	UNM	0 to 255 mA	E = N * 1
Maximum Value for Intake Manifold Absolute Pressure	3	7	8	UNM	0 to 2550 kPa	E = N * 10

This PID contains values to be used by the external test equipment to calculate scaling factors for PIDs that are different from the values in the PID definition tables included in this document. These values are not intended for display to the service technician. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Maximum Value for Equivalence Ratio shall be used by the external test equipment to calculate the scaling per bit of PIDs $24 to $2B, PIDs $34 to $3B, and PID $44.

Maximum Value for Oxygen Sensor Voltage shall be used by the external test equipment to calculate the scaling per bit of PIDs $24 to $2B.

Maximum Value for Oxygen Sensor Current shall be used by the external test equipment to calculate the scaling per bit of PIDs $34 to $3B.

Maximum Value for Intake Manifold Absolute Pressure shall be used by the external test equipment to calculate the scaling per bit of PID $0B.

3.1.57 FULR – PID $0051 – Type of fuel currently being utilized by the vehicle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0051	Type of fuel currently being utilized by the vehicle	1	ENM	N/A	$00= Not available; $01= Gasoline/petrol; $02= Methanol; $03= Ethanol; $04= Diesel; $05= Liquefied Petroleum Gas (LPG); $06= Compressed Natural Gas (CNG); $07= Propane; $08= Battery/electric; $09= Bi-fuel vehicle using gasoline; $0A= Bi-fuel vehicle using methanol; $0B= Bi-fuel vehicle using ethanol; $0C= Bi-fuel vehicle using LPG; $0D= Bi-fuel vehicle using CNG; $0E= Bi-fuel vehicle using propane; $0F= Bi-fuel vehicle using battery; $10=Bi-fuel vehicle using battery and combustion engine; $11=Hybrid vehicle using gasoline engine; $12=Hybrid vehicle using gasoline engine on ethanol; $13=Hybrdi vehicle using diesel engine; $14=Hybrid vehicle using battery and combustion engine; $15=Hybrid vehicle using battery and combustion engine; $16=Hybrid vehicle in regeneration mode; $17-$FF=ISO/SAE reserved

This PID contains the type of fuel currently being utilized by the vehicle. This PID is only required for Bi-Fuel capable vehicles (i.e. CNG/LPG ‘alternative’ fuel system in addition to a ‘primary’ gasoline fuel system). This PID should only be supported for Bi-Fuel applications. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.58 E85R – PID $0052 – Alcohol Fuel Percentage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0052	Alcohol Fuel Percentage	1	UNM	0 - 99.609375 %	E = N * 100 / 256

PID $0052 shall indicate the percentage of alcohol contained in ethanol or methanol fuels, if utilized. Alcohol percentage can be determined using a sensor or can be inferred by the fuel control software. If alcohol percentage is determined using a sensor, PID $0052 must display the undefaulted representation of that sensor. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.59 BSER – PID $005B – Hybrid Battery Pack Remaining Charge

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$005B	Hybrid Battery Pack Remaining Charge	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the percent remaining charge for the hybrid battery pack. If the data used to compute remaining charge is obtained from a serial data signal, then, if the signal is not received, the value of PID $005B shall be set to $00=0%. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.60 EOTR – PID $005C – Engine Oil Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$005C	Engine Oil Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

Engine Oil Temperature shall display engine oil temperature, if utilized by the control module strategy. EOT may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

3.1.61 FULR – PID $005D– Fuel Injection Timing

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$005D	Fuel Injection Timing	2	SNM	-210 to 301.992	E = N * 1/128 with 0 at 26880

This PID shall indicate start of the main fuel injection relative to Top Dead Center (TDC). It shall display the start of the main fuel injection of the last injector that has been driven. Positive degrees indicate Before TDC, negative degrees indicate After TDC. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.62 FULR – PID $005E– Engine Fuel Rate

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$005E	Engine Fuel Rate	2	UNM	0 to 3276.75 L/h	E = N * 0.05

This PID shall indicate the amount of fuel consumed by engine per unit of time in liters per hour. Engine Fuel Rate shall be calculated as the sum of the fuel rate over the last 1000 milliseconds prior to the time when Engine Fuel Rate is updated.. Engine Fuel Rate shall be updated at a minimum rate of 1000 milliseconds. The PID shall indicate zero L/h when the engine is not running. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.63 ODAR – PID $0060 – Parameter Identifiers Supported - $61 to $80

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0060	Parameter Identifiers Supported - $61 to $80	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PID $61 Supported	0	7	1	BLN	N/A	$1=True; $0=False
PID $62 Supported	0	6	1	BLN	N/A	$1=True; $0=False
PID $63 Supported	0	5	1	BLN	N/A	$1=True; $0=False
PID $64 Supported	0	4	1	BLN	N/A	$1=True; $0=False
PID $65 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PID $66 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PID $67 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PID $68 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PID $69 Supported	1	7	1	BLN	N/A	$1=True; $0=False
PID $6A Supported	1	6	1	BLN	N/A	$1=True; $0=False
PID $6B Supported	1	5	1	BLN	N/A	$1=True; $0=False
PID $6C Supported	1	4	1	BLN	N/A	$1=True; $0=False
PID $6D Supported	1	3	1	BLN	N/A	$1=True; $0=False
PID $6E Supported	1	2	1	BLN	N/A	$1=True; $0=False
PID $6F Supported	1	1	1	BLN	N/A	$1=True; $0=False
PID $70 Supported	1	0	1	BLN	N/A	$1=True; $0=False
PID $71 Supported	2	7	1	BLN	N/A	$1=True; $0=False
PID $72 Supported	2	6	1	BLN	N/A	$1=True; $0=False
PID $73 Supported	2	5	1	BLN	N/A	$1=True; $0=False
PID $74 Supported	2	4	1	BLN	N/A	$1=True; $0=False
PID $75 Supported	2	3	1	BLN	N/A	$1=True; $0=False
PID $76 Supported	2	2	1	BLN	N/A	$1=True; $0=False
PID $77 Supported	2	1	1	BLN	N/A	$1=True; $0=False
PID $78 Supported	2	0	1	BLN	N/A	$1=True; $0=False
PID $79 Supported	3	7	1	BLN	N/A	$1=True; $0=False
PID $7A Supported	3	6	1	BLN	N/A	$1=True; $0=False
PID $7B Supported	3	5	1	BLN	N/A	$1=True; $0=False
PID $7C Supported	3	4	1	BLN	N/A	$1=True; $0=False
PID $7D Supported	3	3	1	BLN	N/A	$1=True; $0=False
PID $7E Supported	3	2	1	BLN	N/A	$1=True; $0=False
PID $7F Supported	3	1	1	BLN	N/A	$1=True; $0=False
PID $80 Supported	3	0	1	BLN	N/A	$1=True; $0=False

This PID indicates which legislated PIDs, within the range of $61 to $80, are supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.64 ETQR – PID $0062 – Actual Engine – Percent Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0062	Actual Engine - Percent Torque	1	UNM	-125-130%	E = N * 1-125

This PID contains the current indicated torque (torque developed in the cylinders) as a percent of engine reference torque. Indicated Torque is defined as the sum of Net Brake Torque and Friction Torque (as defined in PID $008E).. The engine percent torque value will not be less than zero and it includes the torque developed in the cylinders required to overcome friction.

3.1.65 ECPR – PID $0063 – Engine Reference Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0063	Engine Reference Torque	2	UNM	0-65535 Nm	E = N * 1

This PID contains the engine reference torque, which is the 100% reference value for all defined indicated engine torque parameters. It is only defined once and doesn't change if a different engine torque map becomes valid.

3.1.66 ENGR – PID $0065 – Auxiliary Inputs/Outputs

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0065	Auxiliary Inputs/Outputs	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	4			
Glow Plug Lamp Status Supported	0	3	1	BLN	N/A	$1=True; $0=False
Manual Trans Neutral Gear Status Supported	0	2	1	BLN	N/A	$1=True; $0=False
Auto Trans Neutral Drive Status Supported	0	1	1	BLN	N/A	$1=True; $0=False
Power Take Off (PTO) Status Supported	0	0	1	BLN	N/A	$1=True; $0=False
Reserved	1	7	4			
Glow Plug Lamp Commanded On	1	3	1	BLN	N/A	$1=True; $0=False
Manual Transmission in Gear	1	2	1	BLN	N/A	$1=True; $0=False
Automatic Transmission in Gear	1	1	1	BLN	N/A	$1=True; $0=False
Power Take Off (PTO) Active	1	0	1	BLN	N/A	$1=True; $0=False

This PID contains several signals associated with auxiliary inputs and outputs. The content of each of these signals is described below. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Glow Plug Lamp Status Supported indicates whether the Glow Plug Lamp Commanded On information is utilized in this application.

Manual Trans Neutral Gear Status Supported indicates whether the Manual Transmission in Gear information is utilized in this application.

Auto Trans Neutral Drive Status Supported indicates whether the Automatic Transmission in Gear information is utilized in this application.

Power Take Off (PTO) Status Supported indicates whether the Power Take Off (PTO) Active information is utilized in this application.

Glow Plug Lamp Commanded On indicates the commanded state of the glow plug or wait-to-start lamp output.

Manual Transmission in Gear indicates whether a manually shifted transmission is in gear with the clutch pedal released. A value of false is provided when the transmission is in neutral via the clutch depressed and/or the transmission in neutral gear.

Automatic Transmission in Gear indicates whether an automatically shifted transmission is in a forward (Drive) or reverse gear. A value of false is provided when the transmission is in Park or Neutral

Power Take Off (PTO) Active indicates the operational state of the power take-off system.

3.1.67 MAFR – PID $0066 – Mass Air Flow Sensor

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0066	Mass Air Flow Sensor	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
MAF Sensor B supported	0	1	1	BLN	N/A	$1=True; $0=False
MAF Sensor A supported	0	0	1	BLN	N/A	$1=True; $0=False
Mass Air Flow Sensor A	1	7	16	UNM	0 - 2047.96875 g/s	E = N / 32
Mass Air Flow Sensor B	3	7	16	UNM	0 - 2047.96875 g/s	E = N / 32

This PID contains the supported status and sensor values for up to two Mass Air Flow (MAF) sensors. MAF A shall display the airflow rate as measured by a vehicle that utilizes a MAF sensor or an equivalent source. If the engine is off and the ignition is on, the actual sensor value reading shall be reported. If the actual sensor reading can not be reported, the MAF value shall be reported as 0.00 g/s. Engines that utilize two MAF sensors shall use MAF A and MAF B, which shall be associated with engine bank 1 and engine bank 2, respectively. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.68 EITR – PID $0068 – Intake Air Temperature Sensor

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0068	Intake Air Temperature Sensor	7	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
IAT Bank 2, Sensor 3 supported	0	5	1	BLN	N/A	$1=True; $0=False
IAT Bank 2, Sensor 2 supported	0	4	1	BLN	N/A	$1=True; $0=False
IAT Bank 2, Sensor 1 supported	0	3	1	BLN	N/A	$1=True; $0=False
IAT Bank 1, Sensor 3 supported	0	2	1	BLN	N/A	$1=True; $0=False
IAT Bank 1, Sensor 2 supported	0	1	1	BLN	N/A	$1=True; $0=False
IAT Bank 1, Sensor 1 supported	0	0	1	BLN	N/A	$1=True; $0=False
Intake Air Temperature Bank 1, Sensor 1	1	7	8	UNM	-40 to +215 °C	E = N * 1 - 40
Intake Air Temperature Bank 1, Sensor 2	2	7	8	UNM	-40 to +215 °C	E = N * 1 - 40
Intake Air Temperature Bank 1, Sensor 3	3	7	8	UNM	-40 to +215 °C	E = N * 1 - 40
Intake Air Temperature Bank 2, Sensor 1	4	7	8	UNM	-40 to +215 °C	E = N * 1 - 40
Intake Air Temperature Bank 2, Sensor 2	5	7	8	UNM	-40 to +215 °C	E = N * 1 - 40
Intake Air Temperature Bank 2, Sensor 3	6	7	8	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the undefaulted intake air temperature measured by up to six sensors as follows. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

IAT Bank 1, Sensor 1 shall display intake air temperature, if utilized by the control module strategy. IAT may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

IAT Bank 1, Sensor 2 shall display intake air temperature, if utilized by the control module strategy.

IAT Bank 1, Sensor 3 shall display intake air temperature, if utilized by the control module strategy.

IAT Bank 2, Sensor 1 shall display intake air temperature, if utilized by the control module strategy.

IAT Bank 2, Sensor 2 shall display intake air temperature, if utilized by the control module strategy.

IAT Bank 2, Sensor 3 shall display intake air temperature, if utilized by the control module strategy.

3.1.69 EGRR– PID $0069 – Commanded EGR and EGR Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0069	Commanded EGR and EGR Error	7	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
EGR B Error Supported	0	5	1	BLN	N/A	$1=True; $0=False
Actual EGR B Duty Cycle/Position Supported	0	4	1	BLN	N/A	$1=True; $0=False
Commanded EGR B Duty Cycle/Position Supported	0	3	1	BLN	N/A	$1=True; $0=False
EGR A Error Supported	0	2	1	BLN	N/A	$1=True; $0=False
Actual EGR A Duty Cycle/Position Supported	0	1	1	BLN	N/A	$1=True; $0=False
Commanded EGR A Duty Cycle/Position Supported	0	0	1	BLN	N/A	$1=True; $0=False
Commanded EGR A Duty Cycle/Position	1	0	8	UNM	0 to 100 %	E = N * 100 / 255
Actual EGR A Duty Cycle/Position	2	0	8	UNM	0 to 100 %	E = N * 100 / 255
EGR A Error	3	0	8	UNM	-100 to 99.21875 %	E = N * 100 / 128-100
Commanded EGR B Duty Cycle/Position	4	0	8	UNM	0 to 100 %	E = N * 100 / 255
Actual EGR B Duty Cycle/Position	5	0	8	UNM	0 to 100 %	E = N * 100 / 255
EGR B Error	6	0	8	UNM	-100 to 99.21875 %	E = N * 100 / 128-100

This PID indicates the commanded EGR valve and relative position for up to two EGR valves on a compression ignition engine. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.70 AICR – PID $006A – Commanded Diesel Intake Air Flow Control and Relative Intake Air Flow Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$006A	Commanded Diesel Intake Air Flow Control and Relative Intake Air Flow Position	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Relative Intake Air Flow B Position supported	0	3	1	BLN	N/A	$1=True; $0=False
Commanded Intake Air Flow B Control supported	0	2	1	BLN	N/A	$1=True; $0=False
Relative Intake Air Flow A Position supported	0	1	1	BLN	N/A	$1=True; $0=False
Commanded Intake Air Flow A Control supported	0	0	1	BLN	N/A	$1=True; $0=False
Commanded Intake Air Flow A Control	1	7	8	UNM	0 to 100 %	E = N * 100 / 255
Relative Intake Air Flow A Position	2	7	8	UNM	0 to 100 %	E = N * 100 / 255
Commanded Intake Air Flow B Control	3	7	8	UNM	0 to 100 %	E = N * 100 / 255
Relative Intake Air Flow B Position	4	7	8	UNM	0 to 100 %	E = N * 100 / 255

This PID indicates the commanded intake airflow and relative intake airflow position for up to two intake airflow valves on a compression ignition engine. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.71 CETR – PID $006B – Exhaust Gas Recirculation Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$006B	Exhaust Gas Recirculation Temperature	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
EGR Temperature Sensor C (Bank 1, Sensor 2) Wide Range supported	0	5	1	BLN	N/A	$1=True; $0=False
EGR Temperature Sensor A (Bank 1, Sensor 1) Wide Range supported	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			
Exhaust Gas Recirculation Temp Sensor A (Bank 1, Sensor 1)	1	7	8	UNM	-40 to 980 °C	E = N * 4 - 40
Exhaust Gas Recirculation Temp Sensor C (Bank 1, Sensor 2)	2	7	8	UNM	-40 to 980 °C	E = N * 4 - 40
Reserved	3	7	8			
Reserved	4	7	8			

This PID indicates the intended temperature range and scaling factor for the temperature sensors being used to determine the cooling effectiveness of the EGR gas heat exchanger. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

EGR Temperature Sensor C (Bank 1, Sensor 2) Wide Range supported indicates the need to use a conversion equation that has a maximum temperature of 980 °C.

EGR Temperature Sensor A (Bank 1, Sensor 1) Wide Range supported indicates the need to use a conversion equation that has a maximum temperature of 980 °C.

Exhaust Gas Recirculation Temp Sensor A (Bank 1, Sensor 1) will sense gas temperature from the hot side of a heat exchanger and would need a range up to 980 °C.

Exhaust Gas Recirculation Temp Sensor C (Bank 1, Sensor 2) will sense exhaust gas temperature from the cold side of a heat exchanger and would need a range up to 980 °C.

3.1.72 FHPR – PID $006D – Fuel Pressure Control System

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$006D	Fuel Pressure Control System	11	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Fuel Temperature B supported	0	5	1	BLN	N/A	$1=True; $0=False
Fuel Rail Pressure B supported	0	4	1	BLN	N/A	$1=True; $0=False
Commanded Fuel Rail Pressure B supported	0	3	1	BLN	N/A	$1=True; $0=False
Fuel Temperature A supported	0	2	1	BLN	N/A	$1=True; $0=False
Fuel Rail Pressure A supported	0	1	1	BLN	N/A	$1=True; $0=False
Commanded Fuel Rail Pressure A supported	0	0	1	BLN	N/A	$1=True; $0=False
Commanded Fuel Rail Pressure A	1	7	16	UNM	0 to 655350 kPa	E = N * 10
Fuel Rail Pressure A	3	7	16	UNM	0 to 655350 kPa	E = N * 10
Fuel Rail Temperature A	5	7	8	UNM	-40 to +215 °C	E = N * 1 - 40
Commanded Fuel Rail Pressure B	6	7	16	UNM	0 to 655350 kPa	E = N * 10
Fuel Rail Pressure B	8	7	16	UNM	0 to 655350 kPa	E = N * 10
Fuel Rail Temperature B	10	7	8	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains fuel pressure control system information.

Commanded Fuel Rail Pressure A shall display commanded fuel rail pressure when the reading is referenced to atmosphere (gage pressure), if utilized by the control module strategy.

Fuel Rail Pressure A shall display fuel rail pressure when the reading is referenced to atmosphere (gage pressure), if utilized by the control module strategy.

Fuel Rail Temperature A shall display fuel rail temperature, if utilized by the control module strategy.

Commanded Fuel Rail Pressure B shall display commanded fuel rail pressure when the reading is referenced to atmosphere (gage pressure), if utilized by the control module strategy.

Fuel Rail Pressure B shall display fuel rail pressure when the reading is referenced to atmosphere (gage pressure), if utilized by the control module strategy.

Fuel Rail Temperature B shall display fuel rail temperature, if utilized by the control module strategy.

3.1.73 MAPR - $006F - Turbocharger Compressor Inlet Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$006F	Turbocharger Compressor Inlet Pressure	3	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	4			
Turbocharger Compressor Inlet Pressure Sensor B Wide Range Supported	0	3	1	BLN	N/A	$1=True; $0=False
Turbocharger Compressor Inlet Pressure Sensor A Wide Range Supported	0	2	1	BLN	N/A	$1=True; $0=False
Turbocharger Compressor Inlet Pressure Sensor B Supported	0	1	1	BLN	N/A	$1=True; $0=False
Turbocharger Compressor Inlet Pressure Sensor A Supported	0	0	1	BLN	N/A	$1=True; $0=False
Turbocharger Compressor Inlet Pressure Sensor A	1	7	8	UNM	0 to 255 kPa or 0 to 2040 kPa	E = N or E = N * 8
Turbocharger Compressor Inlet Pressure Sensor B	2	7	8	UNM	0 to 255 kPa or 0 to 2040 kPa	E = N or E = N * 8

This PID contains the supported status and the turbocharger inlet pressure for up to two charger systems. The turbocharger inlet pressure value shall be measured by a sensor.
Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Turbocharger Compressor Inlet Pressure Sensor B Wide Range supported indicates if a Wide Range Sensor B (0 to 2040kPa) is present in this application. (Not used in GMPT S/W)

Turbocharger Compressor Inlet Pressure Sensor A Wide Range supported indicates if a Wide Range Sensor A (0 to 2040kPa) is present in this application. (Not used in GMPT S/W)

Turbocharger Compressor Inlet Pressure Sensor B supported indicates if a Sensor B (0 to 255kPa) is present in this application.

Turbocharger Compressor Inlet Pressure Sensor A supported indicates if a Sensor A (0 to 255kPa) is present in this application.

Turbocharger Compressor Inlet Pressure Sensor A indicates the raw signal of the sensor A which is present in this application.

Turbocharger Compressor Inlet Pressure Sensor B indicates the raw signal of the sensor B which is present in this application.

3.1.74 AICR– PID $70 – Boost Pressure Control

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0070	Boost Pressure Control	10	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Boost Pressure B Control Status Supported	0	5	1	BLN	N/A	$1=True; $0=False
Boost Pressure Sensor B Supported	0	4	1	BLN	N/A	$1=True; $0=False
Commanded Boost Pressure B Supported	0	3	1	BLN	N/A	$1=True; $0=False
Boost Pressure A Control Status Supported	0	2	1	BLN	N/A	$1=True; $0=False
Boost Pressure Sensor A Supported	0	1	1	BLN	N/A	$1=True; $0=False
Commanded Boost Pressure A Supported	0	0	1	BLN	N/A	$1=True; $0=False
Turbocharger/Supercharger A Commanded Boost Pressure	1	0	16	UNM	0 to 2047.96875 kPa	E = N * 0.03125
Turbocharger/Supercharger A Actual Boost Pressure	3	0	16	UNM	0 to 2047.96875 kPa	E = N * 0.03125
Turbocharger/Supercharger B Commanded Boost Pressure	5	0	16	UNM	0 to 2047.96875 kPa	E = N * 0.03125
Turbocharger/Supercharger B Actual Boost Pressure	7	0	16	UNM	0 to 2047.96875 kPa	E = N * 0.03125
Reserved	8	7	4			
Boost Pressure B Control Status	8	3	2	ENM	N/A	$0 = reserved, not defined $1 = Open Loop (no fault present) $2 = Closed Loop (no fault present) $3 = Fault present (boost data unreliable)
Boost Pressure A Control Status	8	1	2	ENM	N/A	$0 = reserved, not defined $1 = Open Loop (no fault present) $2 = Closed Loop (no fault present) $3 = Fault present (boost data unreliable)

This PID indicates the commanded boost pressure, the relative actual boost pressure and the boost pressure control status for up to two Boost controls on a compression ignition engine. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.75 VGTR– PID $71 – Variable Geometry Turbo (VGT) Control

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0071	Variable Geometry Turbo (VGT) Control	6	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
VGT B Control Status Supported	0	5	1	BLN	N/A	$1=True; $0=False
VGT B Position supported	0	4	1	BLN	N/A	$1=True; $0=False
Commanded VGT B Position Supported	0	3	1	BLN	N/A	$1=True; $0=False
VGT A Control Status Supported	0	2	1	BLN	N/A	$1=True; $0=False
VGT A Position Supported	0	1	1	BLN	N/A	$1=True; $0=False
Commanded VGT A Position Supported	0	0	1	BLN	N/A	$1=True; $0=False
Commanded Variable Geometry Turbo A Position	1	0	8	UNM	0 to 100 %	E = N * 100 / 255
Variable Geometry Turbo A Position	2	0	8	UNM	0 to 100 %	E = N * 100 / 255
Commanded Variable Geometry Turbo B Position	3	0	8	UNM	0 to 100 %	E = N * 100 / 255
Variable Geometry Turbo B Position	4	0	8	UNM	0 to 100 %	E = N * 100 / 255
Reserved	5	7	4			
VGT B Control Status	5	3	2	ENM	N/A	$00 = reserved, not defined $01 = Open Loop (no fault present) $02 = Closed Loop (no fault present) $03 = Fault present (boost data unreliable)
VGT A Control Status	5	1	2	ENM	N/A	$00 = reserved, not defined $01 = Open Loop (no fault present) $02 = Closed Loop (no fault present) $03 = Fault present (boost data unreliable)

This PID indicates the commanded VGT actuator, the relative position and the control status for up to two VGTs on a compression ignition engine. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.76 BSTR - $0072 - Wastegate Control

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0072	Wastegate Control	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	4			
Wastegate B Position Supported	0	3	1	BLN	N/A	$1=True; $0=False
Commanded Wastegate B Position Supported	0	2	1	BLN	N/A	$1=True; $0=False
Wastegate A Position Supported	0	1	1	BLN	N/A	$1=True; $0=False
Commanded Wastegate A Position Supported	0	0	1	BLN	N/A	$1=True; $0=False
Commanded Wastegate A Position	1	7	8	UNM	0 to 100 %	E = N * 100 / 255
Wastegate A Position	2	7	8	UNM	0 to 100 %	E = N * 100 / 255
Commanded Wastegate B Position	3	7	8	UNM	0 to 100 %	E = N * 100 / 255
Wastegate B Position	4	7	8	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the supported status, commanded position and actual position for up to two wastegates. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Wastegate B Position Supported indicates if a WG B is present in this application and the information of the position is supported according to the signal description.

Commanded Wastegate B Position Supported indicates if a WG B is present in this application and the information of the commanded WG position is supported according the signal description.

Wastegate A Position Supported indicates if a WG A is present in this application and the information of the position is supported according to the signal description.

Commanded Wastegate A Position Supported indicates if a WG A is present in this application and the information of the commanded WG position is supported according the signal description.

Commanded Wastegate A/B Position shall display as a percent. It shall be normalized to the maximum wastegate commanded output control parameter. Wastegate systems use a variety of methods to control wastegate position, hence boost pressure.

- If an on/off solenoid is used, the commanded WG position shall display 0% when the WG is commanded off (allow full boost), 100% when the WG system is commanded on (dump boost).
- If a vacuum solenoid is duty cycled, the WG duty cycle from 0 to 100% shall be displayed.
- If a linear or stepper motor valve is used, the fully closed position (full boost) shall be displayed as 0%, the fully open position (dump boost) shall be displayed as 100%. Intermediate positions shall be displayed as a percent of the full-open position.
- Any other actuation method shall be normalized to display 0% when the WG is commanded off and 100% when the WG is commanded on.

Wastegate A/B Position shall display as a percent. It shall be normalized to the maximum wastegate commanded output control parameter. Wastegate systems use a variety of methods to control wastegate position, hence boost pressure.

- If an on/off solenoid is used, The WG actual position shall display 0% when the WG is commanded off (allow full boost), 100% when the WG system is commanded on (dump boost).
- If a vacuum solenoid is duty cycled, the WG duty cycle from 0 to 100% shall be displayed.
- If a linear or stepper motor valve is used, the fully closed position (full boost) shall be displayed as 0%, the fully open position (dump boost) shall be displayed as 100%. Intermediate positions shall be displayed as a percent of the full-open position.
- Any other actuation method shall be normalized to display 0% when the WG is commanded off and 100% when the WG is commanded on.

3.1.77 BSTR - $0074 - Turbocharger RPM

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0074	Turbocharger RPM	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	6			
Turbo B RPM Supported	0	1	1		N/A	$1=True; $0=False
Turbo A RPM Supported	0	0	1		N/A	$1=True; $0=False
Turbocharger A RPM	1	7	16	UNM	0 to 655350 RPM	E = N * 10
Turbocharger B RPM	3	7	16	UNM	0 to 655350 RPM	E = N * 10

This PID contains the supported status and turbocharger speed for up to two charger systems. The turbocharger speed value may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.
Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Turbo B RPM Supported indicates if a turbo charger B is present and the signal is supported.

Turbo A RPM Supported indicates if a turbo charger A is present and the signal is supported.

Turbocharger A RPM indicates the inferred turbocharger A speed if it’s present in this application.

Turbocharger B RPM indicates the inferred turbocharger B speed if it’s present in this application.

3.1.78 BSTR - $0075 - Turbocharger A Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0075	Turbocharger A Temperature	7	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	4			
Turbo A Turbine Outlet Temperature Supported	0	3	1	BLN	N/A	$1=True; $0=False
Turbo A Turbine Inlet Temperature Supported	0	2	1	BLN	N/A	$1=True; $0=False
Turbo A Compressor Outlet Temperature Supported	0	1	1	BLN	N/A	$1=True; $0=False
Turbo A Compressor Inlet Temperature Supported	0	0	1	BLN	N/A	$1=True; $0=False
Turbocharger A Compressor Inlet Temperature	1	7	8	UNM	-40 to 215 °C	E = N - 40
Turbocharger A Compressor Outlet Temperature	2	7	8	UNM	-40 to 215 °C	E = N - 40
Turbocharger A Turbine Inlet Temperature	3	7	16	UNM	-40 to 6513.5 °C	E = (N * 0.1) - 40
Turbocharger A Turbine Outlet Temperature	5	7	16	UNM	-40 to 6513.5 °C	E = (N * 0.1) - 40

This PID contains the supported status and inlet/outlet turbocharger A temperature. The turbocharger A temperature value may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.
Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Turbo A Turbine Outlet Temperature Supported indicates if a turbo A is present and the signal is supported in this application.
Turbo A Turbine Inlet Temperature Supported indicates if turbo A is present and the signal is supported in this application.

Turbo A Compressor Outlet Temperature Supported indicates if turbo A is present and the signal is supported in this application.

Turbo A Compressor Inlet Temperature Supported indicates if turbo A is present and the signal is supported in this application.

Turbocharger A Compressor Inlet Temperature indicates the measured temperature signal if present this application.

Turbocharger A Compressor Outlet Temperature indicates the inferred temperature outlet signal. It's mainly calculated by 'Boyle's law' and the compressor inlet temperature signal.

Turbocharger A Turbine Inlet Temperature indicates the inferred temperature signal based on the exhaust temperature and pressure model.

Turbocharger A Turbine Outlet Temperature indicates the inferred temperature signal based on the exhaust temperature and pressure model.

3.1.79 BSTR - $0076 - Turbocharger B Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0076	Turbocharger B Temperature	7	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	4			
Turbo B Turbine Outlet Temperature Supported	0	3	1	BLN	N/A	$1=True; $0=False
Turbo B Turbine Inlet Temperature Supported	0	2	1	BLN	N/A	$1=True; $0=False
Turbo B Compressor Outlet Temperature Supported	0	1	1	BLN	N/A	$1=True; $0=False
Turbo B Compressor Inlet Temperature Supported	0	0	1	BLN	N/A	$1=True; $0=False
Turbocharger B Compressor Inlet Temperature	1	7	8	UNM	-40 to 215 °C	E = N - 40
Turbocharger B Compressor Outlet Temperature	2	7	8	UNM	-40 to 215 °C	E = N - 40
Turbocharger B Turbine Inlet Temperature	3	7	16	UNM	-40 to 6513.5 °C	E = (N * 0.1) - 40
Turbocharger B Turbine Outlet Temperature	5	7	16	UNM	-40 to 6513.5 °C	E = (N * 0.1) - 40

This PID contains the supported status and inlet/outlet turbocharger B temperature. The turbocharger B temperature value may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Turbo B Turbine Outlet Temperature Supported indicates if turbo B is present and the signal is supported in this application.

Turbo B Turbine Inlet Temperature Supported indicates if turbo B is present and the signal is supported in this application.

Turbo B Compressor Outlet Temperature Supported indicates if turbo B is present and the signal is supported in this application.

Turbo B Compressor Inlet Temperature Supported indicates if turbo B is present and the signal is supported in this application.

Turbocharger B Compressor Inlet Temperature indicates the measured temperature signal if present this application.

Turbocharger B Compressor Outlet Temperature indicates the inferred temperature outlet signal. It's mainly calculated by 'Boyle's law' and the compressor inlet temperature signal.

Turbocharger B Turbine Inlet Temperature indicates the inferred temperature signal based on the exhaust temperature and pressure model.

Turbocharger B Turbine Outlet Temperature indicates the inferred temperature signal based on the exhaust temperature and pressure model.

3.1.80 CITR– PID $0077 – Charge Air Cooler Temperature (CACT)

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0077	Charge Air Cooler Temperature (CACT)	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	4			
CACT Bank 2, Sensor 2 Supported	0	3	1	BLN	N/A	$1=True; $0=False
CACT Bank 2, Sensor 1 Supported	0	2	1	BLN	N/A	$1=True; $0=False
CACT Bank 1, Sensor 2 Supported	0	1	1	BLN	N/A	$1=True; $0=False
CACT Bank 1, Sensor 1 Supported	0	0	1	BLN	N/A	$1=True; $0=False
Charge Air Cooler Temperature Bank 1, Sensor 1	1	7	8	UNM	-40 to 215 °C	E = N * 1 - 40
Charge Air Cooler Temperature Bank 1, Sensor 2	2	7	8	UNM	-40 to 215 °C	E = N * 1 - 40
Charge Air Cooler Temperature Bank 2, Sensor 1	3	7	8	UNM	-40 to 215 °C	E = N * 1 - 40
Charge Air Cooler Temperature Bank 2, Sensor 2	4	7	8	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains signal associated with the charge air cooler temperature sensors. The content of each of these signals is described below.

CACT Bank 2, Sensor 2 Supported indicates bank 2 is configured with a second sensor that measures the temperature of the charge air cooler.

CACT Bank 2, Sensor 1 Supported indicates bank 2 is configured with a sensor that measures the temperature of the charge air cooler.

CACT Bank 1, Sensor 2 Supported indicates bank 1 is configured with a second sensor that measures the temperature of the charge air cooler.

CACT Bank 1, Sensor 1 Supported indicates bank 1 is configured with a sensor that measures the temperature of the charge air cooler.

Charge Air Cooler Temperature Bank 1, Sensor 1 contains the undefaulted charge air cooler temperature measured by a sensor located first in the charge air cooler stream on bank 1 in relationship to any other charge air cooler temperature sensors in the system.

Charge Air Cooler Temperature Bank 1, Sensor 2 contains the undefaulted charge air cooler temperature measured by a sensor located second in the charge air cooler stream on bank 1 in relationship to any other charge air cooler temperature sensors in the system.

Charge Air Cooler Temperature Bank 2, Sensor 1 contains the undefaulted charge air cooler temperature measured by a sensor located first in the charge air cooler stream on bank 2 in relationship to any other charge air cooler temperature sensors in the system.

Charge Air Cooler Temperature Bank 2, Sensor 2 contains the undefaulted charge air cooler temperature measured by a sensor located second in the charge air cooler stream on bank 2 in relationship to any other charge air cooler temperature sensors in the system.

3.1.81 EGTR – PID $0078 – Exhaust Gas Temperature (EGT) Bank 1 Sensors 1 - 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0078	Exhaust Gas Temperature (EGT) Bank 1 Sensors 1 - 4	9	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
EGT Bank 1, Sensor 4 Supported	0	3	1	BLN	N/A	$1=True; $0=False
EGT Bank 1, Sensor 3 Supported	0	2	1	BLN	N/A	$1=True; $0=False
EGT Bank 1, Sensor 2 Supported	0	1	1	BLN	N/A	$1=True; $0=False
EGT Bank 1, Sensor 1 Supported	0	0	1	BLN	N/A	$1=True; $0=False
Exhaust Gas Temperature Bank 1, Sensor 1	1	7	16	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40
Exhaust Gas Temperature Bank 1, Sensor 2	3	7	16	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40
Exhaust Gas Temperature Bank 1, Sensor 3	5	7	16	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40
Exhaust Gas Temperature Bank 1, Sensor 4	7	7	16	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the undefaulted exhaust gas temperature measured by up to four sensors as follows. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

EGT Bank 1, Sensor 4 Supported indicates the system utilizes an exhaust gas temperature sensor in the Bank 1 Sensor 4 position.

EGT Bank 1, Sensor 3 Supported indicates the system utilizes an exhaust gas temperature sensor in the Bank 1 Sensor 3 position.

EGT Bank 1, Sensor 2 Supported indicates the system utilizes an exhaust gas temperature sensor in the Bank 1 Sensor 2 position.

EGT Bank 1, Sensor 1 Supported indicates the system utilizes an exhaust gas temperature sensor in the Bank 1 Sensor 1 position.

Exhaust Gas Temperature Bank 1, Sensor 1 shall display exhaust gas temperature for bank 1, sensor 1. Exhaust gas temperature may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

Exhaust Gas Temperature Bank 1, Sensor 2 shall display exhaust gas temperature for bank 1, sensor 2. Exhaust gas temperature may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

Exhaust Gas Temperature Bank 1, Sensor 3 shall display exhaust gas temperature for bank 1, sensor 3. Exhaust gas temperature may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

Exhaust Gas Temperature Bank 1, Sensor 4 shall display exhaust gas temperature for bank 1, sensor 4. Exhaust gas temperature may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

3.1.82 EGPR – PID $007A – Diesel Particulate Filter (DPF) Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$007A	Diesel Particulate Filter (DPF) Bank 1	7	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
DPF Bank 1 Outlet Pressure Supported	0	2	1	BLN	N/A	$1=True; $0=False
DPF Bank 1 Inlet Pressure Supported	0	1	1	BLN	N/A	$1=True; $0=False
DPF Bank 1 Delta Pressure Supported	0	0	1	BLN	N/A	$1=True; $0=False
Diesel Particulate Filter Bank 1 Delta Pressure	1	7	16	SNM	-327.68 to 327.67 kPa	E = N * 0.01

Diesel Particulate Filter Bank 1 Inlet Pressure	3	7	16	UNM	0 to 655.35 kPa	E = N * 0.01
Diesel Particulate Filter Bank 1 Outlet Pressure	5	7	16	UNM	0 to 655.35 kPa	E = N * 0.01

This PID contains the unfiltered diesel particulate filter pressure measured by a sensor or multiple sensors. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

DPF Bank 1 Outlet Pressure Supported indicates the system utilizes a pressure sensor on the outlet of the Bank 1 diesel particulate filter.

DPF Bank 1 Inlet Pressure Supported indicates the system utilizes a pressure sensor on the inlet of the Bank 1 diesel particulate filter.

DPF Bank 1 Delta Pressure Supported indicates the system utilizes a single pressure sensor that measures the differential pressure between the inlet and the outlet of the Bank 1 diesel particulate filter.

Diesel Particulate Filter Bank 1 Delta Pressure shall display DPF Bank 1 delta pressure measured by a sensor.

Diesel Particulate Filter Bank 1 Inlet Pressure shall display DPF Bank 1 inlet pressure measured by a sensor.

Diesel Particulate Filter Bank 1 Outlet Pressure shall display DPF Bank 1 outlet pressure measured by a sensor.

3.1.83 EPSR – PID $007F – Engine Run Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$007F	Engine Run Time	13	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Total Run Time With PTO Active supported	0	2	1	BLN	N/A	$1=True; $0=False
Total Idle Run Time supported	0	1	1	BLN	N/A	$1=True; $0=False
Total Engine Run Time supported	0	0	1	BLN	N/A	$1=True; $0=False
Total Engine Run Time	1	7	32	UNM	0 to 4,294,967,295 s	E = N * 1
Total Idle Run Time	5	7	32	UNM	0 to 4,294,967,295 s	E = N * 1
Total Run Time With PTO Active	9	7	32	UNM	0 to 4,294,967,295 s	E = N * 1

This PID contains information for total engine run time, run time at idle, and run time with PTO active. Each timer is described below. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Total Engine Run Time shall display the total engine run time for the life of vehicle. Total Engine Run Time shall increment while the engine is running. It shall freeze if the engine stalls. Total Engine Run Time shall never be reset to zero.

Total Idle Run Time shall display the total engine idle time for the life of vehicle. Total Idle Run Time shall increment while the engine is running (engine speed > warmed up idle speed in drive - 150 rpm) at closed throttle/closed pedal, and vehicle speed is less than 1 mph. It shall freeze if the engine stalls or the engine is no longer at idle. Total Idle Run Time shall never be reset to zero.
Total Run Time With PTO Active shall display the total engine run time with PTO engaged for the life of vehicle. Total Run Time With PTO Active shall increment while the engine is running with PTO engaged. It shall freeze if the engine stalls. Total Run Time With PTO Active shall never be reset to zero.

3.1.84 ODAR – PID $0080 – Parameter Identifiers Supported - $81 to $A0

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0080	Parameter Identifiers Supported - $81 to $A0	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PID $81 Supported	0	7	1	BLN	N/A	$1=True; $0=False
PID $82 Supported	0	6	1	BLN	N/A	$1=True; $0=False
PID $83 Supported	0	5	1	BLN	N/A	$1=True; $0=False
PID $84 Supported	0	4	1	BLN	N/A	$1=True; $0=False
PID $85 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PID $86 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PID $87 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PID $88 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PID $89 Supported	1	7	1	BLN	N/A	$1=True; $0=False
PID $8A Supported	1	6	1	BLN	N/A	$1=True; $0=False
PID $8B Supported	1	5	1	BLN	N/A	$1=True; $0=False
PID $8C Supported	1	4	1	BLN	N/A	$1=True; $0=False
PID $8D Supported	1	3	1	BLN	N/A	$1=True; $0=False
PID $8E Supported	1	2	1	BLN	N/A	$1=True; $0=False
PID $8F Supported	1	1	1	BLN	N/A	$1=True; $0=False
PID $90 Supported	1	0	1	BLN	N/A	$1=True; $0=False
PID $91 Supported	2	7	1	BLN	N/A	$1=True; $0=False
PID $92 Supported	2	6	1	BLN	N/A	$1=True; $0=False
PID $93 Supported	2	5	1	BLN	N/A	$1=True; $0=False
PID $94 Supported	2	4	1	BLN	N/A	$1=True; $0=False
PID $95 Supported	2	3	1	BLN	N/A	$1=True; $0=False
PID $96 Supported	2	2	1	BLN	N/A	$1=True; $0=False
PID $97 Supported	2	1	1	BLN	N/A	$1=True; $0=False
PID $98 Supported	2	0	1	BLN	N/A	$1=True; $0=False
PID $99 Supported	3	7	1	BLN	N/A	$1=True; $0=False
PID $9A Supported	3	6	1	BLN	N/A	$1=True; $0=False
PID $9B Supported	3	5	1	BLN	N/A	$1=True; $0=False
PID $9C Supported	3	4	1	BLN	N/A	$1=True; $0=False
PID $9D Supported	3	3	1	BLN	N/A	$1=True; $0=False
PID $9E Supported	3	2	1	BLN	N/A	$1=True; $0=False
PID $9F Supported	3	1	1	BLN	N/A	$1=True; $0=False
PID $A0 Supported	3	0	1	BLN	N/A	$1=True; $0=False

This PID indicates which legislated PIDs, within the range of $81 to $A0, are supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.85 AECR – PID $0081 - Engine Run Time for AECD #1 - #5

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0081	Engine Run Time for AECD #1 - #5	41	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Total run time with EI-AECD #5 active supported	0	4	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #4 active supported	0	3	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #3 active supported	0	2	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #2 active supported	0	1	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #1 active supported	0	0	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #1 Timer 1 active	1	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #1 Timer 2 active	5	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #2 Timer 1 active	9	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #2 Timer 2 active	13	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #3 Timer 1 active	17	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #3 Timer 2 active	21	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #4 Timer 1 active	25	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #4 Timer 2 active	29	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #5 Timer 1 active	33	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #5 Timer 2 active	37	7	32	UNM	0-4,294,967,295 sec	E = N * 1

AECDx_TIME1 shall display the total engine run time with Emission Increasing Auxiliary Emission Control Device #x active. AECDx_TIME1 shall increment while the engine is running with EI-AECD #x active. It shall freeze if the engine stalls. For EI-AECDs requiring only a single timer, Timer 1 shall be used to report the total engine run time for the EI-AECD. For EI-AECDs requiring two timers, Timer 1 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness up to but not including 75 percent of the maximum reduced emission control effectiveness of that EI-AECD.

AECDx_TIME2 shall display the total engine run time with Emission Increasing Auxiliary Emission Control Device #x active. AECDx_TIME2 shall increment while the engine is running with EI-AECD #x active. It shall freeze if the engine stalls. For EI-AECDs requiring only a single timer, Timer 2 shall be reported as "Not Available" using the value 4,294,967,295 seconds. For EI-AECDs requiring two timers, Timer 2 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness of 75 percent or more of the maximum reduced emission control effectiveness of that EI-AECD.

NOTE: Each number shall be reset to zero only when a non-volatile memory reset occurs (e.g., reprogramming event). Numbers may not be reset to zero under any other circumstances including when a scan tool (generic or enhanced) command to clear fault codes or reset KAM is received. If any of the individual counters reach the maximum value, all counters shall be divided by tw before any are incremented again to avoid overflow problems.

3.1.86 AECR – PID $0082 - Engine Run Time for AECD #6 - #10

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0082	Engine Run Time for AECD #6 - #10	41	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Total run time with EI-AECD #10 active supported	0	4	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #9 active supported	0	3	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #8 active supported	0	2	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #7 active supported	0	1	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #6 active supported	0	0	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #6 Timer 1 active	1	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #6 Timer 2 active	5	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #7 Timer 1 active	9	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #7 Timer 2 active	13	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #8 Timer 1 active	17	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #8 Timer 2 active	21	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #9 Timer 1 active	25	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #9 Timer 2 active	29	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #10 Timer 1 active	33	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #10 Timer 2 active	37	7	32	UNM	0-4,294,967,295 sec	E = N * 1

AECDx_TIME1 shall display the total engine run time with Emission Increasing Auxiliary Emission Control Device #x active. AECDx_TIME1 shall increment while the engine is running with EI-AECD #x active. It shall freeze if the engine stalls. For EI-AECDs requiring only a single timer, Timer 1 shall be used to report the total engine run time for the EI-AECD. For EI-AECDs requiring two timers, Timer 1 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness up to but not including 75 percent of the maximum reduced emission control effectiveness of that EI-AECD.

AECDx_TIME2 shall display the total engine run time with Emission Increasing Auxiliary Emission Control Device #x active. AECDx_TIME2 shall increment while the engine is running with EI-AECD #x active. It shall freeze if the engine stalls. For EI-AECDs requiring only a single timer, Timer 2 shall be reported as "Not Available" using the value 4,294,967,295 seconds. For EI-AECDs requiring two timers, Timer 2 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness of 75 percent or more of the maximum reduced emission control effectiveness of that EI-AECD.

3.1.87 NOXR – PID $0083– NOx Sensor

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0083	NOx Sensor	9	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved (bits shall be reported as ‘0’)	0	7	1			
Reserved (bits shall be reported as ‘0’)	0	6	1			
Reserved (bits shall be reported as ‘0’)	0	5	1			
Reserved (bits shall be reported as ‘0’)	0	4	1			
NOx Sensor Concentration Bank 2 Sensor 2 Supported	0	3	1	BLN	N/A	$1=True; $0=False
NOx Sensor Concentration Bank 2 Sensor 1 Supported	0	2	1	BLN	N/A	$1= True; $0=False
NOx Sensor Concentration Bank 1 Sensor 2 Supported	0	1	1	BLN	N/A	$1= True; $0=False
NOx Sensor Concentration Bank 1 Sensor 1 Supported	0	0	1	BLN	N/A	$1= True; $0=False
NOx Sensor Concentration Bank 1 Sensor 1	1	7	16	UNM	0 to 65535 ppm	E = N * 1
NOx Sensor Concentration Bank 1 Sensor 2	3	7	16	UNM	0 to 65535 ppm	E = N * 1
NOx Sensor Concentration Bank 2 Sensor 1	5	7	16	UNM	0 to 65535 ppm	E = N * 1
NOx Sensor Concentration Bank 2 Sensor 2	7	7	16	UNM	0 to 65535 ppm	E = N * 1

This PID shall indicate the information of NOx sensors supported in the exhaust stream and the amount of raw ppm measurement coming via CAN by NOx sensor 1/NOx sensor 2. The PID shall indicate 0 ppm when NOx sensors are not supported. If the data used to compute NOx concentration is obtained from a serial data signal, then, if the signal is not received, the value of PID $0083 NOx concentrations shall be set to $FFFF = 65535. . Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.88 SCRR – PID $0085 – NOx Control System

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0085	NOx Control System	10	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Minutes Run By The Engine While NOx Warning Mode Is Activated Supported	0	3	1	BLN	N/A	$1=True; $0=False
Reagent Tank Level Supported	0	2	1	BLN	N/A	$1=True; $0=False
Average Demanded Reagent Consumption Supported	0	1	1	BLN	N/A	$1=True; $0=False
Average Reagent Consumption Supported	0	0	1	BLN	N/A	$1=True; $0=False
Average Reagent Consumption	1	7	16	UNM	0 to 327.675 l/h	E = N * 0.005
Average Demanded Reagent Consumption	3	7	16	UNM	0 to 327.675 l/h	E = N * 0.005
Reagent Tank Level	5	7	8	UNM	0 to 100 %	E = N * 100 / 255
Total Run Time By The Engine While NOx Warning Mode Is Activated	6	7	32	UNM	0 to 4,294,967,295 sec	E = N * 1

This PID contains information for total engine run time while NOx warning mode is activated, reagent tank level, average demanded reagent consumption and average reagent consumption. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.89 SOTR – PID $0086 – Particulate Matter (PM) Sensor Data

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0086	Particulate Matter (PM) Sensor Data	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
PM Sensor Mass Concentration Bank 2 Sensor 1 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PM Sensor Mass Concentration Bank 1 Sensor 1 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PM Sensor Mass Concentration Bank 1 Sensor 1	1	7	16	UNM	0 to 819.1875 mg/m^3	E = N * 0.0125
PM Sensor Mass Concentration Bank 2 Sensor 1	3	7	16	UNM	0 to 819.1875 mg/m^3	E = N * 0.0125

PM Sensor Mass Concentration Bank 2 Sensor 1 Supported indicates the Bank 2 is configured with a particulate matter sensor which supports soot mass concentration measurement.

PM Sensor Mass Concentration Bank 1 Sensor 1 Supported indicates the Bank 1 is configured with a particulate matter sensor which supports soot mass concentration measurement.

PM Sensor Mass Concentration Bank 1 Sensor 1 contains the soot mass concentration measured by particulate matter sensor installed on Bank 1.

PM Sensor Mass Concentration Bank 2 Sensor 1 contains the soot mass concentration measured by particulate matter sensor installed on Bank 2.

3.1.90 MAPR – PID $0087 – Intake Manifold Absolute Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0087	Intake Manifold Absolute Pressure	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Intake Manifold Absolute Pressure B supported	0	1	1	BLN	N/A	$1=True; $0=False
Intake Manifold Absolute Pressure A supported	0	0	1	BLN	N/A	$1=True; $0=False
Intake Manifold Absolute Pressure A	1	7	16	UNM	0 - 2047.96875 kPa	E = N / 32
Intake Manifold Absolute Pressure B	3	7	16	UNM	0 - 2047.96875 kPa	E = N / 32

This PID contains the unfiltered engine intake manifold absolute pressure (MAP) as measured by up to two sensors. If a vehicle uses both a MAP and MAF sensor, both the MAP and MAF PIDs shall be supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.91 SCRR – PID $0088 – SCR Inducement System

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0088	SCR Inducement System	13	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
SCR Inducement System Active	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
SCR Inducement System Actual State NOx Emissions Too High	0	3	1	BLN	N/A	$1=True; $0=False
SCR Inducement System Actual State Deviation of Reagent Consumption	0	2	1	BLN	N/A	$1=True; $0=False
SCR Inducement System Actual State Incorrect Reagent	0	1	1	BLN	N/A	$1=True; $0=False
SCR Inducement System Actual State Reagent Level Too Low	0	0	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 20K History (10,000 – 20,000 km) NOx Emissions Too High	1	7	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 20K History (10,000 – 20,000 km) Deviation of Reagent Consumption	1	6	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 20K History (10,000 – 20,000 km) Incorrect Reagent	1	5	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 20K History (10,000 – 20,000 km) Reagent Level Too Low	1	4	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 10K History (0 – 10,000 km) NOx Emissions Too High	1	3	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 10K History (0 – 10,000 km) Deviation of Reagent Consumption	1	2	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 10K History (0 – 10,000 km) Incorrect Reagent	1	1	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 10K History (0 – 10,000 km) Reagent Level Too Low	1	0	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 40K History (30,000 – 40,000 km) NOx Emissions Too High	2	7	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 40K History (30,000 – 40,000 km) Deviation of Reagent Consumption	2	6	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 40K History (30,000 – 40,000 km) Incorrect Reagent	2	5	1	BLN	N/A	$1=True; $0=False
SCR Inducement System 40K History (30,000 – 40,000 km) Reagent Level Too Low	2	4	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 30K History (20,000 – 30,000 km) NOx Emissions Too High	2	3	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 30K History (20,000 – 30,000 km) Deviation of Reagent Consumption	2	2	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 30K History (20,000 – 30,000 km) Incorrect Reagent	2	1	1	BLN	N/A	$1=True; $0=False
SCR Inducement System State 30K History (20,000 – 30,000 km) Reagent Level Too Low	2	0	1	BLN	N/A	$1=True; $0=False

Distance Travelled While Inducement System Active In Current 10K Block (0 – 10,000 km)	3	7	16	UNM	0 to 65535 km	E = N * 1
Distance Travelled In Current 10 K Block (0 - 10,000 km Block)	5	7	16	UNM	0 to 65535 km	E = N * 1
Distance Travelled While Inducement System Active In 20K Block (10,000 – 20,000 km)	7	7	16	UNM	0 to 65535 km	E = N * 1
Distance Travelled While Inducement System Active In 30K Block (20,000 – 30,000 km)	9	7	16	UNM	0 to 65535 km	E = N * 1
Distance Travelled While Inducement System Active In 40K Block (30,000 – 40,000 km)	11	7	16	UNM	0 to 65535 km	E = N * 1

This PID contains information for SCR inducement system actual state, SCR inducement system state 10K history (0 – 10,000 km), SCR inducement system state 20K history (10,000 – 20,000 km), SCR inducement system state 30K history (20,000 – 30,000 km), SCR inducement system state 40K history (30,000 – 40,000 km), distance travelled while inducement system active in current 10K block (0 – 10,000 km), distance travelled in current 10 K block (0 - 10,000 km block), distance travelled while inducement system active in 20K block (10,000 – 20,000 km), distance travelled while inducement system active in 30K block (20,000 – 30,000 km) and distance travelled while inducement system active in 40K block (30,000 – 40,000 km). Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

MODIFICATION FOR NA APPLICATIONS:

- Status Bytes 0 – 2:
    - **Reagent Level Too Low** – This state will be true when the customer is at final inducement speed limited to 4 MPH. Rationale for selecting final inducement is that this is the last stage before vehicle is out of DEF after which the vehicle will exceed emissions.
    - **Incorrect Reagent** – This state will be true as soon as poor DEF quality is detected and will stay active until good DEF quality is detected.
    - **Deviation of Reagent Consumption** – Not supported as it is not required for OBD II
    - **NOx Emissions Too High** – This state will be true as long as the Tamper Warning Strategy is active. Tamper faults are defined as disconnects of sensors/actuators that would prevent the SCR system from dosing.
    - **Inducement System Active** – This state is true anytime one of the above conditions is true.

3.1.92 AECR – PID $0089 - Engine Run Time for AECD #11 - #15

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0089	Engine Run Time for AECD #11 - #15	41	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Total run time with EI-AECD #15 active supported	0	4	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #14 active supported	0	3	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #13 active supported	0	2	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #12 active supported	0	1	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #11 active supported	0	0	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #11 Timer 1 active	1	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #11 Timer 2 active	5	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #12 Timer 1 active	9	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #12 Timer 2 active	13	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #13 Timer 1 active	17	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #13 Timer 2 active	21	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #14 Timer 1 active	25	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #14 Timer 2 active	29	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #15 Timer 1 active	33	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #15 Timer 2 active	37	7	32	UNM	0-4,294,967,295 sec	E = N * 1

AECDx_TIME1 shall display the total engine run time with Emission Increasing Auxiliary Emission Control Device #x active. AECDx_TIME1 shall increment while the engine is running with EI-AECD #x active. It shall freeze if the engine stalls. For EI-AECDs requiring only a single timer, Timer 1 shall be used to report the total engine run time for the EI-AECD. For EI-AECDs requiring two timers, Timer 1 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness up to but not including 75 percent of the maximum reduced emission control effectiveness of that EI-AECD.

AECDx_TIME2 shall display the total engine run time with Emission Increasing Auxiliary Emission Control Device #x active. AECDx_TIME2 shall increment while the engine is running with EI-AECD #x active. It shall freeze if the engine stalls. For EI-AECDs requiring only a single timer, Timer 2 shall be reported as "Not Available" using the value 4,294,967,295 seconds. For EI-AECDs requiring two timers, Timer 2 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness of 75 percent or more of the maximum reduced emission control effectiveness of that EI-AECD.

NOTE: Each number shall be reset to zero only when a non-volatile memory reset occurs (e.g., reprogramming event). Numbers may not be reset to zero under any other circumstances including when a scan tool (generic or enhanced) command to clear fault codes or reset KAM is received. If any of the individual counters reach the maximum value, all counters shall be divided by tw before any are incremented again to avoid overflow problems.

3.1.93 AECR – PID $008A - Engine Run Time for AECD #15 - #20

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$008A	Engine Run Time for AECD #15 - #20	41	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Total run time with EI-AECD #20 active supported	0	4	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #19 active supported	0	3	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #18 active supported	0	2	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #17 active supported	0	1	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #16 active supported	0	0	1	BLN	N/A	$1=True; $0=False
Total run time with EI-AECD #16 Timer 1 active	1	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #16 Timer 2 active	5	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #17 Timer 1 active	9	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #17 Timer 2 active	13	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #18 Timer 1 active	17	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #18 Timer 2 active	21	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #19 Timer 1 active	25	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #19 Timer 2 active	29	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #20 Timer 1 active	33	7	32	UNM	0-4,294,967,295 sec	E = N * 1
Total run time with EI-AECD #20 Timer 2 active	37	7	32	UNM	0-4,294,967,295 sec	E = N * 1

AECDx_TIME1 shall display the total engine run time with Emission Increasing Auxiliary Emission Control Device #x active. AECDx_TIME1 shall increment while the engine is running with EI-AECD #x active. It shall freeze if the engine stalls. For EI-AECDs requiring only a single timer, Timer 1 shall be used to report the total engine run time for the EI-AECD. For EI-AECDs requiring two timers, Timer 1 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness up to but not including 75 percent of the maximum reduced emission control effectiveness of that EI-AECD.

AECDx_TIME2 shall display the total engine run time with Emission Increasing Auxiliary Emission Control Device #x active. AECDx_TIME2 shall increment while the engine is running with EI-AECD #x active. It shall freeze if the engine stalls. For EI-AECDs requiring only a single timer, Timer 2 shall be reported as "Not Available" using the value 4,294,967,295 seconds. For EI-AECDs requiring two timers, Timer 2 shall report the total engine hours when the EI-AECD is commanding reduced emission control effectiveness of 75 percent or more of the maximum reduced emission control effectiveness of that EI-AECD.

NOTE: Each number shall be reset to zero only when a non-volatile memory reset occurs (e.g., reprogramming event). Numbers may not be reset to zero under any other circumstances including when a scan tool (generic or enhanced) command to clear fault codes or reset KAM is received. If any of the individual counters reach the maximum value, all counters shall be divided by tw before any are incremented again to avoid overflow problems.

3.1.94 DPFR – PID $008B – Diesel Aftertreatment Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$008B	Diesel Aftertreatment Status	7	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Average Distance Between DPF Regens Supported	0	6	1	BLN	N/A	$1=True; $0=False
Average Time Between DPF Regens Supported	0	5	1	BLN	N/A	$1=True; $0=False
Normalized Trigger for DPF Regen Supported	0	4	1	BLN	N/A	$1=True; $0=False
NOx Adsorber Desulfurization in Progress Supported	0	3	1	BLN	N/A	$1=True; $0=False
NOx Adsorber Regen in Progress Supported	0	2	1	BLN	N/A	$1=True; $0=False
Diesel Particulate Filter (DPF) Regen Type Supported	0	1	1	BLN	N/A	$1=True; $0=False
Diesel Particulate Filter (DPF) Regen in Progress Supported	0	0	1	BLN	N/A	$1=True; $0=False
Reserved	1	7	1			
Reserved	1	6	1			
Reserved	1	5	1			
Reserved	1	4	1			
NOx Adsorber Desulfurization in Progress	1	3	1	BLN	N/A	$1=True; $0=False
NOx Adsorber Regen in Progress	1	2	1	BLN	N/A	$1=True; $0=False
Diesel Particulate Filter (DPF) Regen Type	1	1	1	ENM	N/A	$0=Passive DPF Regen; $1=Active DPF Regen
Diesel Particulate Filter (DPF) Regen in Progress	1	0	1	BLN	N/A	$1=True; $0=False
Normalized Trigger for DPF Regen	2	7	8	UNM	0 to 100%	E = N * 100 / 255
Average Time Between DPF Regens	4	7	16	UNM	0 to 65535 min	E = N * 1
Average Distance Between DPF Regens	6	7	16	UNM	0 to 65535 km	E = N * 1

This PID contains several signals associated with the diesel aftertreatment system. The content of each of these signals is described below. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

Average Distance Between DPF Regens Supported indicates the control system uses the Average Distance Between DPF Regens data parameter.

Average Time Between DPF Regens Supported indicates the control system uses the Average Time Between DPF Regens data parameter.

Normalized Trigger for DPF Regen Supported indicates the control system uses the Normalized Trigger for DPF Regen data parameter.

NOx Adsorber Desulfurization in Progress Supported indicates the control system uses the NOx Adsorber Desulfurization in Progress data parameter.

NOx Adsorber Regen in Progress Supported indicates the control system uses the NOx Adsorber Regen in Progress data parameter.

Diesel Particulate Filter (DPF) Regen Type Supported indicates the control system uses the Diesel Particulate Filter (DPF) Regen Type data parameter.

Diesel Particulate Filter (DPF) Regen in Progress Supported indicates the control system uses the Diesel Particulate Filter (DPF) Regen in Progress data parameter.

NOx Adsorber Desulfurization in Progress indicates the NOx adsorber desulfurization process is active.

NOx Adsorber Regen in Progress indicates the NOx adsorber desorption process is active.

Diesel Particulate Filter (DPF) Regen Type indicates whether a passive or active DPF regeneration is taking place.

Diesel Particulate Filter (DPF) Regen in Progress indicates the ECU has entered a special mode to increase the temperature in the exhaust system to burn the soot in the DPF. The ECU can make the decision either by algorithm, operator interaction or tool request.

Normalized Trigger for DPF Regen shall indicate the normalized DPF loading, time, distance, drive cycles or other criteria before the next DPF regen where 0% means the DPF is clean (a complete regen just occurred) and 100% means the DPF is ready to be regenerated. When there are multiple criteria to trigger a regen, the one that is closest to triggering the regen shall be displayed.
Note: A commanded DPF regen (Service $08, TID $02) shall be reflected in this data.

Average Time Between DPF Regens shall indicate the EWMA filtered time between successful, active triggered DPF regens. The weighting factor shall be chosen to produce a representative value after 6 regen cycles (~0.5).
Note: A commanded DPF regen (Service $08, TID $02) shall be reflected in this data.

Average Distance Between DPF Regens shall indicate the EWMA filtered distance between successful, active triggered DPF regens. The weighting factor shall be chosen to produce a representative value after 6 regen cycles (~0.5)
Note: A commanded DPF regen (Service $08, TID $02) shall be reflected in this data.

3.1.95 OXYR – PID $008C – Oxygen Sensors Concentration

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$008C	Oxygen Sensors Concentration	17	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
O2 Sensor Lambda Bank 2 Sensor 2 Supported	0	7	1	BLN	N/A	$1=True; $0=False
O2 Sensor Lambda Bank 2 Sensor 1 Supported	0	6	1	BLN	N/A	$1=True; $0=False
O2 Sensor Lambda Bank 1 Sensor 2 Supported	0	5	1	BLN	N/A	$1=True; $0=False
O2 Sensor Lambda Bank 1 Sensor 1 Supported	0	4	1	BLN	N/A	$1=True; $0=False
O2 Sensor Concentration Bank 2 Sensor 2 Supported	0	3	1	BLN	N/A	$1=True; $0=False
O2 Sensor Concentration Bank 2 Sensor 1 Supported	0	2	1	BLN	N/A	$1=True; $0=False
O2 Sensor Concentration Bank 1 Sensor 2 Supported	0	1	1	BLN	N/A	$1=True; $0=False
O2 Sensor Concentration Bank 1 Sensor 1 Supported	0	0	1	BLN	N/A	$1=True; $0=False
O2 Sensor Concentration Bank 1 Sensor 1	1	7	16	UNM	0 – 100 %	E = N * 100 / 65535
O2 Sensor Concentration Bank 1 Sensor 2	3	7	16	UNM	0 – 100 %	E = N * 100 / 65535
O2 Sensor Concentration Bank 2 Sensor 1	5	7	16	UNM	0 – 100 %	E = N * 100 / 65535
O2 Sensor Concentration Bank 2 Sensor 2	7	7	16	UNM	0 – 100 %	E = N * 100 / 65535
O2 Sensor Lambda Bank 1 Sensor 1	9	7	16	UNM	0 – 7.99 λ	E = N * 8 / 65535
O2 Sensor Lambda Bank 1 Sensor 2	11	7	16	UNM	0 – 7.99 λ	E = N * 8 / 65535
O2 Sensor Lambda Bank 2 Sensor 1	13	7	16	UNM	0 – 7.99 λ	E = N * 8 / 65535
O2 Sensor Lambda Bank 2 Sensor 2	15	7	16	UNM	0 – 7.99 λ	E = N * 8 / 65535

This PID contains several signals associated with linear or wide-ratio oxygen sensors on compression ignition vehicles that can be standalone sensors or part of the NOx sensor. The oxygen sensor outputs can be Lambda (typically 0 to 4 for a compression ignition engine) and/or oxygen concentration (typically 0 to 25%). The content of each of these signals is described below. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

O2 Sensor Lambda Bank 2 Sensor 2 Supported indicates if lambda measurement is supported at indicated position.

O2 Sensor Lambda Bank 2 Sensor 1 Supported indicates if lambda measurement is supported at indicated position.

O2 Sensor Lambda Bank 1 Sensor 2 Supported indicates if lambda measurement is supported at indicated position.

O2 Sensor Lambda Bank 1 Sensor 1 Supported indicates if lambda measurement is supported at indicated position.

O2 Sensor Concentration Bank 2 Sensor 2 Supported indicates if oxygen measurement is supported at indicated position.

O2 Sensor Concentration Bank 2 Sensor 1 Supported indicates if oxygen measurement is supported at indicated position.

O2 Sensor Concentration Bank 1 Sensor 2 Supported indicates if oxygen measurement is supported at indicated position.

O2 Sensor Concentration Bank 1 Sensor 1 Supported indicates if oxygen measurement is supported at indicated position.

O2 Sensor Concentration Bank 1 Sensor 1 indicates oxygen concentration at indicated location when the corresponding bit indicates that it is supported.

O2 Sensor Concentration Bank 1 Sensor 2 indicates oxygen concentration at indicated location when the corresponding bit indicates that it is supported.

O2 Sensor Concentration Bank 2 Sensor 1 indicates oxygen concentration at indicated location when the corresponding bit indicates that it is supported.

O2 Sensor Concentration Bank 2 Sensor 2 indicates oxygen concentration at indicated location when the corresponding bit indicates that it is supported.

O2 Sensor Lambda Bank 1 Sensor 1 indicates lambda value at indicated location when the corresponding bit indicates that it is supported.

O2 Sensor Lambda Bank 1 Sensor 2 indicates lambda value at indicated location when the corresponding bit indicates that it is supported.

O2 Sensor Lambda Bank 2 Sensor 1 indicates lambda value at indicated location when the corresponding bit indicates that it is supported.

O2 Sensor Lambda Bank 2 Sensor 2 indicates lambda value at indicated location when the corresponding bit indicates that it is supported.

3.1.96 ETQR – PID $008E – Engine Friction – Percent Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$008E	Engine Friction - Percent Torque	1	UNM	-125-130%	E = N * 1-125

This PID contains the friction torque of the engine. Friction Torque is the torque required to drive the engine alone as “fully equipped”. The data is transmitted as friction torque as a percent of engine reference torque. The friction percent torque value will not be less than zero.

NOTE:
A “fully equipped” engine is an engine equipped with accessories necessary to perform its intended service. This includes, but is not restricted to, the basic engine, including fuel, oil, and cooling pumps, plus intake air system, exhaust system, cooling system, alternator, starter, emissions, and noise control. Accessories which are not necessary for the operation of the engine, but may be engine mounted, are not considered part of a fully equipped engine. These items include, but are not restricted to, power steering pump systems, vacuum pumps, and compressor systems for air conditioning, brakes, and suspensions.

3.1.97 SOTR – PID $008F – Particulate Matter (PM) Sensor Output

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$008F	Particulate Matter (PM) Sensor Output	7	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
PM Sensor Signal Bank 2 Sensor 1 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PM Sensor Operating Status Bank 2 Sensor 1 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PM Sensor Signal Bank 1 Sensor 1 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PM Sensor Operating Status Bank 1 Sensor 1 Supported	0	0	1	BLN	N/A	$1=True; $0=False
Reserved	1	7	1			
Reserved	1	6	1			
Reserved	1	5	1			
Reserved	1	4	1			
Reserved	1	3	1			
Reserved	1	2	1			
PM Sensor Bank 1 Sensor 1 Regeneration Active	1	1	1	BLN	N/A	$1=True; $0=False
PM Sensor Bank 1 Sensor 1 Measurement Active	1	0	1	BLN	N/A	$1=True; $0=False
PM Sensor Normalized Output Value Bank 1 Sensor 1	2	7	16	UNM	-327.68 to 327.67 %	E = N * 0.01 – 327.68
Reserved	4	7	1			
Reserved	4	6	1			
Reserved	4	5	1			
Reserved	4	4	1			
Reserved	4	3	1			
Reserved	4	2	1			
PM Sensor Bank 2 Sensor 1 Regeneration Active	4	1	1	BLN	N/A	$1=True; $0=False
PM Sensor Bank 2 Sensor 1 Measurement Active	4	0	1	BLN	N/A	$1=True; $0=False
PM Sensor Normalized Output Value Bank 2 Sensor 1	5	7	16	UNM	-327.68 to 327.67 %	E = N * 0.01 – 327.68

PM Sensor Signal Bank 2 Sensor 1 Supported indicates the Bank 2 is configured with a particulate matter sensor which is based on soot collective measurement principle.

PM Sensor Operating Status Bank 2 Sensor 1 Supported indicates the Bank 2 is configured with a particulate matter sensor which supports different operating statuses.

PM Sensor Signal Bank 1 Sensor 1 Supported indicates the Bank 1 is configured with a particulate matter sensor which is based on soot collective measurement principle.

PM Sensor Operating Status Bank 1 Sensor 1 Supported indicates the Bank 1 is configured with a particulate matter sensor which supports different operating statuses.

PM Sensor Bank 1 Sensor 1 Regeneration Active indicates that the sensor is under regeneration process.

PM Sensor Bank 1 Sensor 1 Measurement Active indicates that the sensor is actively measuring soot.

PM Sensor Normalized Output Value Bank 1 Sensor 1 indicates the soot load level in respect to sensor regeneration need. 100% represents soot level where sensor regeneration is expected to start based on sensor design. 0% represents a sensor which is fully cleaned and regenerated.

PM Sensor Bank 2 Sensor 1 Regeneration Active indicates that the sensor is under regeneration process.

PM Sensor Bank 2 Sensor 1 Measurement Active indicates that the sensor is actively measuring soot.

PM Sensor Normalized Output Value Bank 2 Sensor 1 indicates the soot load level in respect to sensor regeneration need. 100% represents soot level where sensor regeneration is expected to start based on sensor design. 0% represents a sensor which is fully cleaned and regenerated.

3.1.98 EGTR – PID $0098 – Exhaust Gas Temperature (EGT) Bank 1 Sensors 5 - 8

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$0098	Exhaust Gas Temperature (EGT) Bank 1 Sensors 5 - 8	9	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
EGT Bank 1, Sensor 8 Supported	0	3	1	BLN	N/A	$1=True; $0=False
EGT Bank 1, Sensor 7 Supported	0	2	1	BLN	N/A	$1=True; $0=False
EGT Bank 1, Sensor 6 Supported	0	1	1	BLN	N/A	$1=True; $0=False
EGT Bank 1, Sensor 5 Supported	0	0	1	BLN	N/A	$1=True; $0=False
Exhaust Gas Temperature Bank 1, Sensor 5	1	7	16	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40
Exhaust Gas Temperature Bank 1, Sensor 6	3	7	16	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40
Exhaust Gas Temperature Bank 1, Sensor 7	5	7	16	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40
Exhaust Gas Temperature Bank 1, Sensor 8	7	7	16	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the undefaulted exhaust gas temperature measured by up to four sensors as follows. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

EGT Bank 1, Sensor 8 Supported indicates the system utilizes an exhaust gas temperature sensor in the Bank 1 Sensor 8 position.

EGT Bank 1, Sensor 7 Supported indicates the system utilizes an exhaust gas temperature sensor in the Bank 1 Sensor 7 position.

EGT Bank 1, Sensor 6 Supported indicates the system utilizes an exhaust gas temperature sensor in the Bank 1 Sensor 6 position.

EGT Bank 1, Sensor 5 Supported indicates the system utilizes an exhaust gas temperature sensor in the Bank 1 Sensor 5 position.

Exhaust Gas Temperature Bank 1, Sensor 5 shall display exhaust gas temperature for bank 1, sensor 5. Exhaust gas temperature may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

Exhaust Gas Temperature Bank 1, Sensor 6 shall display exhaust gas temperature for bank 1, sensor 6. Exhaust gas temperature may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

Exhaust Gas Temperature Bank 1, Sensor 7 shall display exhaust gas temperature for bank 1, sensor 7. Exhaust gas temperature may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

Exhaust Gas Temperature Bank 1, Sensor 8 shall display exhaust gas temperature for bank 1, sensor 8. Exhaust gas temperature may be obtained directly from a sensor, or may be inferred by the control strategy using other sensor inputs.

3.1.99 FULR – PID $009D– Engine Fuel Rate

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$009D	Engine Fuel Rate	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Rate	0	7	16	UNM	0 to 1310.7 g/s	E = N * 0.02
Vehicle Fuel Rate	2	7	16	UNM	0 to 1310.7 g/s	E = N * 0.02

This PID contains several signals:

- Fuel Rate that is the amount of fuel consumed only by the engine per unit of time (does not include fuel injected directly into the aftertreatment system), in grams per 1 second.
- Vehicle Fuel Rate that is the amount of fuel consumed by the engine and fuel injected directly into the aftertreatment system per unit of time, in grams per 1 second.

This PID shall indicate zero g/s when the engine is not running. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.100 EXFR – PID $009E – Engine Exhaust Flow Rate

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$009E	Engine Exhaust Flow Rate	2	UNM	0 to 13107.0 kg/h	E = 0.2 * N

This PID contains the measured/calculated exhaust gas mass flow upstream of the aftertreatment system in kilograms per hour. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5. The engine exhaust flow rate shall be calculated as the average exhaust flow rate over the last 1000 milliseconds prior to the time when it is updated, multiplied by 3600 in order to convert the units from seconds to hours. This PID shall be updated at a minimum rate of 1000 milliseconds.
NOTE: This PID shall indicate zero kg/h when the engine is not running.

3.1.101ODAR – PID $00A0 – Parameter Identifiers Supported - $A1 to $C0

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$00A0	Parameter Identifiers Supported - $A1 to $C0	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PID $A1 Supported	0	7	1	BLN	N/A	$1=True; $0=False
PID $A2 Supported	0	6	1	BLN	N/A	$1=True; $0=False
PID $A3 Supported	0	5	1	BLN	N/A	$1=True; $0=False
PID $A4 Supported	0	4	1	BLN	N/A	$1=True; $0=False
PID $A5 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PID $A6 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PID $A7 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PID $A8 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PID $A9 Supported	1	7	1	BLN	N/A	$1=True; $0=False
PID $AA Supported	1	6	1	BLN	N/A	$1=True; $0=False
PID $AB Supported	1	5	1	BLN	N/A	$1=True; $0=False
PID $AC Supported	1	4	1	BLN	N/A	$1=True; $0=False
PID $AD Supported	1	3	1	BLN	N/A	$1=True; $0=False
PID $AE Supported	1	2	1	BLN	N/A	$1=True; $0=False
PID $AF Supported	1	1	1	BLN	N/A	$1=True; $0=False
PID $B0 Supported	1	0	1	BLN	N/A	$1=True; $0=False
PID $B1 Supported	2	7	1	BLN	N/A	$1=True; $0=False
PID $B2 Supported	2	6	1	BLN	N/A	$1=True; $0=False
PID $B3 Supported	2	5	1	BLN	N/A	$1=True; $0=False
PID $B4 Supported	2	4	1	BLN	N/A	$1=True; $0=False
PID $B5 Supported	2	3	1	BLN	N/A	$1=True; $0=False
PID $B6 Supported	2	2	1	BLN	N/A	$1=True; $0=False
PID $B7 Supported	2	1	1	BLN	N/A	$1=True; $0=False
PID $B8 Supported	2	0	1	BLN	N/A	$1=True; $0=False
PID $B9 Supported	3	7	1	BLN	N/A	$1=True; $0=False
PID $BA Supported	3	6	1	BLN	N/A	$1=True; $0=False
PID $BB Supported	3	5	1	BLN	N/A	$1=True; $0=False
PID $BC Supported	3	4	1	BLN	N/A	$1=True; $0=False
PID $BD Supported	3	3	1	BLN	N/A	$1=True; $0=False
PID $BE Supported	3	2	1	BLN	N/A	$1=True; $0=False
PID $BF Supported	3	1	1	BLN	N/A	$1=True; $0=False
PID $C0 Supported	3	0	1	BLN	N/A	$1=True; $0=False

This PID indicates which legislated PIDs, within the range of $A1 to $C0, are supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.102FULR – PID $00A2 – Cylinder Fuel Rate

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$00A2	Cylinder Fuel Rate	2	UNM	0 to 2047.96875 mg/stroke	E = N * 0.03125

This PID displays the amount of fuel injected into the cylinder for combustion during the most recent intake stroke of the engine. When the engine is not running, the PID displays zero mg/stroke. Additional PID information and usage examples are detailed in SAE J1979.

3.1.103ODAR – PID $00C0 – Parameter Identifiers Supported - $C1 to $E0

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$00C0	Parameter Identifiers Supported - $C1 to $E0	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PID $C1 Supported	0	7	1	BLN	N/A	$1=True; $0=False
PID $C2 Supported	0	6	1	BLN	N/A	$1=True; $0=False
PID $C3 Supported	0	5	1	BLN	N/A	$1=True; $0=False
PID $C4 Supported	0	4	1	BLN	N/A	$1=True; $0=False
PID $C5 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PID $C6 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PID $C7 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PID $C8 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PID $C9 Supported	1	7	1	BLN	N/A	$1=True; $0=False
PID $CA Supported	1	6	1	BLN	N/A	$1=True; $0=False
PID $CB Supported	1	5	1	BLN	N/A	$1=True; $0=False
PID $CC Supported	1	4	1	BLN	N/A	$1=True; $0=False
PID $CD Supported	1	3	1	BLN	N/A	$1=True; $0=False
PID $CE Supported	1	2	1	BLN	N/A	$1=True; $0=False
PID $CF Supported	1	1	1	BLN	N/A	$1=True; $0=False
PID $D0 Supported	1	0	1	BLN	N/A	$1=True; $0=False
PID $D1 Supported	2	7	1	BLN	N/A	$1=True; $0=False
PID $D2 Supported	2	6	1	BLN	N/A	$1=True; $0=False
PID $D3 Supported	2	5	1	BLN	N/A	$1=True; $0=False
PID $D4 Supported	2	4	1	BLN	N/A	$1=True; $0=False
PID $D5 Supported	2	3	1	BLN	N/A	$1=True; $0=False
PID $D6 Supported	2	2	1	BLN	N/A	$1=True; $0=False
PID $D7 Supported	2	1	1	BLN	N/A	$1=True; $0=False
PID $D8 Supported	2	0	1	BLN	N/A	$1=True; $0=False
PID $D9 Supported	3	7	1	BLN	N/A	$1=True; $0=False
PID $DA Supported	3	6	1	BLN	N/A	$1=True; $0=False
PID $DB Supported	3	5	1	BLN	N/A	$1=True; $0=False
PID $DC Supported	3	4	1	BLN	N/A	$1=True; $0=False
PID $DD Supported	3	3	1	BLN	N/A	$1=True; $0=False
PID $DE Supported	3	2	1	BLN	N/A	$1=True; $0=False
PID $DF Supported	3	1	1	BLN	N/A	$1=True; $0=False
PID $E0 Supported	3	0	1	BLN	N/A	$1=True; $0=False

This PID indicates which legislated PIDs, within the range of $C1 to $E0, are supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.1.104ODAR – PID $00E0 – Parameter Identifiers Supported - $E1 to $FF

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$00E0	Parameter Identifiers Supported - $E1 to $FF	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PID $E1 Supported	0	7	1	BLN	N/A	$1=True; $0=False
PID $E2 Supported	0	6	1	BLN	N/A	$1=True; $0=False
PID $E3 Supported	0	5	1	BLN	N/A	$1=True; $0=False
PID $E4 Supported	0	4	1	BLN	N/A	$1=True; $0=False
PID $E5 Supported	0	3	1	BLN	N/A	$1=True; $0=False
PID $E6 Supported	0	2	1	BLN	N/A	$1=True; $0=False
PID $E7 Supported	0	1	1	BLN	N/A	$1=True; $0=False
PID $E8 Supported	0	0	1	BLN	N/A	$1=True; $0=False
PID $E9 Supported	1	7	1	BLN	N/A	$1=True; $0=False
PID $EA Supported	1	6	1	BLN	N/A	$1=True; $0=False
PID $EB Supported	1	5	1	BLN	N/A	$1=True; $0=False
PID $EC Supported	1	4	1	BLN	N/A	$1=True; $0=False
PID $ED Supported	1	3	1	BLN	N/A	$1=True; $0=False
PID $EE Supported	1	2	1	BLN	N/A	$1=True; $0=False
PID $EF Supported	1	1	1	BLN	N/A	$1=True; $0=False
PID $F0 Supported	1	0	1	BLN	N/A	$1=True; $0=False
PID $F1 Supported	2	7	1	BLN	N/A	$1=True; $0=False
PID $F2 Supported	2	6	1	BLN	N/A	$1=True; $0=False
PID $F3 Supported	2	5	1	BLN	N/A	$1=True; $0=False
PID $F4 Supported	2	4	1	BLN	N/A	$1=True; $0=False
PID $F5 Supported	2	3	1	BLN	N/A	$1=True; $0=False
PID $F6 Supported	2	2	1	BLN	N/A	$1=True; $0=False
PID $F7 Supported	2	1	1	BLN	N/A	$1=True; $0=False
PID $F8 Supported	2	0	1	BLN	N/A	$1=True; $0=False
PID $F9 Supported	3	7	1	BLN	N/A	$1=True; $0=False
PID $FA Supported	3	6	1	BLN	N/A	$1=True; $0=False
PID $FB Supported	3	5	1	BLN	N/A	$1=True; $0=False
PID $FC Supported	3	4	1	BLN	N/A	$1=True; $0=False
PID $FD Supported	3	3	1	BLN	N/A	$1=True; $0=False
PID $FE Supported	3	2	1	BLN	N/A	$1=True; $0=False
PID $FF Supported	3	1	1	BLN	N/A	$1=True; $0=False
Reserved	3	0	1			

This PID indicates which legislated PIDs, within the range of $E1 to $FF, are supported. Additional PID information and usage examples are detailed in SAE J1979 and ISO 15031-5.

3.2 Corporate Standard PIDs

The corporate standard PIDs below are also specified within [5]. In the event of a conflict between this document and [5], the former shall take precedence.

3.2.1 RBSR – PID $12DF – Secondary Processor Reset Source

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$12DF	Secondary Processor Reset Source	1	ENM	N/A	$00=Power up $03=Internal watchdog $04=Unhandled exception $05= Secondary forced a main running reset $06=Main experienced a running reset. $07-FF=Controller-specific exception

This PID contains the reset source for the latest reset in the secondary processor.

3.2.2 CCR – PID $1528 – Air Conditioning Inhibit Reason History 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1529	Air Conditioning Inhibit Reason History 1	1	ENM	N/A	$00=None; $01=AC Protection – High Side Fluid Pressure; $02=AC Protection – Engine Speed; $03=AC Protection – Battery Voltage; $04=Engine Stall Prevention; $05=Engine Speed Stabilization; $06=Wide Open Throttle; $07=Vehicle Launch Performance; $08=High Coolant Temperature; $09=AC Relay Fault $0A = Hybrid Request $0B= AutoStop Request $0C=High Power Electronics Coolant Temperature; $0D=High Transmission Fluid Temperature; $0E=Full Pedal Power Limit Applied; $0F=Airbag deployed; $10=Battery Power Limit Exceeded; $11=Powertrain Diagnostics; $12=AC Compressor Fault; $13=Serial Communication Fault

This PID contains one of the last two reasons that powertrain has inhibited the air conditioning compressor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. If the most recent reason in the history buffer is the same as the current reason, the history buffer is not updated. If powertrain has already commanded inhibit of the air conditioning compressor for a reason other than a platform request, and a new inhibit reason exists, the new reason is placed in the history buffer.

3.2.3 ACCR – PID $1529 – Air Conditioning Inhibit Reason History 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1529	Air Conditioning Inhibit Reason History 2	1	ENM	N/A	$00=None; $01=AC Protection – High Side Fluid Pressure; $02=AC Protection – Engine Speed; $03=AC Protection – Battery Voltage; $04=Engine Stall Prevention; $05=Engine Speed Stabilization; $06=Wide Open Throttle; $07=Vehicle Launch Performance; $08=High Coolant Temperature; $09=AC Relay Fault $0A = Hybrid Request $0B= AutoStop Request $0C=High Power Electronics Coolant Temperature; $0D=High Transmission Fluid Temperature; $0E=Full Pedal Power Limit Applied; $0F=Airbag deployed; $10=Battery Power Limit Exceeded; $11=Powertrain Diagnostics; $12=AC Compressor Fault; $13=Serial Communication Fault

This PID contains one of the last two reasons that powertrain has inhibited the air conditioning compressor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. If the most recent reason in the history buffer is the same as the current reason, the history buffer is not updated. If powertrain has already commanded inhibit of the air conditioning compressor for a reason other than a platform request, and a new inhibit reason exists, the new reason is placed in the history buffer.

3.2.4 IFRR – PID $169E- Filtered MAF Sensor B Residual

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$169E	Filtered MAF Sensor B Residual	2	SNM	-512 - 511.984375 g/s	E = N / 64

This PID contains the filtered MAF sensor B residual (the difference between the mass airflow sensor B and the mass airflow predicted from the throttle model) used within the intake flow rationality diagnostic.

MAF sensor B is associated with engine bank 2.

3.2.5 TFTR – PID $1940 – Transmission Oil Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1940	Transmission Oil Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the undefaulted, unfiltered transmission oil / fluid temperature measured by a sensor. In automatic transmission applications, this PID is based upon the transmission oil temperature sensor input to the TCM. In manual transmission applications with a transmission oil temperature sensor, this PID is based upon the transmission oil temperature sensor input to the ECM. (This PID will display a defaulted temperature value when populated by serial data. This only applies to controllers not directly wired to a sensor.)

3.2.6 TISR – PID $1941 – Transmission Input Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1941	Transmission Input Speed	2	UNM	0 - 8191.875 rpm	E = N * 0.125

This PID contains the transmission input speed (the speed at the input of the gear box) determined from the speed measured by the transmission input speed (TIS) sensor with the application of factors to account for the location of the sensor (e.g., before the sprocket, after the overdrive, etc.). The source of this processed speed can change when faults are detected, so that a speed will be provided as long as another source is available, such as transmission output speed.

3.2.7 TOSR – PID $1942 – Transmission Output Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1942	Transmission Output Speed	2	UNM	0 - 8191.875 rpm	E = N * 0.125

This PID contains the transmission output speed (the speed at the output of the gear box) determined from the speed measured by the transmission output speed (TOS) sensor with the application of factors to account for the location of the sensor (e.g., after the gear box, after the final chain, after the transfer case, after the final drive, etc.). The source of this processed speed can change when faults are detected, so that a speed will be provided as long as another source is available, such as ABS wheel speeds. In applications that include both a transmission output speed sensor and a transfer case speed sensor, this PID will apply specifically to the output of the transmission.

3.2.8 SPRR – PID $194B – Transmission 4-5 Upshift Time Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$194B	Transmission 4-5 Upshift Time Error	1	SNM	-3.2 - 3.175 s	E = N * 0.025

This PID contains the difference between the desired and actual shift times for the most recent transmission 4-5 upshift, which is used to modify the transmission 4-5 upshift pressure adapts.

3.2.9 SHTR – PID $194C – Transmission 4-5 Upshift Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$194C	Transmission 4-5 Upshift Time	1	UNM	0 - 6.375 s	E = N * 0.025

This PID contains the duration of the ratio change portion of the most recent transmission 4-5 upshift.

3.2.10 TRGR – PID $194F – Transmission PRNDL Range

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$194F	Transmission PRNDL Range	1	ENM	N/A	$00=Illegal; $01=Park; $02=Reverse; $04=Neutral; $06=Drive 6; $08=Drive 5; $10=Drive 4; $20=Drive 3; $40=Drive 2; $80=Drive 1; $81=Drive 7; $82=Drive 8; $FF=Unknown

This PID contains the transmission range determined directly from the decoding of the PRNDL A, B, C, P and S inputs that are based upon either an internal mode switch (IMS) or a neutral safety backup (NSBU) switch.

3.2.11 TRGR – PID $1951 – Transmission Range Input Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1951	Transmission Range Input Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PRNDL Inputs Illegal	0	7	1	BLN	N/A	$1=True; $0=False
PRNDL P Input Active	0	6	1	BLN	N/A	$1=True; $0=False
PRNDL C Input Active	0	5	1	BLN	N/A	$1=True; $0=False
PRNDL B Input Active	0	4	1	BLN	N/A	$1=True; $0=False
PRNDL A Input Active	0	3	1	BLN	N/A	$1=True; $0=False
PSM C Input Active	0	2	1	BLN	N/A	$1=True; $0=False
PSM B Input Active	0	1	1	BLN	N/A	$1=True; $0=False
PSM A Input Active	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains several signals associated with the status of the transmission range inputs. The content of each of these signals is described below.

PSM A Input Active, PSM B Input Active, and PSM C Input Active indicate the states of the pressure switch manifold (PSM) inputs. In drive reverse indicator (DRI) applications, only the PSM A Input Active and PSM B Input Active signals are supported, and a value of False will be provided for the signal PSM C Input Active. These bits only apply to freewheel automatic transmissions and do not apply to clutch-to-clutch transmissions which only use internal mode switch (IMS) hardware.

PRNDL A Input Active, PRNDL B Input Active, PRNDL C Input Active, and PRNDL P Input Active indicate the states of the PRNDL A, B, C, and P inputs, respectively, that are based upon either an internal mode switch (IMS) or a neutral safety backup (NSBU) switch.

PRNDL Inputs Illegal indicates that the decoding of the PRNDL A, B, C, and P inputs resulted in an illegal transmission range.

3.2.12 TRTR – PID $195B – Diagnostic Transmission Ratio

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$195B	Diagnostic Transmission Ratio	2	UNM	0 - 7.999877929688	E = N / 8192

This PID contains the undefaulted ratio of transmission input speed (the speed at the input of the gear box) to transmission output speed (the speed at the output of the gear box). If the transmission input speed (TIS) sensor is located after the overdrive, the overdrive is not included in this ratio.

3.2.13 FWDR – PID $195D – Transfer Case Ratio

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$195D	Transfer Case Ratio	2	UNM	0 - 15.99975585938	E = N / 4096

This PID contains the estimated transfer case ratio for applications that are equipped with a transfer case capable of providing range selection (e.g., four wheel drive low).

3.2.14 TCCR – PID $1970 – Torque Converter Clutch PWM Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1970	Torque Converter Clutch PWM Duty Cycle	1	UNM	0 - 99.609375 %	E = N * 100 / 256

This PID contains the commanded duty cycle for the torque converter clutch (TCC) PWM output determined as a function of TCC capacity.

3.2.15 TPCR – PID $1972 – Transmission Line Pressure PWM Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1972	Transmission Line Pressure PWM Duty Cycle	1	UNM	0 - 99.609375 %	E = N * 100 / 256

This PID contains the commanded duty cycle for the transmission line pressure PWM output determined as a function of the error between the desired and actual transmission line pressure currents.

3.2.16 TCCR – PID $1975 – Transmission Component Slipping Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1975	Transmission Component Slipping Counter	1	UNM	0 - 255 counts	E = N * 1

This PID contains the number of times that excessive torque converter clutch (TCC) slip has been detected when TCC is commanded on (on mode or lock on mode) during the current test interval for the transmission component slipping diagnostic (DTC P1870).

3.2.17 TCCR – PID $1991 – Torque Converter Clutch Slip

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1991	Torque Converter Clutch Slip	2	SNM	-4096 - 4095.875 rpm	E = N * 0.125

This PID contains the torque converter clutch (TCC) slip determined as the difference between the speed at the input of the torque converter (engine speed) and the speed at the output of the torque converter (turbine speed).

3.2.18 SHTR – PID $1992 – Transmission Upshift Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1992	Transmission Upshift Time	1	UNM	0 - 6.375 s	E = N * 0.025

This PID contains the duration of the ratio change portion of the most recent transmission upshift.

3.2.19 SHTR – PID $1993 – Transmission 1-2 Upshift Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1993	Transmission 1-2 Upshift Time	1	UNM	0 - 6.375 s	E = N * 0.025

This PID contains the duration of the ratio change portion of the most recent transmission 1-2 upshift.

3.2.20 SHTR – PID $1994 – Transmission 2-3 Upshift Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1994	Transmission 2-3 Upshift Time	1	UNM	0 - 6.375 s	E = N * 0.025

This PID contains the duration of the ratio change portion of the most recent transmission 2-3 upshift.

3.2.21 SHTR – PID $1995 – Transmission 3-4 Upshift Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1995	Transmission 3-4 Upshift Time	1	UNM	0 - 6.375 s	E = N * 0.025

This PID contains the duration of the ratio change portion of the most recent transmission 3-4 upshift.

3.2.22 SPRR – PID $1996 – Transmission Upshift Time Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1996	Transmission Upshift Time Error	1	SNM	-3.2 - 3.175 s	E = N * 0.025

This PID contains the difference between the desired and actual shift times for the most recent transmission upshift, which is used to modify the transmission upshift pressure adapts.

3.2.23 SPRR – PID $1997 – Transmission 1-2 Upshift Time Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1997	Transmission 1-2 Upshift Time Error	1	SNM	-3.2 - 3.175 s	E = N * 0.025

This PID contains the difference between the desired and actual shift times for the most recent transmission 1-2 upshift, which is used to modify the transmission 1-2 upshift pressure adapts.

3.2.24 SPRR – PID $1998 – Transmission 2-3 Upshift Time Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1998	Transmission 2-3 Upshift Time Error	1	SNM	-3.2 - 3.175 s	E = N * 0.025

This PID contains the difference between the desired and actual shift times for the most recent transmission 2-3 upshift, which is used to modify the transmission 2-3 upshift pressure adapts.

3.2.25 SPRR – PID $1999 – Transmission 3-4 Upshift Time Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1999	Transmission 3-4 Upshift Time Error	1	SNM	-3.2 - 3.175 s	E = N * 0.025

This PID contains the difference between the desired and actual shift times for the most recent transmission 3-4 upshift, which is used to modify the transmission 3-4 upshift pressure adapts.

3.2.26 SHPR/HSER – PID $199A – Transmission Commanded Gear or Mode

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$199A	Transmission Commanded Gear or Mode	1	ENM	N/A	$00=Neutral; $01=First Gear; $02=Second Gear; $03=Third Gear; $04=Fourth Gear; $05=Fifth Gear; $06=Sixth Gear; $07=Seventh Gear; $08=Eighth Gear; … $41=EV1; $42=EV2; $43=EV3; $44=EV4

This PID contains the currently commanded transmission gear or mode. The enumeration reported by this PID shall change state at the beginning of a transmission gear or mode shift event.

3.2.27 SPRR – PID $199B – Transmission Upshift Pressure Adapt Cell Index

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$199B	Transmission Upshift Pressure Adapt Cell Index	1	UNM	0 - 255	E = N * 1

This PID contains the index of the entry within the array of transmission upshift pressure adapts that is closest to the shift torque for the most recent transmission upshift.

3.2.28 TPCR – PID $199E– Transmission Line Pressure Actual Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$199E	Transmission Line Pressure Actual Current	1	UNM	0 - 4.98046875 A	E = N * 5 / 256

This PID contains the actual (measured) transmission line pressure current.

3.2.29 TPCR – PID $199F – Transmission Line Pressure Desired Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$199F	Transmission Line Pressure Desired Current	1	UNM	0 - 4.98046875 A	E = N * 5 / 256

This PID contains the desired transmission line pressure current determined from the desired transmission line pressure and other criteria.

3.2.30 TRTR/HSER – PID $19A1 – Transmission Gear Ratio

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$19A1	Transmission Gear Ratio	1	UNM	0 - 3.984375	E = N / 64

This PID contains the ratio of transmission input speed (the speed at the input of the gear box) to transmission output speed (the speed at the output of the gear box).

3.2.31 TBNR – PID $19D4 – Turbine Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$19D4	Turbine Speed	2	UNM	0 - 8191.875 rpm	E = N * 0.125

This PID contains the turbine speed (the speed at the output of the torque converter and the input of the sprocket) determined from the division of the transmission input speed (the speed at the output of sprocket and the input to the gear box) by the sprocket ratio (the ratio of the number of teeth of the drive sprocket to the number of teeth of the driven sprocket). If the transmission is not equipped with an input chain (e.g., a rear wheel drive transmission), the sprocket ratio is unity.

3.2.32 TRTR – PID $1A01 – Torque Converter Speed Ratio

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1A01	Torque Converter Speed Ratio	2	UNM	0 - 7.999877929688	E = N / 8192

This PID contains the ratio of the speed at the output of the torque converter (turbine speed) to the speed at the input of the torque converter (engine speed).

3.2.33 TCCR – PID $1A18 – Torque Converter Clutch Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1A18	Torque Converter Clutch Pressure	2	UNM	0 - 262140 kPa	E = N * 4

This PID contains the desired torque converter clutch (TCC) pressure based upon the current operating mode of the electronically controlled converter clutch (ECCC) algorithm.

3.2.34 TRTR – PID $1A1F – Transmission Clutch Slip

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1A1F	Transmission Clutch Slip	2	SNM	-4096.0 - 4095.875 rpm	E = N * 0.125

This PID contains the transmission clutch slip determined as the difference between transmission input speed (the speed at the input of the gear box) and the product of transmission output speed (the speed at the output of the gear box) and the calibrated gear box ratio for the current commanded gear. If the transmission input speed (TIS) sensor is located after the overdrive, and the current commanded gear is fourth gear, the transmission clutch slip is determined as the difference between the speed after the overdrive and the product of the transmission output speed and the calibrated gear box ratio for third gear.

3.2.35 TCCR – PID $1A26 – Torque Converter Clutch Slip Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1A26	Torque Converter Clutch Slip Error	2	SNM	-4096 - 4095.875 rpm	E = N * 0.125

This PID contains the torque converter clutch (TCC) slip error determined as the difference between the absolute value of TCC slip and the TCC reference slip.

3.2.36 ETQR – PID $1A2D – Engine Actual Steady State Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1A2D	Engine Actual Steady State Torque	2	SNM	-8192 - 8191.75 Nm	E = N * 0.25

This PID contains the engine torque based upon the airflow through the engine and the torque losses due to spark retard and fuel shutoff. Since this is a steady state engine torque, it does not include the inertial component of engine torque.

3.2.37 TRTR – PID $1A88 – Transmission Modeled Gear Ratio

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1A88	Transmission Modeled Gear Ratio	2	UNM	0 - 15.99975585938	E = N / 4096

This PID contains the ratio of the modeled speed at the input of the gear box (difference between engine speed and the modeled torque converter slip multiplied by the sprocket ratio) to transmission output speed (the speed at the output of the gear box) for transmissions that do not have a transmission input speed (TIS) sensor.

3.2.38 EROR – PID $ 1AFE– Powertrain Relay Voltage 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1AFE	Powertrain Relay Voltage 2	1	UNM	0 to 25.5V	E = N * 0.1

This PID contains the voltage from the second input to the controller from the powertrain relay.

3.2.39 AXPR – PID $1C18 – Auxiliary Transmission Pump Relay Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1C18	Auxiliary Transmission Pump Relay Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Auxiliary Transmission Pump Relay Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Auxiliary Transmission Pump Relay Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Auxiliary Transmission Pump Relay Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Auxiliary Transmission Pump Relay Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the auxiliary transmission pump relay output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2798, P2796, and P2799, respectively.

3.2.40 LUBR – PID $1C1A – Variable Displacement Oil Pump Passive Diagnostic Test Events

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1C1A	Variable Displacement Oil Pump Passive Diagnostic Test Events	2	UNM	0 - 65535 counts	E = N * 1

This PID contains the number of times that passive test has been activated (i.e., has transitioned from one control state to the other control state when the diagnostic is enabled) during the current key cycle.

3.2.41 EOPR – PID $1C1B – Filtered Engine Oil Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1C1B	Filtered Engine Oil Pressure	1	UNM	0 to 1020 kPa	E = N * 4

This PID contains the filtered engine oil gauge pressure used by several control and diagnostic functions within the ECM.

3.2.42 VSDR – PID $1C47 – Low Voltage Circuit Commanded Voltage Setpoint

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1C47	Low Voltage Circuit Commanded Voltage Setpoint	1	UNM	0 to 25.5 V	E = N * 0.1

This PID contains the target voltage for the low voltage circuit (buck mode).

3.2.43 EODR – PID $1D41 – Event Rolling Counter Highway

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1D41	Event Rolling Counter Highway	1	UNM	0 - 255 counts	E = N * 1

This PID contains a rolling count of oil level debounce events for the highway calibration.

3.2.44 EODR – PID $1D42 – Event Rolling Counter Country Road

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1D42	Event Rolling Counter Country Road	1	UNM	0 - 255 counts	E = N * 1

This PID contains a rolling count of oil level debounce events for the country road calibration.

3.2.45 EODR – PID $1D43 – Event Rolling Counter City

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1D43	Event Rolling Counter City	1	UNM	0 - 255 counts	E = N * 1

This PID contains a rolling count of oil level debounce events for the city calibration.

3.2.46 EODR – PID $1D44 – Distance ECU at last Oil Level Warning

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1D44	Distance ECU at last Oil Level Warning	2	UNM	0 - 327675 km	E = N * 5

This PID contains the accumulated distance driven (odometer reading) at the last Oil Level Warning.

3.2.47 EOPR – PID $1D45 – Distance ECU at last Oil Pressure Warning

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1D45	Distance ECU at last Oil Pressure Warning	2	UNM	0 - 327675 km	E = N * 5

This PID contains the accumulated distance driven (odometer reading) at the last Oil Pressure Warning.

3.2.48 ARPR -- PID $1F02 – Alternative Fuel Rail Absolute Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F02	Alternative Fuel Rail Absolute Pressure	2	UNM	0 to 2047.96875 kPa	E = N / 32

This PID provides the undefaulted, unfiltered measured alternative fuel rail pressure at the engine read by an absolute pressure sensor.

3.2.49 ARPR -- PID $1F03 – Alternative Fuel Rail Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F03	Alternative Fuel Rail Pressure	2	UNM	0 to 2047.96875 kPa	E = N / 32

This PID provides the fuel rail pressure calculated as the difference between alternative fuel rail absolute pressure and ambient air pressure (barometric pressure). (Gauge = Absolute - Ambient)

3.2.50 ARPR -- PID $1F04 – Alternative Fuel Rail Pressure A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F04	Alternative Fuel Rail Pressure A/D	1	UNM	0 to100% 5 Volts	E = N * 100 / 255

This PID provides the alternative fuel rail pressure sensor analog input as a percentage of its reference voltage.

3.2.51 FISR -- PID $1F05 – Alternative Fuel Shutoff Valve 1 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F05	Alternative Fuel Shutoff Valve 1 Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Shutoff Valve 1 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Alternative Fuel Shutoff Valve 1 Short to Ground Test Status	0	5	1	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Alternative Fuel Shutoff Valve 1 Open Circuit Test Status	0	3	1	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Alternative Fuel Shutoff Valve 1 Short to Power Test Status	0	1	1	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID provides the commanded state of the alternative fuel shutoff valve 1 output (the output is considered to be commanded on if the shutoff valve is enabled while the engine is running or cranking) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0006, P0005, and P0007, respectively. The Test Status bits shall report the fault status directly from the electrical interface for the driver, independent of DTC reporting.

3.2.52 HPSR -- PID $1F06 – Alternative Fuel Shutoff Valve 2 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F06	Alternative Fuel Shutoff Valve 2 Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Shutoff Valve 2 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Alternative Fuel Shutoff Valve 2 Short to Ground Test Status	0	5	1	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Alternative Fuel Shutoff Valve 2 Open Circuit Test Status	0	3	1	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Alternative Fuel Shutoff Valve 2 Short to Power Test Status	0	1	1	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID provides the commanded state of the alternative fuel shutoff valve 2 output (the output is considered to be commanded on if the shutoff valve is enabled while the engine is running or cranking) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2666, P2665, and P2667, respectively. The Test Status bits shall report the fault status directly from the electrical interface for the driver, independent of DTC reporting.

3.2.53 HPSR -- PID $1F07 – Alternative Fuel Shutoff Valve 2 Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F07	Alternative Fuel Shutoff Valve 2 Duty Cycle	1	UNM	0 to100%	E = N * 100 / 255

This PID provides the commanded duty cycle for the alternative fuel shutoff valve 2.

3.2.54 FLVR – PID $1F0B – Alternative Fuel Level Sensor A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F0B	Alternative Fuel Level Sensor A/D	1	UNM	0 to 100 % 5V	E = N * 100 / 255

This PID contains the unfiltered alternative fuel level sensor analog input for the primary alternative fuel tank as a percentage of its reference voltage.

3.2.55 FULR – PID $1F0C – Alternative Fuel Injector Pulse Width Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F0C	Alternative Fuel Injector Pulse Width Bank 1	2	UNM	0 to 65.535 ms	E = N / 1000

This PID contains the alternative fuel injector pulse width (on time) commanded to each cylinder on bank 1.

3.2.56 FULR – PID $1F0D – Alternative Fuel Injector Pulse Width Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$1F0D	Alternative Fuel Injector Pulse Width Bank 2	2	UNM	0 to 65.535 ms	E = N / 1000

This PID contains the alternative fuel injector pulse width (on time) commanded to each cylinder on bank 2.

LAA NOTE – these PIDs are defined and used in existing ‘supplier’ FICM applications, but would not be used in S53 applications in support of a PID design that captures hi and lo side circuit status:

LAA NOTE – request for these PIDs being dropped in support of a PID design that captures hi and lo side circuit status:

LAA NOTE: Alternative Fuel Shorted Circuit DTCs do not yet exist. If the ‘High Side Open’ DTCs listed below cannot be re-assigned, new DTCs will need to be specified. A change request is pending to replace the ‘High Side Open’ reporting requirements to ‘Shorted Circuit’ detection for the S53.

3.2.57 PISR – PID $2039 – Processor Integrity Fault Reason Code (1-32) - Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2039	Processor Integrity Fault Reason Code (1-32) - Current	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
MissingMotorDutyCycle Fault	0	2	1	BLN	N/A	$1=True; $0=False
DMA transfer test	0	1	1	BLN	N/A	$1=True; $0=False
RAM ECC Circuit Test	0	0	1	BLN	N/A	$1=True; $0=False
Flash ECC Circuit Test	1	7	1	BLN	N/A	$1=True; $0=False
Run/Crank Correlation Fault between Main and Secondary Processors	1	6	1	BLN	N/A	$1=True; $0=False
Internal Performance Fault Reported by ETC Subsystem	1	5	1	BLN	N/A	$1=True; $0=False
MainTPU_Flt	1	4	1	BLN	N/A	$1=True; $0=False
External Watchdog Fault	1	3	1	BLN	N/A	$1=True; $0=False
Main Analog-to-Digital Converter Fault	1	2	1	BLN	N/A	$1=True; $0=False
Main Stack Fault	1	1	1	BLN	N/A	$1=True; $0=False
Main Configuration Register Fault	1	0	1	BLN	N/A	$1=True; $0=False
Main ALU Fault	2	7	1	BLN	N/A	$1=True; $0=False
Main ROM Diagnostic Completion Time Fault	2	6	1	BLN	N/A	$1=True; $0=False
Main 12.5 ms Task Sequence Fault	2	5	1	BLN	N/A	$1=True; $0=False
Main Clock Fault	2	4	1	BLN	N/A	$1=True; $0=False
Main Min Learn Corruption Fault	2	3	1	BLN	N/A	$1=True; $0=False
Secondary Detected MAIN Processor Fault	2	2	1	BLN	N/A	$1=True; $0=False
Secondary Configuration Register Fault	2	1	1	BLN	N/A	$1=True; $0=False
Secondary ALU Fault	2	0	1	BLN	N/A	$1=True; $0=False
Main Detected Seed Received In Wrong Order	3	7	1	BLN	N/A	$1=True; $0=False
Main Detected Seed & Key Timeout	3	6	1	BLN	N/A	$1=True; $0=False
Secondary Received Incorrect Keys	3	5	1	BLN	N/A	$1=True; $0=False
Secondary Fails To Take Remedial Action	3	4	1	BLN	N/A	$1=True; $0=False
Secondary Not Running Seed & Key Test	3	3	1	BLN	N/A	$1=True; $0=False
Secondary Stack Fault	3	2	1	BLN	N/A	$1=True; $0=False
Secondary Detected SPI Fault	3	1	1	BLN	N/A	$1=True; $0=False
MAIN Detected SPI Fault	3	0	1	BLN	N/A	$1=True; $0=False

This PID contains the source of the P0606 Diagnostic Trouble Code.

3.2.58 PISR – PID $203A – Processor Security Fault Reason - Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$203A	Processor Security Fault Reason - Current	2	ENM	N/A	$01 = DTMR Parallel Path Fault; $02 = ETMR Parallel Path Fault; $03 = STMR Parallel Path Fault; $04 = CDAR Dual Storage Fault; $05 = CRZR Dual Storage Fault; $06 = ECPR Dual Storage Fault; $07 = SPKR Dual Storage Fault; $08 = EITR Dual Storage Fault; $09 = EPSR Dual Storage Fault; $0A = ETQR Dual Storage Fault; $0B = MAPR After Throttle Pressure Fault; $0C = SPDR Dual Storage Fault; $0D = EOTR Dual Storage Fault; $0E = TPSR Parallel Path Fault; $0F = TPSR Internal Performance Fault; $10 = AAPR Security Fault; $11 = ETMR Desired Torque ETC Fault; $12 = ETMR Rate Limit Hi Fault; $13 = ETMR Rate Limit Lo Fault; $14 = ETMR Proportional Term Fault; $15 = ETMR Integral Term Fault; $16 = ETMR Desired Torque Correlation Fault; $17 = ETMR Desired Throttle Area Fault; $18 = ETMR Torque Sensitivity Fault; $19 = ETMR Base Friction Torque Fault; $1A = ETMR Accessory Drive Friction Fault; $1B = ETMR AC Friction Torque Fault; $1C = ETMR Oil Temperature DeltaTorque Fault; $1D = ETMR Generator Torque Fault; $1E = ETMR Supercharger Friction Torque Fault; $1F = ETMR Torque Error Filter Fault; $20 = ETMR Torque Error Compensation Fault; $21 = ETQR Delta Torque Barometer Compensation Fault; $22 = ETMR Immediate Torque Fault; $23 = STMR Torque Reserve Fault; $24 = STMR AC Torque Load Flt; $25 = TSXR Adjustment For AXLR Fault; $26 = TSXR Engine to Axle Fault; $27 = TSXR Drive Line Loss Fault; $28 = SPKR EQR Offset Fault; $29 = MAPR Engine Vacuum Fault; $2A = MAPR Intake Manifold Pressure Fault; $2B = AXLR Min Axle Capacity Fault

This PID contains the current source of the P16F3 Diagnostic Trouble Code.

3.2.59 PISR – PID $203B – Processor Security Fault History Buffer 0

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$203B	Processor Security Fault History Buffer 0	2	ENM	N/A	$01 = DTMR Parallel Path Fault; $02 = ETMR Parallel Path Fault; $03 = STMR Parallel Path Fault; $04 = CDAR Dual Storage Fault; $05 = CRZR Dual Storage Fault; $06 = ECPR Dual Storage Fault; $07 = SPKR Dual Storage Fault; $08 = EITR Dual Storage Fault; $09 = EPSR Dual Storage Fault; $0A = ETQR Dual Storage Fault; $0B = MAPR After Throttle Pressure Fault; $0C = SPDR Dual Storage Fault; $0D = EOTR Dual Storage Fault; $0E = TPSR Parallel Path Fault; $0F = TPSR Internal Performance Fault; $10 = AAPR Security Fault; $11 = ETMR Desired Torque ETC Fault; $12 = ETMR Rate Limit Hi Fault; $13 = ETMR Rate Limit Lo Fault; $14 = ETMR Proportional Term Fault; $15 = ETMR Integral Term Fault; $16 = ETMR Desired Torque Correlation Fault; $17 = ETMR Desired Throttle Area Fault; $18 = ETMR Torque Sensitivity Fault; $19 = ETMR Base Friction Torque Fault; $1A = ETMR Accessory Drive Friction Fault; $1B = ETMR AC Friction Torque Fault; $1C = ETMR Oil Temperature DeltaTorque Fault; $1D = ETMR Generator Torque Fault; $1E = ETMR Supercharger Friction Torque Fault; $1F = ETMR Torque Error Filter Fault; $20 = ETMR Torque Error Compensation Fault; $21 = ETQR Delta Torque Barometer Compensation Fault; $22 = ETMR Immediate Torque Fault; $23 = STMR Torque Reserve Fault; $24 = STMR AC Torque Load Flt; $25 = TSXR Adjustment For AXLR Fault; $26 = TSXR Engine to Axle Fault; $27 = TSXR Drive Line Loss Fault; $28 = SPKR EQR Offset Fault; $29 = MAPR Engine Vacuum Fault; $2A = MAPR Intake Manifold Pressure Fault; $2B = AXLR Min Axle Capacity Fault

This PID contains one of the last four reasons that P16F3 Diagnostic Trouble Code set. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with buffer 0 containing the most recent reason. If the most recent reason in the history buffer is the same as the current reason, the history buffer is not updated.

3.2.60 PISR – PID $203C – Processor Security Fault History Buffer 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$203C	Processor Security Fault History Buffer 1	2	ENM	N/A	$01 = DTMR Parallel Path Fault; $02 = ETMR Parallel Path Fault; $03 = STMR Parallel Path Fault; $04 = CDAR Dual Storage Fault; $05 = CRZR Dual Storage Fault; $06 = ECPR Dual Storage Fault; $07 = SPKR Dual Storage Fault; $08 = EITR Dual Storage Fault; $09 = EPSR Dual Storage Fault; $0A = ETQR Dual Storage Fault; $0B = MAPR After Throttle Pressure Fault; $0C = SPDR Dual Storage Fault; $0D = EOTR Dual Storage Fault; $0E = TPSR Parallel Path Fault; $0F = TPSR Internal Performance Fault; $10 = AAPR Security Fault; $11 = ETMR Desired Torque ETC Fault; $12 = ETMR Rate Limit Hi Fault; $13 = ETMR Rate Limit Lo Fault; $14 = ETMR Proportional Term Fault; $15 = ETMR Integral Term Fault; $16 = ETMR Desired Torque Correlation Fault; $17 = ETMR Desired Throttle Area Fault; $18 = ETMR Torque Sensitivity Fault; $19 = ETMR Base Friction Torque Fault; $1A = ETMR Accessory Drive Friction Fault; $1B = ETMR AC Friction Torque Fault; $1C = ETMR Oil Temperature DeltaTorque Fault; $1D = ETMR Generator Torque Fault; $1E = ETMR Supercharger Friction Torque Fault; $1F = ETMR Torque Error Filter Fault; $20 = ETMR Torque Error Compensation Fault; $21 = ETQR Delta Torque Barometer Compensation Fault; $22 = ETMR Immediate Torque Fault; $23 = STMR Torque Reserve Fault; $24 = STMR AC Torque Load Flt; $25 = TSXR Adjustment For AXLR Fault; $26 = TSXR Engine to Axle Fault; $27 = TSXR Drive Line Loss Fault; $28 = SPKR EQR Offset Fault; $29 = MAPR Engine Vacuum Fault; $2A = MAPR Intake Manifold Pressure Fault; $2B = AXLR Min Axle Capacity Fault

This PID contains one of the last four reasons that P16F3 Diagnostic Trouble Code set. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with buffer 0 containing the most recent reason. If the most recent reason in the history buffer is the same as the current reason, the history buffer is not updated.

3.2.61 PISR – PID $203D – Processor Security Fault History Buffer 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$203D	Processor Security Fault History Buffer 2	2	ENM	N/A	$01 = DTMR Parallel Path Fault; $02 = ETMR Parallel Path Fault; $03 = STMR Parallel Path Fault; $04 = CDAR Dual Storage Fault; $05 = CRZR Dual Storage Fault; $06 = ECPR Dual Storage Fault; $07 = SPKR Dual Storage Fault; $08 = EITR Dual Storage Fault; $09 = EPSR Dual Storage Fault; $0A = ETQR Dual Storage Fault; $0B = MAPR After Throttle Pressure Fault; $0C = SPDR Dual Storage Fault; $0D = EOTR Dual Storage Fault; $0E = TPSR Parallel Path Fault; $0F = TPSR Internal Performance Fault; $10 = AAPR Security Fault; $11 = ETMR Desired Torque ETC Fault; $12 = ETMR Rate Limit Hi Fault; $13 = ETMR Rate Limit Lo Fault; $14 = ETMR Proportional Term Fault; $15 = ETMR Integral Term Fault; $16 = ETMR Desired Torque Correlation Fault; $17 = ETMR Desired Throttle Area Fault; $18 = ETMR Torque Sensitivity Fault; $19 = ETMR Base Friction Torque Fault; $1A = ETMR Accessory Drive Friction Fault; $1B = ETMR AC Friction Torque Fault; $1C = ETMR Oil Temperature DeltaTorque Fault; $1D = ETMR Generator Torque Fault; $1E = ETMR Supercharger Friction Torque Fault; $1F = ETMR Torque Error Filter Fault; $20 = ETMR Torque Error Compensation Fault; $21 = ETQR Delta Torque Barometer Compensation Fault; $22 = ETMR Immediate Torque Fault; $23 = STMR Torque Reserve Fault; $24 = STMR AC Torque Load Flt; $25 = TSXR Adjustment For AXLR Fault; $26 = TSXR Engine to Axle Fault; $27 = TSXR Drive Line Loss Fault; $28 = SPKR EQR Offset Fault; $29 = MAPR Engine Vacuum Fault; $2A = MAPR Intake Manifold Pressure Fault; $2B = AXLR Min Axle Capacity Fault

This PID contains one of the last four reasons that P16F3 Diagnostic Trouble Code set. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with buffer 0 containing the most recent reason. If the most recent reason in the history buffer is the same as the current reason, the history buffer is not updated.

3.2.62 PISR – PID $203E – Processor Security Fault History Buffer 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$203E	Processor Security Fault History Buffer 3	2	ENM	N/A	$01 = DTMR Parallel Path Fault; $02 = ETMR Parallel Path Fault; $03 = STMR Parallel Path Fault; $04 = CDAR Dual Storage Fault; $05 = CRZR Dual Storage Fault; $06 = ECPR Dual Storage Fault; $07 = SPKR Dual Storage Fault; $08 = EITR Dual Storage Fault; $09 = EPSR Dual Storage Fault; $0A = ETQR Dual Storage Fault; $0B = MAPR After Throttle Pressure Fault; $0C = SPDR Dual Storage Fault; $0D = EOTR Dual Storage Fault; $0E = TPSR Parallel Path Fault; $0F = TPSR Internal Performance Fault; $10 = AAPR Security Fault; $11 = ETMR Desired Torque ETC Fault; $12 = ETMR Rate Limit Hi Fault; $13 = ETMR Rate Limit Lo Fault; $14 = ETMR Proportional Term Fault; $15 = ETMR Integral Term Fault; $16 = ETMR Desired Torque Correlation Fault; $17 = ETMR Desired Throttle Area Fault; $18 = ETMR Torque Sensitivity Fault; $19 = ETMR Base Friction Torque Fault; $1A = ETMR Accessory Drive Friction Fault; $1B = ETMR AC Friction Torque Fault; $1C = ETMR Oil Temperature DeltaTorque Fault; $1D = ETMR Generator Torque Fault; $1E = ETMR Supercharger Friction Torque Fault; $1F = ETMR Torque Error Filter Fault; $20 = ETMR Torque Error Compensation Fault; $21 = ETQR Delta Torque Barometer Compensation Fault; $22 = ETMR Immediate Torque Fault; $23 = STMR Torque Reserve Fault; $24 = STMR AC Torque Load Flt; $25 = TSXR Adjustment For AXLR Fault; $26 = TSXR Engine to Axle Fault; $27 = TSXR Drive Line Loss Fault; $28 = SPKR EQR Offset Fault; $29 = MAPR Engine Vacuum Fault; $2A = MAPR Intake Manifold Pressure Fault; $2B = AXLR Min Axle Capacity Fault

This PID contains one of the last four reasons that P16F3 Diagnostic Trouble Code set. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with buffer 0 containing the most recent reason. If the most recent reason in the history buffer is the same as the current reason, the history buffer is not updated.

3.2.63 FULR – PID $203F – Instantaneous Fuel Consumption

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$203F	Instantaneous Fuel Consumption	1	UNM	0 – 127.5 l/h	E = N / 2

This PID indicates the instantaneous fuel consumption rate of the engine in liters per hour.

3.2.64 EOTR – PID $2040 – Configuration Selected Engine Oil Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2040	Configuration Selected Engine Oil Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the engine oil temperature currently selected by EOTR and used by various algorithms. Depending on the application and/or diagnostic status of the inputs to EOTR, this value will match either the Estimated Engine Oil Temperature or Engine Oil Temperature from Sensor.

3.2.65 VSPR – PID $2049 – ECU Odometer Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2049	ECU Odometer Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
ECU Odometer Supported	0	7	1	BLN	N/A	$1=True; $0=False
ECU Odometer Programmed	0	6	1	BLN	N/A	$1=True; $0=False
ECU Odometer VIN Programmed	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains status bits associated with the ECU odometer strategy. The content of each of these signals is described below.

ECU Odometer Supported indicates if the ECU supports the odometer strategy as detailed in the “Global A Electrical Architecture Vehicle Odometer Sense & Display” functional requirements specification.

ECU Odometer Programmed indicates if the ECU odometer value (DID $DF) has been programmed. This bit will be used as part of the odometer strategy logic to determine if the odometer VIN value needs to be programmed.

ECU Odometer VIN Programmed indicates if the ECU odometer VIN value has been programmed. The ECU odometer VIN is stored in DID $28, which is separate from the standard DID $90 VIN value. This bit will be used as part of the odometer strategy logic to determine if the odometer VIN value needs to be programmed.

3.2.66 RBSR – PID $204A – Main Processor Total Running Resets

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$204A	Main Processor Total Running Resets	2	UNM	0 – 65,535	E = N * 1

This PID contains the total number of running resets that have occurred since last reprogramming event. This PID is intended for use by engineering only. This PID should not be displayed on any off-board device designed for Service. This PID shall be included in the Controller State of Health Freeze Frame.

3.2.67 RBSR – PID $204B – Main Processor Maximum Running Resets Between Power-up Resets

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$204B	Main Processor Maximum Running Resets Between Power-up Resets	2	UNM	0 – 65,535	E = N * 1

This PID contains the maximum number of running resets between power up resets, and since last reprogramming event. This PID is intended for use by engineering only. This PID should not be displayed on any off-board device designed for Service. This PID shall be included in the Controller State of Health Freeze Frame.

3.2.68 TOSR – PID $204C – Replicated Transmission Output Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$204C	Replicated Transmission Output Speed	2	UNM	0 - 8191.875 rpm	E = N * 0.125

This PID contains the Replicated Transmission Output Speed (Replicated TOS) received from the Transmission Control Module. Replicated TOS results from hardware in the TCM that “re-transmits” the Transmission Output Speed Sensor pulses to the ECM.

3.2.69 TRQR – PID $204D – Transmission Torque Request Limited Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$204D	Transmission Torque Request Limited Reason	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Minimum Idle Area	0	7	1	BLN	N/A	$1=True; $0=False
Spark Advance	0	6	1	BLN	N/A	$1=True; $0=False
Throttle Position Security	0	5	1	BLN	N/A	$1=True; $0=False
Engine Minimum Torque	0	4	1	BLN	N/A	$1=True; $0=False
Throttle Arbitration	0	3	1	BLN	N/A	$1=True; $0=False
Spark Arbitration	0	2	1	BLN	N/A	$1=True; $0=False
Fuel Arbitration	0	1	1	BLN	N/A	$1=True; $0=False
Reserved	0	0	1			

This PID indicates the existing condition in the ECM causing the transmission torque request to be limited.

3.2.70 TOSR – PID $2050 – Transfer Case Output Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2050	Transfer Case Output Speed	2	UNM	0 - 8191.875 rpm	E = N * 0.125

This PID contains the transfer case output speed (the speed at the output of the transfer case) determined from the speed measured by the transfer case speed sensor (TCSS).

3.2.71 KNKR – PID $2051 – Knock System Status Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2051	Knock System Status Sensor 1	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Knock Circuit Test Status	0	7	2	ENM	N/A	$0=Passed; $1=Failed; $2=Incomplete
Knock Sensor 1 Circuit Fault	0	5	2	ENM	N/A	$0=Passed; $1=Failed; $2=Incomplete
Knock Sensor 1 Signal Low	0	3	2	ENM	N/A	$0=Passed; $1=Failed; $2=Incomplete
Knock Sensor 1 Signal High	0	1	2	ENM	N/A	$0=Passed; $1=Failed; $2=Incomplete

This PID contains several signals associated with the status of the knock system. The content of each of these signals is described below.

Knock Circuit Test Status indicates the current status of fault conditions within the electronic circuits used to perform knock detection.

Knock Sensor 1 Circuit Fault indicates knock sensor 1 circuit faults other than stuck low/high.

Knock Sensor 1 Signal Low indicates the knock sensor 1 signal low fault test status.

Knock Sensor 1 Signal High indicates the knock sensor 1 signal high fault test status.

3.2.72 KNKR – PID $2052 – Knock System Status Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2052	Knock System Status Sensor 2	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Knock Sensor 2 Circuit Fault	0	5	2	ENM	N/A	$0=Passed; $1=Failed; $2=Incomplete
Knock Sensor 2 Signal Low	0	3	2	ENM	N/A	$0=Passed; $1=Failed; $2=Incomplete
Knock Sensor 2 Signal High	0	1	2	ENM	N/A	$0=Passed; $1=Failed; $2=Incomplete

This PID contains several signals associated with the status of the knock system. The content of each of these signals is described below.

Knock Sensor 2 Circuit Fault indicates knock sensor 2 circuit faults other than stuck low/high.

Knock Sensor 2 Signal Low indicates the knock sensor 2 signal low fault test status.

Knock Sensor 2 Signal High indicates the knock sensor 2 signal high fault test status.

3.2.73 PDAR – PID $2053 – Intake Manifold Runner Control Circuit Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2053	Intake Manifold Runner Control Circuit Bank 1	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Intake Manifold Runner Control Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Intake Manifold Runner Control Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Manifold Runner Control Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Manifold Runner Control Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the Intake Manifold Runner Control output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTC P2008. (Intake Manifold Runner Control (IMRC) is the SAE recommended term for any device that controls charge motion. These devices may be unofficially referred to as swirl control, charge motion control, or port deactivation.) Note: Duty cycle controlled IMRCs are considered commanded “on” when the commanded duty cycle is greater than 0%.

3.2.74 PDAR – PID $2054 – Intake Manifold Runner Control Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2054	Intake Manifold Runner Control Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Intake Manifold Runner Control Position Bank 1	0	7	2	ENM	N/A	$0=Open; $1=Intermediate; $2=Closed
Reserved	0	5	2			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains several signals associated with the status of the intake manifold runner control system. The content of each of these signals is described below. (Intake Manifold Runner Control (IMRC) is the SAE recommended term for any device that controls charge motion. These devices may be unofficially referred to as swirl control, charge motion control, or port deactivation.)

Intake Manifold Tuning Runner Control Position Bank 1 indicates whether the intake manifold tuning vane is in an open, closed, or intermediate position based upon the feedback from the intake manifold runner control sensor on bank 1.

3.2.75 PDAR – PID $2055 – Intake Manifold Runner Control Sensor A/D Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2055	Intake Manifold Runner Control Sensor A/D Bank 1	1	UNM	0 - 100 % 5V	E = N * 100 / 255

This PID contains the Bank 1 Intake Manifold Runner Control position sensor analog input as a percentage of its reference voltage. (Intake Manifold Runner Control (IMRC) is the SAE recommended term for any device that controls charge motion. These devices may be unofficially referred to as swirl control, charge motion control, or port deactivation.)

3.2.76 VLTR – PID $2056 – Reference Voltage C

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2056	Reference Voltage C	1	UNM	0 - 255 counts	E = N * 1

This PID contains the Vref 3 value from the MCP.

3.2.77 VLTR – PID $2057 – Reference Voltage D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2057	Reference Voltage D	1	UNM	0 - 255 counts	E = N * 1

This PID contains the Vref 4 value from the MCP.

3.2.78 THMR - PID $2058 – Engine Coolant Thermostat Heater Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2058	Engine Coolant Thermostat Heater Duty Cycle	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the commanded duty cycle for the engine coolant thermostat.

3.2.79 THMR - PID $2059 – Engine Coolant Thermostat Heater Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2059	Engine Coolant Thermostat Heater Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Engine Coolant Thermostat Heater Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Engine Coolant Thermostat Heater Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Engine Coolant Thermostat Heater Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Engine Coolant Thermostat Heater Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the engine coolant thermostat heater output (the output is considered to be commanded on if its duty cycle is not zero) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTC P0597.

3.2.80 THMR - PID $205A – Radiator Coolant Temperature Sensor

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$205A	Radiator Coolant Temperature Sensor	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the undefaulted radiator coolant temperature measured by a sensor located in the coolant outlet of the radiator.

3.2.81 THMR - PID $205B – Desired Engine Coolant Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$205B	Desired Engine Coolant Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the desired engine coolant temperature in degrees Celsius.

3.2.82 CRZR – PID $205D – Cruise Inhibit Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion

$205D	Cruise Inhibit Reason	1	ENM	N/A	$00=Brake Pedal Applied; $01=Brake Pedal Before Cruise; $02=Cancel Switch Active; $03=Clutch Pedal Applied; $04=Coast Disengage; $05=Coast Below Low Speed; $06=ETC Inhibit; $07=High Acceleration; $08=High Deceleration; $09=High Vehicle Speed; $0A=Illegal Cruise Mode; $0B=Low Vehicle Speed; $0C=None; $0D=Cruise Switch Off; $0E=Speed Difference High; $0F=Invalid Cruise Tap Down; $10=Control Module Inhibit; $11=R/A and S/C Switches Pressed Together; $12=TCS Active; $13=Speed Difference Low; $14=Device Control; $15=Memory Error; $16=Engine Run Time; $17=Engine Speed; $18=DTC Active; $19=Engine Overspeed Protection Active; $1A=First Gear Disable; $1B=Accelerator Pedal Override; $1C=Low Ignition Voltage; $1D=Transmission Not in Gear; $1E=Vehicle Overspeed Protection Active; $1F=Analog Cruise Switch Invalid; $20=High Ignition Voltage; $21=Non Driven Wheel Speed Greater; $22=Driven Wheel Speed Greater; $23=Cruise Execution Error; $24=Cruise Switch Serial Communication Fault; $25=Engine Metal Overtemperature Protection Active; $26=VSES Active; $27=Rear Axle Low; $28=Transmission Gear Fault; $29=Brake Signal Serial Communication Fault; $2A=Brake Pedal Fault; $2B=Park Brake Active; $2C=ACC Inhibit; $2D=ACC Automatic Braking Failed; $2E=Vehicle Speed Control Type Error; $2F=ACC Command Serial Communication Fault; $30=Power Take Off Active; $31=ECM Running Reset;

					$32=Process Fault; $33=Brake Home Position Not Learned; $34=Rate Limit Fault; $35=4WD Low Disable; $36=Gear Change Without Clutch Apply; $37=Brake Pedal Pressure Detected; $38=Speed Limiter/Warning Turned On; $39=Continuous Torque Monitoring Failure; $3A = Enhanced Cruise Auto Braking Failed; $3B = Enhanced Cruise Auto Brake Serial Data Fault; $3C = Hill Descent Active; $3D = Hill Descent Serial Data Fault; $3E = Hybrid Cruise Disengage; $4A = Hybrid Torque Data

This PID contains the current reason that powertrain is not engaging cruise control.

3.2.83 EOSR – PID $205E – Engine Speed Peak Value at Overspeed Condition

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$205E	Engine Speed Peak Value at Overspeed Condition	2	UNM	0 - 16383.75 rpm	E = N * 0.25

This PID contains the peak engine speed observed during all overspeed conditions. This value should be stored even during an un-debounced event, even though the time limit for the overspeed condition detection function may not have expired.

3.2.84 EOSR – PID $205F – Engine Speed at Overspeed Condition Detection

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$205F	Engine Speed at Overspeed Condition Detection	2	UNM	0 - 16383.75 rpm	E = N * 0.25

This PID contains the engine speed observed upon initial detection of an overspeed condition. This value will be updated upon each subsequent overspeed condition.

3.2.85 EOSR – PID $2060 – Vehicle Speed at Overspeed Condition Detection

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2060	Vehicle Speed at Overspeed Condition Detection	1	UNM	0 - 255 kph	E = N * 1

This PID contains the vehicle speed observed upon initial detection of an overspeed condition. This value will be updated upon each subsequent overspeed condition.

3.2.86 EOSR – PID $2061 – Accelerator Pedal Position at Overspeed Condition Detection

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2061	Accelerator Pedal Position at Overspeed Condition Detection	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the absolute accelerator pedal position observed upon initial detection of an overspeed condition. This value will be updated upon each subsequent overspeed condition.

3.2.87 EOSR – PID $2062 – Engine Oil Temperature at Overspeed Condition Detection

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2062	Engine Oil Temperature at Overspeed Condition Detection	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the engine oil temperature observed upon initial detection of an overspeed condition. This value will be updated upon each subsequent overspeed condition.

3.2.88 EOSR – PID $2063 – Transmission Actual Gear at Overspeed Condition Detection

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2063	Transmission Actual Gear at Overspeed Condition Detection	1	ENM	N/A	$00=First Gear Active; $01=Second Gear Active; $02=Third Gear Active; $03=Fourth Gear Active; $04=Fifth Gear Active; $05=Sixth Gear Active; $06=Neutral $07=Reverse Gear Active; $08=Park Active; $09=Seventh Gear Active; $0A=Eighth Gear Active; $11=Illegal/Invalid; $FF=TransNoFailSoft

This PID contains the transmission actual gear observed upon initial detection of an overspeed condition. This value will be updated upon each subsequent overspeed condition.

3.2.89 EOSR – PID $2064 – Intake Mass Airflow at Overspeed Condition Detection

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2064	Intake Mass Airflow at Overspeed Condition Detection	2	UNM	0 - 655.35 g/s	E = N * 0.01

This PID contains the intake mass airflow observed upon initial detection of an overspeed condition. This value will be updated upon each subsequent overspeed condition.

3.2.90 EOSR – PID $2065 – Absolute Vehicle Odometer Reading at Overspeed Condition Detection

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2065	Absolute Vehicle Odometer Reading at Overspeed Condition Detection	2	UNM	0 - 327675 km	E = N * 5

This PID contains the absolute vehicle odometer reading observed upon initial detection of an overspeed condition. This value will be updated upon each subsequent overspeed condition.

3.2.91 TPSR – PID $2066 – Learned Airflow Variation Algorithm Compensation

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2066	Learned Airflow Variation Algorithm Compensation	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the percent compensation of the Learned Airflow Variation Algorithm. This value indicates the current amount of airflow compensation in response to engine intake airflow restrictions (i.e. throttle body coking). It can also be used as feedback for the Learned Airflow Adapts Reset Trigger Device Control.

3.2.92 CRZR – PID $2067 – Cruise Control Status 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2067	Cruise Control Status 2	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Cruise Control Enabled	0	7	1	BLN	N/A	$1=True; $0=False
Speed Limiter/Warning Active	0	6	1	BLN	N/A	$1=True; $0=False
Speed Limiter/Warning On Switch Active	0	5	1	BLN	N/A	$1=True; $0=False
Cruise Control On Switch Both States Latched	0	4	1	BLN	N/A	$1=True; $0=False
Cruise Secondary Switch Both States Latched	0	3	1	BLN	N/A	$1=True; $0=False
Powertrain Brake Request Active	0	2	1	BLN	N/A	$1=True; $0=False
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains signals associated with the status of the cruise control system. The content of each of these signals is described below.

Cruise Control Enabled is a mirror of the GMLAN signal “Cruise Control Enabled”. Cruise Control Enabled is transmitted by the ECM. The data value of this signal shall be set to “True” when all the conditions for allowing cruise are present (no inhibit conditions exist that would prevent cruise engagement) and the Cruise Control Switch Status: On Switch Active signal is “True”. Note that Cruise Control Enabled is non-latching, i.e., any temporary inhibit conditions (braking, de-clutching, etc) as well as permanent failures, would set the signal to “False”. Applications, which do not support Cruise Control, shall transmit this signal with a value of “False”.

Speed Limiter/Warning Active indicates that the speed limiter/warning system is currently operating.

Speed Limiter/Warning On Switch Active indicates the state of the speed limiter/warning on switch. In GMLAN cruise control electrical architectures, this signal is based upon the state of the speed limiter/warning on switch received over serial communication from the platform electronics. This is the latched state for a momentary speed limiter/warning on switch where each apply and release of the momentary switch toggles the state of Speed Limiter/Warning On Switch Active.

Cruise Control On Switch Both States Latched indicates true once both the on and off switch states are observed and will remain latched for the current ignition cycle. This bit will reset to false once the power mode transitions to off.

Cruise Secondary Switch Both States Latched indicates true once the increase high or decrease high switch states are observed and will remain latched for the current ignition cycle. This bit will reset to false once the power mode transitions to off.

Powertrain Brake Request Active indicates true when Powertrain commands the brake system to provide automatic braking. This occurs under conditions where the powertrain is not able to attain a desired vehicle deceleration level using solely powertrain delivered torque,

3.2.93 COTR – PID $2068 – Absolute Controller Operation Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2068	Absolute Controller Operation Time	4	UNM	0 – 4,294,967,295 s	E = N * 1

This PID contains the total controller operation time during the life of the applicable electronic control unit (ECU).

3.2.94 FADR – PID $2069 – Discrete Fan System Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2069	Discrete Fan System Speed	1	ENM	N/A	$00=Fan Off; $01=Fan system speed 1 active; $02=Fan system speed 2 active; $03=Fan system speed 3 active; $04=Fan system speed 4 active; $05=Fan system speed 5 active; $06=Fan system speed 6 active; $07=Fan system speed 7 active

This PID contains the current discrete cooling fan system speed. This is in terms of fan state (for example "off, low, medium, or high") rather than rotational speed. An increase in the system speed enumeration value corresponds to an increase in actual system speed. The speed numbers do not necessarily reflect the binary state of the cooling fan drivers. For a fan system with a small number of possible speeds, the additional defined enumerations will not apply. For example, in a two-speed fan system, $02 will equate to "high" and enumerations $03 - $07 will not be used.

3.2.95 EMDR – PID $206A – Engine Operating Mode Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$206A	Engine Operating Mode Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Engine Running	0	7	1	BLN	N/A	$1=True; $0=False
Engine Cranking	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains the status information for EMDR.

Engine Running indicates the status of the engine running flag within the ECM software.

Engine Cranking indicates the status of the engine cranking flag within the ECM software.

3.2.96 VSDR – PID $206B – Generator Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$206B	Generator Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Generator Failed Indication Request Active	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains signals associated with the status of the generator and APM. The content of each of these signals is described below.

Generator Failed Indication Request Active is a mirror of the GMLAN signal "Generator Failed".

3.2.97 CATR – PID $206C – Catalyst Monitor Diagnostic Status Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$206C	Catalyst Monitor Diagnostic Status Bank 1	1	ENM	N/A	$00=No Decision; $01=Failed; $02=Passed;

This PID indicates the status of the catalyst monitor diagnostic for bank 1. Multiple tests may run before the diagnostic reports a "Pass" or "Fail" and will be tracked by PID $1354 "Catalyst Monitor Diagnostic Test Counter Bank 1". When this PID reports a "Pass" or "Fail", the diagnostic is complete for the current trip and will not attempt any further tests on bank 1.

3.2.98 CATR – PID $206D – Catalyst Monitor Diagnostic Status Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$206D	Catalyst Monitor Diagnostic Status Bank 2	1	ENM	N/A	$00=No Decision; $01=Failed; $02=Passed;

This PID indicates the status of the catalyst monitor diagnostic for bank 2. Multiple tests may run before the diagnostic reports a "Pass" or "Fail" and will be tracked by PID $1355 "Catalyst Monitor Diagnostic Test Counter Bank 2". When this PID reports a "Pass" or "Fail", the diagnostic is complete for the current trip and will not attempt any further tests on bank 2.

3.2.99 CATR – PID $206E – Catalyst Monitor Diagnostic Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$206E	Catalyst Monitor Diagnostic Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Catalyst Monitor Test Running	0	7	1	BLN	N/A	$1=True; $0=False
Off Idle Conditions Met	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID indicates the status of the catalyst monitor diagnostic.

Catalyst Monitor Test Running indicates when the catalyst monitor diagnostic is actively running a test. This condition will only be true when the vehicle is at idle. When Catalyst Monitor Test Running transitions from "True" to "False" with the vehicle at idle, PIDs $1354 & $1355 "Catalyst Monitor Diagnostic Test Counter Bank 1 & 2" can be referenced along with PIDs $206C & $206D "Catalyst Monitor Diagnostic Status Bank 1 & 2" to determine the results of the test.

Off Idle Conditions Met indicates if the off idle conditions for the catalyst monitor diagnostic have been met. Once the Off Idle Conditions Met bit equals "True", the vehicle can return to idle and allow the catalyst monitor diagnostic test to run.

3.2.100RPPR – PID $2070 – Reduced Engine Power Indication History

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2070	Reduced Engine Power Indication History	1	ENM	N/A	$00=None; $01=ETC Fault; $02=EV Fan Fault; $03=Engine Metal Ovrtemp; $04=Engine Oil Overtemp; $05=OnStar Speed Limiting; $06=Engine Calibration Monitor Fault; $07=Poor Engine Oil Quality Due to Failed DPF Regeneration; $08=High Pressure Fuel System Failure; $09=Boost Pressure Out of range; $0A= Intake Air Flow Ratio Out of Range; $0B=Diesel Particulate Filter System Failure; $0C=NOx Reductant System Failure; $0D=NOx Reductant Fluid Level Low or Empty; $0E=Manual Transmission Oil Overtemp (E41332); $0F=Engine Coolant Overtemp; $10=Fuel Overtemp; $11=Intake Air Overtemp; $12=NOx Reductant Fluid Poor Quality; $13=NOx Reductant System Tampering; $14= Battery Power Low; $15= Motor Inverter ;Overtemp $16= Propulsion Charging Unavailable; $17= Oil Pump Control Fault; $18=Crank Sensor Fault; ~~$19=CMP Actuators Disabled;~~ $1A=Hydraulic or HV Pump Failure; $1B=Manual Transmission Automatically Shut Down Request; $1C= Piston Cooling Control Fault; $1D= Charger Air Limit Active; $1E= Torque Control Input Fault

This PID contains the last reason that Powertrain has illuminated the Reduced Engine Power (REP) indication.

3.2.101EITR – PID $2072 – Intake Air Temperature 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2072	Intake Air Temperature 3	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the undefaulted intake air temperature (IAT) measured by the third IAT sensor in the intake air stream.

3.2.102AFIR – PID $207B – Cold Start Fuel Reservoir Pump Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$207B	Cold Start Fuel Reservoir Pump Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Cold Start Fuel Reservoir Pump Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Cold Start Fuel Reservoir Pump Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Cold Start Fuel Reservoir Pump Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Cold Start Fuel Reservoir Pump Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the cold start fuel reservoir pump output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P120D, P120C, and P120E, respectively.

3.2.103AFIR – PID $207C – Cold Start Fuel Injector Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$207C	Cold Start Fuel Injector Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Cold Start Fuel Injector Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Cold Start Fuel Injector Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Cold Start Fuel Injector Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Cold Start Fuel Injector Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the cold start fuel injector output (the output is considered to be commanded on if the injector is enabled while the engine is running or cranking) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P120A, P0213, and P120B, respectively.

3.2.104AFIR – PID $207D – Cold Start Fuel Injector Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$207D	Cold Start Fuel Injector Duty Cycle	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the commanded duty cycle for the cold start fuel injector output.

3.2.105PMDR – PID $207E – System Power Mode

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$207E	System Power Mode	1	ENM	N/A	$00=Off; $01=Accessory; $02=Run; $03=Crank Request

This PID contains the state of the System Power Mode. This signal is based upon the state of the system power mode received over serial communication from the platform electronics. If the serial data signal not received, the PID will display $00 = Off.

3.2.106FULR – PID $207F – Fuel Injector 1 Circuit Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$207F	Fuel Injector 1 Circuit Status	1	ENM	N/A	$00=No Fault is present; $01=Short circuit from injector high side to ground; $02=Short circuit from injector high side to system voltage; $03=Short circuit from injector low side to ground; $04=Short circuit from injector low side to system voltage; $05=Short circuit from injector high side to low side; $06=Injector open circuit; $07=Fault Indeterminate

This PID contains the status of the fuel injector 1 output faults for high side short to ground and short to power, low side short to ground and short to power, high side to low side short, and open circuit. This information is used to report DTCs P2147, P2148, P0261, P0262, P1248, and P0201, respectively.

3.2.107FULR – PID $2080 – Fuel Injector 2 Circuit Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2080	Fuel Injector 2 Circuit Status	1	ENM	N/A	$00=No Fault is present; $01=Short circuit from injector high side to ground; $02=Short circuit from injector high side to system voltage; $03=Short circuit from injector low side to ground; $04=Short circuit from injector low side to system voltage; $05=Short circuit from injector high side to low side; $06=Injector open circuit; $07=Fault Indeterminate

This PID contains the status of the fuel injector 2 output faults for high side short to ground and short to power, low side short to ground and short to power, high side to low side short, and open circuit. This information is used to report DTCs P2150, P2151, P0264, P0265, P1249, and P0202, respectively.

3.2.108FULR – PID $2081 – Fuel Injector 3 Circuit Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2081	Fuel Injector 3 Circuit Status	1	ENM	N/A	$00=No Fault is present; $01=Short circuit from injector high side to ground; $02=Short circuit from injector high side to system voltage; $03=Short circuit from injector low side to ground; $04=Short circuit from injector low side to system voltage; $05=Short circuit from injector high side to low side; $06=Injector open circuit; $07=Fault Indeterminate

This PID contains the status of the fuel injector 3 output faults for high side short to ground and short to power, low side short to ground and short to power, high side to low side short, and open circuit. This information is used to report DTCs P2153, P2154, P0267, P0268, P124A, and P0203, respectively.

3.2.109FULR – PID $2082 – Fuel Injector 4 Circuit Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2082	Fuel Injector 4 Circuit Status	1	ENM	N/A	$00=No Fault is present; $01=Short circuit from injector high side to ground; $02=Short circuit from injector high side to system voltage; $03=Short circuit from injector low side to ground; $04=Short circuit from injector low side to system voltage; $05=Short circuit from injector high side to low side; $06=Injector open circuit; $07=Fault Indeterminate

This PID contains the status of the fuel injector 4 output faults for high side short to ground and short to power, low side short to ground and short to power, high side to low side short, and open circuit. This information is used to report DTCs P2156, P2157, P0270, P0271, P124B, and P0204, respectively.

3.2.110FULR – PID $2083 – Fuel Injector 5 Circuit Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2083	Fuel Injector 5 Circuit Status	1	ENM	N/A	$00=No Fault is present; $01=Short circuit from injector high side to ground; $02=Short circuit from injector high side to system voltage; $03=Short circuit from injector low side to ground; $04=Short circuit from injector low side to system voltage; $05=Short circuit from injector high side to low side; $06=Injector open circuit; $07=Fault Indeterminate

This PID contains the status of the fuel injector 5 output faults for high side short to ground and short to power, low side short to ground and short to power, high side to low side short, and open circuit. This information is used to report DTCs P216B, P216C, P0273, P0274, P124C, and P0205, respectively.

3.2.111FULR – PID $2084 – Fuel Injector 6 Circuit Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2084	Fuel Injector 6 Circuit Status	1	ENM	N/A	$00=No Fault is present; $01=Short circuit from injector high side to ground; $02=Short circuit from injector high side to system voltage; $03=Short circuit from injector low side to ground; $04=Short circuit from injector low side to system voltage; $05=Short circuit from injector high side to low side; $06=Injector open circuit; $07=Fault Indeterminate

This PID contains the status of the fuel injector 6 output faults for high side short to ground and short to power, low side short to ground and short to power, high side to low side short, and open circuit. This information is used to report DTCs P216E, P216F, P0276, P0277, P124D, and P0206, respectively.

3.2.112FULR – PID $2085 – Fuel Injector 7 Circuit Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2085	Fuel Injector 7 Circuit Status	1	ENM	N/A	$00=No Fault is present; $01=Short circuit from injector high side to ground; $02=Short circuit from injector high side to system voltage; $03=Short circuit from injector low side to ground; $04=Short circuit from injector low side to system voltage; $05=Short circuit from injector high side to low side; $06=Injector open circuit; $07=Fault Indeterminate

This PID contains the status of the fuel injector 7 output faults for high side short to ground and short to power, low side short to ground and short to power, high side to low side short, and open circuit. This information is used to report DTCs P217B, P217C, P0279, P0280, P124E, and P0207, respectively.

3.2.113FULR – PID $2086 – Fuel Injector 8 Circuit Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2086	Fuel Injector 8 Circuit Status	1	ENM	N/A	$00=No Fault is present; $01=Short circuit from injector high side to ground; $02=Short circuit from injector high side to system voltage; $03=Short circuit from injector low side to ground; $04=Short circuit from injector low side to system voltage; $05=Short circuit from injector high side to low side; $06=Injector open circuit; $07=Fault Indeterminate

This PID contains the status of the fuel injector 8 output faults for high side short to ground and short to power, low side short to ground and short to power, high side to low side short, and open circuit. This information is used to report DTCs P217E, P217F, P0282, P0283, P124F, and P0208, respectively.

3.2.114FHPR – PID $2087 – Fuel Pressure Regulator 1 Enable Circuit (Low side)

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2087	Fuel Pressure Regulator 1 Enable Circuit (Low side)	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Pressure Regulator 1 Commanded On (Low side)	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Fuel Pressure Regulator 1 Short to Ground Test Status (Low side)	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Fuel Pressure Regulator 1 Open Circuit Test Status (Low side)	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Fuel Pressure Regulator 1 Short to Power Test Status (Low side)	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the fuel pressure regulator 1 low side output (the output is considered to be commanded on if the regulator is enabled while the engine is running or cranking) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTC P0091, P0090, and P0092, respectively.

3.2.115FHPR – PID $2088 – Fuel Pressure Regulator 1 Control Circuit (High side)

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2088	Fuel Pressure Regulator 1 Control Circuit (High side)	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Pressure Regulator 1 Commanded On (High side)	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Fuel Pressure Regulator 1 Short to Ground Test Status (High side)	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Fuel Pressure Regulator 1 Open Circuit Test Status (High side)	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Fuel Pressure Regulator 1 Short to Power Test Status (High side)	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the fuel pressure regulator 1 high side output (the output is considered to be commanded on if the regulator is enabled while the engine is running or cranking) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P00C9, P00C8, and P00CA respectively.

3.2.116FHPR – PID $2089 – Fuel Pressure Regulator 1 Pump Control Angle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2089	Fuel Pressure Regulator 1 Pump Control Angle	2	UNM	0 – 655.35 (degrees crankshaft)	E = N / 100

This PID contains the control angle of the pump in terms of crank shaft degrees before Top Dead Center of the pump lobe.

3.2.117MAPR – PID $208A – Extended Range Intake Manifold Absolute Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$208A	Extended Range Intake Manifold Absolute Pressure	2	UNM	0 - 2047.96875 kPa	E = N / 32

This PID contains the unfiltered engine intake manifold absolute pressure (MAP) for applications with MAP values greater than 255 kPa. In these instances PID $000B does not have sufficient range. This value mirrors legislated PID $0087, bytes 1 & 2.

3.2.118MAPR – PID $208B – Extended Range Intake Manifold Vacuum Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$208B	Extended Range Intake Manifold Vacuum Pressure	2	SNM	-1024 to +1023.96875 kPa	E = N / 32

This PID contains the engine intake manifold vacuum pressure calculated as the difference between ambient air pressure (barometric pressure) and intake manifold absolute pressure (MAP). This PID is the extended range version of PID $2001 and is used with applications that require PID $208A. (Vacuum = Ambient - MAP)

3.2.119SPKR – PID $208C – Ignition Module Power Supply Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$208C	Ignition Module Power Supply Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Ignition Module Supply Input Active Bank 1	0	7	1	BLN	N/A	$1=True; $0=False
Ignition Module Supply Input Active Bank 2	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains signals associated with the status of the power supply for the ignition modules on each bank. The content of each of these signals is described below.

Ignition Module Supply Input Active Bank 1 indicates that the feedback hardwire input to the controller for the ignition module power supply is active on the bank containing cylinder 1.

Ignition Module Supply Input Active Bank 2 indicates that the feedback hardwire input to the controller for the ignition module power supply is active on the bank that does not contain cylinder 1.

3.2.120RBSR – PID $208D – Main Processor Running Reset Reason History 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$208D	Main Processor Running Reset Reason History 1	1	ENM	N/A	$00=No Running Reset; $01=Reserved; $02=External Watchdog Timer; $03=Internal Watchdog Timer; $04=Unidentified unhandled interrupt or exception; $05=A Running Reset invoked by the Secondary Processor (if present); $06=Reserved; $07=Requested by GMPT software via FORCE_RUNNING_RESET interface; $08=Uncorrectable Flash Memory error (ECC error); $09=eTPU Signature error; $0A=Request Response Security Monitor excessive error count; $0B=Request Response Security Monitor RAM test failure; $0C=Request Response Security Monitor ROM test failure; $0D=Uncorrectable RAM Memory error (ECC error); $0E=Uncorrectable Cache Memory error (Parity error); $0F to $FF=Controller-specific fault

This PID contains one of the last four reasons for a main processor running reset. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Main Processor Running Reset Source Address History x” PID of the same number “x”.

3.2.121RBSR – PID $208E – Main Processor Running Reset Reason History 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$208E	Main Processor Running Reset Reason History 2	1	ENM	N/A	$00=No Running Reset; $01=Reserved; $02=External Watchdog Timer; $03=Internal Watchdog Timer; $04=Unidentified unhandled interrupt or exception; $05=A Running Reset invoked by the Secondary Processor (if present); $06=Reserved; $07=Requested by GMPT software via FORCE_RUNNING_RESET interface; $08=Uncorrectable Flash Memory error (ECC error); $09=eTPU Signature error; $0A=Request Response Security Monitor excessive error count; $0B=Request Response Security Monitor RAM test failure; $0C=Request Response Security Monitor ROM test failure; $0D=Uncorrectable RAM Memory error (ECC error); $0E=Uncorrectable Cache Memory error (Parity error); $0F to $FF=Controller-specific fault

This PID contains one of the last four reasons for a main processor running reset. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Main Processor Running Reset Source Address History x” PID of the same number “x”.

3.2.122RBSR – PID $208F – Main Processor Running Reset Reason History 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$208F	Main Processor Running Reset Reason History 3	1	ENM	N/A	$00=No Running Reset; $01=Reserved; $02=External Watchdog Timer; $03=Internal Watchdog Timer; $04=Unidentified unhandled interrupt or exception; $05=A Running Reset invoked by the Secondary Processor (if present); $06=Reserved; $07=Requested by GMPT software via FORCE_RUNNING_RESET interface; $08=Uncorrectable Flash Memory error (ECC error); $09=eTPU Signature error; $0A=Request Response Security Monitor excessive error count; $0B=Request Response Security Monitor RAM test failure; $0C=Request Response Security Monitor ROM test failure; $0D=Uncorrectable RAM Memory error (ECC error); $0E=Uncorrectable Cache Memory error (Parity error); $0F to $FF=Controller-specific fault

This PID contains one of the last four reasons for a main processor running reset. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Main Processor Running Reset Source Address History x” PID of the same number “x”.

3.2.123RBSR – PID $2090 – Main Processor Running Reset Reason History 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2090	Main Processor Running Reset Reason History 4	1	ENM	N/A	$00=No Running Reset; $01=Reserved; $02=External Watchdog Timer; $03=Internal Watchdog Timer; $04=Unidentified unhandled interrupt or exception; $05=A Running Reset invoked by the Secondary Processor (if present); $06=Reserved; $07=Requested by GMPT software via FORCE_RUNNING_RESET interface; $08=Uncorrectable Flash Memory error (ECC error); $09=eTPU Signature error; $0A=Request Response Security Monitor excessive error count; $0B=Request Response Security Monitor RAM test failure; $0C=Request Response Security Monitor ROM test failure; $0D=Uncorrectable RAM Memory error (ECC error); $0E=Uncorrectable Cache Memory error (Parity error); $0F to $FF=Controller-specific fault

This PID contains one of the last four reasons for a main processor running reset. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Main Processor Running Reset Source Address History x” PID of the same number “x”.

3.2.124RBSR – PID $2091 – Main Processor Running Reset Source Address History 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2091	Main Processor Running Reset Source Address History 2	4	UNM	0 - $FFFFFFFF	E = N

This PID contains one of the last four reset source addresses for a running reset in the main processor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Main Processor Running Reset Reason History x” PID of the same number “x”.

3.2.125RBSR – PID $2092 – Main Processor Running Reset Source Address History 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2092	Main Processor Running Reset Source Address History 3	4	UNM	0 - $FFFFFFFF	E = N

This PID contains one of the last four reset source addresses for a running reset in the main processor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Main Processor Running Reset Reason History x” PID of the same number “x”.

3.2.126RBSR – PID $2093 – Main Processor Running Reset Source Address History 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2093	Main Processor Running Reset Source Address History 4	4	UNM	0 - $FFFFFFFF	E = N

This PID contains one of the last four reset source addresses for a running reset in the main processor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Main Processor Running Reset Reason History x” PID of the same number “x”.

3.2.127RBSR – PID $2094 – Secondary Processor Running Reset Reason History 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2094	Secondary Processor Running Reset Reason History 1	1	ENM	N/A	$00=No Running Reset; $01=Reserved; $02=Reserved; $03=Internal Watchdog Timer; $04=Unidentified unhandled interrupt or exception; $05=Secondary Processor invoking the Force Main Reset signal; $06=A Main Processor Running Reset; $07=Requested by GMPT software via FORCE_RUNNING_RESET interface; $08 - $FF=Controller-specific fault

This PID contains one of the last four reasons for a secondary processor running reset. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Secondary Processor Running Reset Source Address History x” PID of the same number “x”.

3.2.128RBSR – PID $2095 – Secondary Processor Running Reset Reason History 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2095	Secondary Processor Running Reset Reason History 2	1	ENM	N/A	$00=No Running Reset; $01=Reserved; $02=Reserved; $03=Internal Watchdog Timer; $04=Unidentified unhandled interrupt or exception; $05=Secondary Processor invoking the Force Main Reset signal; $06=A Main Processor Running Reset; $07=Requested by GMPT software via FORCE_RUNNING_RESET interface; $08 - $FF=Controller-specific fault

This PID contains one of the last four reasons for a secondary processor running reset. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Secondary Processor Running Reset Source Address History x” PID of the same number “x”.

3.2.129RBSR – PID $2096 – Secondary Processor Running Reset Reason History 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2096	Secondary Processor Running Reset Reason History 3	1	ENM	N/A	$00=No Running Reset; $01=Reserved; $02=Reserved; $03=Internal Watchdog Timer; $04=Unidentified unhandled interrupt or exception; $05=Secondary Processor invoking the Force Main Reset signal; $06=A Main Processor Running Reset; $07=Requested by GMPT software via FORCE_RUNNING_RESET interface; $08 - $FF=Controller-specific fault

This PID contains one of the last four reasons for a secondary processor running reset. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Secondary Processor Running Reset Source Address History x” PID of the same number “x”.

3.2.130RBSR – PID $2097 – Secondary Processor Running Reset Reason History 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2097	Secondary Processor Running Reset Reason History 4	1	ENM	N/A	$00=No Running Reset; $01=Reserved; $02=Reserved; $03=Internal Watchdog Timer; $04=Unidentified unhandled interrupt or exception; $05=Secondary Processor invoking the Force Main Reset signal; $06=A Main Processor Running Reset; $07=Requested by GMPT software via FORCE_RUNNING_RESET interface; $08 - $FF=Controller-specific fault

This PID contains one of the last four reasons for a secondary processor running reset. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Secondary Processor Running Reset Source Address History x” PID of the same number “x”.

3.2.131RBSR – PID $2098 – Secondary Processor Running Reset Source Address History 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2098	Secondary Processor Running Reset Source Address History 2	2	UNM	0 - $FFFF	E = N

This PID contains one of the last four reset source addresses for a running reset in the secondary processor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Secondary Processor Running Reset Reason History x” PID of the same number “x”.

3.2.132RBSR – PID $2099 – Secondary Processor Running Reset Source Address History 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2099	Secondary Processor Running Reset Source Address History 3	2	UNM	0 - $FFFF	E = N

This PID contains one of the last four reset source addresses for a running reset in the secondary processor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Secondary Processor Running Reset Reason History x” PID of the same number “x”.

3.2.133RBSR – PID $209A – Secondary Processor Running Reset Source Address History 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$209A	Secondary Processor Running Reset Source Address History 4	2	UNM	0 - $FFFF	E = N

This PID contains one of the last four reset source addresses for a running reset in the secondary processor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the "Secondary Processor Running Reset Reason History x" PID of the same number "x".

3.2.134E85R – PID $209B – Recommended Maximum Ethanol Concentration When Recommended Ethanol Concentration Exceeded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$209B	Recommended Maximum Ethanol Concentration When Recommended Ethanol Concentration Exceeded	1	UNM	0 - 99.609375 %	E = N * 100 / 256

This PID indicates the recommended fuel alcohol percentage for vehicle conditions at the most recent time that the Recommended Ethanol Concentration Exceeded bit became true. This PID value will latch until a mode $04 code clear is issued to the controller. If a Recommended Ethanol Concentration Exceeded event has not taken place (either initially or after a mode $04), this PID shall default to zero.

3.2.135E85R – PID $209C – Currently Recommended Maximum Ethanol Concentration

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$209C	Currently Recommended Maximum Ethanol Concentration	1	UNM	0 - 99.609375 %	E = N * 100 / 256

This PID indicates the recommended maximum fuel alcohol percentage for current vehicle conditions.

3.2.136E85R – PID $209D – Actual Ethanol Composition When Recommended Ethanol Concentration Exceeded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$209D	Actual Ethanol Composition When Recommended Ethanol Concentration Exceeded	1	UNM	0 - 99.609375 %	E = N * 100 / 256

This PID indicates the actual fuel alcohol percentage at the most recent time that the Recommended Ethanol Concentration Exceeded bit became true. In vehicles with an ethanol concentration sensor, this will be a measured value. In vehicles without a physical sensor, this will be a calculated value. This PID will latch until a mode $04 code clear is issued to the controller. If a Recommended Ethanol Concentration Exceeded event has not taken place (either initially or after a mode $04), this PID shall default to zero.

3.2.137E85R – PID $209E – Ambient Air Temperature When Recommended Ethanol Concentration Exceeded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$209E	Ambient Air Temperature When Recommended Ethanol Concentration Exceeded	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID indicates the estimated ambient air temperature at the most recent time that the Recommended Ethanol Concentration Exceeded bit became true. This PID value will latch until a mode $04 code clear is issued to the controller. If a Recommended Ethanol Concentration Exceeded event has not taken place (either initially or after a mode $04), this PID shall default to zero.

3.2.138E85R – PID $209F – Ignition Cycles Since Recommended Ethanol Concentration Exceeded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$209F	Ignition Cycles Since Recommended Ethanol Concentration Exceeded	1	UNM	0 – 255 counts	E = N * 1

This PID indicates the number of ignitions cycles since the most recent time that the Recommended Ethanol Concentration Exceeded bit became true. This PID value will reset when a mode $04 code clear is issued to the controller. If a Recommended Ethanol Concentration Exceeded event has not taken place (either initially or after a mode $04), this PID shall default to zero.

3.2.139E85R – PID $20A0 – Distance Since Recommended Ethanol Concentration Exceeded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A0	Distance Since Recommended Ethanol Concentration Exceeded	2	UNM	0 - 327675 km	E = N * 5

This PID indicates the accumulated distance driven since the most recent time that the Recommended Ethanol Concentration Exceeded bit became true. This PID value will reset when a mode $04 code clear is issued to the controller. If a Recommended Ethanol Concentration Exceeded event has not taken place (either initially or after a mode $04), this PID shall default to zero.

3.2.140E85R – PID $20A1 – Fuel Consumed Since Recommended Ethanol Concentration Exceeded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A1	Fuel Consumed Since Recommended Ethanol Concentration Exceeded	1	UNM	0 – 255 liters	E = N * 1

This PID indicates the amount of fuel consumed since the most recent time that the Recommended Ethanol Concentration Exceeded bit became true. This PID value will reset when a mode $04 code clear is issued to the controller. If a Recommended Ethanol Concentration Exceeded event has not taken place (either initially or after a mode $04), this PID shall default to zero.

3.2.141E85R – PID $20A2 – Refueling Events Since Recommended Ethanol Concentration Exceeded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A2	Refueling Events Since Recommended Ethanol Concentration Exceeded	1	UNM	0 – 255 counts	E = N * 1

This PID indicates the number of times the vehicle has been refueled since the most recent time that the Recommended Ethanol Concentration Exceeded bit became true. This PID value will reset when a mode $04 code clear is issued to the controller. If a Recommended Ethanol Concentration Exceeded event has not taken place (either initially or after a mode $04), this PID shall default to zero.

3.2.142E85R – PID $20A3 – Fuel Composition Sensor Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A3	Fuel Composition Sensor Frequency	1	UNM	0 – 255 Hz	E = N * 1

This PID contains the frequency of the fuel composition sensor input. This value can be decoded to determine fuel composition or sensor fault status, depending on the value.

3.2.143RBSR – PID $20A4 – Secondary Processor Running Reset Source Address History 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A4	Secondary Processor Running Reset Source Address History 1	2	UNM	0 - $FFFF	E = N

This PID contains one of the last four reset source addresses for a running reset in the secondary processor. A FIFO history buffer, which retains its contents over a battery disconnect, is used to store these reasons with reason 1 containing the most recent reason. This PID is directly associated with the “Secondary Processor Running Reset Reason History x” PID of the same number “x”.

3.2.144FPMR – PID $20A5 – Fuel Pump Disable Monitor Test Result

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A5	Fuel Pump Disable Monitor Test Result	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Pump Disable Monitor Off Test Failed	0	7	1	BLN	N/A	$1=True; $0=False
Fuel Pump Disable Monitor On Test Failed	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains information about the Fuel Pump Disable Monitor Test Result. If the corresponding test fails, the PID value is set to 1. The PID is used in addition to PIDs $2037 & $2039 to indicate the reason why the DTC P0606 is set. When the DTC P0606 is cleared the information in this PID will be cleared as well.

3.2.145MAFR – PID $20A6 – MAF Sensor B Mass Airflow

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A6	MAF Sensor B Mass Airflow	2	UNM	0 - 655.35 g/s	E = N * 0.01

This PID contains the mass airflow measured by MAF sensor B. MAF sensor B is associated with engine bank 2.

3.2.146MAFR – PID $20A7 – Mass Airflow Sensor B Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A7	Mass Airflow Sensor B Frequency	2	UNM	0 - 65535 Hz	E = N * 1

This PID contains the frequency of the mass airflow (MAF) sensor B input. MAF sensor B is associated with engine bank 2.

3.2.147AAPR – PID $20A8 – Barometric Pressure Sensor B A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20A8	Barometric Pressure Sensor B A/D	1	UNM	0 - 100 % 5V	E = N * 100 / 255

This PID contains the barometric pressure analog input of sensor B as a percentage of its reference voltage.

3.2.148AAPR – PID $20AA – Barometric Pressure Sensor B

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20AA	Barometric Pressure Sensor B	1	UNM	0 - 255 kPa	E = N * 1

This PID contains the air pressure value read by sensor B. While this pressure is effectively the ambient air (barometric) pressure, if this sensor is located after the intake air filter, actual barometric pressure may be calculated to account for a pressure drop across the filter.

3.2.149FULR – PID $20AB – Fuel Injection Mode Active

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20AB	Fuel Injection Mode Active	1	ENM	N/A	$00= Normal combustion (C1 combustion); $01= Homogenous charge combustion (C2 combustion); $02= Particulate filter (DPF) high O2 concentration regeneration; $03= Small Quantity Adjustment Mode $04= Particulate filter (DPF) low O2 concentration regeneration $05= Lean NOx Trap (LNT) NOx purging combustion; $06= Lean NOx Trap (LNT) SOx purging lean combustion; $07= Lean NOx Trap (LNT) SOx purging rich combustion; $08= Inhomogeneous combustion (C3 combustion); $09=Selective catalyst reductant (SCR) heating Mode 1; $0A=Selective catalyst reductant (SCR) heating Mode 2; $0B= Park/Neutral gear Particulate filter (DPF P/N) $0C= Rich Idle Particulate filter (DPF) $0D= Particulate filter (DPF) engine protection High O2 mode $0E= Particulate filter (DPF) engine protection Low O2 mode; $0F= Lean NOx Trap (LNT) Engine Protection; $10= Selective catalyst reductant (SCR) heating Mode 3; $11= Idle Injection Learning combustion mode; $12= Hydrocarbon (HC) unloading in driving conditions combustion; $13= Hydrocarbon (HC) unloading in park/neutral conditions combustion mode; $14= The Selective catalyst reductant (SCR) Service Warm Up Combustion mode; $15= The Selective catalyst reductant (SCR) Service Check Combustion mode $16= Sulfur (SOx) unloading combustion mode; $17-$FF=Reserved

This PID indicates which of the injection modes that currently is active.

Normal combustion (C1 combustion) indicates that a “conventional” inhomogeneous combustion is being used.
Homogenous charge combustion (C2 combustion) indicates that a partly homogenous combustion, controlled using a pressure sensor is being used.
Particulate filter (DPF) high O2 concentration regeneration indicates that DPF regeneration with high oxygen content in the exhaust is being used.
Small Quantity Adjustment Mode indicates that the ECU is executing a learning routine of the pilot quantity in order to adjust the injector characteristic in the pilot area. During this learning the injection pattern is changed as imposed by the learning strategy.
Particulate filter (DPF) low O2 concentration regeneration indicates that DPF regeneration with low oxygen content in the exhaust is being used. Due to Low Oxygen content chemical exothermy is released gradually, so avoiding thermal stress on substrate.
Lean NOx Trap (LNT) NOx purging combustion indicates that rich mixture combustion with the purpose of LNT cleaning is being used. DeNOx regeneration is done using rich mixture with a duration of some seconds.
Lean NOx Trap (LNT) SOx purging lean combustion indicates that LNT desulphation in lean mixture conditions is being used.
Lean NOx Trap (LNT) SOx purging rich combustion indicates that LNT desulphation in rich mixture conditions is being used. Less frequently than DeNOx but with a duration of some minutes, a DeSOx phase needs to be carried on to get rid of sulphur deposition from LNT; DeSOx is actuated by means of wobbling between rich/lean combustion phases performed at high temperature.
Inhomogeneous combustion (C3 combustion) indicates that an improved driveability mode is active. The C3 combustion mode optimizes emissions, fuel consumption and engine noise to provide improved driveability.
Selective catalyst reductant (SCR) heating Mode 1 indicates that combustion oriented to get the SCR catalyst to an operating temperature which grants the desired NOx conversion efficiency.
Selective catalyst reductant (SCR) heating Mode 2 indicates that combustion oriented to get the SCR catalyst to an operating temperature which grants the desired NOx conversion efficiency.
Park/Neutral gear Particulate filter (DPF P/N) combustion mode has been set-up for specific automatic-transmission applications in which a DPF regeneration is kept on even if in Park/Neutral gear.
Rich Idle Particulate filter (DPF) regeneration mode is used to protect the component against engine idling with the trap is full of soot. Post and HC injections are not active in this mode
Particulate filter (DPF) engine protection High O2 mode is performed in the high load area while high O2 concentration mode is active.
Particulate filter (DPF) engine protection Low O2 mode is performed in the high load area while low O2 concentration mode is active.
Lean NOx Trap (LNT) Engine Protection indicates that LNT desulphation in lean mixture is used while the engine is working in the high load area.
Selective catalyst reductant (SCR) heating Mode 3 indicates that combustion oriented to get the SCR catalyst to an operating temperature which grants the desired NOx conversion efficiency.
Idle Injection Learning combustion mode is used in idle to correct (with a learning routine) the small injection quantities in diesel engines.
Hydrocarbon (HC) unloading in driving conditions combustion mode is used to unload the HC stored by one of the after-treatment devices (Diesel Oxidation Catalyst, Selective catalyst reductant, Particulate filter) in case the HC amount is too high.
Hydrocarbon (HC) unloading in park/neutral conditions combustion mode is used to unload the HC stored by one of the after-treatment devices (Diesel Oxidation Catalyst, Selective catalyst reductant, Particulate filter) in case the HC amount is too high.
The Selective catalyst reductant (SCR) Service Warm Up Combustion mode is used to in order to quickly increase the SCR upstream temperature preparing the system to perform the SCR Service Bay Test.
The Selective catalyst reductant (SCR) Service Check Combustion mode is used during the SCR Service Bay Test in order to increase the engine out NOx concentration over a minimum desired value keeping the SCR upstream temperature into the desired range.
Sulfur (SOx) unloading combustion mode is used to unload the SOx stored by the Selective catalyst reductant (SCR) in case the SOx amount is too high.

3.2.150FULR – PID $20AC – Total Injection Time Cylinder 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20AC	Total Injection Time Cylinder 1	2	UNM	0 to 65.535 ms	E = N / 1000

This PID contains the total injection time of cylinder 1. Total injection time includes all pilot injections, main injections and post injections.

3.2.151FULR – PID $20AD – Total Injection Time Cylinder 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20AD	Total Injection Time Cylinder 2	2	UNM	0 to 65.535 ms	E = N / 1000

This PID contains the total injection time of cylinder 2. Total injection time includes all pilot injections, main injections and post injections.

3.2.152FULR – PID $20AE – Total Injection Time Cylinder 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20AE	Total Injection Time Cylinder 3	2	UNM	0 to 65.535 ms	E = N / 1000

This PID contains the total injection time of cylinder 3. Total injection time includes all pilot injections, main injections and post injections.

3.2.153FULR – PID $20AF – Total Injection Time Cylinder 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20AF	Total Injection Time Cylinder 4	2	UNM	0 to 65.535 ms	E = N / 1000

This PID contains the total injection time of cylinder 4. Total injection time includes all pilot injections, main injections and post injections.

3.2.154FULR – PID $20B4 – Start of Main Injection Cylinder 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20B4	Start of Main Injection Cylinder 1	2	SNM	-3276.8 to +3276.7 °CS	E = N / 10

This PID contains the start point of the main injection in degrees crank shaft (°CS) relative to Top Dead Center (TDC) of cylinder 1. If no start of Main Injection Cylinder 1 is present, the value shall be set to +3276.7. Positive values are before TDC.

3.2.155FULR – PID $20B5 – Start of Main Injection Cylinder 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20B5	Start of Main Injection Cylinder 2	2	SNM	-3276.8 to +3276.7 °CS	E = N / 10

This PID contains the start point of the main injection in degrees crank shaft (°CS) relative to Top Dead Center (TDC) of cylinder 2. If no start of Main Injection Cylinder 2 is present, the value shall be set to +3276.7. Positive values are before TDC.

3.2.156FULR – PID $20B6 – Start of Main Injection Cylinder 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20B6	Start of Main Injection Cylinder 3	2	SNM	-3276.8 to +3276.7 °CS	E = N / 10

This PID contains the start point of the main injection in degrees crank shaft (°CS) relative to Top Dead Center (TDC) of cylinder 3. If no start of Main Injection Cylinder 3 is present, the value shall be set to +3276.7. Positive values are before TDC.

3.2.157FULR – PID $20B7 – Start of Main Injection Cylinder 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20B7	Start of Main Injection Cylinder 4	2	SNM	-3276.8 to +3276.7 °CS	E = N / 10

This PID contains the start point of the main injection in degrees crank shaft (°CS) relative to Top Dead Center (TDC) of cylinder 4. If no start of Main Injection Cylinder 4 is present, the value shall be set to +3276.7. Positive values are before TDC.

3.2.158FULR – PID $20BC –Fuel Injection Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20BC	Fuel Injection Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Pilot Injection Active	0	7	1	BLN	N/A	$1=True; $0=False
Main injection Active	0	6	1	BLN	N/A	$1=True; $0=False
Post Injection Active	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains several signals associated with the status of the Fuel injection system. The content of each of these signals is described below.

Pilot Injection Active indicates that pilot injections are active. (Does not indicate individual pilot injection pulses, rather that the fuel control system is using pilot injections.) This information is set true if at least one pilot injection is active.

Main Injection Active indicates that main injection is active. (Does not indicate individual main injection pulses, rather that the fuel control system is using main injections.)

Post Injection Active indicates that post injections is active. (Does not indicate individual post injection pulses, rather that the fuel control system is using post injections.) On a Diesel vehicle Post injection is used for Particulate Filter regeneration.

3.2.159FADR – PID $20BD – Small Quantity Learning Inhibit Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20BD	Small Quantity Learning Inhibit Reason	1	ENM	N/A	$00=No Abort Reason Detected; $01=Baro Pressure Too Low; $02=Transmission must be in Park or Neutral; $03=Vehicle speed not zero; $04=Accelerator pedal not released; $05=Powertrain relay voltage out of range; $06=F-terminal value out of range; $07=Engine speed deviation above threshold; $08=Common rail pressure deviation (between actual and target) is too big; $09=Cylinder unbalanced; $0A=Torque derivative too high; $0B=Diesel Particulate Filter Regeneration is Active; $0C=Engine Coolant Temperature Out of Range (different range between Service and DVT procedure); $0D=Intake Air temperature out of range (different range between Service and DVT procedure); $0E=Fuel temperature out of range (different range between Service and DVT procedure); $0F=Fan switched on; $10=High Pressure Pump offset learning not performed; $11=EIA code not programmed; $12=Engine Must be Running; $13=Fault Active; $14=Engine Runtime Too Short; $15= AT Fluid Temperature Out of Range; $16= Generator Type not Programmed; $17= Clutch Pedal Depressed; $18= Cabin Heater On; $19 to $FF = Reserved

This PID contains the current reason preventing the SQL (Small Quantity Learning) procedure from initiating.

3.2.160FADR – PID $20BE – Small Quantity Learning Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20BE	Small Quantity Learning Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Small Quantity Learning Status	0	7	2	ENM	N/A	$0=SQL not active; $1=SQL active; $2=SQL failed; $3=SQL successfully ended
Small Quantity Learning Learned Fuel Pressure Level	0	5	2	ENM	N/A	$0=Level 0; $1=Level 1; $2=Level 2; $3=Level 3
Small Quantity Learning/Idle Injection Learning Performed	0	3	1	BLN	N/A	$1=True; $0=False
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID holds the current status of the Small Quantity Learning (SQL) algorithm. The content of each of these signals is described below.

Small Quantity Learning Status indicates the status of SQL only when triggered via CPID in end of line or at service. The information is updated only when the CPID is triggered and it's maintained until the end of the driving cycle.

Small Quantity Learning Learned Fuel Pressure Level indicates the pressure levels in which the SQL has learned. The information is updated only when the SQL is triggered via CPID and is maintained until the end of the driving cycle.

Small Quantity Learning/Idle Injection Learning Performed indicates if the SQL triggered at service or SQL end of line and idle injection learning have been performed successfully on all the pressure levels.

3.2.161PMDR – PID $20D0 – Power Mode Input Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20D0	Power Mode Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Accessory Input Active	0	7	1	BLN	N/A	$1=True; $0=False
Run/Crank Input Active	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains signals associated with the status of hardwire vehicle power mode inputs. The content of each of these signals is described below.

Accessory Input Active indicates that the accessory hardwire input to the controller is active in applications using a power mode electrical architecture from PPEI 2.x or later. In applications using a power mode electrical architecture that precedes PPEI 2.x, this signal indicates that the unlock hardwire input to the controller is active.

Run/Crank Input Active indicates that the run/crank hardwire input to the controller is active. For hardware that measures input voltage, this bit shall represent an input voltage within a calibrated range. For hardware using discrete inputs, this input will represent voltage present.

3.2.162APSR – PID $20D1 – Accelerator Pedal Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20D1	Accelerator Pedal Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Accelerator Pedal Position Sensors 1 and 2 Disagree	0	7	1	BLN	N/A	$1=True; $0=False
APP Sensor 1 Out of Range	0	6	1	BLN	N/A	$1=True; $0=False
APP Sensor 2 Out of Range	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains signals associated with the status of the accelerator pedal control system. The content of each of these signals is described below.

Accelerator Pedal Position Sensors 1 and 2 Disagree indicates that the (corrected) accelerator pedal positions 1 and 2 are currently out of correlation, or that a fault has been detected (and latched) based upon the (corrected) accelerator pedal positions 1 and 2 being out of correlation.

APP Sensor 1 Out of Range indicates that the sensed accelerator pedal position 1 sensor value is currently out of range or that a fault has been detected (and latched) based upon the sensed accelerator pedal position 1 sensor value being out of range

APP Sensor 2 Out of Range indicates that the sensed accelerator pedal position 2 sensor value is currently out of range or that a fault has been detected (and latched) based upon the sensed accelerator pedal position 2 sensor value being out of range

3.2.163BRKR – PID $20D2 – Brake Pedal Position Sensor A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20D2	Brake Pedal Position Sensor A/D	1	UNM	0 - 100 % 5V	E = N * 100 / 255

This PID contains the brake pedal position as indicated by the analog sensor as a percentage of its reference voltage.

3.2.164BRKR – PID $20D3 – Brake Pedal Released Learned Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20D3	Brake Pedal Released Learned Position	1	UNM	0 - 100 % 5V	E = N * 100 / 255

This PID contains the position learned for the brake pedal position sensor when the brake pedal is fully released.

3.2.165BRKR – PID $20D4 – Brake Pedal Displacement

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20D4	Brake Pedal Displacement	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the displacement of the brake pedal (corrected and normalized based upon its learned released position), where 0% corresponds to the brake pedal being released and 100% corresponds to the brake pedal being fully applied. Note: 100% brake pedal position may not be achievable since the brake pedal movement will be limited by hydraulic pressure within the brake system.

3.2.166FULR – PID $20D5 – Fuel Injection Controller Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20D5	Fuel Injection Controller Status	1	ENM	N/A	$00=Not present – injection controller not implemented; $01=Uninitialized – injection controller present but not initialized for operation; $02=Ready – injection controller present and ready for operation; $03-$FF=Failed – injection controller internal fault (implementation-defined detail status)

This PID contains the status of the fuel injector driver circuit controller located in the ECM. The status values of Uninitialized and Failed are monitored to report a fail of DTC P062B.

3.2.167FULR – PID $20D6 – Injector Driver Boost Supply Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20D6	Injector Driver Boost Supply Voltage	2	UNM	0 – 6553.5 V	E = N * 0.1

This PID contains the injector driver circuit boost supply voltage. The boost voltage is monitored and a value that is too high or too low will lead to report a fail of DTC P062B.

3.2.168TPSR – PID $20D9 – Intake Air Flow Valve Actuator Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20D9	Intake Air Flow Valve Actuator Control Circuit.	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Intake Air Flow Valve Control circuits shorted.	0	7	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Air Flow Valve Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Air Flow Valve Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Air Flow Valve Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the status of the associated short to ground (DTC P02E2), open circuit (DTC P02E0), short to power (DTC P02E3) and shorted control circuit to control circuit (DTC P122C) tests of the Intake Air Flow Valve actuator.

3.2.169TPSR – PID $20DB – Desired Intake Air Flow Valve Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20DB	Desired Intake Air Flow Valve Position	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the desired Intake Air Flow Valve position from Intake Air Flow Valve position sensor. A value of 0% indicates that the intake air flow valve is commanded closed, and a value of 100% indicates that the intake air flow valve is commanded to the maximum open position.

3.2.170TPSR – PID $20DC – Actual Intake Air Flow Valve Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20DC	Actual Intake Air Flow Valve Position	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the displacement determined from Intake Air Flow Valve position sensor (corrected based upon its learned minimum position). A value of 0% indicates that the Intake Air Flow Valve is at its learned closed position, and a value of 100% indicates the Intake Air Flow Valve is at its maximum open position.

3.2.171TPSR - PID $20DD – Intake Air Flow Valve Position Sensor A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20DD	Intake Air Flow Valve Position Sensor A/D	1	UNM	0 to 100 % 5V	E = N * 100 / 255

This PID contains the Intake Air Flow Valve position sensor analog input as a percentage of its reference voltage.

3.2.172TPSR - PID $20DE – Intake Air Flow Valve Position Sensor Learned Minimum

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20DE	Intake Air Flow Valve Position Sensor Learned Minimum	1	UNM	0 to 100 % 5V	E = N * 100 / 255

This PID contains the minimum position learned for Intake Air Flow Valve position sensor when the Intake Air Flow Valve is closed.

3.2.173EGRR – PID $20DF – Exhaust Gas Recirculation H-Bridge Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20DF	Exhaust Gas Recirculation H-Bridge Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Exhaust Gas Recirculation H-Bridge Control circuits shorted.	0	7	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Exhaust Gas Recirculation H-Bridge Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Exhaust Gas Recirculation H-Bridge Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Exhaust Gas Recirculation H-Bridge Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the control circuit status of the exhaust gas recirculation (EGR) output short to ground, open circuit, short to power, and high side to low side short tests used to report DTCs P0489, P0403, P0490, and P1407, respectively.

3.2.174FFMR - PID $20E2 – Water in Fuel Detection Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20E2	Water in Fuel Detection Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Water in Fuel Indication On	0	7	1	BLN	N/A	$1=True; $0=False
Water in Fuel Detected	0	6	1	BLN	N/A	$1=True; $0=False

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Water in Fuel Detected Debounced	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains status bits associated with water in fuel detection. The content of each of these signals is described below.

Water in Fuel Indication On indicates that the water in fuel indication is being requested on over serial communication in order to inform the driver of a water in fuel condition. This signal is based upon either a water in fuel switch or water in fuel sensor.

Water in Fuel Detected indicates the undebounced determination of the water in fuel input to the ECM. In electrical architectures using a water in fuel sensor, this signal is based upon the water in fuel signal passing a threshold. In electrical architectures using a discrete water in fuel switch, this signal is based upon the state of the water in fuel switch.

Water in Fuel Detected Debounced indicates the debounced determination of the water in fuel input to the ECM. In electrical architectures using a water in fuel sensor, this signal is based upon the water in fuel signal passing a threshold. In electrical architectures using a discrete water in fuel switch, this signal is based upon the state of the water in fuel switch.

3.2.175CEBR – PID $20E4 – Exhaust Gas Recirculation Cooler Bypass Valve Actuator Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20E4	Exhaust Gas Recirculation Cooler Bypass Valve Actuator Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the commanded duty cycle for the EGR Cooler Bypass Valve actuator output.

3.2.176CEBR – PID $20E5 - Exhaust Gas Recirculation Cooler Bypass Valve Actuator Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20E5	EGR Cooler Bypass Valve Actuator Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
EGR Cooler Bypass Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
EGR Cooler Bypass Valve Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
EGR Cooler Bypass Valve Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
EGR Cooler Bypass Valve Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the EGR Cooler Bypass valve output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P245C, P245A, P245D, respectively.

3.2.177SWCR - PID $20E6 – Intake Manifold Runner Control Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20E6	Intake Manifold Runner Control Duty Cycle	1	UNM	0 - 100%	E = N * 100 / 255

This PID contains the commanded duty cycle for the Intake Manifold Runner Control. (Intake Manifold Runner Control (IMRC) is the SAE recommended term for any device that controls charge motion. These devices may be unofficially referred to as swirl control, charge motion control, or port deactivation.)

3.2.178SWCR - PID $20E7 – Intake Manifold Runner Control Diagnostic Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20E7	Intake Manifold Runner Control Diagnostic Status	1	ENM	N/A	$01=Supply Voltage Not Correct; $02=No Error Detected; $03=Variable Swirl Actuator Damage (Generic HW Fault; $04=Stuck Open; $05= Stuck Closed; $06=Stuck in Intermediate Position; $07=Supplier-specific Generic Fault; $08=Wrong PWM Frequency or Duty-cycle on Command Line; $09= Error PWM Invalid; $0A to $FF Reserved;

This PID contains the diagnostic status for the Intake Manifold Runner Control (IMRC) system. This information relates to a continuously controlled (duty cycle) IMRC system. (Intake Manifold Runner Control is the SAE recommended term for any device that controls charge motion. These devices may be unofficially referred to as swirl control, charge motion control, or port deactivation.)

3.2.179FHPR – PID $20E8 – Fuel Rail Over Pressure Event Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20E8	Fuel Rail Over Pressure Event Counter	1	UNM	0 to 255 counts	E = N * 1

This PID indicates whether an over pressure in the fuel rail pressure system has been detected and, if >0, the number of times it has been detected. The counter will freeze once it reaches its maximum value. If no over pressure events are detected, this value will default to 0.

3.2.180FHPR – PID $20E9 – Fuel Rail Over Pressure Maximum Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20E9	Fuel Rail Over Pressure Maximum Pressure	2	UNM	0 to 655350 kPa	E = N * 10

This PID holds the highest pressure reached under any of the detected over pressure events. If no over pressure events are detected, this value will default to 0.

3.2.181FHPR – PID $20EA – Fuel Rail Over Pressure Maximum Pressure Duration

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20EA	Fuel Rail Over Pressure Maximum Pressure Duration	2	UNM	0 to 65535 sec	E = N / 100

This PID holds the longest duration of the over pressure condition under any of the detected over pressure events. If no over pressure events are detected, this value will default to 0.

3.2.182FHPR – PID $20EB – High Pressure Fuel Pump Learning Abort Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20EB	High Pressure Fuel Pump Learning Abort Reason	1	ENM	N/A	$00=No Abort Reason Detected; $01= Engine speed not idle or Vehicle speed not zero; $02= Common rail pressure deviation (between actual and target) is too big; $03= Fault on rail pressure system; $04 to $FF = Reserved

This PID holds the most recent abort reason for the high pressure fuel pump learning algorithm.

3.2.183FHPR – PID $20EC – High Pressure Fuel Pump Learning Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20EC	High Pressure Fuel Pump Learning Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
High Pressure Fuel Pump Learning Status	0	7	2	ENM	N/A	$0=Not learned; $1=In progress; $2=Completed; $3=Fail
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID holds the current status of the high pressure fuel pump learning algorithm.

3.2.184FFMR – PID $20ED – Fuel Filter Heater Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20ED	Fuel Filter Heater Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Filter Heater Commanded On	0	7	1	BLN	N/A	$1= True; $0= False
Reserved	0	6	1			
Fuel Filter Heater Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Fuel Filter Heater Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Fuel Filter Heater Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the fuel filter heater along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2688, P2687, P2689, respectively.

3.2.185MEMR – PID $20EE – Engine Upgrade Protection Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20EE	Engine Upgrade Protection Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Engine Upgrade Protection Supported	0	7	1	BLN	N/A	$1=True; $0=False
Engine Performance Identifier Programmed	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains status bits associated with the Engine Upgrade Protection strategy. The content of each of these signals is described below.

Engine Upgrade Protection Supported indicates if the ECU supports the PPEI engine upgrade protection strategy.

Engine Performance Identifier Programmed indicates if the engine performance identifier value (DID $27) has been programmed. This bit will be used as part of the engine upgrade prevention strategy logic to determine if the engine performance identifier value needs to be programmed.

3.2.186CRZR – PID $20EF – Cruise/Speed Limiter Raw Switch States

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20EF	Cruise/Speed Limiter Raw Switch States	1	ENM	N/A	$00=Indeterminate; $01=No Activation; $02=Resume; $03=Set; $04=SL/SW On/Off; $05=Cruise On/Off; $06=Cancel; $07=Fault Low; $08=Fault High; $09=Between Ranges / Invalid

This PID contains the cruise and speed limiter raw switch states as received by the ECM. The switch states are used to determine the modes for the cruise and speed limiter functions and apply to architectures using momentary (non-latching) switches. The various states are described as follows:

Indeterminate indicates that the platform module is currently not able to determine the state of the input.

No Activation indicates that no switch is active. Note: this is also the default PID value if a valid serial data signal has not been received.

Resume indicates that the Resume switch is active.

Set indicates that the Set switch is active

SL/SW On/Off indicates that the Speed Limiter/Speed Warning On/Off request switch is active. Note: this reflects the current state of the actual momentary switch and not the latched state of the SL/SW On/Off switch.

Cruise On/Off indicates that the Cruise On/Off switch is active. Note: this reflects the current state of the actual momentary switch and not the latched state of the Cruise On/Off switch.

Cancel indicates that the Cancel switch is active

Fault Low indicates that the switch input voltage ratio is in a range associated with an open circuit or a short to ground. If this condition persists continuously for longer than a calibratable period, a DTC will set.

Fault High indicates that the switch input voltage ratio is in a range associated with a short to power. If this condition persists continuously for longer than a calibratable period, a DTC will set.

Between Ranges / Invalid indicates that the switch input is between valid voltage ratio ranges. If this condition persists continuously for longer than a calibratable period, a DTC will set.

3.2.187FULR – PID $20F0 – Injection Controller Monitoring Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20F0	Injection Controller Monitoring Status	1	ENM	N/A	$00=No Fault Active; $01=Boost Voltage Too Low; $02=Boost Voltage Too High; $03=Controller Status Failed

This PID indicates the status of the injection controller monitoring used to set DTC P062B or the boost voltage error (boost voltage is too high or too low) used to set DTC P062D.

3.2.188FULR – PID $20FB – Number of Actuated Post-Injections

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20FB	Number of Effectively Actuated Post-Injections	1	UNM	0 to 255 Counts	E = N * 1

This PID contains the number of post injections actuated during a combustion cycle.

3.2.189OXYR – PID $20FC – Oxygen Sensor Bank 1 Sensor 1 Heating Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20FC	Oxygen Sensor Bank1 Sensor 1 Heating Status	1	ENM	N/A	$00=Invalid; $01=Operating Temperature Not Reached; $02=Operating Temperature Reached

This PID contains the operating status of the oxygen sensor bank 1 sensor 1 heater.

3.2.190OXYR – PID $20FD – Exhaust Gas Oxygen Concentration Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20FD	Exhaust Gas Oxygen Concentration Sensor 1	2	UNM	-12 to +21.68499 %	E = N * 0.000514 - 12

This PID contains the concentration of oxygen in the exhaust gas measured by oxygen sensor 1.

3.2.191FULR – PID $20FE – Post-Injection Quantity

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20FE	Post-Injection quantity	2	UNM	0 to 200 mm^3/str	E = N * 200 / 65535

This PID contains the total fuel quantity injected with post injections during a combustion cycle.

3.2.192AAPR – PID $20FF – Barometric Pressure Correlation Diagnostic Estimate

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$20FF	Barometric Pressure Estimate	1	UNM	0 - 255 kPa	E = N * 1

This PID contains an estimate of ambient air (barometric) pressure, used to correlate with sensed ambient air pressure in the Barometric Pressure Correlation Diagnostic. In applications without a barometric pressure sensor, the value of this PID is the same as the value of PID $0033.

3.2.193EITR – PID $2300 – Humidity Sensor Intake Air Temperature Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2300	Humidity Sensor Intake Air Temperature Frequency	1	UNM	0 to 510 Hz	E = N * 2

This PID contains the frequency of the humidity sensor's intake air temperature (IAT) sensor input.

3.2.194HUMR – PID $2301 – Relative Humidity

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2301	Relative Humidity	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the undefaulted relative humidity of the intake air as measured by a humidity sensor.

3.2.195HUMR – PID $2302 – Relative Humidity Sensor Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2302	Relative Humidity Sensor Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the raw duty cycle signal from the relative humidity sensor electronics, which represents the relative humidity of the intake air.

3.2.196HUMR – PID $2303 – Weight per cent Water in Air

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2303	Weight per cent Water in Air	1	UNM	0 to 5 %	E = N * 5 / 255

This PID contains the calculated value of weight per cent water (specific humidity) in the intake air.

3.2.197CCLR – PID $2305 – Combustion Operation Mode

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2305	Combustion Operation Mode	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
C1 combustion active	0	7	1	BLN	N/A	$1=True; $0=False
C2 combustion active	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains the combustion operation mode used on the vehicle.

C1 combustion active indicates that a “conventional” inhomogeneous combustion is being used.

C2 combustion active indicates that a partly homogenous combustion, controlled using a pressure sensor, is being used.

3.2.198DSCR – PID $2306 – Permanent DTC Drive Cycle Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2306	Permanent DTC Drive Cycle Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Idle Time Complete	0	7	1	BLN	N/A	$1=True; $0=False
Propulsion System Active Time Complete	0	6	1	BLN	N/A	$1=True; $0=False
Vehicle Speed Time/Load Complete	0	5	1	BLN	N/A	$1=True; $0=False
Standard Drive Cycle Complete	0	4	1	BLN	N/A	$1=True; $0=False
RPM/Load Complete	0	3	1	BLN	N/A	$1=True; $0=False
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains signals associated with the clearing of permanent DTCs (mode $0A). The content of each of these signals is described below.

Idle Time Complete indicates that the internal combustion engine (ICE) idle time requirement is complete. (30 seconds of contiguous time after engine start with foot off accelerator pedal and vehicle speed = 0. This bit is active in the ECM only.

Propulsion System Active Time Complete indicates that the vehicle propulsion system has been active for the required amount of time. (10 minutes) This bit is active in the ECM only.

Vehicle Speed/Load Time Complete indicates that the vehicle speed or load has been above a calibratable threshold for five minutes. This bit is active in the ECM only.

Standard Drive Cycle Complete means that all required components of the standard drive cycle have completed since a mode $04 was last issued. This bit is non-volatile and may be set even if the other three components have not taken place on the current key cycle. This bit is active in all OBD modules.

RPM/Load Time Complete indicates that the rpm or load has been above a calibratable threshold for five minutes. This bit is active in the ECM only.

3.2.199DSCR – PID $2307 – Permanent DTC Drive Cycle Idle Complete Timer

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2307	Permanent DTC Drive Cycle Idle Complete Timer	2	UNM	0 to 65535 s	E = N * 1

This PID contains the accumulated amount of contiguous time after the first internal combustion engine (ICE) start in which both pedal position and vehicle speed are zero. If this timer reaches 30 seconds, then the Idle Time Complete bit in PID $2306 will be set to “True”.

3.2.200DSCR – PID $2308 – Permanent DTC Drive Cycle Propulsion Active Timer

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2308	Permanent DTC Drive Cycle Propulsion Active Timer	2	UNM	0 to 65535 s	E = N * 1

This PID contains the total accumulated time in which the propulsion system has been active after the 1st time the internal combustion engine (ICE) is started each key on.

3.2.201DSCR – PID $2309 – Permanent DTC Drive Cycle Vehicle Speed Timer

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2309	Permanent DTC Drive Cycle Vehicle Speed Timer	2	UNM	0 to 65535 s	E = N * 1

This PID contains the total accumulated time in which the vehicle speed has been > 25 mph or the load has been above a calibratable threshold after the 1st time the internal combustion engine (ICE) is started each key on. In instances where Diesel Equipped Medium Duty Vehicles are Dyno Certified and on all Diesel Equipped Heavy Duty Vehicles which are required to be Dyno Certified, the 25 miles per hour criteria, stated above, may be replaced with 1150 rpm.

3.2.202CCLR – PID $230A – Mass Fraction Burned Cylinder 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$230A	Mass Fraction Burned Cylinder 1	1	UNM	-60 to +90 °cs	E = N * 150 / 255 - 60

This PID contains the actual crank shaft angle for cylinder 1 when 50% of the mass fraction is burned. A positive value indicates a position before Top Dead Center (TDC).

3.2.203CCLR – PID $230B – Mass Fraction Burned Cylinder 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$230B	Mass Fraction Burned Cylinder 2	1	UNM	-60 to +90 °cs	E = N * 150 / 255 - 60

This PID contains the actual crank shaft angle for cylinder 2 when 50% of the mass fraction is burned. A positive value indicates a position before Top Dead Center (TDC).

3.2.204CCLR – PID $230C – Mass Fraction Burned Cylinder 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$230C	Mass Fraction Burned Cylinder 3	1	UNM	-60 to +90 °cs	E = N * 150 / 255 - 60

This PID contains the actual crank shaft angle for cylinder 3 when 50% of the mass fraction is burned. A positive value indicates a position before Top Dead Center (TDC).

3.2.205CCLR – PID $230D – Mass Fraction Burned Cylinder 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$230D	Mass Fraction Burned Cylinder 4	1	UNM	-60 to +90 °cs	E = N * 150 / 255 - 60

This PID contains the actual crank shaft angle for cylinder 4 when 50% of the mass fraction is burned. A positive value indicates a position before Top Dead Center (TDC).

3.2.206CCLR – PID $230E – Desired Mass Fraction Burned

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$230E	Desired Mass Fraction Burned	1	UNM	-60 to +90 °cs	E = N * 150 / 255 - 60

This PID contains the desired crank shaft angle position of the combustion. A positive value indicates a position before Top Dead Center (TDC).

3.2.207CCLR – PID $230F – Indicated Mean Effective Pressure Cylinder 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$230F	Indicated Mean Effective Pressure Cylinder 1	1	UNM	-240 to +3840 kPa	E = N * 16 - 240

This PID contains the indicated mean effective pressure detected in cylinder 1.

3.2.208CCLR – PID $2310 – Indicated Mean Effective Pressure Cylinder 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2310	Indicated Mean Effective Pressure Cylinder 2	1	UNM	-240 to +3840 kPa	E = N * 16 - 240

This PID contains the indicated mean effective pressure detected in cylinder 2.

3.2.209CCLR – PID $2311 – Indicated Mean Effective Pressure Cylinder 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2311	Indicated Mean Effective Pressure Cylinder 3	1	UNM	-240 to +3840 kPa	E = N * 16 - 240

This PID contains the indicated mean effective pressure detected in cylinder 3.

3.2.210CCLR – PID $2312 – Indicated Mean Effective Pressure Cylinder 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2312	Indicated Mean Effective Pressure Cylinder 4	1	UNM	-240 to +3840 kPa	E = N * 16 - 240

This PID contains the indicated mean effective pressure detected in cylinder 4.

3.2.211CCLR – PID $2314 – Pressure Sensor Offset Cylinder 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2314	Pressure Sensor Offset Cylinder 1	2	SNM	-2048 to 2047.9375 kPa	E = N / 16

This PID contains the offset of the cylinder pressure sensor in cylinder 1.

3.2.212CCLR – PID $2315 – Pressure Sensor Offset Cylinder 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2315	Pressure Sensor Offset Cylinder 2	2	SNM	-2048 to 2047.9375 kPa	E = N / 16

This PID contains the offset of the cylinder pressure sensor in cylinder 2.

3.2.213CCLR – PID $2316 – Pressure Sensor Offset Cylinder 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2316	Pressure Sensor Offset Cylinder 3	2	SNM	-2048 to 2047.9375 kPa	E = N / 16

This PID contains the offset of the cylinder pressure sensor in cylinder 3.

3.2.214CCLR – PID $2317 – Pressure Sensor Offset Cylinder 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2317	Pressure Sensor Offset Cylinder 4	2	SNM	-2048 to 2047.9375 kPa	E = N / 16

This PID contains the offset of the cylinder pressure sensor in cylinder 4.

3.2.215EOPR – PID $2318 – Distance Since Low Oil Pressure Engine Shutdown

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2318	Distance Since Low Oil Pressure Engine Shutdown	2	UNM	0 to 327675 km	E = N * 5

This PID indicates the accumulated distance driven since the most recent low oil pressure engine shutdown.

3.2.216EOPR – PID $2319 – Low Oil Pressure Engine Shutdown Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2319	Low Oil Pressure Engine Shutdown Counter	1	UNM	0 to 255 counts	E = N * 1

This PID indicates total number of low oil pressure engine shutdown events over the life of the controller.

3.2.217ECTR – PID $231A – Distance Since High Coolant Temperature Engine Shutdown

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$231A	Distance Since High Coolant Temperature Engine Shutdown	2	UNM	0 to 327675 km	E = N * 5

This PID indicates the accumulated distance driven since the most recent high coolant temperature engine shutdown.

3.2.218ECTR – PID $231B – High Coolant Temperature Engine Shutdown Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$231B	High Coolant Temperature Engine Shutdown Counter	1	UNM	0 to 255 counts	E = N * 1

This PID indicates total number of high coolant temperature engine shutdown events over the life of the controller.

3.2.219OXYR – PID $231C – Oxygen Sensor Service Bay Heater Resistance Learn Result

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$231C	Oxygen Sensor Service Bay Heater Resistance Learn Result	1	ENM	N/A	$00=Not complete; $01=Complete; $02=In Progress

This PID contains the results of the oxygen sensor heater resistance learn procedure.

3.2.220OXYR – PID $231D – Oxygen Sensor Service Bay Heater Resistance Learn Abort Reasons

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$231D	Oxygen Sensor Service Bay Heater Resistance Learn Abort Reasons	1	ENM	N/A	$00=Not Aborted; $01=Open Loop; $02=Voltage too high; $03=DTC Present; $04=Coolant Temp; $05=Vehicle Speed; $06=Engine Not Running; $07=Not in Park or Neutral; $08=Limits Out of Range

This PID contains the reason that the oxygen sensor service bay heater resistance learn was aborted.

3.2.221OXYR – PID $231E – Oxygen Sensor Service Bay Response Test Result

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$231E	Oxygen Sensor Service Bay Response Test Result	1	ENM	N/A	$00=Not complete; $01=Complete; $02=In Progress

This PID contains the results of the oxygen sensor response service bay test.

3.2.222OXYR – PID $231F – Oxygen Sensor Service Bay Response Test Abort Reasons

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$231F	Oxygen Sensor Service Bay Response Test Abort Reasons	1	ENM	N/A	$00=Not Aborted; $01=Open Loop; $02=Engine Speed; $03=Voltage Out of Range; $04=Intake Air Temp; $05=DTC Present; $06=Coolant Temp; $07=Vehicle Speed; $08=Engine Not Running; $09=Not in Park or Neutral; $0A=Barometric Pressure Too Low; $0B=Limits Out of Range

This PID contains the reason that the oxygen sensor service bay response test was aborted.

3.2.223OXYR – PID $2320 – High Bias Oxygen Sensor Voltage Bank 1 Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2320	High Bias Oxygen Sensor Voltage Bank 1 Sensor 1	1	UNM	0 to 2.55 V	E = N / 100

This PID contains the oxygen sensor voltage for bank 1 sensor 1.

3.2.224OXYR – PID $2321 – High Bias Oxygen Sensor Voltage Bank 1 Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2321	High Bias Oxygen Sensor Voltage Bank 1 Sensor 2	1	UNM	0 to 2.55 V	E = N / 100

This PID contains the oxygen sensor voltage for bank 1 sensor 2.

3.2.225OXYR – PID $2322 – High Bias Oxygen Sensor Voltage Bank 2 Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2322	High Bias Oxygen Sensor Voltage Bank 2 Sensor 1	1	UNM	0 to 2.55 V	E = N / 100

This PID contains the oxygen sensor voltage for bank 2 sensor 1.

3.2.226OXYR – PID $2323 – High Bias Oxygen Sensor Voltage Bank 2 Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2323	High Bias Oxygen Sensor Voltage Bank 2 Sensor 2	1	UNM	0 to 2.55 V	E = N / 100

This PID contains the oxygen sensor voltage for bank 2 sensor 2.

3.2.227MEMR – PID $2324 – RAM Failure Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2324	RAM Failure Reason	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1	BLN	N/A	
Reserved	0	6	1	BLN	N/A	
Write protected RAM Failure	0	5	1	BLN	N/A	$1=True; $0=False
Dual Store RAM Failure	0	4	1	BLN	N/A	$1=True; $0=False
Secondary Processor RAM Failure	0	3	1	BLN	N/A	$1=True; $0=False
TPU Microcode RAM Failure	0	2	1	BLN	N/A	$1=True; $0=False
Cache RAM Failure	0	1	1	BLN	N/A	$1=True; $0=False
System RAM Failure	0	0	1	BLN	N/A	$1=True; $0=False

This PID indicates the particular area of RAM that caused the RAM fault to be set, as indicated by DTC P0604. More than one area may be indicated if the fault has set multiple times during the current controller power cycle.

3.2.228MEMR – PID $2325 – RAM Fault Instruction Address

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2325	RAM Fault Instruction Address	4	UNM	$00000000 to $FFFFFFFF	E = N * 1

This PID indicates the address of the instruction that was executing when the RAM fault occurred, as indicated by DTC P0604.

3.2.229MEMR – PID $2326 – ROM Fault Address

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2326	ROM Fault Address	4	UNM	$00000000 to $FFFFFFFF	E = N * 1

This PID indicates the address of the failed ROM location, as indicated by DTC P0601.

3.2.230MEMR – PID $2327 – ROM Fault Instruction Address

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2327	ROM Fault Instruction Address	4	UNM	$00000000 to $FFFFFFFF	E = N * 1

This PID indicates the address of the instruction that was executing when the ROM fault occurred, as indicated by DTC P0601.

3.2.231MEMR – PID $2328 – ROM Failure Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2328	ROM Failure Reason	1	ENM	N/A	$00=No ROM Fault; $01=CRC16 Checksum Fault; $02=ROM ECC Fault; $03=Torque Security Calibration Error

This PID indicates the particular area of ROM that caused the ROM fault to be set, as indicated by DTC P0601.

3.2.232MAFR – PID $2329 – Mass Air Flow Sensor Supply Voltage Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2329	Mass Air Flow Sensor Supply Voltage Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Mass Air Flow Sensor Supply Voltage Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Mass Air Flow Sensor Supply Voltage Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Mass Air Flow Sensor Supply Voltage Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Mass Air Flow Sensor Supply Voltage Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the mass air flow sensor supply voltage output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P121B, P121A, and P121C, respectively.

3.2.233EPSR – PID $232A – Crankshaft Position Signal Output Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$232A	Crankshaft Position Signal Output Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Crankshaft Position Signal Output Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Crankshaft Position Signal Output Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Crankshaft Position Signal Output Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Crankshaft Position Signal Output Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the crankshaft position signal output (the output is considered to be commanded on if engine speed is not zero) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2618, P2617, and P2619, respectively.

3.2.234EPSR – PID $232B – Camshaft Position Signal Output Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$232B	Camshaft Position Signal Output Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Camshaft Position Signal Output Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Camshaft Position Signal Output Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Camshaft Position Signal Output Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Camshaft Position Signal Output Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the camshaft position signal output (the output is considered to be commanded on if engine speed is not zero) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2615, P2614, and P2616, respectively.

3.2.235OATR – PID $232C – Outside Air Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$232C	Outside Air Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID displays the undefaulted temperature measured by an outside (ambient) air temperature sensor. This is not necessarily an OBD-compliant sensor. If the data used to compute the outside air temperature is obtained from a serial data signal, then, if the signal is not received, the value of PID $232C shall be set to $00 = -40°C.

3.2.236CCLR – PID $232D – Pressure Sensor Voltage Cylinder 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$232D	Pressure Sensor Voltage Cylinder 1	1	UNM	0 to 100% 5V	E = N * 100 / 255

This PID contains cylinder 1 pressure analogue input as a percentage of its reference voltage. With the engine not running, this value is time based sampled. With engine running, the value is sampled at a fixed angular crank shaft position.

3.2.237CCLR – PID $232E – Pressure Sensor Voltage Cylinder 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$232E	Pressure Sensor Voltage Cylinder 2	1	UNM	0 to 100% 5V	E = N * 100 / 255

This PID contains cylinder 2 pressure analogue input as a percentage of its reference voltage. With the engine not running, this value is time based sampled. With engine running, the value is sampled at a fixed angular crank shaft position.

3.2.238CCLR – PID $232F – Pressure Sensor Voltage Cylinder 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$232F	Pressure Sensor Voltage Cylinder 3	1	UNM	0 to 100% 5V	E = N * 100 / 255

This PID contains cylinder 3 pressure analogue input as a percentage of its reference voltage. With the engine not running, this value is time based sampled. With engine running, the value is sampled at a fixed angular crank shaft position.

3.2.239CCLR – PID $2330 – Pressure Sensor Voltage Cylinder 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2330	Pressure Sensor Voltage Cylinder 4	1	UNM	0 to 100% 5V	E = N * 100 / 255

This PID contains cylinder 4 pressure analogue input as a percentage of its reference voltage. With the engine not running, this value is time based sampled. With engine running, the value is sampled at a fixed angular crank shaft position.

3.2.240FHPR – PID $2331 – High Fuel Pressure Limiter Valve Test Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2331	High Fuel Pressure Limiter Valve Test Status	1	ENM	N/A	$00= Test Not Performed During This Key On; $01= Test Running; $02= Test Passed; $03= Test Failed; $04= Test Aborted; $05 to $FF = Reserved;

This PID indicates the result of the most recent high fuel pressure limiter valve test performed during this key on.

3.2.241FHPR – PID $2332 – High Fuel Pressure Limiter Valve Test Abort Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2332	High Fuel Pressure Limiter Valve Test Abort Reason	1	ENM	N/A	$00= None; $01= Vehicle Speed Detected; $02= DTC Present; $03= Engine Speed Too Low; $04=Voltage Out of Range; $05 to $FF = Reserved;

This PID contains the reason why the high fuel pressure limiter valve test has been aborted.

3.2.242FHPR – PID $2333 – Fuel Rail Pressure Sensor Rationality Diagnostic Test Result

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2333	Fuel Rail Pressure Sensor Rationality Diagnostic Test Result	1	ENM	N/A	$00=No Result; $01=Test Passed; $02=Failed - Stuck; $03=Failed - Offset Low; $04=Failed - Offset High; $05=Failed - Proportional Low; $06=Failed - Proportional High; $07=Relief Valve Offset Learn Occurred This Trip; $08=Relief Valve Offset Not Learned; $09= Failed Dual Sensor Rationality Low; $0A= Failed Dual Sensor Rationality High

This PID contains the result of the fuel rail pressure sensor rationality diagnostic test.

3.2.243FULR – PID $2334 – Fuel Injector Flow Test Abort Reasons

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2334	Fuel Injector Flow Test Abort Reasons	1	ENM	N/A	$00=None; $01=Acceleration Pedal Pressed; $02=Clutch Pedal Must Be Depressed; $03=Pressure Set Point Not Reached During Injector Flow Test Active Phase; $04=RPM Out of Range; $05=DTC Present; $06=Vehicle Speed Has Been Detected; $07=DPF Regeneration is Active (normal or service); $08=SQL Strategy Service Procedure Has Become Active, $09=Coolant Temperature Out of Range; $0A=A/C Must Be Off; $0B=PTO is Enabled; $0C= Transmission Must Be in Park or Neutral; $0D to $FF=Reserved

This PID contains the abort reason for the fuel injector flow test device control.

3.2.244FULR – PID $2335 – Fuel Injector Flow Test Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2335	Fuel Injector Flow Test Status	1	ENM	N/A	$00=Test Not Performed During This Key Cycle; $01=Fuel Pressure Normalization Process Initialized; $02=Cylinder 1 Test Active; $03=Cylinder 2 Test Active; $04=Cylinder 3 Test Active; $05=Cylinder 4 Test Active; $06=Cylinder 5 Test Active; $07=Cylinder 6 Test Active; $08=Cylinder 7 Test Active; $09=Cylinder 8 Test Active; $0A=Injector Flow Test Complete; $0B= Injector Flow Test Aborted; $0C to $FF=Reserved

This PID contains the status of the fuel injector flow test device control. The result of the fuel injector flow test is stored in DID $6C “Direct Injection Fuel Injector Flow Test Results”.

3.2.245VSLR – PID $2336 – Speed Limiter/Warning Disengage Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2336	Speed Limiter/Warning Disengage Reason	1	ENM	N/A	$00=None; $01=Switch Type Error; $02=Switch Serial Data Error; $03=Cancel Switch Active; $04=On/Off Switch Off; $05=Switch Failure; $06=Vehicle Speed Fault; $07=Cruise Control On/Off Switch On; $08 = Transmission Gear Fault; $09 = Hill Descent Active; $0A = Hill Descent Serial Data Fault

This PID contains the most recent reason that the speed limiter/warning system has been disengaged.

3.2.246VSLR – PID $2337 – Speed Limiter/Warning Inhibit Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2337	Speed Limiter/Warning Inhibit Reason	1	ENM	N/A	$00=None; $01=Switch Type Error; $02=Switch Serial Data Error; $03=Cancel Switch Active; $04=On/Off Switch Off; $05=Switch Failure; $06=Vehicle Speed Fault; $07=Cruise Control On/Off Switch On; $08 = Transmission Gear Fault; $09 = Hill Descent Active; $0A = Hill Descent Serial Data Fault

This PID contains the current reason that the speed limiter/warning system has been inhibited from activating.

3.2.247FABR – PID $2338 – Air Fuel Imbalance Monitor Diagnostic Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2338	Air Fuel Imbalance Monitor Diagnostic Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Air Fuel Imbalance Monitor Diagnosing Bank 1	0	7	1	BLN	N/A	$1=True; $0=False
Air Fuel Imbalance Monitor Diagnosing Bank 2	0	6	1	BLN	N/A	$1=True; $0=False
Air Fuel Imbalance Monitor Diagnostic Report Status Bank 1	0	5	2	ENM	N/A	$0=No Decision; $1=Failed; $2=Passed
Air Fuel Imbalance Monitor Diagnostic Report Status Bank 2	0	3	2	ENM	N/A	$0=No Decision; $1=Failed; $2=Passed
Reserved	0	1	1			
Reserved	0	0	1			

This PID displays status information associated with the Air Fuel Imbalance Monitor (AFIM).

Air Fuel Imbalance Monitor Diagnosing Bank 1 and Air Fuel Imbalance Monitor Diagnosing Bank 2 indicate the current state of the AFIM. A value of “True” indicates that the monitor is currently performing an evaluation of the specified bank. A value of “False” indicates that the monitor is NOT currently performing an evaluation of the specified bank.

Air Fuel Imbalance Monitor Diagnostic Report Status Bank 1 indicates the report status of the AFIM diagnostic for bank 1. Multiple tests may run before the diagnostic reports a “Pass” or “Fail”. When this bit reports a “Pass” or “Fail”, the diagnostic is complete for the current trip and will not attempt any further tests on bank 1. Some applications may require multiple trips to make a “Pass” or “Fail” decision after a Code Clear or NVM reset.

Air Fuel Imbalance Monitor Diagnostic Report Status Bank 2 indicates the report status of the AFIM diagnostic for bank 2. Multiple tests may run before the diagnostic reports a “Pass” or “Fail”. When this bit reports a “Pass” or “Fail”, the diagnostic is complete for the current trip and will not attempt any further tests on bank 2. Some applications may require multiple trips to make a “Pass” or “Fail” decision after a Code Clear or NVM reset.

3.2.248FABR – PID $2339 – Air Fuel Imbalance Monitor Test Counter Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2339	Air Fuel Imbalance Monitor Test Counter Bank 1	1	UNM	0 to 255 counts	E = N * 1

This PID contains the number of Air Fuel Imbalance Monitor tests that have been performed since a code clear or NVM reset for bank 1. This counter will reset after a sufficient number of tests have run to allow reporting of a passing or failing condition. Some applications may require multiple trips to make a “Pass” or “Fail” decision after a code clear or NVM reset. Once the maximum allowed number of tests per trip have been completed, this counter will not increment until the next trip.

3.2.249FABR – PID $233A – Air Fuel Imbalance Monitor Test Counter Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$233A	Air Fuel Imbalance Monitor Test Counter Bank 2	1	UNM	0 to 255 counts	E = N * 1

This PID contains the number of Air Fuel Imbalance Monitor tests that have been performed since a code clear or NVM reset for bank 2. This counter will reset after a sufficient number of tests have run to allow reporting of a passing or failing condition. Some applications may require multiple trips to make a “Pass” or “Fail” decision after a code clear or NVM reset. Once the maximum allowed number of tests per trip have been completed, this counter will not increment until the next trip.

3.2.250FABR – PID $233B – Air Fuel Imbalance Monitor Test Counter for Half-Cylinder Mode Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$233B	Air Fuel Imbalance Monitor Test Counter for Half-Cylinder Mode Bank 1	1	UNM	0 to 255 counts	E = N * 1

This PID contains the number of Air Fuel Imbalance Monitor tests that have been performed since a code clear or NVM reset for bank 1 when in half-cylinder (V4) mode. This counter will reset after a sufficient number of tests have run to allow reporting of a passing or failing condition. Some applications may require multiple trips to make a “Pass” or “Fail” decision after a code clear or NVM reset. Once the maximum allowed number of tests per trip have been completed, this counter will not increment until the next trip.

3.2.251FABR – PID $233C – Air Fuel Imbalance Monitor Test Counter for Half-Cylinder Mode Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$233C	Air Fuel Imbalance Monitor Test Counter for Half-Cylinder Mode Bank 2	1	UNM	0 to 255 counts	E = N * 1

This PID contains the number of Air Fuel Imbalance Monitor tests that have been performed since a code clear or NVM reset for bank 2 when in Half-Cylinder (V4) mode. This counter will reset after a sufficient number of tests have run to allow reporting of a passing or failing condition. Some applications may require multiple trips to make a “Pass” or “Fail” decision after a code clear or NVM reset. Once the maximum allowed number of tests per trip have been completed, this counter will not increment until the next trip.

3.2.252FULR – PID $233D – Estimated Fuel Injector Return Fuel Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$233D	Estimated Fuel Injector Return Fuel Temperature	1	UNM	-40 to 215 °C	E = N * 1 – 40

This PID contains the estimated (calculated) fuel temperature in the return line from the fuel injectors to the high pressure fuel pump.

3.2.253EOPR – PID $2344 – Engine Oil Absolute Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2344	Engine Oil Absolute Pressure	1	UNM	0 to 1020 kPa	E = N * 4

This PID contains the undefaulted, unfiltered measured engine oil pressure read by an oil absolute pressure sensor.

3.2.254EOPR – PID $2345 – Engine Oil Gauge Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2345	Engine Oil Gauge Pressure	1	UNM	0 to 1020 kPa	E = N * 4

This PID contains the engine oil gauge pressure calculated as the difference between engine oil absolute pressure and ambient air pressure (barometric pressure). (Gauge = Absolute - Ambient)

3.2.255MSFR – PID $2346 – Misfire Speed/Load History 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2346	Misfire Speed/Load History 1	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Misfire Engine Speed 1	0	7	16	UNM	0 to 12799.8046875 rpm	E = N * 100 / 512
Misfire Load 1	2	7	16	UNM	0 to 1279.98046875 %	E = N * 10 / 512

This PID indicates the specific engine speed and load in which the engine was operating for one of the last five misfire events immediately prior to Misfire Test Failed This Key On (TFTKO) = True. A FIFO history buffer is used to store this information with “Misfire Speed/Load History 1” containing the most recent data.

3.2.256MSFR – PID $2347 – Misfire Speed/Load History 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2347	Misfire Speed/Load History 2	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Misfire Engine Speed 2	0	7	16	UNM	0 to 12799.8046875 rpm	E = N * 100 / 512
Misfire Load 2	2	7	16	UNM	0 to 1279.98046875 %	E = N * 10 / 512

This PID indicates the specific engine speed and load in which the engine was operating for one of the last five misfire events immediately prior to Misfire Test Failed This Key On (TFTKO) = True. A FIFO history buffer is used to store this information with “Misfire Speed/Load History 1” containing the most recent data.

3.2.257MSFR – PID $2348 – Misfire Speed/Load History 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2348	Misfire Speed/Load History 3	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Misfire Engine Speed 3	0	7	16	UNM	0 to 12799.8046875 rpm	E = N * 100 / 512
Misfire Load 3	2	7	16	UNM	0 to 1279.98046875 %	E = N * 10 / 512

This PID indicates the specific engine speed and load in which the engine was operating for one of the last five misfire events immediately prior to Misfire Test Failed This Key On (TFTKO) = True. A FIFO history buffer is used to store this information with “Misfire Speed/Load History 1” containing the most recent data.

3.2.258MSFR – PID $2349 – Misfire Speed/Load History 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2349	Misfire Speed/Load History 4	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Misfire Engine Speed 4	0	7	16	UNM	0 to 12799.8046875 rpm	E = N * 100 / 512
Misfire Load 4	2	7	16	UNM	0 to 1279.98046875 %	E = N * 10 / 512

This PID indicates the specific engine speed and load in which the engine was operating for one of the last five misfire events immediately prior to Misfire Test Failed This Key On (TFTKO) = True. A FIFO history buffer is used to store this information with “Misfire Speed/Load History 1” containing the most recent data.

3.2.259MSFR – PID $234A – Misfire Speed/Load History 5

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$234A	Misfire Speed/Load History 5	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Misfire Engine Speed 5	0	7	16	UNM	0 to 12799.8046875 rpm	E = N * 100 / 512
Misfire Load 5	2	7	16	UNM	0 to 1279.98046875 %	E = N * 10 / 512

This PID indicates the specific engine speed and load in which the engine was operating for one of the last five misfire events immediately prior to Misfire Test Failed This Key On (TFTKO) = True. A FIFO history buffer is used to store this information with “Misfire Speed/Load History 1” containing the most recent data.

3.2.260FPMR/FCBR – PID $234B – Commanded Fuel Pump Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$234B	Commanded Fuel Pump Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the fuel pump commanded PWM duty cycle which is determined by the ECM and delivered to the fuel pump by the FPPM (Fuel Pump Power Module).

3.2.261FRPR – PID $234C Fuel Control Enable Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$234C	Fuel Control Enable Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1	BLN	N/A	$1=True; $0=False
Reserved	0	2	1	BLN	N/A	$1=True; $0=False
Reserved	0	1	1	BLN	N/A	$1=True; $0=False
Fuel Control Enable	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains the state of the Fuel Control Enable signal. When Fuel Control Enable is TRUE, the fuel pump is commanded on, when Fuel Control Enable is FALSE, the fuel pump is commanded off.

3.2.262FPMR/FCBR – PID $234D – Fuel Pump Power Module Output PWM Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$234D	Fuel Pump Power Module Output PWM Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the actual Fuel Pump Power Module output PWM duty cycle sent directly to the fuel pump.

3.2.263FRPR – PID $ 234E Fuel Pump Output Driver Circuit 15A/100A Ground Short Diagnostic Fault Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$ 234E	Fuel Pump Output Driver Circuit 15A/100A Ground Short Diagnostic Fault Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1	BLN	N/A	$1=True; $0=False

Reserved	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1	BLN	N/A	$1=True; $0=False
Fuel Pump Output Driver Latched 100A Circuit Low History	0	3	1	BLN	N/A	$1=True; $0=False
Fuel Pump Output Driver Latched 100A Circuit Low Current	0	2	1	BLN	N/A	$1=True; $0=False
Fuel Pump Output Driver 15A Circuit Low History	0	1	1	BLN	N/A	$1=True; $0=False
Fuel Pump Output Driver 15A Circuit Low Current	0	0	1	BLN	N/A	$1=True; $0=False

This PID is dependent on DTC P0231 - Fuel Pump Output Driver Circuit 15A/100A Ground Short Diagnostics. This PID contains the following signals that indicate the fuel pump output driver faults.

- Fuel Pump Output Driver Latched 100A Circuit Low History – the fault status of the fuel pump output driver .
- Fuel Pump Output Driver Latched 100A Circuit Low Current – the current fault status of the fuel pump output driver latched circuit low (100A) diagnostic.
- Fuel Pump Output Driver 15A Circuit Low History – the fault status of the fuel pump output driver circuit low (15A) diagnostic since the last code clear.
- Fuel Pump Output Driver 15A Circuit Low Current - the current fault status of the fuel pump output driver circuit low (15A) diagnostic.

3.2.264FRPR/FCBR – PID $ 234F – Desired Fuel Supply Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$234F	Desired Fuel Supply Pressure	1	UNM	0 to 765 kPa	E = N * 3

This PID contains the desired fuel supply pressure, defined as the low side fuel pressure supplying either the fuel rail or a high pressure fuel pump.

3.2.265FRPR – PID $2350 – Fuel Pump Duty Cycle Short Term Integral Correction

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2350	Fuel Pump Duty Cycle Short Term Integral Correction	2	UNM	0 to 3.9999389648	E = N / 16384

This PID contains the short term integral correction value for the electronic returnless fuel system (ERFS). This value indicates how the ERFS is adapting the fuel pump duty cycle over the short term.

3.2.266FRPR – PID $2351 – Fuel Pump Duty Cycle Long Term Correction

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2351	Fuel Pump Duty Cycle Long Term Correction	2	UNM	0 to 3.9999389648	E = N / 16384

This PID contains the long term correction value for the electronic returnless fuel system (ERFS). This value indicates how the ERFS is adapting the fuel pump duty cycle over the long term.

3.2.267FRPR/FDBR – PID $2352 – Fuel System Rationality Diagnostic Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2352	Fuel System Rationality Diagnostic Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Previous Test Failed	0	5	1	BLN	N/A	$1=True; $0=False
Low Fuel Level	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1			
Decreasing Pressure Response Active	0	2	1	BLN	N/A	$1=True; $0=False
Test Active	0	1	1	BLN	N/A	$1=True; $0=False
Data Integrity OK	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains signals that are used within the fuel system rationality diagnostic (FSRD).

Previous Test Failed indicates that the previous fuel system rationality diagnostic test failed.

Low Fuel Level indicates that the fuel level is too low for the purpose of executing the fuel system rationality diagnostic.

Decreasing Pressure Response Active indicates that the fuel system response to decreasing fuel rail pressure request is active.

Test Active indicates that the fuel system rationality diagnostic test is active.

Data Integrity OK indicates that no other faults are active that prevent the fuel system rationality diagnostic from executing.

3.2.268FRPR/FDBR – PID $2353 – Fuel System Rationality Diagnostic Fuel Pump Long Term Correction

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2353	Fuel System Rationality Diagnostic Fuel Pump Long Term Correction	2	UNM	0 to 3.9998779296875	E = N / 16384

This PID indicates the long term fuel pump duty cycle correction factor used by the fuel system rationality diagnostic (FSRD). This value is applied to the open-loop estimate for fuel pump duty cycle when open-loop control of the fuel pump is being requested due to a FSRD diagnostic fault.

3.2.269FRPR/FDBR – PID $2354 – Fuel System Rationality Diagnostic Filtered Fuel Pressure Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2354	Fuel System Rationality Diagnostic Filtered Fuel Pressure Error	2	SNM	-512 to + 511.98438 kPa	E = N * 1/64

This PID indicates the filtered fuel pressure error for the fuel system rationality diagnostic (FSRD).

3.2.270FRPR/FDBR – PID $2355 – Fuel System Rationality Diagnostic High Fault Threshold

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2355	Fuel System Rationality Diagnostic High Fault Threshold	2	SNM	-512 to + 511.98438 kPa	E = N * 1/64

This PID contains the high fault threshold for the fuel system rationality diagnostic (FSRD). FSRD will set when the Filtered Fuel Pressure Error is larger than the FSRD High Fault Threshold.

3.2.271FRPR/FDBR – PID $2356 – Fuel System Rationality Diagnostic Low Fault Threshold

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2356	Fuel System Rationality Diagnostic Low Fault Threshold	2	SNM	-512 to + 511.98438 kPa	E = N * 1/64

This PID contains the low fault threshold for the fuel system rationality diagnostic (FSRD). FSRD will set when the Filtered Fuel Pressure Error is less than the FSRD High Fault Threshold.

3.2.272FRPR – PID $2357 – Fuel System Rationality Diagnostic Enable Timer

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2357	Fuel System Rationality Diagnostic Enable Timer	2	UNM	0 to 409.59375 s	E = N / 160

This PID indicates the timer value used to manage the delay before activating the fuel system rationality diagnostic (FSRD) after all enable criteria have been met.

3.2.273FRPR/FCBR – PID $2358 - Fuel Supply Pressure Sensor A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2358	Fuel Supply Pressure Sensor A/D	1	UNM	0 to 100 % 5V	E = N * 100 / 255

This PID contains the fuel supply pressure sensor analog input as a percentage of its reference voltage.

3.2.274FRPR/FDBR – PID $2359 – Fuel Pressure Sensor Stuck Diagnostic Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2359	Fuel Pressure Sensor Stuck Diagnostic Status	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Intrusive Test Required	0	4	1	BLN	N/A	$1=True; $0=False
Intrusive Test Complete	0	3	1	BLN	N/A	$1=True; $0=False
Fuel Pump Control Required	0	2	1	BLN	N/A	$1=True; $0=False
Test Active	0	1	1	BLN	N/A	$1=True; $0=False
Data Integrity OK	0	0	1	BLN	N/A	$1=True; $0=False
Reserved	1	7	1			
Reserved	1	6	1			
Reserved	1	5	1			
Reserved	1	4	1			
Reserved	1	3	1			
Intrusive Test This Trip	1	2	1	BLN	N/A	$1=True; $0=False
Static Fuel Pressure	1	1	1	BLN	N/A	$1=True; $0=False
Fuel System Clamp Active	1	0	1	BLN	N/A	$1=True; $0=False

This PID contains signals that are used within the electronic returnless fuel system (ERFS) fuel pressure sensor stuck (FPSS) diagnostic.

Intrusive Test Required indicates that the Fuel Pressure Sensor Stuck Diagnostic is requesting an intrusive test.

Intrusive Test Complete indicates that the Fuel Pressure Sensor Stuck Diagnostic intrusive test is complete.

Fuel Pump Control Required Fuel Pressure Sensor Stuck Diagnostic is requesting control of the fuel pump for the intrusive test.

Test Active indicates that the Fuel Pressure Sensor Stuck Diagnostic test is active.

Data Integrity OK indicates that no other faults are active that prevent the Fuel Pressure Sensor Stuck Diagnostic from executing.

Intrusive Test This Trip indicates that the Fuel pressure stuck diagnostic intrusive test has been requested at least once this trip.

Static Fuel Pressure indicates insufficient variation in fuel pressure, which indicates that the fuel pressure sensor may be stuck.

Fuel System Clamp Active indicates either short term integral correction, or fuel pump duty cycle is clamped, indicating that the fuel pressure sensor may be stuck.

3.2.275FRPR/FAFR – PID $235A – Fuel Pump Power Module Driver Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$235A	Fuel Pump Power Module Driver Current	1	UNM	0 to 63.75 A	E = N / 4

This PID contains the current (Amps) supplied to the fuel pump from the Fuel Pump Power Module (FPPM).

3.2.276FRPR/FAFR – PID $235B – Fuel Pump Power Module Sensed Battery Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$235B	Fuel Pump Power Module Sensed Battery Voltage	1	UNM	0 to 25.5 V	E = N * 0.1

This PID contains the battery voltage sensed by the fuel pump power module (FPPM).

3.2.277FRPR/FAFR – PID $235C – Fuel Pump Power Module Fuel Pump Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$235C	Fuel Pump Power Module Fuel Pump Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
FPPM Fuel Pump Output Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
FPPM Fuel Pump Output Short to Ground Test Status	0	5	2	ENM	N/A	$0=Passed; $1=Failed; $2=Indeterminate
FPPM Fuel Pump Output Open Circuit Test Status	0	3	2	ENM	N/A	$0=Passed; $1=Failed; $2=Indeterminate
FPPM Fuel Pump Output Short to Power Test Status	0	1	2	ENM	N/A	$0=Passed; $1=Failed; $2=Indeterminate

This PID contains the commanded state of the fuel pump power module (FPPM) output (the output is considered to be commanded on if its duty cycle is not zero) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0231, P023F and P0232, respectively.

3.2.278EVPR – PID $235D – Canister Vent Diurnal Control Valve Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$235D	Canister Vent Diurnal Control Valve Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Canister Vent Diurnal Control Valve Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Canister Vent Diurnal Control Valve Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Canister Vent Diurnal Control Valve Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Canister Vent Diurnal Control Valve Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the DCV output (the canister vent is commanded open when the output is commanded off) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0498, P0449, and P0499, respectively.

3.2.279EVPR – PID $235E – EVAP Leak Check Pump Switching Valve Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$235E	EVAP Leak Check Pump Switching Valve Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
ELCP Switching Valve Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
ELCP Switching Valve Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
ELCP Switching Valve Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
ELCP Switching Valve Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the EVAP Leak Check Pump (ELCP) switching valve output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2419, P2418 and P2420, respectively.

3.2.280EVPR – PID $235F – EVAP Leak Check Pump Absolute Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$235F	EVAP Leak Check Pump Absolute Pressure	2	UNM	0 to 120 kPa	E = N * 120 / 65535

This PID contains the undefaulted EVAP Leak Check Pump (ELCP) absolute pressure measured by a sensor.

3.2.281EVPR – PID $2360 – EVAP Leak Check Pump Absolute Pressure A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2360	EVAP Leak Check Pump Absolute Pressure A/D	1	UNM	0 to 100 % 5V	E = N * 100 / 255

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure analog input as a percentage of its reference voltage.

3.2.282EVPR – PID $2361 – EVAP Leak Check Pump Vacuum Pump Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2361	EVAP Leak Check Pump Vacuum Pump Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
ELCP Vacuum Pump Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
ELCP Vacuum Pump Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
ELCP Vacuum Pump Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
ELCP Vacuum Pump Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the EVAP Leak Check Pump (ELCP) vacuum pump output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2401, P2400 and P2402, respectively.

3.2.283THMR – PID $2362 –Engine Coolant Pump Clutch Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2362	Engine Coolant Pump Clutch Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Engine Coolant Pump Clutch Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Engine Coolant Pump Clutch Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Engine Coolant Pump Clutch Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Engine Coolant Pump Clutch Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the engine coolant pump clutch output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P261C, P261A, and P261D, respectively. When the clutch is commanded on, the coolant pump is not spinning, so coolant flow is stopped.

3.2.284ECTR – PID $2363 – Engine Block Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2363	Engine Block Temperature	1	UNM	-40 to +215°C	E = N * 1 - 40

This PID contains the engine block temperature measured by a sensor.

3.2.285EVPR – PID $236F – EVAP Wakeup Window Attained

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$236F	EVAP Wakeup Window Attained	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Wakeup Needed	0	7	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Window 1	0	6	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Window 2	0	5	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Window 3	0	4	1	BLN	N/A	$1=True; $0=False
Wakeup Report	0	3	1	BLN	N/A	$1=True; $0=False
EVAP Handshake 1	0	2	1	BLN	N/A	$1=True; $0=False
EVAP Handshake 2	0	1	1	BLN	N/A	$1=True; $0=False
EVAP Handshake 3	0	0	1	BLN	N/A	$1=True; $0=False

This PID indicates if the EVAP wake-up window(s) were reached and a status determined. This PID will indicate if a wake-up is needed for the EVAP diagnostic.

3.2.286EVPR – PID $2370 – EVAP Wakeup Requests

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2370	EVAP Wakeup Requests	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
EVAP Wakeup Request Acknowledged Window 1	0	7	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Request Acknowledged Window 2	0	6	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Request Acknowledged Window 3	0	5	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Request Made Window 1	0	4	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Request Made Window 2	0	3	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Request Made Window 3	0	2	1	BLN	N/A	$1=True; $0=False
reserved	0	1	1			
reserved	0	0	1			

This PID indicates if the EVAP wake-up request has been made to the VICM and received from the VICM for a particular wake-up window.

3.2.287EVPR – PID $2371 – EVAP Leak Check Test State

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2371	EVAP Leak Check Test State	1	ENM	N/A	$00=Wait To Begin; $01=Ambient Pressure Check; $02=Fuel Tank Venting; $03=First Reference Check; $04=Check Between DCV and ELCP; $05=EVAP Leak Check; $06=Second reference Check; $07=Final Check; $08=Test Complete; $09=Service Bay Fuel Tank Venting; $0A=Switch Valve Check; $0B=Blockage Check; $0C=Refueling Rationality

This PID contains the current state of the test sequence in the EVAP leak check diagnostic for a system incorporating an EVAP leak check pump.

3.2.288EVPR – PID $2372 – EVAP Leak Check Monitor State

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2372	EVAP Leak Check Monitor State	1	ENM	N/A	$00=Undetermined; $01=Enabled; $02=Disabled; $03=Fast Pass; $04=Service Bay Test

This PID contains the current monitoring state of the EVAP leak check diagnostic for a system incorporating an EVAP leak check pump.

3.2.289EVPR – PID $2373 – EVAP Wakeup Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2373	EVAP Wakeup Time	2	UNM	0 to 65535 s	E = N * 1

This PID indicates the wake-up time that was sent to the VICM.

3.2.290EVPR – PID $2374 – EVAP Wakeup Window Test Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2374	EVAP Wakeup Window Test Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
EVAP Wakeup Window 1 Status	0	7	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
EVAP Wakeup Window 2 Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
EVAP Wakeup Window 3 Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
reserved	0	1	1			
reserved	0	0	1			

This PID indicates the pass/fail status of each wake-up window.

3.2.291EVPR – PID $2375 – EVAP Wakeup Alarm Clock State

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2375	EVAP Wakeup Alarm Clock State	1	ENM	N/A	$00=No Action; $01=Set Alarm; $02=Cancel Alarm

This PID indicates the state of the VICM alarm clock.

3.2.292EVPR – PID $2376 – EVAP Wakeup Alarm Request Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2376	EVAP Wakeup Alarm Request Status	1	ENM	N/A	$00=Inactive; $01=Active; $02=Expired; $03=Failsoft

This PID indicates the status of the VICM alarm request.

3.2.293EVPR – PID $2377 – EVAP Leak Check Pump Absolute Pressure – Barometric Pressure Correlation

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2377	EVAP Leak Check Pump Absolute Pressure – Barometric Pressure Correlation	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change measured at the end of the ELCP pressure sensor to Baro correlation check.

3.2.294EVPR – PID $2378 – EVAP Leak Check Pump Absolute Pressure – Start of EVAP Leak Check Diagnostic

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2378	EVAP Leak Check Pump Absolute Pressure – Start of EVAP Leak Check Diagnostic	2	UNM	0 to 327.675 kPa	E = N * 0.005

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure measured at the start of the EVAP leak check diagnostic.

3.2.295EVPR – PID $2379 – EVAP Leak Check Pump Absolute Pressure – First Reference Orifice Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2379	EVAP Leak Check Pump Absolute Pressure – First Reference Orifice Test	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change measured at the end of the first reference orifice test.

3.2.296EVPR – PID $237A – EVAP Leak Check Pump Absolute Pressure – First Saturation Check

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$237A	EVAP Leak Check Pump Absolute Pressure – First Saturation Check	2	UNM	0 to 327.675 kPa	E = N * 0.005

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure measured near the end of the first reference orifice test in order to check for saturation.

3.2.297EVPR – PID $237B – EVAP Leak Check Pump Absolute Pressure – Second Reference Orifice Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$237B	EVAP Leak Check Pump Absolute Pressure – Second Reference Orifice Test	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change measured at the end of the second reference orifice test.

3.2.298EVPR – PID $237C – EVAP Leak Check Pump Absolute Pressure – Second Saturation Check

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$237C	EVAP Leak Check Pump Absolute Pressure – Second Saturation Check	2	UNM	0 to 327.675 kPa	E = N * 0.005

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure measured near the end of the second reference orifice test in order to check for saturation.

3.2.299EVPR – PID $237D – EVAP Leak Check Pump Absolute Pressure – Leak Between Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$237D	EVAP Leak Check Pump Absolute Pressure – Leak Between Test	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change as measured at the end of the leak between ELCP pump and DCV test.

3.2.300EVPR – PID $237E – EVAP Leak Check Pump Absolute Pressure – EVAP Leak Check

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$237E	EVAP Leak Check Pump Absolute Pressure – EVAP Leak Check	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change measured at the end of the EVAP leak check portion of the test sequence.

3.2.301EVPR – PID $237F – EVAP Leak Check Pump Absolute Pressure – DCV Stuck Closed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$237F	EVAP Leak Check Pump Absolute Pressure – DCV Stuck Closed	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change calculated between the first reference orifice pressure change and the ELCP pressure change shortly after beginning the EVAP leak check portion of the testing sequence. This PID is used to check for a DCV stuck closed if the fuel tank venting test sequence was not run.

3.2.302EVPR – PID $2380 – EVAP Leak Check Pump Absolute Pressure – Final Check Sequence

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2380	EVAP Leak Check Pump Absolute Pressure – Final Check Sequence	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change measured at the end of the final check sequence.

3.2.303EVPR – PID $2381 – EVAP Leak Check Pump Small Leak Threshold

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2381	EVAP Leak Check Pump Small Leak Threshold	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change threshold used for detecting a small leak in the EVAP system.

3.2.304EVPR – PID $2382 – EVAP Leak Check Pump Large Leak Threshold

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2382	EVAP Leak Check Pump Large Leak Threshold	2	SNM	-131.072 to +131.068 kPa	E = N * 0.004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change threshold used for detecting a large leak in the EVAP system.

3.2.305EVPR – PID $2383 – EVAP Leak Check Test Completed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2383	EVAP Leak Check Test Completed	3	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Last EVAP Leak Check Test Successfully Completed	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			
EVAP Leak Check Tests Completed	1	7	16	UNM	0 to 65535 counts	E = N * 1

This PID contains the signals associated with the completion of the EVAP leak check test.

Last EVAP Leak Check Test Successfully Completed indicates if the last EVAP leak check test performed has successfully completed.

EVAP Leak Check Tests Completed contains the number of times that the EVAP leak check tests have completed since a code clear or NVM reset.

3.2.306EVPR – PID $2384 – Purge Valve Flow Test Vacuum Absolute Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2384	Purge Valve Flow Test Vacuum Absolute Pressure	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Purge Valve Flow Test Completed This Trip	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			
Purge Valve Flow Test Vacuum Absolute Pressure	1	7	8	SNM	-12.8 to +12.7 kPa	E = N * 0.1

This PID contains the signals associated with the purge valve flow test vacuum absolute pressure.

Purge Valve Flow Test Completed This Trip indicates if the purge valve flow test has completed this trip.

Purge Valve Flow Test Vacuum Absolute Pressure contains the EVAP Leak Check Pump (ELCP) absolute pressure change measured at the end of the purge valve flow test.

3.2.307EVPR – PID $2385 – Fuel Tank Pressure – Start of EVAP Leak Check Diagnostic

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2385	Fuel Tank Pressure – Start of EVAP Leak Check Diagnostic	2	SNM	-131.072 to +131.068 in H_2O	E = N * 0.004

This PID contains the fuel tank pressure (FTP) measured at the start of the EVAP leak check diagnostic.

3.2.308EVPR – PID $2386 – Fuel Tank Pressure Sensor Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2386	Fuel Tank Pressure Sensor Voltage	1	UNM	0 to 25.5 Volts	E = N * 0. 1

This PID indicates the fuel tank pressure (FTP) sensor voltage captured at the start of the EVAP leak check diagnostic.

3.2.309EVPR – PID $2387 – Fuel Tank Pressure – Leak Between Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2387	Fuel Tank Pressure – Leak Between Test	2	SNM	-131.072 to +131.068 in H_2O	E = N * 0.004

This PID contains the fuel tank pressure (FTP) change measured at the end of the leak between ELCP pump and DCV test.

3.2.310EVPR – PID $2388 – Fuel Tank Pressure – EVAP Leak Check

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2388	Fuel Tank Pressure – EVAP Leak Check	2	SNM	-131.072 to +131.068 in H_2O	E = N * 0.004

This PID contains the fuel tank pressure (FTP) measured at the end of the EVAP leak check portion of the test sequence.

3.2.311EVPR – PID $2389 – Fuel Tank Pressure Sensor Correlation – Propulsion Off

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2389	Fuel Tank Pressure Sensor Correlation – Propulsion Off	3	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Tank Pressure Out of Readable Range	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			
FTP-ELCP Pressure Correlation	1	7	16	UNM	0 to 327.675 in H_2O	E = N * 0.005

This PID contains the signals associated with fuel tank pressure sensor correlation when the propulsion system is off.

Fuel Tank Pressure Out of Readable Range indicates if the FTP sensor is out of readable range and correlation is confirmed with the ELCP pressure sensor.

FTP-ELCP Pressure Correlation contains the average difference of correlation between the fuel tank pressure (FTP) sensor and the EVAP leak check pump (ELCP) pressure sensor.

3.2.312EVPR – PID $238A – Fuel Tank Pressure Sensor Correlation – Propulsion On

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$238A	Fuel Tank Pressure Sensor Correlation – Propulsion On	3	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Tank Pressure Out of Readable Range	0	7	1	BLN	N/A	$1=True; $0=False
FTP-ELCP Correlated This Trip	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			
FTP-ELCP Pressure Correlation	1	7	16	UNM	0 to 327.675 in H_2O	E = N * 0.005

This PID contains the signals associated with fuel tank pressure sensor correlation when the propulsion system is on.

Fuel Tank Pressure Out of Readable Range indicates if the FTP sensor is out of readable range and correlation is confirmed with the ELCP pressure sensor.

FTP-ELCP Correlated This Trip indicates if the correlation test between the fuel tank pressure (FTP) sensor and the EVAP leak check pump (ELCP) pressure sensor has completed this trip.

FTP-ELCP Pressure Correlation contains the average difference of correlation between the fuel tank pressure (FTP) sensor and the EVAP leak check pump (ELCP) pressure sensor.

3.2.313EVPR – PID $238B – Fuel Tank Pressure – Fuel Tank Vented Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$238B	Fuel Tank Pressure – Fuel Tank Vented Test	2	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Tank Test Sequence Check Run	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			
Fuel Tank Pressure – Fuel Vented Test	1	7	8	UNM	0 to 24.90234375 in H_2O	E = N * 25 / 256

This PID contains the signals associated with the check for a stuck diurnal control valve (DCV) during the fuel tank venting test section.

Fuel Tank Test Sequence Run indicates if the fuel tank vented test sequence was used in checking for a stuck closed diurnal control valve (DCV).

Fuel Tank Pressure – Fuel Tank Vented Test_contains the fuel tank pressure (FTP) change when the DCV is commanded open during the fuel tank vented sequence.

3.2.314EVPR – PID $238C – EVAP System Leak Check Timer

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$238C	EVAP System Leak Check Timer	1	UNM	0 to 2550 s	E = N * 10

This PID contains the time elapsed during the EVAP system leak check portion of the test sequence.

3.2.315MAPR – PID $238E – Intake Manifold Gauge Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$238E	Intake Manifold Gauge Pressure	1	SNM	-128 to +127 kPa	E = N * 1

This PID contains the engine intake manifold gauge pressure calculated as the difference between intake manifold absolute pressure (MAP) and ambient air pressure (barometric pressure). This value corresponds to the value measured on a technician's pressure gauge. (Gauge = MAP - Ambient)

3.2.316MAPR – PID $238F – Supercharger Inlet Gauge Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$238F	Supercharger Inlet Gauge Pressure	1	UNM	-255 to 0 kPa	E = N * 1 - 255

This PID contains the supercharger inlet gauge pressure calculated as the difference between supercharger inlet absolute pressure (SCIAP) and ambient air pressure (barometric pressure). This value corresponds to the value measured on a technician's pressure gauge. (Gauge = SCIAP - Ambient)

3.2.317MAPR – PID $2390 – Extended Range Intake Manifold Gauge Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2390	Extended Range Intake Manifold Gauge Pressure	2	SNM	- 1024 to +1023.96875 kPa	E = N / 32

This PID contains the engine intake manifold gauge pressure calculated as the difference between intake manifold absolute pressure (MAP) and ambient air pressure (barometric pressure). This PID is the extended range version of PID $238E and is used with applications that require PID $208A. This value corresponds to the value measured on a technician's pressure gauge. (Gauge = MAP - Ambient)

3.2.318EVPR – PID $2391 – EVAP Leak Check Pump Service Bay Test Result

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2391	EVAP Leak Check Pump Service Bay Test Result	1	ENM	N/A	$00=Not Run; $01=Service Test Passed $02=ELCP Pump Stuck On Test Failed; $03=Reference Orifice High Test Failed; $04=DCV Stuck Open Test Failed; $05=Reference Orifice Performance Test Failed; $06=ELCP Switch Valve Stuck Open Test Failed; $07=Small Leak Test Failed; $08=Large Leak Test Failed; $09=Leak Between ELCP DCV Test Failed; $0A=DCV Stuck Closed Test Failed; $0B=ELCP Pressure Sensor Correlation Test Failed; $0C=FTP Sensor Correlation Test Failed; $0D=ELCP Pump Stuck Off Test Failed; $0E=Reference Orifice Low Test Failed; $0F=Fuel Tank Blockage Test Failed; $10=Fuel Level Refueling Detected; $11=Fuel Level Refueling Rationality Test Failed; $12=Reference Orifice Vacuum Not Stable Test Failed; $13=Vacuum Refueling Detected $14=Vacuum Refueling Rationality Test Failed

This PID contains the result of the EVAP service bay test for a system incorporating an EVAP leak check pump.

3.2.319EVPR – PID $2392 – EVAP Leak Check Pump Service Bay Test Denial Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2392	EVAP Leak Check Pump Service Bay Test Denial Reason	1	ENM	N/A	$00=No Request; $01=Test Not Available; $02=EVAP IM Flag Set; $03=Malfunction Code Present; $04=Propulsion System Active; $05=Refueling Request; $06=Battery Voltage Too Low; $07=IAT Out of Range; $08=Engine Coolant Too High; $09=Soak Conditions Not Met; $0A=Fuel Level Incorrect; $0B=No Denial

This PID contains the reason the EVAP service bay test, for a system incorporating an EVAP leak check pump, was denied.

3.2.320EVPR – PID $2393 – EVAP Leak Check Pump Service Bay Test Abort Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2393	EVAP Leak Check Pump Service Bay Test Abort Reason	1	ENM	N/A	$00=No Abort; $01=Fuel Level Slosh; $02=Vacuum Slosh; $03=Battery Voltage Too Low; $04=Keyed Up During Test; $05=Refueling Request; $06=Propulsion System Active

This PID contains the reason the EVAP service bay test, for a system incorporating an EVAP leak check pump, was aborted.

3.2.321EVPR – PID $2394 – EVAP Window 1 Wakeup Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2394	EVAP Window 1 Wakeup Time	2	UNM	0 to 131070 s	E = N * 2

This PID indicates the time when the first EVAP wakeup has occurred based on the ECM clock.

3.2.322EVPR – PID $2395 – EVAP Window 2 Wakeup Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2395	EVAP Window 2 Wakeup Time	2	UNM	0 to 131070 s	E = N * 2

This PID indicates the time when the second EVAP wakeup has occurred based on the ECM clock.

3.2.323EVPR – PID $2396 – EVAP Window 3 Wakeup Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2396	EVAP Window 3 Wakeup Time	2	UNM	0 to 131070 s	E = N * 2

This PID indicates the time when the third EVAP wakeup has occurred based on the ECM clock.

3.2.324EVPR – PID $2397 – EVAP Wakeup Communication Test Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2397	EVAP Wakeup Communication Test Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
EVAP Wakeup Comm 1 Pass	0	7	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Comm 2 Pass	0	6	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Comm 3 Pass	0	5	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Comm 1 Fail	0	4	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Comm 2 Fail	0	3	1	BLN	N/A	$1=True; $0=False
EVAP Wakeup Comm 3 Fail	0	2	1	BLN	N/A	$1=True; $0=False
reserved	0	1	1			
reserved	0	0	1			

This PID indicates if the EVAP wakeup communication test with the VICM has been passed or failed for a particular wakeup window.

3.2.325EGRR – PID $2398 – EGR Cooler Coolant Pump Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2398	EGR Cooler Coolant Pump Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
EGR Cooler Coolant Pump Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
EGR Cooler Coolant Pump Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
EGR Cooler Coolant Pump Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
EGR Cooler Coolant Pump Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the EGR cooler coolant pump output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P24A9, P24A7, and P24AA respectively.

3.2.326EGRR – PID $2399 – EGR System Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2399	EGR System Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
EGR Cooler Coolant Pump Actual State On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains the EGR system status.

EGR Cooler Coolant Pump Actual State On indicates if the feedback signal from the EGR cooler coolant pump is on.

3.2.327ECTR – PID $23A0 – Cylinder Head Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A0	Cylinder Head Temperature	1	UNM	-40 to +215°C	E = N * 1 - 40

This PID contains the cylinder head temperature measured by a sensor.

3.2.328CRZR – PID $23A1– Cruise Secondary Raw Switch States

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A1	Cruise Secondary Raw Switch States	1	ENM	N/A	$00=Indeterminate; $01=No Activation; $02= Reserved; $03= Gap Down; $04=Gap Up; $05= Speed Increase High; $06= Speed Decrease High; $07=Fault Low; $08=Fault High; $09=Between Ranges / Invalid

This PID contains the cruise secondary switch states as received by the ECM. The switch states are used to determine the modes for the enhanced cruise control functions and apply to architectures using momentary (non-latching) switches. The various states are described as follows:

Indeterminate indicates that the platform module is currently not able to determine the state of the input.

No Activation indicates that no switch is active. Note: this is also the default PID value if a valid serial data signal has not been received.

Reserve red is a currently unassigned switch for future expansion.

Gap Down indicates that the Gap Down switch is active.

Gap Up indicates that the Gap Up switch is active.

Speed Increase High indicates that the Speed Increase High switch is active.

Speed Decrease High indicates that the Speed Decrease High switch is active.

Fault Low indicates that the switch input voltage ratio is in a range associated with an open circuit or a short to ground. If this condition persists continuously for longer than a calibratable period, a DTC will set.

Fault High indicates that the switch input voltage ratio is in a range associated with a short to power. If this condition persists continuously for longer than a calibratable period, a DTC will set.

Between Ranges / Invalid indicates that the switch input is between valid voltage ratio ranges. If this condition persists continuously for longer than a calibratable period, a DTC will set.

3.2.329FRPR – PID $23A2 Estimated Fuel Supply Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A2	Estimated Fuel Supply Pressure	2	UNM	0 – 1024 kPa	E = N *1 / 64

This PID contains the estimated fuel pressure as follows:

PFI (Port Fuel Injected Engines) – estimated pressure in the fuel rail.

SIDI (Spark Ignited Direct Injection) – estimated pressure at the inlet to the high pressure pump.

3.2.330FADR – PID $23A3 – Small Quantity Learning Current Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A3	Small Quantity Learning Current Status	1	ENM	N/A	$00=SQL not active; $01=SQL active; $02=SQL failed; $03=SQL successfully ended; $04 to $FF = Reserved

This PID holds the current status of the Small Quantity Learning (SQL) algorithm.

3.2.331FADR – PID $23A4 – Small Quantity Learning Current Learned Fuel Pressure Level

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A4	Small Quantity Learning Current Learned Fuel Pressure Level	1	ENM	N/A	$00=Level 0; $01=Level 1; $02=Level 2; $03=Level 3; $04=Level 4; $05 to $FF = Reserved

This PID holds the current status of the Small Quantity Learning (SQL) Learned Fuel Pressure Level.

3.2.332RBSR – PID $23A5 – Main Processor Running Reset CPU Identifier History 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A5	Main Processor Running Reset CPU Identifier History 1	1	ENM	N/A	$00=CPU 1 caused reset; $01=CPU 2 caused reset; $02=CPU 3 caused reset; $03=CPU 4 caused reset;

This PID contains one of the last four identifiers to indicate which CPU caused the running reset in a multi-core main processor. If the main processor is a single core processor then this array shall always indicate that CPU 1 caused the reset ($00). A FIFO history buffer, which retains its contents over a battery disconnect, is used to store the CPU ID with history 1 containing the most recent CPU to cause the reset. This PID is directly associated with the “Main Processor Running Reset Reason History x” and “Main Processor Running Reset Source Address History x” PIDs of the same number “x”.

3.2.333RBSR – PID $23A6 – Main Processor Running Reset CPU Identifier History 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A6	Main Processor Running Reset CPU Identifier History 2	1	ENM	N/A	$00=CPU 1 caused reset; $01=CPU 2 caused reset; $02=CPU 3 caused reset; $03=CPU 4 caused reset;

This PID contains one of the last four identifiers to indicate which CPU caused the running reset in a multi-core main processor. If the main processor is a single core processor then this array shall always indicate that CPU 1 caused the reset ($00). A FIFO history buffer, which retains its contents over a battery disconnect, is used to store the CPU ID with history 1 containing the most recent CPU to cause the reset. This PID is directly associated with the “Main Processor Running Reset Reason History x” and “Main Processor Running Reset Source Address History x” PIDs of the same number “x”.

3.2.334RBSR – PID $23A7 – Main Processor Running Reset CPU Identifier History 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A7	Main Processor Running Reset CPU Identifier History 3	1	ENM	N/A	$00=CPU 1 caused reset; $01=CPU 2 caused reset; $02=CPU 3 caused reset; $03=CPU 4 caused reset;

This PID contains one of the last four identifiers to indicate which CPU caused the running reset in a multi-core main processor. If the main processor is a single core processor then this array shall always indicate that CPU 1 caused the reset ($00). A FIFO history buffer, which retains its contents over a battery disconnect, is used to store the CPU ID with history 1 containing the most recent CPU to cause the reset. This PID is directly associated with the “Main Processor Running Reset Reason History x” and “Main Processor Running Reset Source Address History x” PIDs of the same number “x”.

3.2.335RBSR – PID $23A8 – Main Processor Running Reset CPU Identifier History 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23A8	Main Processor Running Reset CPU Identifier History 4	1	ENM	N/A	$00=CPU 1 caused reset; $01=CPU 2 caused reset; $02=CPU 3 caused reset; $03=CPU 4 caused reset;

This PID contains one of the last four identifiers to indicate which CPU caused the running reset in a multi-core main processor. If the main processor is a single core processor then this array shall always indicate that CPU 1 caused the reset ($00). A FIFO history buffer, which retains its contents over a battery disconnect, is used to store the CPU ID with history 1 containing the most recent CPU to cause the reset. This PID is directly associated with the “Main Processor Running Reset Reason History x” and “Main Processor Running Reset Source Address History x” PIDs of the same number “x”.

3.2.336TSER – PID $23AB – Dominant Clutch Controlled

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23AB	Dominant Clutch Controlled	1	ENM	N/A	$00=Both Off; $01=Clutch 1 Active; $02=Clutch 2 Active

The Dominant Clutch Controlled indicates which of the two clutches are currently being controlled, if either.

3.2.337OXYR – PID $23AC – Oxygen Sensor Bank 1 Sensor 2 Heating Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23AC	Oxygen Sensor Bank1 Sensor 2 Heating Status	1	ENM	N/A	$00=Invalid; $01=Operating Temperature Not Reached; $02=Operating Temperature Reached

This PID contains the operating status of the oxygen sensor bank 1 sensor 2 heater.

3.2.338OXYR – PID $23AD – Exhaust Gas Oxygen Concentration Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23AD	Exhaust Gas Oxygen Concentration Sensor 2	2	UNM	-12 to +21.68499 %	E = N * 0.000514 - 12

This PID contains the concentration of oxygen in the exhaust gas measured by oxygen sensor 2.

3.2.339EVPR – PID $23AF – EVAP Leak Check Pump ELCP Baro Correlation Result Ambient Check

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23AF	EVAP Leak Check Pump ELCP Baro Correlation Result Ambient Check	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the result for the ELCP pressure sensor correlation to Barometric pressure in the ambient check phase for a system incorporating an EVAP Leak Check Pump.

3.2.340EVPR – PID $23B0 – EVAP Leak Check Pump ELCP Baro Correlation Result Final Check

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B0	EVAP Leak Check Pump ELCP Baro Correlation Result Final Check	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the result for the ELCP pressure sensor correlation to Barometric pressure in the final check phase for a system incorporating an EVAP Leak Check Pump.

3.2.341EVPR – PID $23B1 – EVAP Leak Check Pump Barometric Pressure – Start of EVAP Leak Check Diagnostic

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B1	EVAP Leak Check Pump Barometric Pressure - Start of EVAP Leak Check Diagnostic	2	UNM	0 to 327.675 kPa	E = N * .005

This PID contains the barometric pressure measured at the start of the EVAP leak check diagnostic for a system incorporating an EVAP Leak Check Pump.

3.2.342EVPR – PID $23B2 – EVAP Leak Check Pump Fuel Level – Start of EVAP Leak Check Diagnostic

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B2	EVAP Leak Check Pump Fuel Level – Start of EVAP Leak Check Diagnostic	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the fuel level at the start of the EVAP leak check diagnostic for a system incorporating an EVAP Leak Check Pump.

3.2.343EVPR – PID $23B3 – Conventional EVAP With Pump Reference Orifice Saturation – First Reference Orifice Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B3	Conventional EVAP With Pump Reference Orifice Saturation – First Reference Orifice Test	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the reference orifice saturation value measured during the first reference orifice check for a system incorporating an EVAP Leak Check Pump on a conventional EVAP system.

3.2.344EVPR – PID $23B4 – Conventional EVAP With Pump Reference Orifice Saturation – Second Reference Orifice Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B4	Conventional EVAP With Pump Reference Orifice Saturation – Second Reference Orifice Test	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the reference orifice saturation value measured during the second reference orifice check for a system incorporating an EVAP Leak Check Pump on a conventional EVAP system.

3.2.345EVPR – PID $23B5 – Conventional EVAP With Pump Absolute Pressure - Fuel Tank Blockage Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B5	Conventional EVAP With Pump Absolute Pressure - Fuel Tank Blockage Test	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the EVAP Leak Check Pump absolute pressure sensor change as measured at the end of the blockage test for a system incorporating an EVAP Leak Check Pump on a conventional EVAP system.

3.2.346EVPR – PID $23B6 – EVAP Leak Check Pump Absolute Pressure – ELCP Switch Valve Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B6	EVAP Leak Check Pump Absolute Pressure – ELCP Switch Valve Test	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change measured at the end of the switch valve test for a system incorporating an EVAP Leak Check Pump.

3.2.347EVPR – PID $23B7 – Conventional EVAP With Pump Fuel Tank Blockage Threshold

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B7	Conventional EVAP With Pump Fuel Tank Blockage Threshold	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the EVAP Leak Check Pump (ELCP) absolute pressure change threshold used for the fuel tank blockage test on systems incorporating an EVAP Leak Check Pump on conventional EVAP systems.

3.2.348EVPR – PID $23B8 – Conventional EVAP With Pump Fuel Tank Pressure – Fuel Tank Blockage Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B8	Conventional EVAP With Pump Fuel Tank Pressure – Fuel Tank Blockage Test	2	SNM	-131.07 to +131.07 in H_2O)	E = N * .004

This PID contains the fuel tank pressure (FTP) change measured at the end of the fuel tank blockage test for a system incorporating an EVAP Leak Check Pump on a conventional EVAP system.

3.2.349EVPR – PID $23B9 – Conventional EVAP With Pump - Fuel Tank Pressure Sensor Correlation – Propulsion Off

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23B9	Conventional EVAP With Pump - Fuel Tank Pressure Sensor Correlation – Propulsion Off	3	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
FTP Correlation Leak Check End	0	7	1	BLN	N/A	$1=True; $0=False
FTP Correlation Time End	0	6	1	BLN	N/A	$1=True; $0=False
FTP Correlation Sensor Saturation End	0	5	1	BLN	N/A	$1=True; $0=False
FTP Correlation Minimum Test Time Not Met	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1		N/A	
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			
FTP-ELCP Pressure Correlation	1	7	16	UNM	0 to 327.675 in H_2O	E = N * .005

This PID contains the signals associated with fuel tank pressure sensor correlation when the propulsion system is off for conventional EVAP systems with an EVAP Leak Check Pump.

FTP Correlation Leak Check End indicates the FTP sensor correlation test ended when the EVAP leak check completed.
FTP Correlation Time End indicates the FTP sensor correlation test ended based on test time.
FTP Correlation Saturation End indicates the FTP sensor correlation test ended because the FTP sensor reached its unreadable range.
FTP Correlation Minimum Test Time Not Met indicates there was not enough time to run the FTP sensor correlation test.
FTP-ELCP Pressure Correlation contains the average difference of correlation between the fuel tank pressure (FTP) sensor and the EVAP leak check pump (ELCP) pressure sensor.

[add to freeze frame data PID$23B9]

3.2.350EVPR – PID $23BA – Conventional EVAP With Pump Fuel Tank Blockage Location

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23BA	Conventional EVAP With Pump Fuel Tank Blockage Location	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Tank Blockage Location is Indeterminate	0	7	1	BLN	N/A	$1=True; $0=False
Fuel Tank Blockage Location is Between the ELCP Vacuum Pump and FTP Sensor	0	6	1	BLN	N/A	$1=True; $0=False
Fuel Tank Blockage Location is Between the FTP Sensor and Fuel Tank	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains the signals associated with the fuel tank blockage location for conventional EVAP systems with an EVAP Leak Check Pump.

Fuel Tank Blockage Location is Indeterminate indicates the location of the fuel tank blockage can't be determined.
Fuel Tank Blockage Location is Between the ELCP Vacuum Pump and FTP Sensor indicates the location of the fuel tank blockage is somewhere between the ELCP vacuum pump and the FTP sensor location.
Fuel Tank Blockage Location is Between the FTP Sensor and Fuel Tank indicates the location of the fuel tank blockage is somewhere between the FTP sensor location and the fuel tank.

3.2.351EVPR – PID $23BB – Conventional EVAP With Pump Refueling Detection Rationality

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23BB	Conventional EVAP With Pump Refueling Detection Rationality	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Level Refueling Detected	0	7	1	BLN	N/A	$1=True; $0=False
Vacuum Refueling Detected	0	6	1	BLN	N/A	$1=True; $0=False
Refueling Rationality Check Completed	0	5	1	BLN	N/A	$1=True; $0=False
Refueling Event Detected is Irrational	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains the signals associated with the refueling detection rationality for conventional EVAP systems with an EVAP Leak Check Pump.

Fuel Level Refueling Detected indicates that an apparent refueling event was detected based on a change in the fuel level.
Vacuum Refueling Detected indicates that an apparent refueling event was detected based on a change in the fuel tank vacuum.
Refueling Rationality Check Completed indicates the refueling rationality check completed and the apparent refueling event was determined to be either rational or irrational.
Refueling Event Detected is Irrational indicates whether the refueling event that was detected is irrational.

3.2.352EVPR – PID $23BC – EVAP Leak Check Pump Barometric Pressure Maximum Increase

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23BC	EVAP Leak Check Pump Barometric Pressure Maximum Increase	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the maximum amount the barometric pressure increased during the EVAP leak check diagnostic on systems incorporating an EVAP Leak Check Pump.

3.2.353EVPR – PID $23BD – EVAP Leak Check Pump Barometric Pressure Maximum Decrease

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23BD	EVAP Leak Check Pump Barometric Pressure Maximum Decrease	2	SNM	-131.07 to +131.07 kPa	E = N * .004

This PID contains the maximum amount the barometric pressure decreased during the EVAP leak check diagnostic on systems incorporating an EVAP Leak Check Pump.

3.2.354EVPR – PID $23BE – Conventional EVAP With Pump – EVAP Leak Check Test

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23BE	Conventional EVAP With Pump – EVAP Leak Check Test	3	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Small Leak Vacuum Target Met	0	7	1	BLN	N/A	$1=True; $0=False
EVAP Leak Check Ended Due to Peak Vacuum Reached	0	6	1	BLN	N/A	$1=True; $0=False
EVAP Leak Check Analysis Complete	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			
EVAP Leak Check Test Count	1	7	16	UNM	0 to 65535 counts	E = N * 1

This PID contains the signals associated with the analysis of the EVAP leak check test for a conventional EVAP system with an EVAP Leak Check Pump.

Small Leak Vacuum Target Met indicates the vacuum target was reached during the EVAP leak check test.
EVAP Leak Check Ended Due to Peak Vacuum Reached indicates the vacuum during the EVAP leak check reached a stabilized value and the test ended.
EVAP Leak Check Analysis Complete indicates the small/large leak DTCs were updated and the test counter incremented the last time the EVAP leak check diagnostic ran.

EVAP Leak Check Test Count contains the number of times the small/large leak tests have completed since a code clear or NVM reset.

3.2.355FHPR – PID $23BF – Fuel Rail Pressure Sensor Adaptation

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23BF	Fuel Rail Pressure Sensor Adaptation	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Service Mode Adaptation Active	0	2	1	BLN	N/A	$1=True; $0=False
In Plant Adaptation Active	0	1	1	BLN	N/A	$1=True; $0=False
Fuel Rail Pressure Sensor First Time Adaptation Complete	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains several signals associated with the status of the fuel rail pressure sensor adaptation status.

Fuel Rail Pressure Sensor First Time Adaptation Complete indicates that the first time large step adaptation has been completed.

3.2.356FHPR – PID $23C0 – Fuel Rail Pressure Sensor 1 Adaptation

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C0	Fuel Rail Pressure Sensor 1 Adaptation	2	SNM	-32767 to 32768 kPa	E = N * 10

This PID displays the adaptation pressure for fuel rail pressure sensor 1.

3.2.357TTMR– PID $23C1 – Clutch Temperature Over Hot Level3 Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C1	Clutch Temperature Over Hot Level3 Counter	2	UNM	0 to 65535 Counts	E = N

This PID contains the counts of clutch temperature over a certain severe hot temperature (hot level3 determined by calibration and hysteresis). This PID may be reset by Transmission over Temperature Counters Reset Trigger Device Control.

3.2.358TTMR– PID $23C2 – Transmission Oil Temperature over Hot Level3 Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C2	Transmission Oil Temperature Over Hot Level3 Counter	2	UNM	0 to 65535 Counts	E = N

This PID contains the counts of transmission oil temperature over a certain severe hot temperature (hot level3 determined by calibration and hysteresis). This PID may be reset by Transmission over Temperature Counters Reset Trigger Device Control.

3.2.359TTMR– PID $23C3 – Electric Pump Temperature Over Hot Level3 Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C3	Electric Pump Temperature Over Hot Level3 Counter	2	UNM	0 to 65535 Counts	E = N

This PID contains the counts of electric pump temperature over a certain severe hot temperature (hot level3 determined by calibration and hysteresis). This PID may be reset by Transmission over Temperature Counters Reset Trigger Device Control.

3.2.360TTMR– PID $23C4 – Pump Inverter Temperature Over Hot Level3 Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C4	Pump Inverter Temperature Over Hot Level3 Counter	2	UNM	0 to 65535 Counts	E = N

This PID contains the counts of pump inverter temperature over a certain severe hot temperature (hot level3 determined by calibration and hysteresis). This PID may be reset by Transmission over Temperature Counters Reset Trigger Device Control.

3.2.361RBSR – PID $23C5 – Spurious Reset at PowerUp Reset

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C5	Spurious Reset at PowerUp Reset	2	UNM	0 to 65535 counts	E = N * 1

This PID is to allow the QIS system to read all spurious reset at power up reset.

3.2.1 LTCR – PID $23C6– DCT EOL and Service Fast Learn Abort Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C6	DCT EOL and Service Fast Learn Abort Reason	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Enable Conditions Abort Reason	0	7	4	ENM	N/A	$0=No Abort Reason Detected; $1=Transmission Oil Temperature Too Low; $2=Transmission Oil Temperature Too High; $3=Transmission Output Speed Too High; $4=Transmission Output Speed Defaulted; $5=Accumulator Pressure Too Low; $6=Brake Pedal Not Applied; $7=Engine Coolant Temperature Too Low; $8=Engine Coolant Temperature Too High; $9=Transmission Not in Park
Shift Forks Neutral and Engaged Positions Learn Abort Reason	0	3	4	ENM	N/A	$0=No Abort Reason Detected; $1=Shift Forks 1 & 4 Validity Check Failed; $2=Shift Forks 2 & 3 Validity Check Failed; $3=Shift Forks not in Neutral after Shift Forks 1 & 4 Learn; $4=Shift Forks not in Neutral after Shift Forks 2&3 Learn
Shift Forks Not In Neutral During Clutch Learn Abort Reason	1	7	4	ENM	N/A	$0=No Abort Reason Detected; $1=Shift Forks Not in Neutral
Clutch Disengaged Position Learn Abort Reason	1	3	4	ENM	N/A	$0=No Abort Reason Detected; $1=Validity Check Failed
Clutch Null Point Learn Abort Reason	2	7	4	ENM	N/A	$0=No Abort Reason Detected; $1= 1st capture failed; $2= 2nd capture failed; $3= 3rd capture failed; $4= 4th capture failed
Clutch Initial Apply Position (Kiss Point) and Shift Forks Synchronize Positions Learn Abort Reason	2	3	4	ENM	N/A	$0=No Abort Reason Detected; $1=Clutch Initial Apply Position (Kiss Point) not detected on all test cycles; $2=Shift Fork control to Neutral failed on gear shaft 1; $3=Shift Fork control to Neutral failed on gear shaft 2; $4=Shift Fork synchronize position detection failed on gear shaft 1; $5=Shift Fork synchronize position detection failed on gear shaft 2

Clutch Flow Point Learn Abort Reason	3	7	4	ENM	N/A	$0=No Abort Reason Detected; $1=Increasing position learn timeout; $2=Decreasing position learn timeout; $3=Overall test timeout
Accumulator Precharge Pressure Learn Abort Reason	3	3	4	ENM	N/A	$0=No Abort Reason Detected; $1=Pressure bleed timeout $2=Pressure recovery timeout $3=Precharge pressure learn timeout

This PID contains the reason that the dual clutch transmission (DCT) end-of-line (EOL) and Service Fast Learn procedure has been aborted.

3.2.2 TNSR – PID $23C7 – Odd Speed Sensor Raw Sensed Internal Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C7	Odd Speed Sensor Raw Sensed Internal Speed (RPM)	2	UNM	0 to 8192 rpm	E = N * 0.125

This PID contains the Odd Speed Sensor Raw Sensed Internal Speed

3.2.3 TNSR – PID $23C8 – Even Speed Sensor Raw Sensed Internal Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C8	Even Speed Sensor Raw Sensed Internal Speed (RPM)	2	UNM	0 to 8192 rpm	E = N * 0.125

This PID contains the Even Speed Sensor Raw Sensed Internal Speed

3.2.4 VLTR – PID $23C9 – Buffered 9 Volt Supply Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23C9	Buffered 9 Volt Supply Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
V9B1 Signal Out of Range	0	7	1	BLN	N/A	$1=True; $0=False
V9B2 Signal Out of Range	0	6	1	BLN	N/A	$1=True; $0=False
Reserved	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1	BLN	N/A	$1=True; $0=False
Reserved	0	2	1	BLN	N/A	$1=True; $0=False
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains signals associated with the status of the Buffered 9 Volt Supplies.

V9B1 Signal Out of Range indicates that the buffered 9 volt supply 1 reading (within the main processor) is currently out of range. This parameter must not latch or include any MCP V9B information.

V9B2 Signal Out of Range indicates that the buffered 9 volt supply 2 reading (within the main processor) is currently out of range. This parameter must not latch or include any MCP V9B information.

3.2.5 FOPR – PID $23DA – Flow Control Solenoid 7 Commanded Flow

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23DA	Flow Control Solenoid 7 Commanded Flow	1	SNM	-6 to 5.953125 L/m	E = N * 3 / 64

This PID contains the flow command for fork shift rail 1 solenoid command.

PID returns variable VeFOPR_dV_SlndCmndShfRail1

3.2.6 FOPR – PID $23DB – Flow Control Solenoid 8 Commanded Flow

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23DB	Flow Control Solenoid 8 Commanded Flow	1	SNM	-6 to 5.953125 L/m	E = N * 3 / 64

This PID contains the flow command for fork shift rail 2 solenoid command.

PID returns variable VeFOPR_dV_SlndCmndShfRail2

3.2.7 BRPR – PID $23EC - Pressure Control Solenoid 1 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23EC	Pressure Control Solenoid 1 Commanded Pressure	2	UNM	0 to 8191.875 kPa	E = N / 8

This PID contains the commanded pressure for pressure control solenoid 1.

3.2.8 BRPR – PID $23ED – Pressure Control Solenoid 2 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23ED	Pressure Control Solenoid 2 Commanded Pressure	2	UNM	0 to 8191.875 kPa	E = N / 8

This PID contains the commanded pressure for pressure control solenoid 2.

3.2.9 FHPR – PID $23F0 – Fuel Rail Pressure Sensor Adaptation Reasons For Test Disablement Part 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23F0	Fuel Rail Pressure Sensor Adaptation Reasons For Test Disablement Part 1	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
reserved	0	7	1	BLN	N/A	$1=True; $0=False
Normal control duration too short	0	6	1	BLN	N/A	$1=True; $0=False
Predicted fuel temperature out of test range	0	5	1	BLN	N/A	$1=True; $0=False
Engine run time too short	0	4	1	BLN	N/A	$1=True; $0=False
Requested pressure too low	0	3	1	BLN	N/A	$1=True; $0=False
Feed pressure too low during test	0	2	1	BLN	N/A	$1=True; $0=False
Engine is not rotating	0	1	1	BLN	N/A	$1=True; $0=False
Feed pressure system not valid	0	0	1	BLN	N/A	$1=True; $0=False
Device Control - Pump Circuit Check	1	7	1	BLN	N/A	$1=True; $0=False
Device Control - commanded pressure	1	6	1	BLN	N/A	$1=True; $0=False
Injector flow test is active	1	5	1	BLN	N/A	$1=True; $0=False
Feed pump is off	1	4	1	BLN	N/A	$1=True; $0=False
Fuel level too low	1	3	1	BLN	N/A	$1=True; $0=False
Green engine mode enabled	1	2	1	BLN	N/A	$1=True; $0=False
Time after drive test too short	1	1	1	BLN	N/A	$1=True; $0=False
reserved	1	0	1	BLN	N/A	$1=True; $0=False
Clutch switch/sensor fault	2	7	1	BLN	N/A	$1=True; $0=False
Induction air temp too low	2	6	1	BLN	N/A	$1=True; $0=False
Ambient air pressure too low	2	5	1	BLN	N/A	$1=True; $0=False
Ambient air pressure update error	2	4	1	BLN	N/A	$1=True; $0=False
Fault Pending - fuel rail pressure sensor circuit	2	3	1	BLN	N/A	$1=True; $0=False
Fault Pending - high pressure pump circuit	2	2	1	BLN	N/A	$1=True; $0=False
MfrsEnblCounter is not zero	2	1	1	BLN	N/A	$1=True; $0=False
Engine not synced	2	0	1	BLN	N/A	$1=True; $0=False
reserved	3	7	1	BLN	N/A	$1=True; $0=False
reserved	3	6	1	BLN	N/A	$1=True; $0=False
reserved	3	5	1	BLN	N/A	$1=True; $0=False
Idle Test Failed	3	4	1	BLN	N/A	$1=True; $0=False
Sensor proportion high test failed	3	3	1	BLN	N/A	$1=True; $0=False
Sensor stuck test failed	3	2	1	BLN	N/A	$1=True; $0=False
All cylinders are not being fueled	3	1	1	BLN	N/A	$1=True; $0=False
Voltage too low	3	0	1	BLN	N/A	$1=True; $0=False

This PID contains the current reasons for test disablement of the fuel rail pressure sensor first time large adaptation.

3.2.10 FHPR – PID $23F1 – Fuel Rail Pressure Sensor Adaptation Reasons For Test Disablement Part 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23F1	Fuel Rail Pressure Sensor Adaptation Reasons For Test Disablement Part 2	4	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Vehicle Speed too high	0	7	1	BLN	N/A	$1=True; $0=False
Not in closed loop fuel control	0	6	1	BLN	N/A	$1=True; $0=False
Clutch in between top and bottom of travel	0	5	1	BLN	N/A	$1=True; $0=False
Not in Idle speed control	0	4	1	BLN	N/A	$1=True; $0=False
Accelerator pedal not at zero for long enough time	0	3	1	BLN	N/A	$1=True; $0=False
Engine in catalyst light off mode	0	2	1	BLN	N/A	$1=True; $0=False
Test in suspend state	0	1	1	BLN	N/A	$1=True; $0=False
Total engine run time after green engine mode too short	0	0	1	BLN	N/A	$1=True; $0=False
reserved	1	7	1	BLN	N/A	$1=True; $0=False
reserved	1	6	1	BLN	N/A	$1=True; $0=False
reserved	1	5	1	BLN	N/A	$1=True; $0=False
reserved	1	4	1	BLN	N/A	$1=True; $0=False
reserved	1	3	1	BLN	N/A	$1=True; $0=False
DTC disable	1	2	1	BLN	N/A	$1=True; $0=False
Feed pressure spike	1	1	1	BLN	N/A	$1=True; $0=False
Other test has purge priority	1	0	1	BLN	N/A	$1=True; $0=False
reserved	2	7	1	BLN	N/A	$1=True; $0=False
reserved	2	6	1	BLN	N/A	$1=True; $0=False
reserved	2	5	1	BLN	N/A	$1=True; $0=False
reserved	2	4	1	BLN	N/A	$1=True; $0=False
reserved	2	3	1	BLN	N/A	$1=True; $0=False
reserved	2	2	1	BLN	N/A	$1=True; $0=False
reserved	2	1	1	BLN	N/A	$1=True; $0=False
reserved	2	0	1	BLN	N/A	$1=True; $0=False
reserved	3	7	1	BLN	N/A	$1=True; $0=False
reserved	3	6	1	BLN	N/A	$1=True; $0=False
reserved	3	5	1	BLN	N/A	$1=True; $0=False
reserved	3	4	1	BLN	N/A	$1=True; $0=False
reserved	3	3	1	BLN	N/A	$1=True; $0=False
reserved	3	2	1	BLN	N/A	$1=True; $0=False
reserved	3	1	1	BLN	N/A	$1=True; $0=False
reserved	3	0	1	BLN	N/A	$1=True; $0=False

This PID contains the current reasons for test disablement of the fuel rail pressure sensor first time large adaptation.

3.2.11 FHPR – PID $23F2 – Fuel Rail Pressure Sensor Adaptation Test Status

This PID contains the status of the fuel rail pressure sensor first time large learn test.

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$23F2	Fuel Rail Pressure Sensor Adaptation Test Status	1	ENM	N/A	$00 = Not attempted due to disablement; $01 = Test in Progress; $02 = Test Complete – Passed; $03 = Test Complete – Failed; $04 = Aborted; $05 = Reserved; $06 = Reserved; $07 = Reserved; $08 = Reserved; $09 = Reserved; $0A = Abort - Feed pressure system not valid; $0B = Abort - Engine is not rotating; $0C = Abort - Feed pressure too low during test; $0D = Abort - Requested pressure too low; $0E = Abort - Engine run time too short; $0F = Abort - Predicted fuel temperature out of test range; $10 = Abort - Normal control duration too short; $11 = Abort - reserved; $12 = Abort - reserved; $13 = Abort - Time after drive test too short; $14 = Abort - Green engine mode enabled; $15 = Abort - Fuel level too low; $16 = Abort - Feed pump is off; $17 = Abort - Injector flow test is active; $18 = Abort - Device Control - commanded pressure; $19 = Abort - Device Control - Pump Circuit Check; $1A = Abort - Engine not synced; $1B = Abort - MfrsEnblCounter is not zero; $1C = Abort - Fault Pending - high pressure pump circuit; $1D = Abort - Fault Pending - fuel rail pressure sensor circuit; $1E = Abort - Ambient air pressure update error; $1F = Abort - Ambient air pressure too low; $20 = Abort - Induction air temp too low; $21 = Abort - Clutch switch/sensor fault; $22 = Abort - Voltage too low; $23 = Abort - All cylinders are not being fueled; $24 = Abort - Sensor stuck test failed; $25 = Abort - Sensor proportion high test failed; $26 = Abort - Idle Test Failed; $27 = Abort - reserved; $28 = Abort - reserved; $29 = Abort - reserved; $2A = Abort - Total engine run time after green engine mode too short; $2B = Abort - Test in suspend state; $2C = Abort - Engine in catalyst light off mode; $2D = Abort - Accelerator pedal not at zero for long enough time; $2E = Abort - Not in Idle speed control; $2F = Abort - Clutch in between top and bottom of travel; $30 = Abort - Not in closed loop fuel control; $31 = Abort - Vehicle Speed too high; $32 = Abort - Other test has purge priority; $33 = Abort - Feed pressure spike; $34 = Abort - DTC disable; $35-$FF=Reserved

This PID contains the status of the fuel rail pressure sensor first time large learn test

3.2.12 ENER/HSWR – PID $240C – Hood Switch Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$240C	Hood Switch Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Hood Status Signal Invalid	0	7	1	BLN	N/A	$1=True; $0=False
Hood Status	0	6	2	ENM	N/A	$0=Closed; $1=Ajar; $2=Open
Discrete Hood Switch Input 1 Active	0	4	1	BLN	N/A	$1=True; $0=False
Discrete Hood Switch Input 2 Active	0	3	1	BLN	N/A	$1=True; $0=False
Reserved	0	2	1			
Reserved	0	1	1			
Hood Position Switch Both States Latched	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains statuses associated with the hood switch and may apply to either discrete or analog hood switches based on the application.

Hood Status Signal Invalid indicates that the hood switch is indicating an invalid state. For an analog switch, this indicates that the voltage observed is between or outside valid ranges.

Hood Status indicates the hood position determined by the algorithm based on information from the hood switch or switches.

Discrete Hood Switch Input 1 Active and Discrete Hood Switch Input 2 Active apply to applications using two discrete switch inputs. These bits will be populated when this type of switch is directly connected to the controller supporting this PID. Each bit indicates that the raw hood switch hardwire input to the controller is active.

Hood Position Switch Both States Latched indicates true once both the on and off switch states are observed and will remain latched for the current ignition cycle. This bit will reset to false once the power mode transitions to off. (This bit applies to either a discrete or analog system)

3.2.13 THMR – PID $241C – Heater Core Pump Relay Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$241C	Heater Core Pump Relay Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Heater Core Pump Relay Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Heater Core Pump Relay Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Heater Core Pump Relay Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Heater Core Pump Relay Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the heater core pump relay output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs B269C, B269A, and B269D, respectively.

3.2.14 ETQR – PID $2427 – Engine Airflow Steady State Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2427	Engine Airflow Steady State Torque	1	UNM	-848 - 1192 Nm	E = N * 8 - 848

This PID contains the Engine Airflow Steady State Torque value sent from the ECM to the HCP.

3.2.15 AXLR – PID $2428 – Commanded Predicted Axle Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2428	Commanded Predicted Axle Torque	2	UNM	-22,534 – 43,001 Nm	E = N * 1 – 22,534

This PID contains the Commanded Predicted Axle Torque value sent from the ECM to HCP.

3.2.16 AXLR – PID $2429 – Axle Torque Actual

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2429	Axle Torque Actual	2	UNM	-22,534 – 43,001 Nm	E = N * 1 – 22,534

This PID contains the Axle Torque Actual value sent from the HCP to the ECM and ABS module.

3.2.17 BRKR – PID $242C – Driver Intended Total Brake Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$242C	Driver Intended Total Brake Torque	2	UNM	0 - 32,767.5 Nm	E = N / 2

This PID contains the Driver Intended Total Brake Torque value sent from the ABS module to the HCP. This PID will be defaulted to zero Nm if serial data signal is not being received.

3.2.18 AXLR – PID $242D – Commanded Immediate Axle Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$242D	Commanded Immediate Axle Torque	2	UNM	-22,534 – 43,001 Nm	E = N * 1 - 22,534

This PID contains the Commanded Immediate Axle Torque value sent from the ECM to HCP.

3.2.19 HSER – PID $2432 – Engine Crank Speed Commanded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2432	Engine Crank Speed Commanded	1	UNM	0 – 2040 rpm	E = N * 8

This PID indicates the commanded cranking speed for the internal combustion engine.

3.2.20 DTRR – PID $2434 – Driver Intended Axle Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2434	Driver Intended Axle Torque	2	UNM	-22,534 – 43,001 Nm	E = N * 1 – 22,534

This PID contains the commanded driver intended axle torque after response filtering, clunk, etc. This is an input to the start of torque arbitration.

3.2.21 PTAR – PID $2437 – Commanded Predicted Engine Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2437	Commanded Predicted Engine Torque	1	UNM	-848 to +1192 Nm	E = N * 8 – 848

This PID contains the Commanded Predicted Engine Torque value, which is the torque request for the slow response air path (control of the throttle). Based on current conditions the hybrid controller determines the optimum engine torque, which it sends as a request to the Engine Controller engine torque arbitration. This is the requested torque that the system will attain if there are not any engine torque intervention requestors. In a conventional engine vehicle, this value resides within the ECM. In a 2-mode hybrid vehicle, this value is sent from the HCP to the ECM.

Note: This contains the same data as the requested predicted axle torque converted to predicted engine torque (reference PID $2428).

3.2.22 PTAR – PID $2438 – Commanded Immediate Engine Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2438	Commanded Immediate Engine Torque	1	UNM	-848 to +1192 Nm	E = N * 8 – 848

This PID contains the Commanded Immediate Engine Torque value which is the fast response torque request for spark control (retardation). Based on current conditions, the hybrid controller determines the optimum engine torque, which it sends as a request to the engine controller's engine torque arbitration. During normal operating conditions (no spark retardation), the value of this PID is identical to PID $2437 "Commanded Predicted Engine Torque". While spark retardation is active, this PID shows values lower than PID $2437.In a conventional engine vehicle, this value resides within the ECM. In a 2-mode hybrid vehicle, this value is sent from the HCP to the ECM.

Note: This contains the same data as the requested immediate axle torque converted to immediate engine torque (reference PID $242D).

3.2.23 ETQR – PID $2448 – Engine Actual Steady State Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2448	Engine Actual Steady State Torque	1	UNM	-848 to +1192 Nm	E = N * 8 - 848

This PID contains the Engine Actual Steady State Torque value. In a conventional engine vehicle, this value resides within the ECM. In a 2-mode hybrid vehicle, this value is sent from the ECM to the HCP.

Note: This is the similar to the requested engine airflow torque but includes the effects of immediate torque requests (reference PID $2427). It is also the same as PID $1A2D with different scaling.

3.2.24 ETQR – PID $2449 – Torque Based Calculated Engine Load

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2449	Torque Based Calculated Engine Load	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains a percent engine load that should have the following characteristics defined by SAE J1979:

- Reaches 100% when Driver Commanded Torque is 100% while at any altitude or temperature.
- Indicates percent of peak available torque during normal non-faulted conditions.
- For spark ignited engines, linear correlated with vacuum at MBT spark and stoichiometry.
- Reads 0% with the engine off and ignition on.

Note: Hybrid Controls can independently control final axle torque. This PID contains the alternate calculated percent engine load in a torque control based system that could be used for PID $0004 per J1979.

3.2.25 ETQR – PID $244A – Torque Based Absolute Engine Load

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$244A	Torque Based Absolute Engine Load	2	UNM	0 to 400 %	E = N * 400 / 65535

This PID contains an absolute engine load that should have the following characteristics defined by SAE J1979:

- Ranges from 0% to 95% for naturally aspirated engines, 0% to 400% for boosted engines
- Linear correlated with engine indicated and brake torque
- Peak value correlates with volumetric efficiency at Wide Open Throttle (WOT) at Standard Temperature and Pressure (STP)
- Indicates percent of absolute torque of engine given its current operating limits vs. STP

Note: This PID contains the alternate calculated percent absolute engine load in a torque control based system that could be used for PID $0043 per J1979.

3.2.26 ECPR – PID $244B – Peak Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$244B	Peak Torque	1	UNM	-848 to +1192 Nm	E = N * 8 - 848

This PID contains the peak indicated torque at the current engine speed, standard temperature and pressure, and will include torque reached during power enrichment.

3.2.27 ECPR – PID $244C – Air Density Scalar

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$244C	Air Density Scalar	1	UNM	0 to 2	E = N * 2 / 255

This PID contains a multiplier to account for changes in air density from STP conditions due to effects of air temperature and air pressure.

3.2.28 ECPR – PID $244D – Maximum Predicted Engine Torque Capacity

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$244D	Maximum Predicted Engine Torque Capacity	1	UNM	-848 to +1192 Nm	E = N * 8 - 848

This PID contains the maximum torque the engine can produce. It is determined assuming maximum airflow, current engine speed and current engine operating conditions. It takes into account any factors that would limit the available torque from the engine. A positive value represents the engine is providing torque. A negative value represents the engine is providing engine braking.

3.2.29 ECPR – PID $244E – Minimum Engine Off Torque Capacity

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$244E	Minimum Engine Off Torque Capacity	1	UNM	-848 to +1192 Nm	E = N * 8 - 848

This PID contains a value that is determined assuming no combustion, minimum airflow, current engine speed and current engine operating conditions.

3.2.30 ECPR – PID $244F – Minimum Predicted Engine Run Torque Capacity

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$244F	Minimum Predicted Engine Run Torque Capacity	1	UNM	-848 to +1192 Nm	E = N * 8 - 848

This PID contains a value represents the minimum combustion engine torque limit.

3.2.31 ECPR – PID $2450 – Engine Torque Capability Scalar

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2450	Engine Torque Capability Scalar	1	UNM	0 to 2	E = N * 2 / 255

This PID contains the modifier to the engine torque capability for fault remedial action. It will read 1 when no remedial actions are taken place.

3.2.32 SPDR – PID $2451 – Zero Pedal Engine Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2451	Zero Pedal Engine Torque	1	UNM	-848 to +1192 Nm	E = N * 8 - 848

This PID contains the Zero Pedal Engine Torque value, which is the minimum limit for the immediate torque domain. Arbitrated torque requestors are clamped at this limit.

3.2.33 HSER – PID $2453 – Transmission Input Speed Acceleration Profile Desired

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2453	Transmission Input Speed Acceleration Profile Desired	2	SNM	-32768 - 32767 rpm/s	E = N * 1

This PID contains the desired input speed acceleration profile for a hybrid transmission.

3.2.34 TRQR – PID $245B – Commanded Indicated Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$245B	Commanded Indicated Torque	1	UNM	-848 to 1192 Nm	E = N * 8 - 848

This PID contains the commanded indicated torque that is being converted into a fuel mass request.

3.2.35 TRQR – PID $245C – Commanded Predicted Indicated Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$245C	Commanded Predicted Indicated Torque	1	UNM	-848 to 1192 Nm	E = N * 8 - 848

This PID contains the commanded predicted indicated torque without active damping and smoke limitation.

3.2.36 TRQR – PID $245D – Commanded Fuel Mass

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$245D	Commanded Fuel Mass	1	UNM	0 to 255 mg	E = N * 1

This PID contains the commanded fuel mass, converted from torque.

3.2.37 TRQR – PID $245E – Fuel Mass Target

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$245E	Fuel Mass Target	1	UNM	0 to 255 mg	E = N * 1

This PID contains the fuel mass target value, converted from torque, without active damping and smoke limitation.

3.2.38 ECPR – PID $245F – Maximum Torque Limit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$245F	Maximum Torque Limit	1	UNM	-848 to 1192 Nm	E = N * 8 - 848

This PID contains the maximum torque limit calculated from smoke fuel limitation.

3.2.39 ETQR – PID $2460 – Total System Friction Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2460	Total System Friction Torque	1	UNM	-848 to 1192 Nm	E = N * 8 - 848

This PID contains the total friction torque in the system. The value includes both internal engine losses and external loads.

3.2.40 ETQR – PID $2461 – Actual Delivered Crankshaft Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2461	Actual Delivered Crankshaft Torque	1	UNM	-848 to 1192 Nm	E = N * 8 - 848

This PID contains the actual delivered crankshaft torque, converted from fuel mass.

3.2.41 HSWR – PID $247A – Analog Hood Position Switch A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$247A	Analog Hood Position Switch A/D	1	UNM	0 - 100 % V ref	E = N * 100 / 255

This PID contains the hood position switch analog input as a percentage of the reference voltage. This PID will function for both 5V and 12V reference hood switch applications.

3.2.42 DTRR – PID $247C – Raw Driver Intended Axle Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$247C	Raw Driver Intended Axle Torque	2	UNM	0 to 65,535 Nm	E = N * 1

This PID contains the raw driver intended axle torque without any filtering or idle contribution.

3.2.43 VMMR – PID $2498 – Time Since Last Engine Maintenance Mode Completion

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2498	Time Since Last Engine Maintenance Mode Completion	2	UNM	0 to 65535 days	E = N * 1

This PID indicates the elapsed time since the most recent completion of Engine Maintenance Mode. If an Engine Maintenance Mode event has not taken place, this PID shall default to zero. This value will only be accurate after the propulsion system is activated and VMMR receives the necessary time signal. The PID will hold a previous value until an updated value is available.

3.2.44 VMMR – PID $2499 – Time Since Last Fuel Maintenance Mode Completion

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2499	Time Since Last Fuel Maintenance Mode Completion	2	UNM	0 to 65535 days	E = N * 1

This PID indicates the elapsed time since the most recent completion of Fuel Maintenance Mode. If a Fuel Maintenance Mode event has not taken place, this PID shall default to zero. This value will only be accurate after the propulsion system is activated and VMMR receives the necessary time signal. The PID will hold a previous value until an updated value is available.

3.2.45 VMMR – PID $249A – Engine Maintenance Mode Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$249A	Engine Maintenance Mode Counter	1	UNM	0 to 255 counts	E = N * 1

This PID indicates the cumulative number of engine maintenance mode events over the life of the controller.

3.2.46 VMMR – PID $249B – Fuel Maintenance Mode Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$249B	Fuel Maintenance Mode Counter	1	UNM	0 to 255 counts	E = N * 1

This PID indicates the cumulative number of fuel maintenance mode events over the life of the controller.

3.2.47 VMMR – PID $249C – Estimated Time Remaining Until Next Engine Maintenance Mode

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$249C	Estimated Time Remaining Until Next Engine Maintenance Mode	1	UNM	0 to 255 days	E = N * 1

This PID indicates the estimated time remaining until the next engine maintenance mode. This value will only be accurate after the propulsion system is activated and VMMR receives the necessary time signal. The PID will hold a previous value until an updated value is available.

3.2.48 VMMR – PID $249D – Estimated Time Remaining Until Next Fuel Maintenance Mode

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$249D	Estimated Time Remaining Until Next Fuel Maintenance Mode	2	UNM	0 to 65535 days	E = N * 1

This PID indicates the estimated time remaining until the next fuel maintenance mode. This value will only be accurate after the propulsion system is activated and VMMR receives the necessary time signal. The PID will hold a previous value until an updated value is available.

3.2.49 VMMR – PID $249E – Maintenance Mode Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$249E	Maintenance Mode Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Engine Maintenance Mode Active	0	7	1	BLN	N/A	$1=True; $0=False
Fuel Maintenance Mode Active	0	6	1	BLN	N/A	$1=True; $0=False
Last Engine Maintenance Mode Incomplete	0	5	1	BLN	N/A	$1=True; $0=False
Engine Maintenance Mode Completed This Ignition Cycle	0	4	1	BLN	N/A	$1=True; $0=False
Engine Maintenance Mode Disabled Due to Low Fuel Level	0	3	1	BLN	N/A	$1=True; $0=False
Fuel Maintenance Mode Disabled Due to Low Fuel Level	0	2	1	BLN	N/A	$1=True; $0=False
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains status bits associated with Engine Maintenance Mode (EMM) and Fuel Maintenance Mode (FMM). The content of each of these signals is described below.

Engine Maintenance Mode Active indicates that an EMM is currently active.

Fuel Maintenance Mode Active indicates that an FMM is currently active.

Last Engine Maintenance Mode Incomplete indicates that the last EMM event was interrupted and did not run to completion.

Engine Maintenance Mode Completed This Ignition Cycle indicates that an EMM has run to completion during the current ignition cycle.

Engine Maintenance Mode Disabled Due to Low Fuel Level indicates that an EMM is currently disabled due to a low fuel tank level. This bit will set when the fuel level is low, EMM is desired and the low fuel delay time has not elapsed.

Fuel Maintenance Mode Disabled Due to Low Fuel Level indicates that an FMM is currently disabled due to a low fuel tank level. This bit will set when the fuel level is low, FMM is desired and the low fuel delay time has not elapsed.

3.2.50 VMMR - PID $249F – Engine Maintenance Mode Percent Complete

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$249F	Engine Maintenance Mode Percent Complete	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains a percent complete indication of a currently running engine maintenance mode (EMM). Once the EMM is complete (PID = 100%), the PID value will latch for the current key cycle.

3.2.51 FLVR – PID $24A0 – Time Since Last Refuel Event

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24A0	Time Since Last Refuel Event	2	UNM	0 to 65535 days	E = N * 1

This PID indicates the elapsed time since the most recent refueling event.

3.2.52 AXLR – PID $24BE – Regenerative Braking Axle Torque Capability

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24BE	Regenerative Braking Axle Torque Capability	1	UNM	0 to 8160 Nm	E = N * 32

This PID contains the Regenerative Braking Axel Torque Capability value sent from the HCP to the ABS module.

3.2.53 CSLR – PID $24C7 – Clutch 1 Max Slip Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24C7	Clutch 1 Max Slip Speed	1	UNM	0 to 2040 rpm	E = N * 8

This variable records the maximum slip speed observed across an off-going Clutch 1. It gets reset to zero when the Clutch State is Locked.

3.2.54 CSLR – PID $24C8 – Clutch 2 Max Slip Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24C8	Clutch 2 Max Slip Speed	1	UNM	0 to 2040 rpm	E = N * 8

This variable records the maximum slip speed observed across an off-going Clutch 2. It gets reset to zero when the Clutch State is Locked.

3.2.55 CSLR – PID $24C9 – Clutch 3 Max Slip Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24C9	Clutch 3 Max Slip Speed	1	UNM	0 to 2040 rpm	E = N * 8

This variable records the maximum slip speed observed across an off-going Clutch 3. It gets reset to zero when the Clutch State is Locked.

3.2.56 CSLR – PID $24CA – Clutch 4 Max Slip Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24CA	Clutch 4 Max Slip Speed	1	UNM	0 to 2040 rpm	E = N * 8

This variable records the maximum slip speed observed across an off-going Clutch 4. It gets reset to zero when the Clutch State is Locked.

3.2.57 HSER – PID $24CB – Transmission Shift Range State

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24CB	Transmission Shift Range State	1	ENM	N/A	$00= Range St 0; $01= Range St 1; $02= Range St 2; $03= Range St 3: $04= Range St 4; $05= Range St 5; $06= Range St 6; $07= Range St 7; $08= Range St 8; $09= Range St 9; $0A= Range St 10; $0B= Range St 11; $0C= Range St 12; $0D= Range St 13; $0E= Range St 14; $0F= Range St 15; $10= Range St 16; $11= Range St 17; $12= Range St 18; $13= Range St 19; $14= Range St 20; $15= Range St 21; $16= Range St 22; $17= Range St 23; $18= Range St 24; $19= Range St 25; $1A= Range St 26; $1B= Range St 27; $1C= Range St 28; $1D= Range St 29; $1E= Range St 30; $1F= Range St 31; $20= Range St 32; $21= Range St 33; $22= Range St 34; $23= Range St 35; $24= Range St 36; $25= Range St 37; $26= Range St 38; $27= Range St 39; $28= Range St 40; $29= Range St 41; $2A= Range St 42; $2B= Range St 43; $2C= Range St 44; $FF= Invalid Range St;

Transmission Shift Range State, which represents an internal state denoting the specific shift operations being performed.

3.2.58 CSLR – PID $24CD– Clutch 5 Max Slip Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24CD	Clutch 5 Max Slip Speed	1	UNM	0-2040 RPM	E = N * 8

This variable records the maximum slip speed observed across an off-going Clutch 5. It gets reset to zero when the Clutch State is locked.

3.2.59 MEMR – PID $24CE – Main Processor NVM Reset Data

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$ 24CE	Main Processor NVM Reset Data	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Static NVM Error	0	7	1	ENM	N/A	$0=No error present; $1=Error present;
Dynamic NVM Error	0	6	1	ENM	N/A	$0=No error present; $1=Error present;
Shutdown Finished	0	5	1	ENM	N/A	$0=Shutdown not finished; $1=Shutdown finished;
BINVDM ECC Error	0	4	1	ENM	N/A	$0=No error present $1=Error present
Static NVM Default Status	0	3	1	ENM	N/A	$0=Static NVM not defaulted; $1=Static NVM defaulted;
Cumulative NVM Default Status	0	2	1	ENM	N/A	$0=Cumulative NVM not defaulted; $1=Cumulative NVM defaulted;
Adaptable NVM Default Status	0	1	1	ENM	N/A	$0=Adaptable NVM not defaulted; $1=Adaptable NVM defaulted;
Preserved NVM Default Status	0	0	1	ENM	N/A	$0=Preserved NVM not defaulted; $1=Preserved NVM defaulted;

This PID contains several parameters helpful in diagnosing main processor NVM resets.

3.2.60 HRTR – PID $24E8 – Fuel Rail Temperature 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24E8	Fuel Rail Temperature 1	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the undefaulted fuel rail temperature measured by the first sensor located on the fuel rail. If the data used to compute the temperature is obtained from a serial data signal, then, if the signal is not received, the value shall be set to -40 °C.

3.2.61 HRTR – PID $24E9 – Fuel Rail Temperature 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24E9	Fuel Rail Temperature 2	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains theundefaulted fuel rail temperature measured by the second sensor located on the fuel rail. . If the data used to compute the temperature is obtained from a serial data signal, then, if the signal is not received, the value shall be set to -40 °C.

3.2.62 HRTR – PID $24EA– Power-Up Fuel Rail Temperature 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24EA	Power-Up Fuel Rail Temperature 1	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the fuel rail temperature at power-up (Run/Crank transition from low to high) from the first sensor located on the fuel rail.

3.2.63 HRTR – PID $24EB– Power-Up Fuel Rail Temperature 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24EB	Power-Up Fuel Rail Temperature 2	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the fuel rail temperature at power-up (Run/Crank transition from low to high) from the second sensor located on the fuel rail.

3.2.64 AAPR – PID $24EC – Barometric Pressure Sensor C A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24EC	Barometric Pressure Sensor C A/D	1	UNM	0 to 100% 5V	E = N * 100 / 255

This PID contains the barometric pressure analog input of sensor C as a percentage of its reference voltage.

3.2.65 AAPR – PID $24ED – Barometric Pressure Sensor C

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24ED	Barometric Pressure Sensor C	1	UNM	0 to 255 kPa	E = N * 1

This PID contains the air pressure value read by sensor C. While this pressure is effectively the ambient air (barometric) pressure, if this sensor is located after the intake air filter, actual barometric pressure may be calculated to account for a pressure drop across the filter.

3.2.66 MSFR – PID $24F0 – Crank Angle Sensing Error Learn Status 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24F0	Crank Angle Sensing Error Learn Status 2	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Airflow Not Met	0	7	1	BLN	N/A	$1=True; $0=False
Airflow Not Stable	0	6	1	BLN	N/A	$1=True; $0=False
reserved	0	5	1	BLN	N/A	$1=True; $0=False
RPM Not Stable	0	4	1	BLN	N/A	$1=True; $0=False
Engine RPM Too High	0	3	1	BLN	N/A	$1=True; $0=False
Engine RPM Too Low	0	2	1	BLN	N/A	$1=True; $0=False
Insufficient Data Collected	0	1	1	BLN	N/A	$1=True; $0=False
Reserved				=		

This PID contains several signals associated with the status of the crank angle sensing error (CASE) learn algorithm. The content of each of these signals is described below.

Airflow Not Met indicates that the engine did not reach the minimum airflow indicative of a closed throttle.

Airflow Not Stable indicates that once the engine reached the minimum airflow enablement value, the airflow was not consistent throughout the learn.

RPM Not Stable indicates that once the enablement engine speed has been met, the engine speed was not stable throughout the learn.

Engine RPM Too High indicates that once the engine reached the target engine speed for fuel cut, or in the case of a strong hybrid the unfueled engine has been spun to the target engine speed by the electric motor, the engine speed high limit was reached before sufficient engine data has been collected.

Engine RPM Too Low indicates that once the engine reached the target engine speed for fuel cut, or in the case of a strong hybrid the unfueled engine has been spun to the target engine speed by the electric motor, the engine speed low limit was reached before sufficient engine data has been collected.

Insufficient Data Collected indicates that the CASE learn terminated before a predetermined amount of data samples was collected in order to calculate an accurate CASE factor.

3.2.67 MSFR – PID $24F1 – CASE Opposing Factor Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24F1	CASE Opposing Factor Error	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Cylinder 1 and its Paired Cylinder Out of Range	0	7	1	BLN	N/A	$1=True; $0=False
Cylinder 2 and its Paired Cylinder Out of Range	0	6	1	BLN	N/A	$1=True; $0=False
Cylinder 3 and its Paired Cylinder Out of Range	0	5	1	BLN	N/A	$1=True; $0=False
Cylinder 4 and its Paired Cylinder Out of Range	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains the results of the Opposing Cylinder Compensation Factors Out of Range segment of the crank angle sensing error (CASE) learn algorithm. The content of each of these signals is described below. On engines in which there are less than 8 cylinders, not all pairs will be used. It is understood that if cylinders 1 and 4 constitute a firing pair in a 4-cylinder engine, then only Cylinder 1 and its Paired Cylinder Out of Range will be supported.

Cylinder 1 and its Paired Cylinder Out of Range indicates that variation between pairs of cylinders is excessive on cylinder 1 and the cylinder that is at TDC at the same time as cylinder 1.

Cylinder 2 and its Paired Cylinder Out of Range indicates that variation between pairs of cylinders is excessive on cylinder 2 and the cylinder that is at TDC at the same time as cylinder 2.

Cylinder 3 and its Paired Cylinder Out of Range indicates that variation between pairs of cylinders is excessive on cylinder 3 and the cylinder that is at TDC at the same time as cylinder 3.

Cylinder 4 and its Paired Cylinder Out of Range indicates that variation between pairs of cylinders is excessive on cylinder 4 and the cylinder that is at TDC at the same time as cylinder 4.

3.2.68 PHSR – PID $24FA – Intake Camshaft Actuator Park Lock Control Circuit Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24FA	Intake Camshaft Actuator Park Lock Control Circuit Bank 1	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Intake Camshaft Actuator Park Lock Bank 1 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Intake Camshaft Actuator Park Lock Bank 1 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Camshaft Actuator Park Lock Bank 1 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Camshaft Actuator Park Lock Bank 1 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the intake camshaft actuator park lock output for bank 1 along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P25CB, P25CA, and P25CC, respectively.

3.2.69 PHSR – PID $24FB – Intake Camshaft Actuator Park Lock Control Circuit Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24FB	Intake Camshaft Actuator Park Lock Control Circuit Bank 2	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Intake Camshaft Actuator Park Lock Bank 2 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Intake Camshaft Actuator Park Lock Bank 2 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Camshaft Actuator Park Lock Bank 2 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Intake Camshaft Actuator Park Lock Bank 2 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the intake camshaft actuator park lock output for bank 2 along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P25CE, P25CD, and P25CF, respectively.

3.2.70 PHSR – PID $24FC – Exhaust Cam Phase Angle Actual Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24FC	Exhaust Cam Phase Angle Actual Bank 1	1	SNM	-50.1960784314° to 49.8039215686°	E = N * 100 / 255

This PID contains the position of the exhaust camshaft for bank 1 in terms of degrees of camshaft rotation (retard) from the park position (a value of zero represents the park position).

3.2.71 PHSR – PID $24FD – Intake Cam Phase Angle Actual Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24FD	Intake Cam Phase Angle Actual Bank 1	1	SNM	-50.1960784314° to 49.8039215686°	E = N * 100 / 255

This PID contains the position of the intake camshaft for bank 1 in terms of degrees of camshaft rotation (advance) from the park position (a value of zero represents the park position).

3.2.72 PHSR – PID $24FE – Exhaust Cam Phase Angle Actual Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24FE	Exhaust Cam Phase Angle Actual Bank 2	1	SNM	-50.1960784314° to 49.8039215686°	E = N * 100 / 255

This PID contains the position of the exhaust camshaft for bank 2 in terms of degrees of camshaft rotation (retard) from the park position (a value of zero represents the park position).

3.2.73 PHSR – PID $24FF – Intake Cam Phase Angle Actual Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$24FF	Intake Cam Phase Angle Actual Bank 2	1	SNM	-50.1960784314° to 49.8039215686°	E = N * 100 / 255

This PID contains the position of the intake camshaft for bank 2 in terms of degrees of camshaft rotation (advance) from the park position (a value of zero represents the park position).

3.2.74 PHSR – PID $2500 – Exhaust Cam Phase Angle Desired Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2500	Exhaust Cam Phase Angle Desired Bank 1	1	SNM	-50.1960784314° to 49.8039215686°	E = N * 100 / 255

This PID contains the set point for the position of the exhaust camshaft for bank 1 in terms of degrees of camshaft rotation (retard) from the park position.

3.2.75 PHSR – PID $2501 – Intake Cam Phase Angle Desired Bank 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2501	Intake Cam Phase Angle Desired Bank 1	1	SNM	-50.1960784314° to 49.8039215686°	E = N * 100 / 255

This PID contains the set point for the position of the intake camshaft for bank 1 in terms of degrees of camshaft rotation (advance) from the park position.

3.2.76 PHSR – PID $2502 – Exhaust Cam Phase Angle Desired Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2502	Exhaust Cam Phase Angle Desired Bank 2	1	SNM	-50.1960784314° to 49.8039215686°	E = N * 100 / 255

This PID contains the set point for the position of the exhaust camshaft for bank 2 in terms of degrees of camshaft rotation (retard) from the park position.

3.2.77 PHSR – PID $2503 – Intake Cam Phase Angle Desired Bank 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2503	Intake Cam Phase Angle Desired Bank 2	1	SNM	-50.1960784314° to 49.8039215686°	E = N * 100 / 255

This PID contains the set point for the position of the intake camshaft for bank 2 in terms of degrees of camshaft rotation (advance) from the park position.

3.2.78 PHSR – PID $2504 –Intake Camshaft Actuator Park Lock Bank 1 Pin Test Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2504	Intake Camshaft Actuator Park Lock Bank 1 Pin Test Status	1	ENM	N/A	$00=Test Not Performed This Key Cycle; $01=Test Running; $02=Test Aborted; $03=Test Passed; $04=Test Failed $05 to $FF = Reserved;

This PID contains the status of the intake camshaft actuator park lock pin test for bank 1.

3.2.79 PHSR – PID $2505 – Intake Camshaft Actuator Park Lock Bank 1 Pin Test Abort Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2505	Intake Camshaft Actuator Park Lock Bank 1 Pin Test Abort Reason	1	ENM	N/A	$00=No Abort Reason Detected; $01= DTC Present; $02= Engine Speed Low; $03= Oil Temp Cold; $04= Incorrect Power Mode; $05 to $FF = Reserved

This PID contains the reason why the Intake Camshaft Actuator Park Lock Pin Test for Bank 1 was aborted.

3.2.80 PHSR – PID $2506 –Intake Camshaft Actuator Park Lock Bank 2 Pin Test Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2506	Intake Camshaft Actuator Park Lock Bank 2 Pin Test Status	1	ENM	N/A	$00=Test Not Performed During This Key Cycle; $01=Test Running; $02=Test Aborted; $03=Test Passed; $04=Test Failed $05 to $FF = Reserved;

This PID contains the status of the intake camshaft actuator park lock pin test for Bank 2.

3.2.81 PHSR – PID $2507 – Intake Camshaft Actuator Park Lock Bank 2 Pin Test Abort Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2507	Intake Camshaft Actuator Park Lock Bank 2 Pin Test Abort Reason	1	ENM	N/A	$00=No Abort Reason Detected; $01= DTC Present; $02= Engine Speed Low; $03= Oil Temp Cold; $04= Incorrect Power Mode; $05 to $FF = Reserved

This PID contains the reason why the Intake Camshaft Actuator Park Lock Pin Test for Bank 2 was aborted.

3.2.82 OPCR – PID $27F1 – Piston Cooling Oil Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$27F1	Piston Cooling Oil Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Piston Cooling Oil Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Piston Cooling Oil Control Circuit Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Piston Cooling Oil Control Circuit Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Piston Cooling Oil Control Circuit Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the piston cooling oil control output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P25AA, P25A9, P25AB, respectively.

3.2.83 OPCR – PID $27F2 – Piston Cooling Oil Control Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$27F2	Piston Cooling Oil Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Piston Cooling Oil Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Piston Cooling Oil Measured Pressure High	0	6	1	BLN	N/A	$1=True; $0=False
Piston Cooling Oil Rationality Enabled	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains several signals associated with the status of the piston cooling oil system. The content of each of these signals is described below.

Piston Cooling Oil Commanded On indicates that the control state of piston cooling jet control valve from the ECM is commanded open allowing oil to flow into the piston cooling jet gallery.

Piston Cooling Oil Measured Pressure High indicates that the oil pressure switch undebounced input to the ECM indicates that a high engine oil pressure condition exists in the piston cooling jet gallery.

Piston Cooling Oil Rationality Test Enabled indicates that the operating conditions have been meet to allow the rationality diagnostic for the piston cooling jet control to execute.

3.2.84 EPSR – PID $27F3 – Engine Synchronization State

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$27F3	Engine Synchronization State	1	ENM	N/A	$00= No Activity $01= Activity Detected $02= Crank Only Sync $03= Complete Engine Sync $04= Cam Backup Sync $05= Remembered Sync $06= Verify Sync

This PID contains the synchronization state of the engine……...

State	Description
No Activity	No valid edges observed for either the crankshaft or primary camshaft sensor.
Activity Detected	Valid edges observed for the crankshaft and/or primary camshaft sensor, but neither sensor has been synchronized (yet). Usually seen briefly during initial startup.
Crank Only Sync	The engine is synchronized based on the crankshaft position sensor, but not the primary camshaft position sensor. This may occur briefly during startup if the crankshaft achieves synchronization before the camshaft. Otherwise, the camshaft sensor may have a problem (which should be accompanied by a code).
Complete Engine Sync	The engine is synchronized based on both, the crankshaft and primary camshaft positions sensors. This is the normal engine operating state.
Cam Backup Sync	The engine is synchronized without using the crankshaft sensor. The engine is synchronized based on either the primary camshaft sensor, or the primary and secondary camshaft sensors (should be accompanied by a code)
Remembered Sync	The engine is synchronized based on crankshaft and camshaft sensor data stored from the previous engine shutdown. This state is only seen briefly during startup (provided that the applicable feature is enabled).
Verify Sync	The engine is synchronized based on crankshaft and camshaft sensor data maintained throughout an Autostop/Autostart sequence. This state is only seen briefly during an Autostart on Hybrid and Start/Stop applications (provided that the applicable feature is enabled).

3.2.85 TPCR – PID $27FE – Transmission Accumulator Pressure Relief Procedure Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$27FE	Transmission Accumulator Pressure Relief Procedure Status	1	ENM	NA	$00=Not requested; $01= Active; $02= Completed; $03= Failed; $04 to $FF Reserved

3.2.86 TPCR – PID $27FF – Transmission Accumulator Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$27FF	Transmission Accumulator Pressure	2	UNM	0 to 7281.666666666667 kPa	E = N / 9

This PID contains the pressure of the transmission accumulator.

3.2.87 SHPR – PID $2800 – Transmission Shift Solenoid A Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2800	Transmission Shift Solenoid A Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Transmission Shift Solenoid A Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Transmission Shift Solenoid A Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission Shift Solenoid A Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission Shift Solenoid A Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the transmission shift solenoid A output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0973 and P0974.

3.2.88 SHPR – PID $2801 – Transmission Shift Solenoid B Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2801	Transmission Shift Solenoid B Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Transmission Shift Solenoid B Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Transmission Shift Solenoid B Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission Shift Solenoid B Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission Shift Solenoid B Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the transmission shift solenoid B output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0976 and P0977.

3.2.89 SHPR – PID $2802 – Transmission Shift Solenoid C Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2802	Transmission Shift Solenoid C Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Transmission Shift Solenoid C Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Transmission Shift Solenoid C Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission Shift Solenoid C Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission Shift Solenoid C Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the transmission shift solenoid C output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0979 and P0980.

3.2.90 D32R – PID $2803 – Transmission 3-2 Downshift Solenoid Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2803	Transmission 3-2 Downshift Solenoid Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Transmission 3-2 Downshift Solenoid Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Transmission 3-2 Downshift Solenoid Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission 3-2 Downshift Solenoid Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission 3-2 Downshift Solenoid Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the transmission 3-2 downshift solenoid output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0787 and P0788.

3.2.91 HSDR/SHDR – PID $2804 – Shared High Side Driver 1 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2804	Shared High Side Driver 1 Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Shared High Side Driver 1 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Shared High Side Driver 1 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shared High Side Driver 1 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shared High Side Driver 1 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the shared high side driver 1 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0658, P0657, and P0659, respectively. (A shared high side driver serves as a controlled actuator supply voltage for devices with individual low side control.)

3.2.92 HSDR/SHDR – PID $2805 – Shared High Side Driver 2 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2805	Shared High Side Driver 2 Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Shared High Side Driver 2 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Shared High Side Driver 2 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shared High Side Driver 2 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shared High Side Driver 2 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the shared high side driver 2 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2670, P02669, and P02671, respectively. (A shared high side driver serves as a controlled actuator supply voltage for devices with individual low side control.)

3.2.93 TCCR – PID $2806 – Torque Converter Clutch Enable Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2806	Torque Converter Clutch Enable Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Torque Converter Clutch Enable Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Torque Converter Clutch Enable Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Torque Converter Clutch Enable Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Torque Converter Clutch Enable Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the torque converter clutch (TCC) enable output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2769 and P2770.

3.2.94 TCCR – PID $2807 – Torque Converter Clutch Control Solenoid Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2807	Torque Converter Clutch Control Solenoid Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Torque Converter Clutch Control Solenoid Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Torque Converter Clutch Control Solenoid Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Torque Converter Clutch Control Solenoid Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Torque Converter Clutch Control Solenoid Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the torque converter clutch (TCC) control solenoid output (the output is considered to be commanded on if its duty cycle is not zero) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2763 and P2764.

3.2.95 TPCR – PID $2808 – Transmission Line Pressure Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2808	Transmission Line Pressure Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Transmission Line Pressure Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Transmission Line Pressure Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission Line Pressure Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Transmission Line Pressure Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the transmission line pressure output (the output is considered to be commanded on if its duty cycle is not zero) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0960, P0962, and P0963.

3.2.96 TCCR – PID $2809 – Torque Converter Clutch Reference Slip

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2809	Torque Converter Clutch Reference Slip	2	UNM	0 - 8191.875 rpm	E = N * 0.125

This PID contains the torque converter clutch (TCC) reference slip, the desired slip when the electronically controlled converter clutch (ECCC) algorithm is operating in on mode, determined as a function of commanded gear, turbine speed, engine torque, and other parameters.

3.2.97 SHTR – PID $280A – Transmission 5-6 Upshift Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$280A	Transmission 5-6 Upshift Time	1	UNM	0 - 6.375 s	E = N * 0.025

This PID contains the duration of the ratio change portion of the most recent transmission 5-6 upshift.

3.2.98 SPRR – PID $280B – Transmission 5-6 Upshift Time Error

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$280B	Transmission 5-6 Upshift Time Error	1	SNM	-3.2 - 3.175 s	E = N * 0.025

This PID contains the difference between the desired and actual shift times for the most recent transmission 5-6 upshift, which is used to modify the transmission 5-6 upshift pressure adapts.

3.2.99 TCCR – PID $280C – Torque Converter Clutch Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$280C	Torque Converter Clutch Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Torque Converter Clutch Control Solenoid Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Torque Converter Clutch Enable Commanded On	0	6	1	BLN	N/A	$1=True; $0=False
Torque Converter Clutch Release Switch Active	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains several signals associated with the status of the torque converter clutch (TCC) system. The content of each of these signals is described below.

Torque Converter Clutch Control Solenoid Commanded On indicates that the TCC control solenoid output is commanded on (the output is considered to be commanded on if its duty cycle is not zero).

Torque Converter Clutch Enable Commanded On indicates that the TCC enable output is commanded on.

Torque Converter Clutch Release Switch Active indicates the debounced state of the TCC release switch (TCC is not applied when the release switch is active).

3.2.100TFTR – PID $280D – Transmission Control Module Substrate Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$280D	Transmission Control Module Substrate Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the temperature measured by a sensor located on the substrate of the transmission control module, which is used to determine when special actions are required to protect control module components from an overtemperature condition.

3.2.101TOSR – PID $280E – Replicated Transmission Output Speed Output Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$280E	Replicated Transmission Output Speed Output Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Replicated Transmission Output Speed Output Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Replicated Transmission Output Speed Output Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Replicated Transmission Output Speed Output Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Replicated Transmission Output Speed Output Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the replicated transmission output speed (TOS) output (the output is considered to be commanded on if TOS is not zero) along with the status of its associated short to ground, open circuit, and short to power tests used to report DTC P1729.

3.2.102TBPR – PID $280F – Transmission Base Pattern

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$280F	Transmission Base Pattern	1	ENM	N/A	$00=Tap Up/Down Detent; $01=Tap Up/Down; $02=Hot Mode Tap Up/Down; $03=Hot Mode Detent; $04=Hot Mode; $05=Hot Engine Detent; $06=Switch A Detent; $07=Switch A; $08=Switch B Detent; $09=Switch B; $0A=Default Detent; $0B=Default $0C = 4WD Low Detent $0D = Hot Mode Manual Up / Manual Down $0E = Manual Up / Manual Down $0F = Tap Up / Tap Down Cruise

This PID contains the current transmission base pattern determined from the transmission tap up/down status, detent status, hot mode determination, hot engine determination, and driver selected patterns. Calibrations are used to map the switch A and switch B patterns to the appropriate driver selected patterns (e.g., cruise, trailer, sport, winter, etc.) for a particular application.

3.2.103TPSR/DTRR – PID $2810 – Accelerator Effective Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2810	Accelerator Effective Position	1	UNM	0 - 99.609375 %	E = N * 100 / 256

This PID contains the accelerator effective position, which represents the driver's intended request for torque or acceleration. It is determined from the maximum of the driver requests to open the throttle (e.g., accelerator pedal, cruise control, PTO, etc.) and is normalized to 0 to 100 %, where 0 % represents an idle or coast request and 100% represents a request for wide open throttle (WOT). It is the primary indication of accelerator position / throttle position used by transmission control algorithms and, thus, is often referred to as transmission throttle position. In the event that the engine is operating in a forced idle mode due to an accelerator pedal position fault, the accelerator effective position will have a value of 0 %, corresponding to an idle request.

On Heavy Duty Truck applications with ETC and incorporating a maximum vehicle speed within the normal operating region, this signal may be limited by the ECM with a bit selectable option. If the limiting option is selected and the vehicle speed is above the maximum vehicle speed, the Accelerator Effective Position will be replaced by the "Back Calculated Pedal" position calculated for the current operating condition. This option is intended only to be used by heavy duty truck applications to allow proper transmission shifting during vehicle speed governed conditions.

3.2.104TPCR/HACR – PID $2811 – Pressure Control Solenoid 1 Output Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2811	Pressure Control Solenoid 1 Output Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Pressure Control Solenoid 1 Stuck Low Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 1 Stuck High Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 1 Rationality Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the status of the pressure control solenoid (PCS) 1 output's stuck low, stuck high, and rationality tests used to report DTCs P0962, P0963, and P0961, respectively.

3.2.105RSCR/HACR – PID $2812 – Pressure Control Solenoid 2 Output Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2812	Pressure Control Solenoid 2 Output Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Pressure Control Solenoid 2 Stuck Low Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 2 Stuck High Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 2 Rationality Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the status of the pressure control solenoid (PCS) 2 output's stuck low, stuck high, and rationality tests used to report DTCs P0966, P0967, and P0965, respectively.

3.2.106RSCR/HACR – PID $2813 – Pressure Control Solenoid 3 Output Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2813	Pressure Control Solenoid 3 Output Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Pressure Control Solenoid 3 Stuck Low Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 3 Stuck High Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 3 Rationality Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the status of the pressure control solenoid (PCS) 3 output's stuck low, stuck high, and rationality tests used to report DTCs P0970, P0971, and P0969, respectively.

3.2.107RSCR/HACR – PID $2814 – Pressure Control Solenoid 4 Output Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2814	Pressure Control Solenoid 4 Output Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Pressure Control Solenoid 4 Stuck Low Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 4 Stuck High Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 4 Rationality Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the status of the pressure control solenoid (PCS) 4 output's stuck low, stuck high, and rationality tests used to report DTCs P2720, P2721, and P2719, respectively.

3.2.108RSCR/HACR – PID $2815 – Pressure Control Solenoid 5 Output Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2815	Pressure Control Solenoid 5 Output Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Pressure Control Solenoid 5 Stuck Low Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 5 Stuck High Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 5 Rationality Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the status of the pressure control solenoid (PCS) 5 output's stuck low, stuck high, and rationality tests used to report DTCs P2729, P2730, and P2728, respectively.

3.2.109TCCR – PID $2816 – Torque Converter Clutch Pressure Control Solenoid Output Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2816	Torque Converter Clutch Pressure Control Solenoid Output Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Torque Converter Clutch Pressure Control Solenoid Stuck Low Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Torque Converter Clutch Pressure Control Solenoid Stuck High Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Torque Converter Clutch Pressure Control Solenoid Rationality Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the status of the torque converter clutch (TCC) pressure control solenoid (PCS) output's stuck low, stuck high, and rationality tests used to report DTCs P2763, P2764, and P2762, respectively.

3.2.110TPCR/HACR – PID $2817 – Pressure Control Solenoid 1 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2817	Pressure Control Solenoid 1 Commanded Pressure	2	UNM	0 - 4095.9375 kPa	E = N * 0.0625

This PID contains the commanded pressure for pressure control solenoid (PCS) 1.

3.2.111RSCR/HACR – PID $2818 – Pressure Control Solenoid 2 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2818	Pressure Control Solenoid 2 Commanded Pressure	2	UNM	0 - 4095.9375 kPa	E = N * 0.0625

This PID contains the commanded pressure for pressure control solenoid (PCS) 2.

3.2.112RSCR/HACR – PID $2819 – Pressure Control Solenoid 3 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2819	Pressure Control Solenoid 3 Commanded Pressure	2	UNM	0 - 4095.9375 kPa	E = N * 0.0625

This PID contains the commanded pressure for pressure control solenoid (PCS) 3.

3.2.113RSCR/HACR – PID $281A – Pressure Control Solenoid 4 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$281A	Pressure Control Solenoid 4 Commanded Pressure	2	UNM	0 - 4095.9375 kPa	E = N * 0.0625

This PID contains the commanded pressure for pressure control solenoid (PCS) 4.

3.2.114RSCR/HACR/FOPR – PID $281B – Pressure Control Solenoid 5 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$281B	Pressure Control Solenoid 5 Commanded Pressure	2	UNM	0 - 4095.9375 kPa	E = N * 0.0625

This PID contains the commanded pressure for pressure control solenoid (PCS) 5.

3.2.115RSCR – PID $281C – Torque Converter Clutch Pressure Control Solenoid Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$281C	Torque Converter Clutch Pressure Control Solenoid Commanded Pressure	2	UNM	0 - 4095.9375 kPa	E = N * 0.0625

This PID contains the commanded pressure for the torque converter clutch (TCC) pressure control solenoid (PCS).

3.2.116DGSR/HACR – PID $281D – Transmission Fluid Pressure Switch Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$281D	Transmission Fluid Pressure Switch Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Pressure Switch 1 Active	0	7	1	BLN	N/A	$1=True; $0=False
Pressure Switch 2 Active	0	6	1	BLN	N/A	$1=True; $0=False
Pressure Switch 3 Active	0	5	1	BLN	N/A	$1=True; $0=False
Pressure Switch 4 Active	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains the status of the individual transmission fluid pressure switches. An active pressure switch indicates that the associated clutch has been exhausted while an inactive pressure switch indicates that pressure is applied to the associated clutch.

3.2.117RSSR/HACR – PID $281E – Shift Solenoid 1 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$281E	Shift Solenoid 1 Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Shift Solenoid 1 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Shift Solenoid 1 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shift Solenoid 1 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shift Solenoid 1 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the shift solenoid 1 (mode 2 solenoid) output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0973 and P0974.

3.2.118RSSR/HACR – PID $281F – Shift Solenoid 2 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$281F	Shift Solenoid 2 Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Shift Solenoid 2 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Shift Solenoid 2 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shift Solenoid 2 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shift Solenoid 2 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the shift solenoid 2 (mode 3 solenoid) output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0976 and P0977.

3.2.119RSSR – PID $2820 – Transmission Primary Oncoming Clutch

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2820	Transmission Primary Oncoming Clutch	1	ENM	N/A	$01 - C1 Clutch (C1234) $02 - C2 Clutch (C456) $03 - C3 Clutch (C35R) $04 - C4 Clutch (CB26) $05 - C5 Clutch (CBR1) $06 - No Clutch

This PID indicates the transmission primary oncoming clutch (the clutch being filled).

3.2.120CCER – PID $2821 – Transmission Full Feed Fill Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2821	Transmission Full Feed Fill Pressure	2	SNM	-4096 - 4095.875 kPa	E = N * 0.125

The PID contains the commanded transmission full feed clutch fill pressure.

3.2.121TOSR – PID $2822 – TISS/TOSS Regulated Voltage Supply Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2822	TISS/TOSS Regulated Voltage Supply Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
TISS/TOSS Regulated Voltage Supply Status	0	7	1	BLN	N/A	$1=True; $0=False

Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

Indicates the regulated TISS/TOSS supply voltage is outside of the normal operating range as specified in the operating specifications. This PID does not indicate whether the fault is external or internal to the TCM. Measurements are made by TCM internal components.

3.2.122RSCR/LTCR – PID $2823 – Transmission EOL and Service Fast Learn Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion

$2823	Service Fast Learn Status	1	ENM	N/A	$00=Fast Learn Disabled; $01= Fast Learn Completed; $02=Service Brake Must be Applied; $03=Sump Temp Below Operating Range; $04= Sump Temp Above Operating Range; $05=Park/Neutral Must Be Selected; $06=Drive Must Be Selected; $07= Reverse Must Be Selected; $08=Max Throttle/Eng Speed Exceeded; $09=Low Engine Speed Detected; $0A=Max Output Speed Exceeded; $0B= Trans Purge in Progress; $0C=Drive Calibration test in Progress; $0D= Drive Calibration test Complete; $0E=Reverse Calibration test in Progress; $0F=Abort-Test Criteria Not Met; $10=Abort-Test Failed; $11=Vehicle Must Be Stationary in SFL; $12=Park Must Be Selected in SFL; $13=Accumulator Pressure Below Operating Range; $14=Forks Must Be in Neutral Positions; $15=Transmission Output Speed Sensor Fault; $16=Fork Engage/Neutral Position Learn in Progress; $17=Fork Engage/Neutral Position Learn Complete; $18=Fork Synchronize Position Learn in Progress; $19=Fork Synchronize Position Learn Complete; $1A=Clutch Off Position Learn in Progress; $1B=Clutch Off Position Learn Complete; $1C=Clutch Kiss-point Position Learn in Progress; $1D=Clutch Kiss-point Position Learn Complete; $1E=Clutch Pressure vs. Position Learn in Progress; $1F=Clutch Pressure vs. Position Learn Complete; $20=Clutch Flow Solenoid Null Point Learn in Progress; $21=Clutch Flow Solenoid Null Point Learn Complete; $22=Clutch Torque Engagement Position (Bite Point) Learn in Progress; $23=Clutch Torque Engagement Position (Bite Point) Learn Complete; $24=Accumulator Pre-charge Pressure Learn in Progress; $25=Accumulator Pre-charge Pressure Learn Complete; $26=Clutch Flow Point Learn In Progress; $27=Clutch Flow Point Learn Complete;

					$28-$FF=Reserved

This PID contains the status of the Transmission EOL and Service Fast Learn procedure.

3.2.123RSCR/ HACR – PID $2824 – Transmission Cleaning Procedure Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2824	Transmission Cleaning Procedure Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Transmission Cleaning Procedure complete	0	2	1	BLN	N/A	$1=True; $0=False
Transmission Cleaning Procedure aborted	0	1	1	BLN	N/A	$1=True; $0=False
Transmission Cleaning Procedure in progress	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains the status of the transmission cleaning procedure.

3.2.124TFTR – PID $2826 –Transmission Control Module Restart Sensor Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2826	Transmission Control Module Restart Sensor Temperature	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the actual temperature measured by a secondary sensor located on the substrate of the transmission control module, which is used to determine when the transmission control module should restart after a hot mode shutdown.

3.2.125TRGR – PID $2845 – PRNDL Direction Range

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2845	PRNDL Direction Range	1	ENM	N/A	$00=Illegal; $01=Park; $02=Reverse; $03=Neutral; $04=Drive; $05=Transition; $06=Invalid

This PID indicates the transmission range decoded from the Internal Mode Switch (IMS) switch states.

3.2.126TRGR – PID $2846 – PRNDL Direction Encoder Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2846	PRNDL Direction Encoder Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
PRNDL Inputs Illegal	0	7	1	BLN	N/A	$1=True; $0=False

PRNDL D1 Input Active	0	6	1	BLN	N/A	$1=True; $0=False
PRNDL D2 Input Active	0	5	1	BLN	N/A	$1=True; $0=False
PRNDL R1 Input Active	0	4	1	BLN	N/A	$1=True; $0=False
PRNDL R2 Input Active	0	3	1	BLN	N/A	$1=True; $0=False
PRNDL S Input Active	0	2	1	BLN	N/A	$1=True; $0=False
Reserved	0	1	1			
Reserved	0	0	1			

This PID indicates the status of the transmission Internal Mode Switch (IMS) direction contacts. PID $1951 indicates the status of the IMS range contacts.

The IMS assembly is a dual sliding contact switch attached to the control valve body within the transmission. The 9 outputs from the switch indicate which position is selected by the transmission manual shaft. Four outputs (A, B, C, P) are range selection inputs to the transmission control module (TCM). Five outputs (R1, R2, D1, D2, S) are direction selection inputs to the hybrid powertrain control module (HPCM). The input voltage at the modules is high when the switch is open and low when the switch is closed to ground.

3.2.127TOSR – PID $284A – Hybrid Transmission Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$284A	Hybrid Transmission Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Output Shaft Direction	0	7	2	ENM	N/A	$0=Unknown; $1=Invalid; $2=Forward; $3=Reverse
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID indicates status information for the hybrid transmission.

Output Shaft Direction indicates the rotational direction of the output shaft in reference to the resulting direction of the vehicle.

3.2.128CLSR – PID $2855 –Transmission Clutch Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2855	Transmission Clutch Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Transmission C1 Clutch Applied	0	7	1	BLN	N/A	$1=True; $0=False

Transmission C2 Clutch Applied	0	6	1	BLN	N/A	$1=True; $0=False
Transmission C3 Clutch Applied	0	5	1	BLN	N/A	$1=True; $0=False
Transmission C4 Clutch Applied	0	4	1	BLN	N/A	$1=True; $0=False
Transmission C5 Clutch Applied	0	3	1	BLN	N/A	$1=True; $0=False
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID indicates transmission clutch applied status.

3.2.129APSR – PID $2856 – Accelerator Pedal Fully Applied Learned Position Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2856	Accelerator Pedal Fully Applied Learned Position Sensor 1	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the learned maximum (fully applied) accelerator pedal position relating to absolute accelerator pedal position (no correction or normalization based upon learned minimum or maximum positions) from accelerator pedal position sensor 1.

3.2.130APSR – PID $2857 – Accelerator Pedal Fully Applied Learned Position Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2857	Accelerator Pedal Fully Applied Learned Position Sensor 2	1	UNM	0 - 100 %	E = N * 100 / 255

This PID contains the learned maximum (fully applied) accelerator pedal position relating to absolute accelerator pedal position (no correction or normalization based upon learned minimum or maximum positions) from accelerator pedal position sensor 2.

3.2.131HACR – PID $2859 –Pressure Control Solenoid 6 Output Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2859	Pressure Control Solenoid 6 Output Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Pressure Control Solenoid 6 Stuck Low Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 6 Stuck High Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Pressure Control Solenoid 6 Rationality Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the status of the pressure control solenoid (PCS) 6 output's stuck low, stuck high, and rationality tests used to report DTCs P2738, P2739, and P2737, respectively.

3.2.132HACR/FOPR – PID $285A – Pressure Control Solenoid 6 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$285A	Pressure Control Solenoid 6 Commanded Pressure	2	UNM	0 - 4095.9375 kPa	E = N * 0.0625

This PID contains the commanded pressure for pressure control solenoid (PCS) 6.

3.2.133TPCR – PID $2862 – Calculated Transmission Line Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2862	Calculated Transmission Line Pressure	2	UNM	0 – 4095.9375 kPa	E = N / 16

This PID contains the expected real line pressure after accounting for pump loss factors.

3.2.134CLSR – PID $2866 – Transmission Clutch 1 Maximum Energy

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2866	Transmission Clutch 1 Maximum Energy	1	UNM	0 - 51,000 Joules	E = N * 200

This PID contains the calculated clutch remaining energy capability for clutch assisted shifts. This quantity is used to prevent clutch overheating during shift to protect clutch hardware.

3.2.135CLSR – PID $2867 – Transmission Clutch 2 Maximum Energy

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2867	Transmission Clutch 2 Maximum Energy	1	UNM	0 - 51,000 Joules	E = N * 200

This PID contains the calculated clutch remaining energy capability for clutch assisted shifts. This quantity is used to prevent clutch overheating during shift to protect clutch hardware.

3.2.136CLSR – PID $2868 – Transmission Clutch 3 Maximum Energy

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2868	Transmission Clutch 3 Maximum Energy	1	UNM	0 - 51,000 Joules	E = N * 200

This PID contains the calculated clutch remaining energy capability for clutch assisted shifts. This quantity is used to prevent clutch overheating during shift to protect clutch hardware.

3.2.137CLSR – PID $2869 – Transmission Clutch 4 Maximum Energy

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2869	Transmission Clutch 4 Maximum Energy	1	UNM	0 - 51,000 Joules	E = N * 200

This PID contains the calculated clutch remaining energy capability for clutch assisted shifts. This quantity is used to prevent clutch overheating during shift to protect clutch hardware.

3.2.138CLSR – PID $286A – Transmission Clutch 1 Estimated Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$286A	Transmission Clutch 1 Estimated Torque	1	UNM	-5 - 2035 Nm	E = N * 8 - 5

This PID contains the estimated clutch torque capacity based on the clutch pressure model and clutch properties. This quantity is used to prevent clutch torque load from exceeding clutch capacity.

3.2.139CLSR – PID $286B – Transmission Clutch 2 Estimated Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$286B	Transmission Clutch 2 Estimated Torque	1	UNM	-5 - 2035 Nm	E = N * 8 - 5

This PID contains the estimated clutch torque capacity based on the clutch pressure model and clutch properties. This quantity is used to prevent clutch torque load from exceeding clutch capacity.

3.2.140CLSR – PID $286C – Transmission Clutch 3 Estimated Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$286C	Transmission Clutch 3 Estimated Torque	1	UNM	-5 - 2035 Nm	E = N * 8 - 5

This PID contains the estimated clutch torque capacity based on the clutch pressure model and clutch properties. This quantity is used to prevent clutch torque load from exceeding clutch capacity.

3.2.141CLSR – PID $286D – Transmission Clutch 4 Estimated Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$286D	Transmission Clutch 4 Estimated Torque	1	UNM	-5 - 2035 Nm	E = N * 8 - 5

This PID contains the estimated clutch torque capacity based on the clutch pressure model and clutch properties. This quantity is used to prevent clutch torque load from exceeding clutch capacity.

3.2.142CLSR – PID $286E – Transmission Clutch 1 Fill Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$286E	Transmission Clutch 1 Fill Time	1	UNM	0 - 3.1875 s	E = N / 80

This PID contains the estimated clutch fill time based on the clutch flow model and adapted stroke volume. This quantity is used to command clutch fill in expectation of aligning clutch fill and synchronization events.

3.2.143CLSR – PID $286F – Transmission Clutch 2 Fill Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$286F	Transmission Clutch 2 Fill Time	1	UNM	0 - 3.1875 s	E = N / 80

This PID contains the estimated clutch fill time based on the clutch flow model and adapted stroke volume. This quantity is used to command clutch fill in expectation of aligning clutch fill and synchronization events.

3.2.144CLSR – PID $2870 – Transmission Clutch 3 Fill Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2870	Transmission Clutch 3 Fill Time	1	UNM	0 - 3.1875 s	E = N / 80

This PID contains the estimated clutch fill time based on the clutch flow model and adapted stroke volume. This quantity is used to command clutch fill in expectation of aligning clutch fill and synchronization events.

3.2.145CLSR – PID $2871 – Transmission Clutch 4 Fill Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2871	Transmission Clutch 4 Fill Time	1	UNM	0 - 3.1875 s	E = N / 80

This PID contains the estimated clutch fill time based on the clutch flow model and adapted stroke volume. This quantity is used to command clutch fill in expectation of aligning clutch fill and synchronization events.

3.2.146HACR – PID $2872 – Commanded X-Y Valve State

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2872	Commanded X-Y Valve State	1	ENM	N/A	$00=Electrically Variable Trans Low; $01=Electrically Variable Trans High; $02=Transmission Range Low; $03=Transmission Range High; $04=Invalid State – Engine Off

This PID contains the commanded state of the transmission X-Y valve. The actual X-Y valve state is displayed in PID $2873 “X-Y Valve Transition State”.

3.2.147HACR – PID $2873 – X-Y Valve Transition State

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2873	X-Y Valve Transition State	1	ENM	N/A	$00=Electrically Variable Trans Low; $01=Electrically Variable Trans High; $02=Transmission Range Low; $03=Transmission Range High; $04=Invalid State - Engine Off; $05=Transitioning from EVT Low to Range Low; $06=Transitioning from Range Low to EVT Low; $07=Transitioning from Range Low to Range High; $08=Transitioning from Range High to Range Low; $09=Transitioning from Range High to EVT High; $0A=Transitioning from EVT High to Range High; $0B=Transitioning from EVT High to EVT Low; $0C=Transitioning from EVT Low to EVT High; $0D=Valve X Stuck Low; $0E=Valve X Stuck High; $0F=Valve Y Stuck Low; $10=Valve Y Stuck High; $11=Aborting Transition from EVT Low to EVT High; $12=Aborting Transition from Range Low to Range High; $FF=No/Invalid Response from Controller

This PID contains the actual state of the transmission X-Y valve transition.

3.2.148CLSR – PID $2874 – Clutch 1 Accumulator Volume

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2874	Clutch 1 Accumulator Volume	1	UNM	0 – 100 ml	E = N * 100 / 255

This PID contains the estimated clutch 1 piston stroke volume. This quantity is calculated using the clutch flow model.

3.2.149CLSR – PID $2875 – Clutch 2 Accumulator Volume

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2875	Clutch 2 Accumulator Volume	1	UNM	0 – 100 ml	E = N * 100 / 255

This PID contains the estimated clutch 2 piston stroke volume. This quantity is calculated using the clutch flow model.

3.2.150CLSR – PID $2876 – Clutch 3 Accumulator Volume

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2876	Clutch 3 Accumulator Volume	1	UNM	0 – 100 ml	E = N * 100 / 255

This PID contains the estimated clutch 3 piston stroke volume. This quantity is calculated using the clutch flow model.

3.2.151CLSR – PID $2877 – Clutch 4 Accumulator Volume

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2877	Clutch 4 Accumulator Volume	1	UNM	0 – 100 ml	E = N * 100 / 255

This PID contains the estimated clutch 4 piston stroke volume. This quantity is calculated using the clutch flow model.

3.2.152HSER – PID $2879 – Transmission Clutch 1 Torque Command

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2879	Transmission Clutch 1 Torque Command	1	UNM	-5 - 2035 Nm	E = N * 8 - 5

This PID contains the commanded (desired) clutch torque capacity used during shift events and normal operation.

3.2.153HSER – PID $287A – Transmission Clutch 2 Torque Command

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$287A	Transmission Clutch 2 Torque Command	1	UNM	-5 - 2035 Nm	E = N * 8 - 5

This PID contains the commanded (desired) clutch torque capacity used during shift events and normal operation.

3.2.154HSER – PID $287B – Transmission Clutch 3 Torque Command

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$287B	Transmission Clutch 3 Torque Command	1	UNM	-5 - 2035 Nm	E = N * 8 - 5

This PID contains the commanded (desired) clutch torque capacity used during shift events and normal operation.

3.2.155HSER – PID $287C – Transmission Clutch 4 Torque Command

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$287C	Transmission Clutch 4 Torque Command	1	UNM	-5 - 2035 Nm	E = N * 8 - 5

This PID contains the commanded (desired) clutch torque capacity used during shift events and normal operation.

3.2.156TOSR – PID $2888 – Transmission Signed Filtered Output Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2888	Transmission Signed Filtered Output Speed	2	SNM	-8192 – 8191.75 rpm	E = N / 4

This PID contains the directionally signed, stabilized (filtered), default, transmission output speed. This is the speed at the output of the gear box as measured by the transmission output speed (TOS) sensor, with the application of factors to account for the location of the sensor (e.g., after the gear box, after the final chain, after the transfer case, after the final drive, etc.). In applications that include both a transmission output speed sensor and a transfer case speed sensor, this PID will apply specifically to the output of the transmission.

3.2.157TRGR – PID $2889 – Validated (Commanded) Transmission Range State

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2889	Validated (Commanded) Transmission Range State	1	ENM	N/A	$00=Drive 1; $01=Drive 2; $02=Drive 3; $03=Drive 4; $04=Drive 5; $05=Drive 6; $06=Neutral; $07=Reverse; $08=Park; $09=Transition 1; $0A=Transition 2; $0B=Transition 4; $0C=Transition 7; $0D=Transition 8; $0E=Transition 11; $0F=Transition 13; $10=Illegal State; $11=Drive 7; $12=Drive 8

This PID contains the actual transmission range determined by arbitrating between one or more sources of transmission range information. This will be the command to all the control algorithms.

3.2.158HTDR – PID $288A – Transmission Output Torque Commanded

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$288A	Transmission Output Torque Commanded	2	UNM	-22,534 – 43,001 Nm	E = N * 1 – 22,534

This PID contains the commanded transmission output torque.

3.2.159HSDR/SHDR – PID $288C – Shared High Side Driver 3 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$288C	Shared High Side Driver 3 Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Shared High Side Driver 3 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Shared High Side Driver 3 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shared High Side Driver 3 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Shared High Side Driver 3 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the shared high side driver 3 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2685, P2684, and P2686, respectively. (A shared high side driver serves as a controlled actuator supply voltage for devices with individual low side control.)

3.2.160TRGR – PID $289C – Neutral Gear Sensor Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$289C	Neutral Gear Sensor Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the actual duty cycle received from the manual transmission neutral gear sensor.

3.2.161TRGR – PID $289D – Neutral Gear Sensor Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$289D	Neutral Gear Sensor Status	1	ENM	N/A	$00=Neutral; $01=Not neutral; $02=Signal failure; $03=Signal Indeterminate; $04=Sensor not Learned; $05=Reverse; $06 to $FF = Reserved

This PID contains the status of the manual transmission neutral gear sensor.

3.2.162TNSR – PID $289E – Transmission Planetary Gearset Ring Gear Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$289E	Transmission Planetary Gearset Ring Gear Speed	2	UNM	0 to 16383.75 rpm	E = N / 4

This is the undefaulted rotation speed sensed by the Transmission Planetary Gearset Ring Gear Speed Sensor within the 8-speed transmission. The speed sensor monitors ring gear 4 on planetary carrier 1. This component is held stationary in Park, Neutral, Reverse, 1st, and 2nd gear. In the other forward gears, the speed of this component should non-zero when transmission input speed is non-zero.

3.2.163TNSR – PID $289F – Transmission Shaft A Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$289F	Transmission Shaft A Speed	2	UNM	0 to 16383.75 rpm	E = N / 4

This is the undefaulted sensed rotational speed of shaft A on a dual clutch transmission.

3.2.164TNSR – PID $28A0 – Transmission Shaft B Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28A0	Transmission Shaft B Speed	2	UNM	0 to 16383.75 rpm	E = N / 4

This is the undefaulted sensed rotational speed of shaft B on a dual clutch transmission.

3.2.165HSER – PID $28C4 – Transmission Clutch 5 Torque Command

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28C4	Transmission Clutch 5 Torque Command	1	UNM	-5 to +2035 Nm	E = N * 8 - 5

This PID contains the commanded (desired) clutch torque capacity used during shift events and normal operation.

3.2.166CLSR – PID $28C6 – Transmission Clutch 5 Maximum Energy

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28C6	Transmission Clutch 5 Maximum Energy	1	UNM	0 to 51000 Joules	E = N * 200

This PID contains the calculated clutch remaining energy capability for clutch assisted shifts. This quantity is used to prevent clutch overheating during shift to protect clutch hardware.

3.2.167CLSR – PID $28C7 – Transmission Clutch 5 Estimated Torque

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28C7	Transmission Clutch 5 Estimated Torque	1	UNM	-5 to +2035 Nm	E = N * 8 - 5

This PID contains the estimated clutch torque capacity based on the clutch pressure model and clutch properties. This quantity is used to prevent clutch torque load from exceeding clutch capacity.

3.2.168CLSR – PID $28C8 – Transmission Clutch 5 Fill Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28C8	Transmission Clutch 5 Fill Time	1	UNM	0 to 3.1875 s	E = N / 80

This PID contains the estimated clutch fill time based on the clutch flow model and adapted stroke volume. This quantity is used to command clutch fill in expectation of aligning clutch fill and synchronization events.

3.2.169CLSR – PID $28C9 – Transmission Clutch 5 Accumulator Volume

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28C9	Transmission Clutch 5 Accumulator Volume	1	UNM	0 to 100 ml	E = N * 100 / 255

This PID contains the estimated clutch 5 piston stroke volume. This quantity is calculated using the clutch flow model.

3.2.170HSER – PID $28E8 – Transmission Gear or Mode Achieved

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28E8	Transmission Gear or Mode Achieved	1	ENM	N/A	$00=Neutral; $01=First Gear; $02=Second Gear; $03=Third Gear; $04=Fourth Gear; $05=Fifth Gear; $06=Sixth Gear; $07=Seventh Gear; $08=Eighth Gear; … $41=EV1; $42=EV2; $43=EV3; $44=EV4

This PID contains the transmission gear or mode that is defined as currently achieved. This is an actual value (compared with the commanded value in PID $199A) determined by the state of the transmission clutches.

3.2.171HSER – PID $28E9 – Transmission Range Equation

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28E9	Transmission Range Equation	1	ENM	N/A	$00=Range Equation N1; $01= Range Equation N2; $02= Range Equation N3; $03= Range Equation N4: $04= Range Equation N5; $05= Range Equation M1; $06= Range Equation M2; $07= Range Equation M3; $08= Range Equation M4; $09= Range Equation M5; $0A= Range Equation M6; $0B= Range Equation G1; $0C= Range Equation G2; $0D= Range Equation G3; $0E= Range Equation G4

This PID contains the actual active range equation for an Electrically Variable Transmission (EVT). The range equation represents a defined combination of transmission gears, modes, and clutches. The exact combination for each range equation depends on the application.

3.2.172CSLR – PID $28EA – Transmission C1 Clutch Slip

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28EA	Transmission C1 Clutch Slip	2	SNM	-16384 to 16383.5 rpm	E = N * 0.5

This PID contains the transmission C1 clutch slip.

3.2.173CSLR – PID $28EB – Transmission C2 Clutch Slip

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28EB	Transmission C2 Clutch Slip	2	SNM	-16384 to 16383.5 rpm	E = N * 0.5

This PID contains the transmission C2 clutch slip.

3.2.174CSLR – PID $28EC – Transmission C3 Clutch Slip

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28EC	Transmission C3 Clutch Slip	2	SNM	-16384 to 16383.5 rpm	E = N * 0.5

This PID contains the transmission C3 clutch slip.

3.2.175CSLR – PID $28ED – Transmission C4 Clutch Slip

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28ED	Transmission C4 Clutch Slip	2	SNM	-16384 to 16383.5 rpm	E = N * 0.5

This PID contains the transmission C4 clutch slip.

3.2.176CSLR – PID $28EE – Transmission C5 Clutch Slip

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$28EE	Transmission C5 Clutch Slip	2	SNM	-16384 to 16383.5 rpm	E = N * 0.5

This PID contains the transmission C5 clutch slip.

3.2.177TISR – PID $2931 –Raw Transmission Input Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2931	Raw Transmission Input Speed	2	UNM	0 to 8191.875 rpm	E = N * 0.125

For applications that are equipped with a transmission input speed sensor, this PID contains the undefaulted, raw transmission input speed (the speed at the input of the gear box) determined from the speed measured by the transmission input speed (TIS) sensor with the application of factors to account for the location of the sensor (e.g., before the sprocket, after the overdrive, etc.). The source of this raw, undefaulted speed will not change when faults are detected, so this PID will always display the sensed input speed from the input speed sensor.
In applications that are not equipped with a transmission speed sensor, this PID contains the transmission input speed derived from another source.
For example, for the 2-Mode strong hybrid, this PID contains the transmission input speed based on motor speeds.

3.2.178TOSR – PID $2932 – Raw Transmission Output Speed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2932	Raw Transmission Output Speed	2	UNM	0 to 8191.875 rpm	E = N * 0.125

This PID contains the undefaulted, raw transmission output speed (the speed at the output of the gear box) determined from the speed measured by the transmission output speed (TOS) sensor with the application of factors to account for the location of the sensor (e.g., after the gear box, after the final chain, after the transfer case, after the final drive, etc.). The source of this raw, undefaulted speed will not change when faults are detected, so this PID will always display the sensed output speed from the output speed sensor. In applications that include both a transmission output speed sensor and a transfer case speed sensor, this PID will apply specifically to the output of the transmission. In non-TCM controllers, this PID may be populated with raw TOSS information originating from the TCM. This information could be message frame $0C7, or in an ECM provided with a replicated TOSS signal, RTOSS could populate this PID.In all controllers, the output speed location calibration must be properly configured for the PID to report accurately compensated data.

3.2.179TRTR – PID $2933– Overall Undefaulted Transmission Ratio

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2933	Overall Undefaulted Transmission Ratio	2	UNM	0 to 31.9995117188	E = N / 2048

This PID contains the overall undefaulted transmission ratio used primarily for the dual clutch transmission (DCT). The gear set ratios along with the final drive ratio are used to calculate the ratio.

3.2.1 TPCR – PID $2934 – Transmission Accumulator Pressure Restoration Procedure Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2934	Transmission Accumulator Pressure Restoration Procedure Status	1	ENM	NA	$00=Accumulator charging disabled; $01= Accumulator charging enabled; $02 to $FF Reserved

This PID contains the status of the transmission accumulator pressure restoration procedure.

3.2.2 EODR – PID $2935– Engine Oil Level Sensor Type

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2935	Engine Oil Level Sensor Type	1	ENM	N/A	$00=Engine Oil Level Engine Not Running $01=Engine Oil Level Engine Running; $02= Engine Oil Level Not Present;

This PID indicates the presence of Oil Level Sensing.

3.2.3 STRR– PID $2937 – Crank Time For Conventional Vehicles

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2937	Crank Time For Conventional Vehicles	2	UNM	0 to 409.59375 s	E = N * 0.00625

This PID will give the time from when the conventional starter motor is commanded on to when engine first reaches a calibratable rpm. The time is obtained from a timer which will hold the value when the calibratable rpm is reached. The timer gets reset on a new start.

3.2.4 THMR – PID $2940 – Average Engine Power For Flow Rationality Diagnostic Enable

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2940	Average Engine Power For Flow Rationality Diagnostic Enable	1	UNM	0 to 512 kW	E = N * 512 / 255

This PID contains the average engine power which is calculated as an enable criteria for the switchable water pump flow rationality diagnostic. The average engine power must be greater or equal to a minimum threshold and less than or equal to a maximum threshold for the flow rationality diagnostic to be enabled for the engine power condition.

3.2.5 OXYR – PID $2941 – Green O2 Sensor Delay Remaining Percentage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2941	Green O2 Sensor Delay Remaining Percentage	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the green oxygen sensor delay that is remaining as a percentage. 100% indicates that the sensor(s) are completely green. This only occurs when the vehicle is first built from the factory where it will be set to 100%. The PID will count down from 100% to 0% as the green oxygen sensor effect goes away. Each sensor has its own green delay, however this PID will report the maximum green percentage of all sensors. Once the green effect is gone for all sensors, this PID will be equal to 0% for the rest of the life of the vehicle. This PID can never be reset, even if a new oxygen sensor is installed in the vehicle in service. This PID will be set to 0% in service controllers. All pre and post O2 sensors can have a green sensor delay. Certain oxygen sensor performance diagnostics are delayed until this value is equal to 0%.

3.2.6 OXYR – PID $2942 – Green Catalyst O2 Diag Delay Remaining Percentage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2942	Green Catalyst O2 Diag Delay Remaining Percentage	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the green catalyst oxygen sensor diagnostic delay that is remaining as a percentage. 100% indicates that the catalyst(s) are completely green. This only occurs when the vehicle is first built from the factory where it will be set to 100%. The PID will count down from 100% to 0% as the green catalyst outgassing effect goes away. Once the green catalyst effect is gone, this PID will be equal to 0% for the rest of the life of the vehicle. This PID can never be reset, even if a new catalyst is installed on the vehicle in service. This PID will be set to 0% in service controllers. Only post O2 sensors can have a green catalyst O2 diagnostic delay. Certain post-oxygen sensor performance diagnostics are delayed until this value is equal to 0%.

3.2.7 DRRR – PID $2C09– Electronic Transmission Range Selection Request

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$2C09	Electronic Transmission Range Selection Request	2	ENM	N/A	$00 Undefined $01 Park $02 Reverse $03 Neutral $04 Drive $05 Low $06 Manual $07 Null

This PID indicates the electronic transmission range selection (ETRS) request based on the gear selector position. Certain events, such as a Park request while the vehicle is moving, may result in the actual transmission range being different than the ETRS request.

3.2.8 EITR – PID $3031 – Power-Up Intake Air Temperature 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3031	Power-Up Intake Air Temperature 2	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the intake air temperature (measured by a sensor located second in the direction of air flow) recorded at power-up (Run/Crank transition from low to high).

3.2.9 FHPR – PID $3034 – Fuel High Pressure Pump Offset Learned

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3034	Fuel High Pressure Pump Offset Learned	1	SNM	-0.5 to 0.5 A	E = N * 0.5 / 128

This PID contains the learned offset of Fuel High Pressure Pump.

3.2.10 EGTR – PID $303A – Exhaust Gas Temperature Sensor Bank 1 Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$303A	Exhaust Gas Temperature Sensor Bank 1 Sensor 1	2	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the exhaust gas temperature from bank 1 sensor 1.

3.2.11 EGTR – PID $303B – Exhaust Gas Temperature Sensor Bank 1 Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$303B	Exhaust Gas Temperature Sensor Bank 1 Sensor 2	2	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the exhaust gas temperature from bank 1 sensor 2.

3.2.12 CETR – PID $3040 – EGR Cooler Temperature Sensor 1 (Pre Cooler)

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3040	EGR Cooler Temperature Sensor 1 (Pre Cooler)	2	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the undefaulted exhaust gas recirculation cooler temperature measured by a sensor that is located before the inlet of the EGR heat exchanger.

3.2.13 CETR – PID $3041 – EGR Cooler Temperature Sensor 2 (Post Cooler)

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3041	EGR Cooler Temperature Sensor 2 (Post Cooler)	2	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the undefaulted exhaust gas recirculation cooler temperature measured by a sensor that is located on the outlet of the EGR heat exchanger.

3.2.14 FISR -- PID $304F – Alternative Fuel Shutoff Valve 1 Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$304F	Alternative Fuel Shutoff Valve 1 Duty Cycle	1	UNM	0 to100%	E = N * 100 / 255

This PID provides the commanded duty cycle for the alternative fuel shutoff valve 1.

3.2.15 AAPR – PID $3088 – Turbocharger Boost Pressure Sensor A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3088	Turbo Boost Sensor A/D	1	UNM	0 – 100% 5V	E = N * 100 / 255

This PID contains the turbocharger boost pressure sensor analog input as a percentage of its reference voltage. (For turbocharged applications, the Throttle Inlet Air Pressure (TIAP) sensor indicates the boost pressure.)

3.2.16 FHPR – PID $308A – Desired Fuel Rail Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$308A	Desired Fuel Rail Pressure	2	UNM	0 – 655350 kPa	E = N * 10

This PID displays desired fuel rail pressure at the engine.

3.2.17 FHPR – PID $309C –Fuel Rail Pressure A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$309C	Fuel Rail Pressure A/D	1	UNM	0 – 100% 5 V	E = N * 100 / 255

This PID contains the fuel rail pressure analog input as a percentage of its reference voltage.

3.2.18 EGTR – PID $30A9 – Exhaust Gas Temperature Sensor Bank 1 Sensor 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$30A9	Exhaust Gas Temperature Sensor Bank 1 Sensor 3	2	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the exhaust gas temperature bank 1sensor 3.

3.2.19 EGTR – PID $30AA – Exhaust Gas Temperature Sensor Bank 1 Sensor 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$30AA	Exhaust Gas Temperature Sensor Bank 1 Sensor 4	2	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the undefaulted exhaust gas temperature measured by a sensor that is located fourth in the direction of exhaust flow on a single exhaust system or on bank 1 of a dual exhaust system.

3.2.20 EGTR – PID $30AB – Exhaust Gas Temperature Sensor Bank 1 Sensor 5

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$30AB	Exhaust Gas Temperature Sensor Bank 1 Sensor 5	2	UNM	-40 to 6513.5 °C	E = N * 0.1 - 40

This PID contains the undefaulted exhaust gas temperature measured by a sensor that is located fifth in the direction of exhaust flow on a single exhaust system or on bank 1 of a dual exhaust system.

3.2.21 CITR – PID $30AD – Charge Air Cooler Temperature Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$30AD	Charge Air Cooler Temperature Sensor 1	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the undefaulted charge air cooler temperature measured by a sensor located first in the charge air cooler stream in relationship to any other charge air cooler temperature sensors in the system.

3.2.22 CITR – PID $30AE – Charge Air Cooler Temperature Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$30AE	Charge Air Cooler Temperature Sensor 2	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the undefaulted charge air cooler temperature measured by a sensor located second in the charge air cooler stream in relationship to any other charge air cooler temperature sensors in the system.

1.1.1 MAFR- PID $304E – Air Flow Ratio

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$304E	Air Flow Ratio	1	UNM	0 to 1.999965	E = N * 0.007843

This PID contains the ratio of desired intake air flow and the actual intake air flow as determined by the MAF Sensor.

3.2.23 TGRR – PID $3201 – Transmission Forward Measured Gear

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3201	Transmission Forward Measured Gear	1	ENM	N/A	$00=Unknown $01=First Gear; $02=Second Gear; $03=Third Gear; $04=Fourth Gear; $05=Fifth Gear; $06=Sixth Gear; $07=Seventh Gear; $08=Eighth Gear

This PID contains the current measured, or 'active' forward gear in the transmission for all types of transmissions (automatic, manual, or MTA) as interpreted or determined in the Engine Control Module. It is not intended to report gear information for Parallel Hybrid applications.

For traditional automatic transmissions, the 'Unknown' state may be reported when either Park, Reverse, or Neutral range is selected, or when the automatic transmission range sensor is faulty and a forward range cannot be determined. For manual transmissions, the 'Unknown' state may be reported when the input clutch is slipping at a very high rate such that Engine to Vehicle speed ratio does not match any known gear ratios, or the bottom of travel clutch switch is invoked, or Reverse is selected, or vehicle speed is faulty causing the Engine to Vehicle Speed ratio to be indeterminate. For MTA transmissions, the 'Unknown' state may be reported when Park, Reverse, or Neutral is selected, or when the transmission selector system is faulty, or the input clutch control system is faulty.

Otherwise (i.e. First, Second, Third, Fourth, Fifth, Sixth, Seventh, or Eighth Gear), for all types of transmissions, the PID contains the forward measured gear state and is therefore updated upon completion of gear shift events.

3.2.24 EOSR – PID $3210 – Engine Overspeed Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3210	Engine Overspeed Counter	1	UNM	0 - 255 counts	E = N * 1

This PID contains the accumulated number of engine overspeed detection events.

3.2.25 FTSR – PID $3218 – Fuel Temperature 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3218	Fuel Temperature 1	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the fuel temperature 1 value as measured by a sensor. This sensor is typically located on the fuel filter.

3.2.26 VSPR – PID $321B – Vehicle Distance Traveled in Driving Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$321B	Vehicle Distance Traveled in Driving Cycle	2	UNM	0 – 65,535 km	E = N * 1

This PID contains the distance a vehicle has traveled in the current driving cycle.

3.2.27 EOSR – PID $321F – Engine Overspeed Detection Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$321F	Engine Overspeed Detection Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Engine Overspeed Condition Detected	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains the status of the engine overspeed detection algorithm. The content of each of these signals is described below.

Engine Overspeed Condition Detected indicates that an engine overspeed condition has been detected.

3.2.28 EOSR – PID $3220 – Total Duration of Overspeed Conditions

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3220	Total Duration of Overspeed Conditions	2	UNM	0 - 655350 ms	E = N * 10

This PID contains the total accumulated duration of overspeed conditions (accumulated time for engine speed above a calibrated threshold).

3.2.29 APSR – PID $323B – Accelerator Pedal Released Learned Position Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$323B	Accelerator Pedal Released Learned Position Sensor 1	1	UNM	0 - 100 % 5V	E = N * 100 / 255

This PID indicates the accelerator pedal sensor 1 signal at its minimum learned position (pedal released) as a percentage of its reference voltage. (The minimum learned position is automatically updated periodically as determined by an algorithm. This is not a static value from a single learn event.)

3.2.30 APSR – PID $323C – Accelerator Pedal Released Learned Position Sensor 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$323C	Accelerator Pedal Released Learned Position Sensor 2	1	UNM	0 - 100 % 5V	E = N * 100 / 255

This PID indicates the accelerator pedal sensor 2 signal at its minimum learned position (pedal released) as a percentage of its reference voltage. (The minimum learned position is automatically updated periodically as determined by an algorithm. This is not a static value from a single learn event.)

3.2.31 FHPR – PID $3240 – Fuel Rail Actuator Command Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3240	Fuel Rail Actuator Command Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the Fuel Rail Pressure Regulator (located in the fuel rail) duty cycle.

3.2.32 AICR – PID $327B – Intended Intake Manifold Airflow

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$327B	Intended Intake Manifold Airflow	2	UNM	0 to 655.35 g/s	E = N * 0.01

This PID contains the intended intake manifold airflow.

3.2.33 AAPR – PID $3298 – Actual Turbocharger Boost Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3298	Actual Turbocharger Boost Pressure	2	UNM	0 - 655.35 kPa	E = N / 100

This PID contains the undefaulted turbocharger boost pressure as measured by the Throttle Inlet Air Pressure (TIAP) sensor.

3.2.34 EGRR – PID $32A8 - Exhaust Gas Recirculation Actuator Commanded Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$32A8	Exhaust Gas Recirculation Actuator Commanded Duty Cycle	1	SNM	-128 to +127 %	E = N * 1

This PID contains the commanded duty cycle for the exhaust gas recirculation valve actuator output.

3.2.35 TPSR – PID $32A9 – Intake Airflow Valve Control Actuator Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$32A9	Intake Airflow Valve Control Actuator Duty Cycle	1	SNM	-128 to +127 %	E = N * 1

This PID contains the commanded duty cycle for the intake airflow valve output.

3.2.36 CCLR – PID $32AC – Cylinder 1 Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$32AC	Cylinder 1 Pressure	1	UNM	0 to 25500 kPa	E= N * 100

This PID contains the uncorrected pressure for cylinder 1. With the engine not running, this value is time based sampled. With engine running, the value is sampled at a fixed angular crank shaft position.

3.2.37 CCLR – PID $32AD – Cylinder 2 Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$32AD	Cylinder 2 Pressure	1	UNM	0 to 25500 kPa	E= N * 100

This PID contains the uncorrected pressure for cylinder 2. With the engine not running, this value is time based sampled. With engine running, the value is sampled at a fixed angular crank shaft position.

3.2.38 CCLR – PID $32AE – Cylinder 3 Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$32AE	Cylinder 3 Pressure	1	UNM	0 to 25500 kPa	E= N * 100

This PID contains the uncorrected pressure for cylinder 3. With the engine not running, this value is time based sampled. With engine running, the value is sampled at a fixed angular crank shaft position.

3.2.39 CCLR – PID $32AF – Cylinder 4 Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$32AF	Cylinder 4 Pressure	1	UNM	0 to 25500 kPa	E= N * 100

This PID contains the uncorrected pressure for cylinder 4. With the engine not running, this value is time based sampled. With engine running, the value is sampled at a fixed angular crank shaft position.

3.2.40 LUBR – PID $3318 – Variable Displacement Oil Pump Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3318	Variable Displacement Oil Pump Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bit)	Data Type	Range	Conversion
Variable Displacement Oil Pump Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Variable Displacement Oil Pump Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Variable Displacement Oil Pump Open Circuit Test Status	0	4	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Variable Displacement Oil Pump Short to Power Test Status	0	0	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the variable displacement oil pump output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P06DB, P06DA and P06DC, respectively.

3.2.41 CVHR – PID $3319 – Crankcase Heater Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3319	Crankcase Heater Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Crankcase Heater Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Crankcase Heater Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Crankcase Heater Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Crankcase Heater Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the Crankcase Heater output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTC P053B, P053A and P053C, respectively.

3.2.42 PID $333E – Fuel System Priming Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$333E	Fuel System Priming Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Fuel Priming Status	0	7	2	ENM	N/A	$0=Not Primed, $1=Priming In Progress $2=Priming Suspended $3=Priming Complete – okay to crank
Fuel Tank Level Too Low to Prime	0	5	1	BLN	N/A	$1=True; $0=False
AutoPrime Feature Enabled	0	4	1	BLN	N/A	$1=True; $0=False
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID is used to report the plant end-of-line fuel pump priming testing criteria, status and request information. The content of each of these signals is described below.

Fuel Priming Status indicates the status of the plant end of line fuel pump priming routine.

Not Primed – engine and ECM are green and have never been primed.
Priming in progress – Fuel pump priming has been triggered and is in progress.
Priming Suspended – the fuel pump priming has been suspended by interrupting the fuel pump priming via CPID or transition to Off power mode.
Priming complete – fuel pump priming has be successfully completed

Fuel Tank Level Too Low to Prime indicates the fuel level is above the desired (calibratable) value to enable fuel pump priming

AutoPrime Feature Enabled indicates the fuel pump priming will automatically trigger if all the enable criteria is meet: MEC counter is non zero, fuel level is above calibratable desired value, and the vehicle is in the Run power mode

3.2.43 PID $333F – Fuel System Priming Timer

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$333F	Fuel System Priming Timer	1	UNM	0 to 255 Seconds	E=N

This PID is used to report the desired (calibrated time) fuel pump priming duration required to purge air out of the fuel system. This timer indicates the remaining duration.

3.2.44 CITR - PID $3341 – Power-Up Charge Air Cooler Temperature Sensor 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3341	Power-Up Charge Air Cooler Temperature Sensor 1	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the charge air temperature (measured by a sensor that is located first in the charge air cooler stream in relationship to any other charge air cooler temperature sensors in the system.) recorded at power-up.

3.2.45 ARTR -- PID $3347 – Alternative Fuel Rail Temperature

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3347	Alternative Fuel Rail Temperature	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID provides the undefaulted alternative fuel rail temperature measured by a sensor.

3.2.46 EITR – PID $334C – Power-Up Intake Air Temperature 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$334C	Power-Up Intake Air Temperature 3	1	UNM	-40 to +215 °C	E = N * 1 - 40

This PID contains the intake air temperature (measured by a sensor located third in the direction of air flow) recorded at power-up (Run/Crank transition from low to high).

1.2 EGTR – PID $334F – Power-Up Exhaust Gas Temperature 1

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$334F	Power-Up Exhaust Gas Temperature 1	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the exhaust gas temperature (measured by a sensor located first in the direction of exhaust flow) recorded at power-up.

1.3 EGTR – PID $3350 – Power-Up Exhaust Gas Temperature 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3350	Power-Up Exhaust Gas Temperature 2	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the exhaust gas temperature (measured by a sensor located second in the direction of exhaust flow) recorded at power-up.

1.4 EGTR – PID $3351 – Power-Up Exhaust Gas Temperature 3

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3351	Power-Up Exhaust Gas Temperature 3	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the exhaust gas temperature (measured by a sensor located third in the direction of exhaust flow) recorded at power-up.

1.5 EGTR – PID $3369 – Power-Up Exhaust Gas Temperature 4

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3369	Power-Up Exhaust Gas Temperature 4	1	UNM	-40 to 215 °C	E = N * 1 - 40

This PID contains the exhaust gas temperature (measured by a sensor located fourth in the direction of exhaust flow) recorded at power-up.

3.2.47 EVPR – PID $336F – Conventional EVAP With Pump - Fuel Tank Pressure Sensor Correlation – Propulsion On

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$336F	Conventional EVAP With Pump - Fuel Tank Pressure Sensor Correlation – Propulsion On	5	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
FTP Correlation Test Complete	0	7	1	BLN	N/A	$1=True; $0=False
FTP Correlation Soak Time Met	0	6	1	BLN	N/A	$1=True; $0=False
FTP Correlation Window Pass	0	5	1	BLN	N/A	$1=True; $0=False
FTP Correlation Window Fail	0	4	1	BLN	N/A	$1=True; $0=False
FTP Correlation Vacuum Pass	0	3	1	BLN	N/A	$1=True; $0=False
FTP Correlation Vacuum Fail	0	2	1	BLN	N/A	$1=True; $0=False
Reserved	0	1	1			
Reserved	0	0	1			
FTP Correlation Vented Vacuum Result	1	7	16	SNM	-32.7675 to 32.7675 in H_2O	E = N * .001
FTP Correlation Test Time	3	7	8	UNM	0 to 127.5 sec	E = N * 0.5
FTP Correlation Vacuum Result	4	7	8	UNM	-20.0 to 5.5 in H_2O	E = N * 0.1 - 20

This PID contains the signals associated with fuel tank pressure sensor correlation when the propulsion system is on for conventional EVAP systems with an EVAP Leak Check Pump.

FTP (Fuel Tank Pressure) Correlation Test Complete indicates if the FTP sensor correlation diagnostic completed this trip.

FTP Correlation Soak Time Met indicates if the FTP sensor correlation diagnostic soak time conditions were met.

FTP Correlation Window Pass indicates if the FTP sensor correlation diagnostic vented vacuum was within the acceptable window.

FTP Correlation Window Fail indicates if the FTP sensor correlation diagnostic vented vacuum was outside the acceptable window.

FTP Correlation Vacuum Pass indicates if the FTP sensor correlation diagnostic vacuum change was above the threshold.

FTP Correlation Vacuum Fail indicates if the FTP sensor correlation diagnostic vacuum change was below the threshold.

FTP Correlation Vented Vacuum Result contains the result during the vented vacuum window check of the FTP correlation diagnostic.

FTP Correlation Test Time contains the amount of time the vacuum change test ran during the FTP correlation diagnostic.

FTP Correlation Vacuum Result contains the result during for the vacuum change portion of the FTP correlation diagnostic.

3.2.48 FHPR_SIDI – PID $337B – Fuel Rail Pressure 2 A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$337B	Fuel Rail Pressure 2 A/D	1	UNM	0 to 100% 5 V	E = N * 100 / 255

This PID contains the fuel rail pressure 2 analog input as a percentage of its reference voltage.

3.2.49 FHPR_SIDI – PID $337C – Fuel Rail Pressure 2 Relative to Atmosphere

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$337C	Fuel Rail Pressure 2 Relative to Atmosphere	2	UNM	0 to 655350 kPa	E = N * 10

This PID displays undefaulted fuel rail pressure 2 at the engine when the reading is referenced to atmosphere (gauge pressure). This PID is used for the high side pressure for fuel systems using a high pressure fuel pump.

3.2.50 TSCR – PID $3380 – Solenoid 1 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3380	Solenoid 1 Current	2	UNM	0 to 15.9997558594Amps	E = N / 4096

This PID contains the Current Command for Solenoid 1

3.2.51 TSCR – PID $3381 – Solenoid 2 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3381	Solenoid 2 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 2

3.2.52 TSCR – PID $3382 – Solenoid 3 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3382	Solenoid 3 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 3

3.2.53 TSCR – PID $3383 – Solenoid 4 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3383	Solenoid 4 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 4

3.2.54 TSCR – PID $3384 – Solenoid 5 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3384	Solenoid 5 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 5

3.2.55 TSCR – PID $3385 – Solenoid 6 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3385	Solenoid 6 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 6

3.2.56 TSCR – PID $3386 – Solenoid 7 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3386	Solenoid 7 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 7

3.2.57 TSCR – PID $3387 – Solenoid 8 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3387	Solenoid 8 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 8

3.2.58 TSCR – PID $3388 – Solenoid 9 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3388	Solenoid 9 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 9

3.2.59 TSCR – PID $3389 – Solenoid 10 Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3389	Solenoid 10 Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Current Command for Solenoid 10

3.2.60 TSCR – PID $338A – Solenoid 1 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$338A	Solenoid 1 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 1

3.2.61 TSCR – PID $338B – Solenoid 2 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$338B	Solenoid 2 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 2

3.2.62 TSCR – PID $338C – Solenoid 3 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$338C	Solenoid 3 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 3

3.2.63 TSCR – PID $338D – Solenoid 4 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$338D	Solenoid 4 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 4

3.2.64 TSCR – PID $338E – Solenoid 5 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$338E	Solenoid 5 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 5

3.2.65 TSCR – PID $338F – Solenoid 6 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$338F	Solenoid 6 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 6

3.2.66 TSCR – PID $3390 – Solenoid 7 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3390	Solenoid 7 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 7

3.2.67 TSCR – PID $3391 – Solenoid 8 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3391	Solenoid 8 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 8

3.2.68 TSCR – PID $3392 – Solenoid 9 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3392	Solenoid 9 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 9

3.2.69 TSCR – PID $3393 – Solenoid 10 Dither Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3393	Solenoid 10 Dither Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Dither Current Command for Solenoid 10

3.2.70 TSCR – PID $3394 – Solenoid 1 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3394	Solenoid 1 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 1

3.2.71 TSCR – PID $3395 – Solenoid 2 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3395	Solenoid 2 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 2

3.2.72 TSCR – PID $3396 – Solenoid 3 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3396	Solenoid 3 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 3

3.2.73 TSCR – PID $3397 – Solenoid 4 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3397	Solenoid 4 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 4

3.2.74 TSCR – PID $3398 – Solenoid 5 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3398	Solenoid 5 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 5

3.2.75 TSCR – PID $3399 – Solenoid 6 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3399	Solenoid 6 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 6

3.2.76 TSCR – PID $339A – Solenoid 7 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$339A	Solenoid 7 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 7

3.2.77 TSCR – PID $339B – Solenoid 8 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$339B	Solenoid 8 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 8

3.2.78 TSCR – PID $339C – Solenoid 9 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$339C	Solenoid 9 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 9

3.2.79 TSCR – PID $339D – Solenoid 10 Dither Frequency

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$339D	Solenoid 10 Dither Frequency	2	UNM	0 to 8191.875 Hz	E = N / 8

This PID contains the Dither Frequency Command for Solenoid 10

3.2.80 TSCR – PID $339E – Solenoid 1 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$339E	Solenoid 1 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 1 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 1 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 1 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 1 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 1 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0962 P0960, and P0963 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.81 TSCR – PID $339F – Solenoid 2 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$339F	Solenoid 2 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 2 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 2 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 2 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 2 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 2 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0966, P0964, and P0967 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.82 TSCR – PID $33A0 – Solenoid 3 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A0	Solenoid 3 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 3 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 3 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 3 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 3 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 3 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P0970, P0968, and P0971 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.83 TSCR – PID $33A1 – Solenoid 4 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A1	Solenoid 4 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 4 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 4 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 4 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 4 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 4 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2720, P2718, and P2721 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.84 TSCR – PID $33A2 – Solenoid 5 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A2	Solenoid 5 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 5 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 5 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 5 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 5 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 5 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2729, P2727, and P2730 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.85 TSCR – PID $33A3 – Solenoid 6 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A3	Solenoid 6 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 6 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 6 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 6 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 6 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 6 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2738, P2736, and P2739 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.86 TSCR – PID $33A4 – Solenoid 7 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A4	Solenoid 7 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 7 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 7 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 7 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 7 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 7 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2814, P2812, and P2815 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.87 TSCR – PID $33A5 – Solenoid 8 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A5	Solenoid 8 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 8 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 8 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 8 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 8 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 8 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P281D, P281B, and P281E respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.88 TSCR – PID $33A6 – Solenoid 9 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A6	Solenoid 9 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 9 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 9 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 9 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 9 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 9 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P2826, P2824, and P2827 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.89 TSCR – PID $33A7 – Solenoid 10 Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A7	Solenoid 10 Control Circuit	1	PKT	NA	NA

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Solenoid 10 Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Solenoid 10 Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 10 Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Solenoid 10 Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the solenoid 10 output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTCs P282F, P282D, and P2830 respectively. Command is stubbed to CbFALSE when not type discrete - ON/OFF.

3.2.90 TSCR – PID $33A8 – Solenoid 1 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A8	Solenoid 1 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 1

3.2.91 TSCR – PID $33A9 – Solenoid 2 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33A9	Solenoid 2 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 2

3.2.92 TSCR – PID $33AA – Solenoid 3 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33AA	Solenoid 3 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 3

3.2.93 TSCR – PID $33AB – Solenoid 4 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33AB	Solenoid 4 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 4

3.2.94 TSCR – PID $33AC – Solenoid 5 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33AC	Solenoid 5 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 5

3.2.95 TSCR – PID $33AD – Solenoid 6 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33AD	Solenoid 6 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 6

3.2.96 TSCR – PID $33AE – Solenoid 7 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33AE	Solenoid 7 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 7

3.2.97 TSCR – PID $33AF – Solenoid 8 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33AF	Solenoid 8 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 8

3.2.98 TSCR – PID $33B0 – Solenoid 9 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33B0	Solenoid 9 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 9

3.2.99 TSCR – PID $33B1 – Solenoid 10 Average Diagnostic Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33B1	Solenoid 10 Average Diagnostic Current	2	UNM	0 to 15.9997558594 Amps	E = N / 4096

This PID contains the Average Diagnostic Current for Solenoid 10

3.2.100TPCR – PID $33B2 –Transmission Fluid Pressure A/D

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33B2	Transmission Fluid Pressure A/D	1	UNM	0 – 100% 5 V	E = N * 100 / 255

This PID contains the transmission fluid pressure analog input as a percentage of its reference voltage.

3.2.101SBRR – PID $33BA – Sensor Bus Relay Control Circuit

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33BA	Sensor Bus Relay Control Circuit	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Sensor Bus Relay Commanded On	0	7	1	BLN	N/A	$1=True; $0=False
Reserved	0	6	1			
Sensor Bus Relay Short to Ground Test Status	0	5	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Sensor Bus Relay Open Circuit Test Status	0	3	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed
Sensor Bus Relay Short to Power Test Status	0	1	2	ENM	N/A	$0=Failed; $1=Indeterminate; $3=Passed

This PID contains the commanded state of the sensor bus relay output along with the status of its associated short to ground, open circuit, and short to power tests used to report DTC Pxxxx, Pxxxx and Pxxxx, respectively.

3.2.102 FDPR – PID $33BC – Adjusted Fuel Pump Current

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33BC	Adjusted Fuel Pump Current	2	UNM	0 to 20 Amps	E = N * 4 / 13107

The fuel pump current is determined by dividing the output driver's current by the Fuel Pump Duty Cycle and multiplied by gain.
The fuel pump current is then filtered using a first order low pass filter.

3.2.103 FDPR – PID $33BD – Fuel Pump Input Power

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33BD	Fuel Pump Input Power	2	UNM	0 to 400 Watts	E = N * 80 / 13107

The fuel pump input power is determined by multiplying the duty cycle voltage with the fuel pump current.

3.2.104 FDPR – PID $33BE – Fuel Pump Output Power

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33BE	Fuel Pump Output Power	2	UNM	0 to 1200 kPa(l/sec)	E = N * 80 / 4369

The fuel pump output power is determined by multiplying the Fuel Pressure with the fuel flow rate

3.2.105 FDPR – PID $33BF – Estimated Motor Resistance

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33BF	Estimated Motor Resistance	2	UNM	0 to 10 Ohms	E = N * 2 / 13107

The pump motor resistance is estimated using a least square estimation algorithm

3.2.106 FDPR – PID $33C0 – Fuel Pump State Of Health

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C0	Fuel Pump State Of Health	2	UNM	0 to 100 %	E = N * 100 / 65535

Fuel Pump State of Health is determined based on the deviation of the pump speed and the motor resistance from their nominal values and the ratio of the pump pressure to the fuel pressure request.

3.2.107 FOPR – PID $33C1 – Pressure Control Solenoid 5 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C1	Pressure Control Solenoid 5 Commanded Pressure	2	UNM	0 to 8191.875 kPa	E = N / 8

This PID contains the commanded pressure for pressure control solenoid 5.

3.2.108FOPR – PID $33C2 – Pressure Control Solenoid 6 Commanded Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C2	Pressure Control Solenoid 6 Commanded Pressure	2	UNM	0 to 8191.875 kPa	E = N / 8

This PID contains the commanded pressure for pressure control solenoid 6.

3.2.109TRGR – PID $33C3 – Internal Mode Switch Mode "A" Position Sensor Signal Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C3	Internal Mode Switch Mode "A" Position Sensor Signal Voltage	1	UNM	0 to 5V	E = N / 51

This PID contains the internal mode switch voltage for the mode "A" position sensor analog input.

3.2.110TRGR – PID $33C4 – Internal Mode Switch Mode "B" Position Sensor Signal Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C4	Internal Mode Switch Mode "B" Position Sensor Signal Voltage	1	UNM	0 to 5V	E = N / 51

This PID contains the internal mode switch voltage for the mode "B" position sensor analog input.

3.2.111TRGR – PID $33C5 – Internal Mode Switch Mode "C" Position Sensor Signal Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C5	Internal Mode Switch Mode "C" Position Sensor Signal Voltage	1	UNM	0 to 5V	E = N / 51

This PID contains the internal mode switch voltage for the mode "C" position sensor analog input.

3.2.112TRGR – PID $33C6 – Internal Mode Switch Mode "P" Position Sensor Signal Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C6	Internal Mode Switch Mode "P" Position Sensor Signal Voltage	1	UNM	0 to 5V	E = N / 51

This PID contains the internal mode switch voltage for the mode "P" position sensor analog input

3.2.113TRGR – PID $33C7 – Internal Mode Switch Mode "S" Position Sensor Signal Voltage

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C7	Internal Mode Switch Mode "S" Position Sensor Signal Voltage	1	UNM	0 to 5V	E = N / 51

This PID contains the internal mode switch voltage for the mode "S" position sensor analog input.

3.2.114TRGR – PID $33C8 – Internal Mode Switch Mode "A" Position Sensor Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C8	Internal Mode Switch Mode "A" Position Sensor Status	1	ENM	N/A	$00= Indeterminate; $01= Valid High; $02= Valid Low; $03= Invalid High; $04= Invalid Low; $05 to $FF=Reserved

This PID contains the status of the internal mode switch mode "A" position sensor.

3.2.115TRGR – PID $33C9 – Internal Mode Switch Mode "B" Position Sensor Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33C9	Internal Mode Switch Mode "B" Position Sensor Status	1	ENM	N/A	$00= Indeterminate; $01= Valid High; $02= Valid Low; $03= Invalid High; $04= Invalid Low; $05 to $FF=Reserved

This PID contains the status of the internal mode switch mode "B" position sensor.

3.2.116TRGR – PID $33CA – Internal Mode Switch Mode "C" Position Sensor Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33CA	Internal Mode Switch Mode "C" Position Sensor Status	1	ENM	N/A	$00= Indeterminate; $01= Valid High; $02= Valid Low; $03= Invalid High; $04= Invalid Low; $05 to $FF=Reserved

This PID contains the status of the internal mode switch mode "C" position sensor.

3.2.117TRGR – PID $33CB – Internal Mode Switch Mode "P" Position Sensor Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33CB	Internal Mode Switch Mode "P" Position Sensor Status	1	ENM	N/A	$00= Indeterminate; $01= Valid High; $02= Valid Low; $03= Invalid High; $04= Invalid Low; $05 to $FF=Reserved

This PID contains the status of the internal mode switch mode "P" position sensor.

3.2.118TRGR – PID $33CC – Internal Mode Switch Mode "S" Position Sensor Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33CC	Internal Mode Switch Mode "S" Position Sensor Status	1	ENM	N/A	$00= Indeterminate; $01= Valid High; $02= Valid Low; $03= Invalid High; $04= Invalid Low; $05 to $FF=Reserved

This PID contains the status of the internal mode switch mode "S" position sensor.

3.2.119 TRGR – PID $33CD – Internal Mode Switch Input Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33CD	Internal Mode Switch Input Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Internal Mode Switch Mode "S" Input Active	0	4	1	BLN	N/A	$1=True; $0=False
Internal Mode Switch Mode "A" Input Active	0	3	1	BLN	N/A	$1=True; $0=False
Internal Mode Switch Mode "B" Input Active	0	2	1	BLN	N/A	$1=True; $0=False
Internal Mode Switch Mode "C" Input Active	0	1	1	BLN	N/A	$1=True; $0=False
Internal Mode Switch Mode "P" Input Active	0	0	1	BLN	N/A	$1=True; $0=False

This PID contains several signals associated with the status of the internal mode switch inputs. The content of each of these signals is described below.

Internal Mode Switch Mode "A" Input Active, Internal Mode Switch Mode "B" Input Active, Internal Mode Switch Mode "C" Input Active, Internal Mode Switch Mode "P" Input Active and Internal Mode Switch Mode "S" Input Active indicate the states of the Internal Mode Switch Mode A, B, C, P and S inputs, respectively, that are based upon an internal mode switch (IMS).

3.2.120 FADR – PID $33CF – Small Quantity Learning Abort Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33CF	Small Quantity Learning Abort Reason	1	ENM	N/A	$00=No Abort Reason Detected; $01=Baro Pressure Too Low; $02=Transmission must be in Park or Neutral; $03=Vehicle speed not zero; $04=Accelerator pedal not released; $05=Powertrain relay voltage out of range; $06=F-terminal value out of range; $07=Engine speed deviation above threshold; $08=Common rail pressure deviation (between actual and target) is too big; $09=Cylinder unbalanced; $0A=Torque derivative too high; $0B=Diesel Particulate Filter Regeneration is Active; $0C=Engine Coolant Temperature Out of Range (different range between Service and DVT procedure); $0D=Intake Air temperature out of range (different range between Service and DVT procedure); $0E=Fuel temperature out of range (different range between Service and DVT procedure); $0F=Fan switched on; $10=High Pressure Pump offset learning not performed; $11=EIA code not programmed; $12=Engine Must be Running; $13=Fault Active; $14=Engine Runtime Too Short; $15 = AT Fluid Temperature Out of Range; $16= Generator Type not Programmed $17= Clutch Pedal Depressed; $18= Cabin Heater On; $19 to $FF = Reserved

This PID contains the reason that the SQL (Small Quantity Learning) procedure has been aborted.

3.2.122 FADR – PID $33D0 – Small Quantity Adjustment – Min Number of Learning Performed

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D0	Small Quantity Adjustment – Min Number of Learning Performed	2	UNM	0 to 65535	E = N * 1

This PID contains the number of minimum learning performed by SQA.

3.2.123 FADR – PID $33D1 – Idle Injection Learning Total Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D1	Idle Injection Learning Total Time	4	UNM	0 to 214748364.75s	E = N * 0.05

This PID contains information about the time in which the system has been operated in IIL combustion mode.
Note: This PID is intend for use by engineering only.

3.2.124SPMR – PID $33D2 – Gear Position Sensor Circuit "A" Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D2	Gear Position Sensor Circuit "A" Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the actual duty cycle received from the gear position sensor X-axis signal on circuit “A”.

3.2.125SPMR – PID $33D3 – Gear Position Sensor Circuit "B" Duty Cycle

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D3	Gear Position Sensor Circuit "B" Duty Cycle	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the actual duty cycle received from the gear position sensor Y-axis signal on circuit “B”.

3.2.126SPMR – PID $33D4 – Gear Position Sensor Circuit "A" Learned Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D4	Gear Position Sensor Circuit "A" Learned Position	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the learned position duty cycle relating to the gear position sensor X-axis signal on circuit “A”.

3.2.127SPMR – PID $33D5 – Gear Position Sensor Circuit "B" Learned Position

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D5	Gear Position Sensor Circuit "B" Learned Position	1	UNM	0 to 100 %	E = N * 100 / 255

This PID contains the learned position duty cycle relating to the gear position sensor Y-axis signal on circuit “B”.

3.2.128SPMR – PID $33D6 – Manual Transmission Gear Position Sensor

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D6	Manual Transmission Gear Position Sensor	1	ENM	N/A	$00=Unknown; $01=First Gear; $02=Second Gear; $03=Third Gear; $04=Fourth Gear $05=Fifth Gear; $06=Sixth Gear; $07=Seventh Gear; $08=Reserved; $09=Reverse; $0A=Neutral 1-2; $0B=Neutral 3-4; $0C=Neutral 5-6; $0D=Neutral 7-Reverse; $0E=Reserved; $0F=Reserved; $10=Reserved; $11=Not Learned; $12 to $FF = Reserved

This PID contains the location of the manual transmission shift lever as determined by the gear position sensor.

3.2.129SPMR – PID $33D7 – Manual Transmission Gear Position Sensor Learn Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D7	Manual Transmission Gear Position Sensor Learn Status	1	ENM	N/A	$00=Learned Successfully; $01=Not Learned; $02=Phase 1- Neutral 3-4 Learn Complete; $03=Phase 2 - Out of Neutral Learn Complete; $04=Phase 3 - Neutral 3-4 Learn Complete; $05=Learn Aborted; $06 to $FF = Reserved

This PID contains the status of the manual transmission gear position sensor learn.

3.2.130SPMR – PID $33D8 – Manual Transmission Gear Position Sensor Learn Abort Reason

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D8	Manual Transmission Gear Position Sensor Learn Abort Reason	1	ENM	N/A	$00=None; $01=Shifter Not in Neutral 3-4 Gate; $02=Shifter Not Out of Neutral 3-4 Gate; $03=Learn Timer Expired; $04 to $FF = Reserved

This PID contains the reason why the manual transmission gear position sensor learn has been aborted.

3.2.131SPMR – PID $33D9 – Speed Matching System Status

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33D9	Speed Matching System Status	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Speed Matching Mode Driver Request	0	7	1	BLN	N/A	$1=True; $0=False
Speed Matching Mode Active	0	6	1	BLN	N/A	$1=True; $0=False
Speed Matching Mode Unavailable	0	5	1	BLN	N/A	$1=True; $0=False
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This PID contains signals associated with the status of the Speed Matching System. The content of each of these signals is described below.

Speed Matching Mode Driver Request indicates if the driver of the vehicle has requested speed matching by pressing the Speed Matching mode switch.

Speed Matching Mode Active indicates that the system is an active speed matching mode, all conditions for allowing the mode are present (no inhibit conditions exist) and the driver request for the mode is True.

Speed Matching Mode Unavailable indicates that conditions exist preventing the speed matching mode from operating.

3.2.132FCBR - PID $33DB – Open Loop Fuel Pump Duty Cycle Correction

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33DB	Open Loop Fuel Pump Duty Cycle correction	2	UNM	0 to 3.9998779296875	E = N / 16384

The fuel pump duty cycle is determined from an ‘open loop’ lookup (based on fuel flow and requested fuel pressure) corrected by ‘closed loop’ adjustments. This PID contains the value of the correction to the ‘open loop’ fuel pump duty cycle.

3.2.133 FULR – PID $33DC – Fuel System Status 2

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$33DC	Fuel System Status 2	1	PKT	N/A	N/A

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Dynamic Power Enrichment	0	7	1	BLN	N/A	$1=Active; $0=Inactive
Reserved	0	6				
Reserved	0	5				
Reserved	0	4				
Reserved	0	3				
Reserved	0	2				
Reserved	0	1				
Reserved	0	0				

This PID contains several signals associated with the status of the fuel system. The content of each of these signals is described below.

Dynamic Power Enrichment is a function for turbo charged engines which provides a possibility to enrich the A/F ratio when the driver quickly requests much torque in lower engine speeds to speed up the charger turbine and increase the response. It is activated when a calibrated pedal threshold and a calibrated engine load is reached in a calibrated engine speed range. The function is deactivated again when either the pressure ratio over the turbine reaches a calibrated threshold or a calibrated activation duration has passed.

3.2.134EVPR – PID $3A00 – Engine Off Natural Vacuum Phase 1 Result Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A00	Engine Off Natural Vacuum Phase 1 Result Time	2	UNM	0 - 3276.75 s	E = N * 0.05

This PID captures the Engine Off Natural Vacuum (EONV) Phase 1 time result. (Engineering use only)

3.2.135EVPR – PID $3A01 – Engine Off Natural Vacuum Phase 2 End Time

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A01	Engine Off Natural Vacuum Phase 2 End Time	2	UNM	0 - 3276.75 s	E = N * 0.05

This PID captures the time at the end of Engine Off Natural Vacuum (EONV) Phase 2. (Engineering use only)

3.2.136EVPR – PID $3A02 – Engine Off Natural Vacuum EWMA Test Sample Counter

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A02	Engine Off Natural Vacuum EWMA Test Sample Counter	2	UNM	0 – 65535 counts	E = N * 1

This PID keeps track of how many times the 0.020 inch Exponentially Weighted Moving Average (EWMA) parameter has been updated. (Engineering use only)

3.2.137EVPR – PID $3A03 – Accumulated Distance Traveled During Previous Key On Cycles

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A03	Accumulated Distance Traveled During Previous Key On Cycles	4	UNM	0 - 1048575.999755859375 km	E = N / 4096

This PID stores the total accumulated distance occurring with the engine running above a minimum vehicle speed saved at shutdown. (Engineering use only)

3.2.138EVPR – PID $3A04 – Engine Off Natural Vacuum Corrected Phase 1 End Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A04	Engine Off Natural Vacuum Corrected Phase 1 End Pressure	2	SNM	-32 - 31.9990234375 in. H_20	E = N / 1024

This PID indicates the pressure at the end of Phase 1, corrected for fuel volatility. (Engineering use only)

3.2.139EVPR – PID $3A05 – Engine Off Natural Vacuum Phase 1 Result Pressure

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A05	Engine Off Natural Vacuum Phase 1 Result Pressure	2	SNM	-32 - 31.9990234375 in. H_20	E = N / 1024

This PID indicates the pressure at the end of Phase 1. (Engineering use only)

3.2.140EVPR – PID $3A06 – Engine Off Natural Vacuum Phase 2 Result Vacuum

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A06	Engine Off Natural Vacuum Phase 2 Result Vacuum	2	SNM	-32 - 31.9990234375 in. H_20	E = N / 1024

This PID indicates the vacuum level at the end of Phase 2. (Engineering use only)

3.2.141EVPR – PID $3A07 – Engine Running EVAP Diagnostics Fault History

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A07	Engine Running EVAP Diagnostics Fault History	1	ENM	N/A	$00=No Fault; $01=Canister Vent Restriction; $02=Purge Valve Leak; $03=Small Leak; $04=Weak Vacuum; $05=No Test Result

This PID keeps track of the test result from the previous trip where an Engine Running EVAP Diagnostics (EVPD) test was completed. (Engineering use only)

3.2.142EVPR – PID $3A08 – EVPD tank pressure sensor learned offset

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A08	EVPD tank pressure sensor learned offset	2	SNM	-32 to 31.9990234375 V	E = N * 1/1024

This PID indicates the tank pressure sensor offset voltage. It is updated at key on for cold starts when IAT, fuel level percentage and tank pressure sensor voltage are in acceptable ranges.

3.2.143EVPR – PID $3A09 – EONV tank pressure sensor learned offset

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$3A09	EONV tank pressure sensor learned offset	2	UNM	0 to 31.99951171875 V	E = N * 1/2048

3.2.144HSWR – PID $905C – Analog Hood Switch Input

PID	Signal	Length (Bytes)	Data Type	Range	Conversion
$905C	Analog Hood Switch Input	1	ENM	N/A	$00=Closed; $01=Open; $02=Shorted to Ground; $03=Shorted to Battery; $04=Invalid; $05=Indeterminate; $06 to $FF = Reserved

This PID contains hood switch status information.

Closed indicates that the hood switch is in the closed position.

Open indicates that the hood switch is in the open position.

Shorted to Ground indicates that the hood switch has an electrical short circuit to ground. This is value is used by the hood switch diagnostic to determine if the corresponding DTC should be set. This value may be displayed in the PID before the diagnostic actually sets the corresponding DTC.

Shorted to Battery indicates that the hood switch has an electrical short circuit to battery voltage. This is value is used by the hood switch diagnostic to determine if the corresponding DTC should be set. This value may be displayed in the PID before the diagnostic actually sets the corresponding DTC.

Invalid indicates that the voltage observed at the analog hood switch is between or outside valid ranges.

Indeterminate indicates that the hood switch model or algorithm is unable to determine the state of the switch due to problems with the reference voltage, input signal, etc.

3.3 Development PIDs

To be determined in a future version of this document.

4. Data Identifiers (DIDs)

DIDs are divided into several ranges that determine their usage and the ownership of their requirements, as shown in Table 34: DID Ranges
. Definitions for application specific DIDs and corporate standard DIDs are provided in the following sections. Definitions for controller specific DIDs must be provided in the associated controller specific diagnostic CTS.

Range	Usage
$00	Reserved
$01 to $0F	Controller Specific DIDs
$10 to $7F	Application Specific DIDs (see Section 4.1)
$80 to $FF	Corporate Standard DIDs (see Section 4.3)

Table 34: DID Ranges

4.1 Application Specific DIDs

4.1.1 ESNR – DID $17 – Engine Serial Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$17	Engine Serial Number (ESN)	17	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
ESN character position 1	0	7	8	ASCII	N/A	N/A
ESN character position 2	1	7	8	ASCII	N/A	N/A
ESN character position 3	2	7	8	ASCII	N/A	N/A
ESN character position 4	3	7	8	ASCII	N/A	N/A
ESN character position 5	4	7	8	ASCII	N/A	N/A
ESN character position 6	5	7	8	ASCII	N/A	N/A
ESN character position 7	6	7	8	ASCII	N/A	N/A
ESN character position 8	7	7	8	ASCII	N/A	N/A
ESN character position 9	8	7	8	ASCII	N/A	N/A
ESN character position 10	9	7	8	ASCII	N/A	N/A
ESN character position 11	10	7	8	ASCII	N/A	N/A
ESN character position 12	11	7	8	ASCII	N/A	N/A
ESN character position 13	12	7	8	ASCII	N/A	N/A
ESN character position 14	13	7	8	ASCII	N/A	N/A
ESN character position 15	14	7	8	ASCII	N/A	N/A
ESN character position 16	15	7	8	ASCII	N/A	N/A
ESN character position 17	16	7	8	ASCII	N/A	N/A

This DID contains the Engine Serial Number which is both written and read by tooling in the vehicle assembly plant during the production process. If the ESN has not yet been programmed into the ECU's permanent memory, those locations shall be either '$00' or '$FF'.

* When the Manufacturing Enable Counter (MEC) is equal to zero, a DID $17 write may only be performed under two circumstances:

1) When preceded by a ClearDiagnosticInformation ($04) Service during the same ignition cycle
   - or -
2) When the controller has been programmed within the last three ignition cycles

There are no DID $17 write restrictions when the MEC is non-zero.

If the DID value is not programmed, DTC P264F shall be reported once the Manufacturing Enable Counter equals zero.

Example:

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
ESN	0	0	0	0	1	0	Y	D	S	R	1	2	7	3	7	7	2
Hex value	30	30	30	30	31	30	59	44	53	52	31	32	37	33	37	37	32

4.1.2 MEMR – DID $23 – Dependent Secondary Controller 1 End Model Part Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$23	Dependent Secondary Controller 1 End Model Part Number	4	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
End Model Part Number Position 1	0	7	8	UNM	$00 - $FF	E = N
End Model Part Number Position 2	1	7	8	UNM	$00 - $FF	E = N
End Model Part Number Position 3	2	7	8	UNM	$00 - $FF	E = N
End Model Part Number Position 4	3	7	8	UNM	$00 - $FF	E = N

This DID is used to identify the part number that represents the combination of hardware/software/calibrations present in a dependent secondary controller as it is received in the vehicle assembly plant. This part number is also used in service to uniquely identify the combination of hardware/software/calibrations programmed into the dependent secondary controller at the time the part is ordered. Converting the 4 byte USN value to decimal provides the 8 digit part number assigned by the division.

This DID is used within a primary OBD controller to report the end model part number for a dependent secondary controller that is not connected to the GMLAN vehicle bus. The primary OBD controller will initialize this DID to all zeros upon power up, prior to receiving the dependent secondary controller message with the part number. After receiving the part number message, this DID will mirror the value stored in the dependent secondary controller until the next power cycle.

Example:

Position	1	2	3	4	---
EndModelPartNumber	16265965				---
Hex value	00	F8	32	ED	---

4.1.3 MEMR – DID $24 – Dependent Secondary Controller 1 Manufacturing Traceability Characters

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$24	Dependent Secondary Controller 1 Manufacturing Traceability Characters	16	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Component ID character position 1	0	7	8	ASCII	N/A	N/A
Component ID character position 2	1	7	8	ASCII	N/A	N/A
Part Number/Broadcast Code character position 1	2	7	8	ASCII	N/A	N/A
Part Number/Broadcast Code character position 2	3	7	8	ASCII	N/A	N/A
Part Number/Broadcast Code character position 3	4	7	8	ASCII	N/A	N/A
Part Number/Broadcast Code character position 4	5	7	8	ASCII	N/A	N/A
Supplier Code character position	6	7	8	ASCII	N/A	N/A
Traceability Number character position 1	7	7	8	ASCII	N/A	N/A
Traceability Number character position 2	8	7	8	ASCII	N/A	N/A
Traceability Number character position 3	9	7	8	ASCII	N/A	N/A
Traceability Number character position 4	10	7	8	ASCII	N/A	N/A
Traceability Number character position 5	11	7	8	ASCII	N/A	N/A
Traceability Number character position 6	12	7	8	ASCII	N/A	N/A
Traceability Number character position 7	13	7	8	ASCII	N/A	N/A
Traceability Number character position 8	14	7	8	ASCII	N/A	N/A
Traceability Number character position 9	15	7	8	ASCII	N/A	N/A

This DID contains the data necessary to meet component traceability requirements for a dependent secondary controller as specified in GMW 4710GS and GM1737. This DID is used within a primary OBD controller to report traceability information for a dependent secondary controller that is not connected to the GMLAN vehicle bus. The primary OBD controller will initialize this DID to all zeros upon power up, prior to receiving the message with the traceability data. After receiving the message, this DID will mirror the value stored in the dependent secondary controller until the next power cycle.

The length of this DID is variable but shall not exceed 16 characters. The data is stored in ASCII format and is broken up into 4 fields. A description of each field follows:

Component Identifier (Comp Id)

The component Identifier is a 2 byte field (starting at byte 1 of the DID) that is used to indicate the component type. A master list of Component Identifiers can be found in Appendix 1 of GM1737.

Part Number / Broadcast Code (PN / BC)

This is a 4 byte field (starting at byte 3 of the DID) and contains either the ECU Broadcast Code or the last 4 digits of the ECU part number. GM1737 specifies this requirement.

Supplier Code (sup)

This is a one byte field occupying the 7th byte of the DID that identifies the supplier and manufacturing location of the ECU. . The Master List of Supplier Codes is located in Appendix 2 of the GM1737 document.

Traceability Number

This field is used to uniquely identify a specific ECU. The field starts at byte 8 of the DID and is 9 bytes in length. The content is assigned by the module manufacturer and must consist of enough information to uniquely identify an ECU. Examples of the type of data included in this field are Julian Date of production, shift build data, production assembly line information and ECU serial number. If less than 9 characters are required to uniquely identify an ECU, then the trailing characters shall be filled with spaces (0x20).

Example:

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	---
MTC	A	S	7	0	8	1	K	1	5	0	6	1	0	0	0	F	
Hex value	41	53	37	30	38	31	4B	31	35	30	36	31	30	30	30	46	---
Field Desc.	Comp Id		PN / BC				sup	Traceability Number									---

4.1.4 MEMR – DID $27 – Engine Performance Identifier

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$27	Engine Performance Identifier	1	UNM	0 - 510 kW	E = N * 2	R/W*

This DID contains the nominal engine power in kilowatts, rounded up to the next even value (e.g. 147 kW to 148 kW), if the engine data sheet states an odd value. The identifier shall provide the means to prevent a gratis engine power and engine torque upgrade of hardware-identical engines, offered in two or more different performance stages. Engine Performance Identifier is used by the ECM to compare the received value with the internal engine performance, stored in the ECM EEPROM. If the values do not match, the engine power will be reduced and the corresponding DTC will be set.

*In the ECM EEPROM delivery condition for production, the Engine Identifier content shall be $FF. Before security access activation (MEC > 0), a reprogramming shall be possible for an unlimited number of times. After security access activation (MEC = 0), the Engine Identifier shall only be reprogrammable, if the corresponding EEPROM byte contains $FF. In the delivery condition for service, the Engine Identifier in EEPROM shall also contain $FF.

4.1.5 VINR – DID $28 – Odometer Vehicle Identification Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$28	Odometer Vehicle Identification Number	17	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
VIN character position 1	0	7	8	ASCII	N/A	N/A
VIN character position 2	1	7	8	ASCII	N/A	N/A
VIN character position 3	2	7	8	ASCII	N/A	N/A
VIN character position 4	3	7	8	ASCII	N/A	N/A
VIN character position 5	4	7	8	ASCII	N/A	N/A
VIN character position 6	5	7	8	ASCII	N/A	N/A
VIN character position 7	6	7	8	ASCII	N/A	N/A
VIN character position 8	7	7	8	ASCII	N/A	N/A
VIN character position 9	8	7	8	ASCII	N/A	N/A
VIN character position 10	9	7	8	ASCII	N/A	N/A
VIN character position 11	10	7	8	ASCII	N/A	N/A
VIN character position 12	11	7	8	ASCII	N/A	N/A
VIN character position 13	12	7	8	ASCII	N/A	N/A
VIN character position 14	13	7	8	ASCII	N/A	N/A
VIN character position 15	14	7	8	ASCII	N/A	N/A
VIN character position 16	15	7	8	ASCII	N/A	N/A
VIN character position 17	16	7	8	ASCII	N/A	N/A

This DID contains the Odometer Vehicle Identification Number (VIN). The data content shall be specified by the vehicle manufacturer. If the Odometer VIN has not yet been programmed into the ECU's permanent memory, those locations shall be either “$00” or “$FF”.

The Odometer VIN is to be used as part of the odometer strategy as a “write once” DID to store the Vehicle Identification Number (VIN) in the ECM. The odometer strategy is detailed in the “Global A Electrical Architecture Vehicle Odometer Sense & Display” functional requirements specification.

When the Manufacturing Enable Counter (MEC) is non-zero, the controller will populate the value in DID $28 so that it will mirror the value stored in DID $90 “Vehicle Identification Number”. Any read of DID $28 while the MEC is non-zero will show its current contents.

When the MEC is equal to zero and DID $28 has been programmed with a valid VIN (defined as no characters equal to “$00” or “$FF”), the value of DID $28 shall be locked and the “ECU Odometer VIN Programmed” bit in PID $2049 “ECU Odometer Status” shall be set to true.

When the MEC is equal to zero and DID $28 has not been programmed (as in the case of a service controller) an update will be allowed. During an ignition cycle when DID $28 has not been programmed, the controller will populate the value in DID $28 so that it will mirror the value stored in DID $90 throughout the current ignition cycle. Any read of DID $28 during the same ignition cycle as the DID $28 update will report back a value of “$FF” in all bytes. (This behavior is intended to prevent someone from entering a VIN for the purpose of an odometer update and then returning the VIN to an invalid number so as not to lock the DID.) Once a valid VIN is stored in DID $28 and the ignition key is cycled off, the value of DID $28 shall be locked and the “ECU Odometer VIN Programmed” bit in PID $2049 “ECU Odometer Status” shall be set to true. Any read of DID $28 on any subsequent ignition cycles will show the actual contents of the DID.

Example:

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
VIN	W	0	L	0	0	0	0	3	6	T	1	0	0	0	0	8	5
Hex value	57	30	4C	30	30	30	30	33	36	54	31	30	30	30	30	38	35

4.1.6 MEMR – DID $29 – Primary Calibration Flash History

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$29	Primary Calibration Flash History	92	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Number of Entries	0	7	8	UNM	0 – 255	E = N * 1
History Buffer Locked	1	7	1	BLN	N/A	$1=True; $0=False
Reserved	1	6	1			
Reserved	1	5	1			
Reserved	1	4	1			
Reserved	1	3	1			
Reserved	1	2	1			
Reserved	1	1	1			
Reserved	1	0	1			
Position 0 - Part Number	2	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 0 - Calibration Verification Number	6	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 0 - Programming Count	10	7	8	UNM	0 – 255 counts	E = N * 1
Position 1 - Part Number	11	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 1 - Calibration Verification Number	15	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 1 - Programming Count	19	7	8	UNM	0 – 255 counts	E = N * 1
Position 2 - Part Number	20	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 2 - Calibration Verification Number	24	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 2 - Programming Count	28	7	8	UNM	0 – 255 counts	E = N * 1
Position 3 - Part Number	29	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 3 - Calibration Verification Number	33	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 3 - Programming Count	37	7	8	UNM	0 – 255 counts	E = N * 1
Position 4 - Part Number	38	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 4 – Calibration Verification Number	42	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 4 - Programming Count	46	7	8	UNM	0 – 255 counts	E = N * 1
Position 5 - Part Number	47	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 5 - Calibration Verification Number	51	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 5 - Programming Count	55	7	8	UNM	0 – 255 counts	E = N * 1
Position 6 - Part Number	56	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 6 - Calibration Verification Number	60	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 6 - Programming Count	64	7	8	UNM	0 – 255 counts	E = N * 1
Position 7 - Part Number	65	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 7 - Calibration Verification Number	69	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 7 - Programming Count	73	7	8	UNM	0 – 255 counts	E = N * 1
Position 8 – Part Number	74	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 8 - Calibration Verification Number	78	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 8 - Programming Count	82	7	8	UNM	0 – 255 counts	E = N * 1
Position 9 - Part Number	83	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 9 - Calibration Verification Number	87	7	32	UNM	0 – 4,294,967,295	E = N * 1
Position 9 - Programming Count	91	7	8	UNM	0 – 255 counts	E = N * 1

This DID contains the history buffer entries for the primary calibration part in the controller. The first entry tells how many of the 10 possible locations are valid for this particular controller. The History Buffer Locked signal will be set to true when the buffer has been locked against further updates due to detection of a non-production calibration. The remaining entries in the DID contain the flash history for the primary calibration.

4.2 GENR – DID $71 – Small Quantity Learning Vehicle Configuration Data

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$71	Small Quantity Learning Vehicle Configuration Data	1	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Generator Type	0	7	3	ENM	N/A	$0=Type A; $1=Type B; $2=Type C; $3=Type D; $4 - $7=Reserved;
Reserved	0	4	1			
Reserved	0	3	1			
Reserved	0	2	1			
Reserved	0	1	1			
Reserved	0	0	1			

This DID contains signals associated with the Small Quantity Learning (SQL) configuration. The content of each of these signals is described below.

Generator Type indicates which kind of alternator (generator ampere value) is present in the vehicle.

4.2.1 MEMR – DID $75 – First Time Power Up Flag

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$75	First Time Power Up Flag	2	UNM	0 - 65535	E = N * 1	R/W

This DID allows the reset of the single NVM Static variable that serves as the 'first time initialization' flag. During NVM initialization, before anything else takes place, this 'first init' flag will be checked. If the state of the flag indicates that this is the first initialization of the controller then invoke TSKR functions and then set the flag to NOT first initialization state. The states of this flag are encoded to avoid selecting values that might occur naturally in memory (ie, FF, 00, AA, 55). The initial value of this flag is specified within the EBOM.

4.2.2 IDRR – DID $7B – Data Recorder Control

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$7B	Data Recorder Control	32	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Data Recorder Compatibility ID	0	7	16	UNM	0 to 65535	E = N * 1
Maximum Possible Number of Triggers	2	7	8	UNM	0 to 255	E = N * 1
Read Trigger Number for DID $7C	3	7	8	UNM	0 to 255	E = N * 1
Maximum Possible Number of Buffers	4	7	8	UNM	0 to 255	E = N * 1
Read Buffer Number for DID $7D	5	7	8	UNM	0 to 255	E = N * 1
Maximum Possible Number of Data Collection Location IDs	6	7	8	UNM	0 to 255	E = N * 1
Read Data Collection Location ID for DID $7E	7	7	8	UNM	0 to 255	E = N * 1
Maximum Possible Number of DTCs in lists	8	7	8	UNM	0 to 255	E = N * 1
Read DTC List Sequential Entry Number for DID $7F	9	7	8	UNM	0 to 255	E = N * 1
Data Recorder Default Configuration	10	7	1	BLN	N/A	$1=True; $0=False
Data Recorder Active	10	6	1	BLN	N/A	$1=True; $0=False
Data Recorder Configuration Initialized	10	5	1	BLN	N/A	$1=True; $0=False
Data Recorder Data Available	10	4	1	BLN	N/A	$1=True; $0=False
Reserved	10	3	1			
Reserved	10	2	1			
Reserved	10	1	1			
Reserved	10	0	1			
Data Recorder Status	11	7	8	ENM	N/A	$00=Not Configured; $01=Configuration In Process; $02=Configured; $03=Error Detected in Configuration; $04=Active
Data Recorder Configuration Error Status	12	7	16	ENM	N/A	$00=No Configuration Error Exists; $01=Too Many DTCs In DTC List; $02=Too Many DTC Lists; $03=Too Many DTC Triggers; $04=Trigger Task Not Found; $05=Too Many Triggers Assigned to a Task; $06=Too Many Buffer Definitions; $07=Too Many Triggers Defined; $08=Not Enough Data Memory; $09=Trigger Defined With No Action; $0A=Not Enough Parameter Memory; $0B=Parameter ID Not Valid; $0C=Buffer Configuration Error; $0D=PID Returns Too Many Bytes; $0E=Motor Voltage Commands Data Buffer Exceeded; $0F=DTC Trigger Defined In Time Based Loop

Data Recorder Configuration Error Location	14	7	16	UNM	0 to 65535	E = N * 1
Data Buffer Starting Address	16	7	32	UNM	0 to $FFFFFFFF	E = N * 1
Size of Data Buffer	20	7	24	UNM	0 to 16777215 Bytes	E = N * 1
Development Software ID	23	7	32	UNM	0 to $FFFFFFFF	E = N * 1
Development Patch ID	27	7	16	UNM	0 to $FFFFFFFF	E = N * 1
Reserved	29	7	8			
Reserved	30	7	8			
Reserved	31	7	8			

This DID contains control information used to allow DIDs $7C, $7D, $7E, and $7F to access all of the information that is stored in a controller's memory. For example, there may be several trigger configurations defined. By incrementing the Read Trigger Number from 0 to the number of triggers, and calling DID $7C in between, a tool can read the configurations for all of the triggers.

This DID also contains status information for the data recorder in general and can be used to enable and disable the data recorder. After a new configuration is sent from a tool to the controller, this DID can be read and then the exact same data sent back to the controller with the exception of the "Data Recorder Configuration Initialized" bit. If this bit is sent back as "True", the data recorder will initialize and use the configuration that was just sent from the tool.

4.2.3 IDRR – DID $7C – Data Recorder Trigger Configuration

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$7C	Data Recorder Trigger Configuration	32	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Trigger Number	0	7	8	UNM	0 to 255	E = N * 1
Trigger Action	1	7	8	ENM	N/A	$00=No Action; $01=Trigger Data Collection; $02=Re-arm Data Buffer; $03=OR with Another Trigger; $04=AND with Another Trigger; $05=Count Trigger Occurrences Only
Trigger Action Destination Buffer/Trigger	2	7	8	UNM	0 to 255	E = N * 1
Trigger Parameter Class	3	7	8	ENM	N/A	$00=Not Applicable; $01=Address; $02=PID; $03=VID; $04=Any DTC Fault; $05=Specific DTC Fault; $06=DTC List
Trigger Parameter Base Type	4	7	8	ENM	N/A	$00=Not Applicable; $01=Signed Byte; $02=Unsigned Byte; $03=Signed 2 Byte; $04=Unsigned 2 Byte; $05=Signed 4 Byte; $06=Unsigned 4 Byte; $07=Floating Point
Trigger Parameter (Address/PID/DTC/DTC List)	5	7	32	UNM	0 to $FFFFFFFF	E = N * 1
Mask for Trigger Parameter Value	9	7	32	UNM	0 to $FFFFFFFF	E = N * 1
Trigger Comparison Type (Trigger Parameter as compared to the Trigger Comparison Value)	13	7	8	ENM	N/A	$00=Not Applicable; $01=Not Equal; $02=Equal; $03=Greater Than; $04=Greater Than or Equal To; $05=Less Than; $06=Less Than or Equal To
Trigger Comparison Value	14	7	32	UNM	0 to $FFFFFFFF	E = N * 1

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Trigger Evaluation Rate	18	7	8	ENM	N/A	$00=Not Active; $01=DTC Fault Report; $02=Motor Voltage Commands Task; $03=Match Task; $04=Direct Injection Scheduling Task; $05=Combustion Data Sampling Buffer Task; $06=Combustion Data Sampling Buffer 2 Task; $07=Direct Injection Fuel Pump Task; $08=Direct Injection Fuel Pump 2 Task; $09=6.25 ms Hi Priority Task; $0A=MEDRES Task; $0B=LORES Compression Task; $0C=LORES Intake Task; $0D=Fuel Boundary Task; $0E=Combustion Data Sampling Window Task; $0F=Combustion Data Sampling Window 2 Task; $10=Cam Task; $11=Engine Cycle Task; $12=Communications Complete 3 Task; $13=Communications Complete 2 Task; $14=Communications Complete Task; $15=Motor Control 1 Task; $16=3.125 ms Task; $17=6.25 ms Task; $18=Motor Control 2 Task; $19=12.5 ms Task; $1A=25 ms Task; $1B=50 ms Task; $1C=100 ms Task; $1D=250 ms Task; $1E=500 ms Task; $1F=1000 ms Task; $20 = Buffer Full Task; $21=Win Close Task; $22-$FF=Reserved
Trigger Occurrence Count (data not used when written)	19	7	32	UNM	0 to 4294967295	E = N * 1
Reserved	23	7	8			
Reserved	24	7	8			
Reserved	25	7	8			
Reserved	26	7	8			
Reserved	27	7	8			
Reserved	28	7	8			
Reserved	29	7	8			
Reserved	30	7	8			
Reserved	31	7	8			

This DID configures one of the triggers in the data recorder configuration. If this DID is written to, the data recorder will stop recording and the current configuration will be erased in anticipation of a new configuration being written. The trigger that is configured will be determined by the Trigger Number field.

4.2.4 IDRR – DID $7D – Data Recorder Buffer Configuration

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$7D	Data Recorder Buffer Configuration	32	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Buffer Number	0	7	8	UNM	0 to 255	E = N * 1
Buffer Status (data not used when written)	1	7	8	ENM	N/A	$00=Not Used; $01=ConfigError $02=Configured; $03=Armed; $04=Trigger In Process; $05=Triggered, Collecting Post-Trigger Data; $06=Triggered, Data Collection Complete
Buffer Data Collection Rate	2	7	8	ENM	N/A	$00=Not Active, $01=DTC Fault Report; $02=Motor Voltage Commands Task; $03=Match Task; $04=Direct Injection Scheduling Task; $05=Combustion Data Sampling Buffer Task; $06=Combustion Data Sampling Buffer 2 Task; $07=Direct Injection Fuel Pump Task; $08=Direct Injection Fuel Pump 2 Task; $09=6.25 ms Hi Priority Task; $0A=MEDRES Task; $0B=LORES Compression Task; $0C=LORES Intake Task; $0D=Fuel Boundary Task; $0E=Combustion Data Sampling Window Task; $0F=Combustion Data Sampling Window 2 Task; $10=Cam Task; $11=Engine Cycle Task; $12=Communications Complete 3 Task; $13=Communications Complete 2 Task; $14=Communications Complete Task; $15=Motor Control 1 Task; $16=3.125 ms Task; $17=6.25 ms Task; $18=Motor Control 2 Task; $19=12.5 ms Task; $1A=25 ms Task; $1B=50 ms Task; $1C=100 ms Task; $1D=250 ms Task; $1E=500 ms Task; $20 = Buffer Full Task; $21=Win Close Task; $22-$FF=Reserved $1F=1000 ms Task
Buffer Data Samples to Collect	3	7	16	UNM	0 to 65535	E = N * 1
Number of Parameters in each Sample (data not used when written)	5	7	8	UNM	0 to 255	E = N * 1
Number of Bytes per Sample (data not used when written)	6	7	16	UNM	0 to 65535	E = N * 1
Pre-trigger Samples	8	7	16	UNM	0 to 65535	E = N * 1

Re-arm Buffer on Code Clear	10	7	1	BLN	N/A	$1=True; $0=False
Re-arm Buffer on Key Cycle Count	10	6	1	BLN	N/A	$1=True; $0=False
Re-arm Buffer on Trigger Occurrence	10	5	1	BLN	N/A	$1=True; $0=False
Reserved	10	4	1			
Reserved	10	3	1			
Reserved	10	2	1			
Reserved	10	1	1			
Reserved	10	0	1			
Key Cycles Since Trigger (data not used when written)	11	7	8	UNM	0 to 255	E = N * 1
Re-arm Key Cycle Count	12	7	8	UNM	0 to 255	E = N * 1
Buffer Trigger Count (data not used when written)	13	7	8	UNM	0 to 255	E = N * 1
Buffer Starting Byte Offset (data not used when written)	14	7	16	UNM	0 to 65535	E = N * 1
Buffer Trigger Position Offset in Bytes (data not used when written)	16	7	16	UNM	0 to 65535	E = N * 1
Buffer Ending Byte Offset (data not used when written)	18	7	16	UNM	0 to 65535	E = N * 1
Reserved	20	7	8			
Reserved	21	7	8			
Reserved	22	7	8			
Reserved	23	7	8			
Reserved	24	7	8			
Reserved	25	7	8			
Reserved	26	7	8			
Reserved	27	7	8			
Reserved	28	7	8			
Reserved	29	7	8			
Reserved	30	7	8			
Reserved	31	7	8			

This DID configures a buffer in the data recorder when written and returns status information for the buffer when read.

4.2.5 IDRR – DID $7E – Data Collection Parameters Configuration

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$7E	Data Collection Parameters Configuration	15	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Data Collection Parameter Sequential ID (data not used when written)	0	7	8	UNM	0 to 255	E = N * 1
Data Collection Parameter Buffer Number	1	7	8	UNM	0 to 255	E = N * 1
Data Collection Parameter Type	2	7	8	ENM	N/A	$00=Not Used; $01=PID; $02=VID; $03=Address 1 Byte; $04=Address 2 Byte; $05=Address 4 Byte
Data Collection Parameter Address/PID Number	3	7	32	UNM	0 to $FFFFFFFF	E = N * 1
Reserved	7	7	8			
Reserved	8	7	8			
Reserved	9	7	8			
Reserved	10	7	8			
Reserved	11	7	8			
Reserved	12	7	8			
Reserved	13	7	8			
Reserved	14	7	8			

This DID is used to set up a data collection parameter and associate the parameter to be collected with the specified buffer. Both the Data Collection Location ID and the Buffer Number from the control DID are used to specify the desired information. The Data Collection Location ID specifies the sequential number of the parameter within the buffer (i.e. Parameter 1 from Buffer 2). To read out every parameter that has been programmed, the buffer information must be read for each buffer. Within the buffer read DID, there is a field that contains the number of parameters associated with each buffer. This DID will be called once for each defined parameter (the sum of all of the parameters that are associated with each buffer).

4.2.6 IDRR – DID $7F – DTC List Configuration

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$7F	DTC List Configuration	18	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
DTC List Sequential Entry Number (data not used when written)	0	7	8	UNM	0 to 255	E = N * 1
DTC List Number	1	7	8	UNM	0 to 255	E = N * 1
DTC Number	2	7	16	UNM	0 to $FFFF	E = N * 1
Reserved	4	7	8			
Reserved	5	7	8			
Reserved	6	7	8			
Reserved	7	7	8			
Reserved	8	7	8			
Reserved	9	7	8			
Reserved	10	7	8			
Reserved	11	7	8			
Reserved	12	7	8			
Reserved	13	7	8			
Reserved	14	7	8			
Reserved	15	7	8			
Reserved	16	7	8			
Reserved	17	7	8			

This DID is used to set up a list of DTCs that will be monitored by an already defined trigger. If any of the DTCs in the list are reported as failing, the trigger will occur.

4.3 Corporate Standard DIDs

The corporate standard DIDs below are also specified within [1]. In the event of a conflict between this document and [1], the later shall take precedence.

4.3.1 VINR – DID $90 – Vehicle Identification Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$90	VehicleIdentificationNumber (VIN)	17	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
VIN character position 1	0	7	8	ASCII	N/A	N/A
VIN character position 2	1	7	8	ASCII	N/A	N/A
VIN character position 3	2	7	8	ASCII	N/A	N/A
VIN character position 4	3	7	8	ASCII	N/A	N/A
VIN character position 5	4	7	8	ASCII	N/A	N/A
VIN character position 6	5	7	8	ASCII	N/A	N/A
VIN character position 7	6	7	8	ASCII	N/A	N/A
VIN character position 8	7	7	8	ASCII	N/A	N/A
VIN character position 9	8	7	8	ASCII	N/A	N/A
VIN character position 10	9	7	8	ASCII	N/A	N/A
VIN character position 11	10	7	8	ASCII	N/A	N/A
VIN character position 12	11	7	8	ASCII	N/A	N/A
VIN character position 13	12	7	8	ASCII	N/A	N/A
VIN character position 14	13	7	8	ASCII	N/A	N/A
VIN character position 15	14	7	8	ASCII	N/A	N/A
VIN character position 16	15	7	8	ASCII	N/A	N/A
VIN character position 17	16	7	8	ASCII	N/A	N/A

This DID contains the Vehicle Identification Number. The data content shall be specified by the vehicle manufacturer. If the VIN has not yet been programmed into the ECU's permanent memory, those locations shall be either '$00' or '$FF'.

* When the Manufacturing Enable Counter (MEC) is equal to zero, a DID $90 write may only be performed under two circumstances:

1) When preceded by a ClearDiagnosticInformation ($04) Service during the same ignition cycle
   - or -
2) When the controller has been programmed within the last three ignition cycles

There are no DID $90 write restrictions when the MEC is non-zero.

Example:

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
VIN	W	0	L	0	0	0	0	3	6	T	1	0	0	0	0	8	5
Hex value	57	30	4C	30	30	30	30	33	36	54	31	30	30	30	30	38	35

4.3.2 MEMR – DID $92 – System Supplier ID

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$92	System Supplier ID (SYSSUPPID)	9	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
System Supplier ID position 1	0	7	8	ASCII	N/A	N/A
System Supplier ID position 2	1	7	8	ASCII	N/A	N/A
System Supplier ID position 3	2	7	8	ASCII	N/A	N/A
System Supplier ID position 4	3	7	8	ASCII	N/A	N/A
System Supplier ID position 5	4	7	8	ASCII	N/A	N/A
System Supplier ID position 6	5	7	8	ASCII	N/A	N/A
System Supplier ID position 7	6	7	8	ASCII	N/A	N/A
System Supplier ID position 8	7	7	8	ASCII	N/A	N/A
System Supplier ID position 9	8	7	8	ASCII	N/A	N/A

This DID contains SystemSupplierId information which identifies the ECU hardware supplier and the system type (System Group Identifier). The SYSSUPPID also contains a 2 character reserved field for future definition. The SYSSUPPID must be stored in the ECU in a **non-erasable memory area (e.g. boot sector of flash, or EEPROM)**. It shall not be possible to change the SYSSUPPID during a programming session or via normal diagnostic services. The SYSSUPPID consists of a structured sequence of 9 ASCII characters which conform to GM-SPS (GM Service Programming System) requirements.

1. System Supplier Identifier (5 ASCII characters) (e.g. GM Powertrain=GMPT, DELCO=DELCO; SIEMENS=SIEM; BOSCH=BOSCH)
   Note: For GM Powertrain and Siemens, the 5th ASCII character in this field shall be a space ($20).
2. System Group Identifier (2 ASCII characters):
   00=Powertrain (combined ECM/TCM), 01=Engine, 02=Transmission, 03=Chassis, 04=Body, all others reserved by document (RBD).
3. Reserved area (2 ASCII characters)

Example:

Position	1	2	3	4	5	6	7	8	9	---
SYSSUPPID	D	E	L	C	O	0	1	0	0	---

Hex value	57	30	4C	30	30	30	30	33	35	---
Data Type	SystemSupplierIdentifier					System-GroupId		Reserved		---

4.3.3 MEMR – DID $97 – System Name or Engine Type

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$97	SystemNameOrEngineType (SNOET)	≤20	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
System Name or Engine Type character position 1	0	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 2	1	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 3	2	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 4	3	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 5	4	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 6	5	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 7	6	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 8	7	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 9	8	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 10	9	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 11	10	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 12	11	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 13	12	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 14	13	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 15	14	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 16	15	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 17	16	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 18	17	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 19	18	7	8	ASCII	N/A	N/A
System Name or Engine Type character position 20	19	7	8	ASCII	N/A	N/A

This DID contains the SystemNameOrEngineType which identifies the electronic system name (e.g. Automatic Transmission with 2 Ltr engine: GS820 X20XEV) or engine type (X30XE) installed. The data shall always be of type ASCII. The length is variable (depends on length of string) but shall not exceed 20 characters.

Example:

Position	1	2	3	4	5	6	7	8	9	10	11	12	…
SNOET	G	S	8	2	0		X	2	0	X	E	V	…
Hex Value	47	53	38	32	30	20	58	32	30	58	45	56	…

4.3.4 MEMR – DID $98 – Repair Shop Code or Tester Serial Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$98	RepairShopCodeOrTesterSerialNumber (RSCOTSN)	10	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Repair Shop or Tester Serial # character position 1	0	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 2	1	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 3	2	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 4	3	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 5	4	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 6	5	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 7	6	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 8	7	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 9	8	7	8	ASCII	N/A	N/A
Repair Shop or Tester Serial # character position 10	9	7	8	ASCII	N/A	N/A

This DID contains the RepairShopCodeOrTesterSerialNumber which identifies the dealer's code or the tester's serial number. This ASCII string is programmed into the ECU's memory during the last programming session (SPS) at the dealer site.

Example:

Position	1	2	3	4	5	6	7	8	9	10	---
RSCOSN	S	N	1	2	3	4	5	6	7	8	---
Hex Value	58	31	31	32	33	34	35	36	37	38	---

4.3.5 MEMR – DID $99 – Programming Date

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$99	ProgrammingDate (PD)	4	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Programming Date position 1	0	7	8	BCD	0 - 99	N/A
Programming Date position 2	1	7	8	BCD	0 - 99	N/A
Programming Date position 3	2	7	8	BCD	0 - 99	N/A
Programming Date position 4	3	7	8	BCD	0 - 99	N/A

This DID contains the ProgrammingDate of the last programming event (SPS) at the dealer site. This BCD value was programmed into the ECU's memory during the last programming session (SPS).

Example:

Position	1	2	3	4	Description
PD (BCD)	20	00	01	22	2000 = Year; 01 = Month; 22 = Day
Hex Value	20	00	01	22	

4.3.6 MEMR – DID $9A – Diagnostic Data Identifier

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$9A	DiagnosticDataIdentifier (DDI)	2	PKT	N/A	N/A	R/W*

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Diagnostic Data ID Digit 1	0	7	8	UNM	0 - $FF	E = N
Diagnostic Data ID Digit 2	1	7	8	UNM	0 - $FF	E = N

This DID contains the DiagnosticDataIdentifier which identifies a supplier and system specific diagnostic data stream and shall be used by diagnostic testers to interpret a diagnostic data stream. The data format of the DDI is defined as two (2) bytes where the most significant byte of the DDI identifies the system by a system code and the least significant byte identifies the system data stream version number. For an SPS_TYPE_B or SPS_TYPE_C ECU, the value of the least significant byte of the DDI shall always be $00 (provided that the DID is supported in a node which is not completely programmed).

Example:

Position	1	2	Description
DDI	02	03	System Code e.g. $02 = Simtec MS56.5
Hex Value	02	03	Diagnostic Data Stream Version Code e.g. $03 = version 03

4.3.7 MEMR – DID $9B – XML Configuration Compatibility Identifier

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$9B	XML Configuration Compatibility Identifier	2	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
XML Configuration Compatibility Identifier Position 1	0	7	8	UNM	$00 - $FF	E = N * 1
XML Configuration Compatibility Identifier Position 2	1	7	8	UNM	$00 - $FF	E = N * 1

This DID contains the XmlConfigurationCompatibilityIdentifier (XMLCCID). This DID shall be supported if an ECU performs option configuration via a released XML file. The tool must read this DID and ensure that the data read matches the value of the XMLCCID embedded in the XML file before performing any DID writes dictated by the XML configuration file. This ensures that the configuration data within the XML file is compatible with the software/hardware within the ECU.

Example:

Position	1	2	---
XMLCCID	44166		---
Hex Value	AC	86	---

4.3.8 MEMR – DID $9C – XML Data File Part Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$9C	XML Data File Part Number	4	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
XML Data File Part Number Position 1	0	7	8	UNM	$00 - $FF	E = N * 1
XML Data File Part Number Position 2	1	7	8	UNM	$00 - $FF	E = N * 1
XML Data File Part Number Position 3	2	7	8	UNM	$00 - $FF	E = N * 1
XML Data File Part Number Position 4	3	7	8	UNM	$00 - $FF	E = N * 1

This DID is used to identify the part number of the released XML data file (XMLDFPN) that is used to configure an ECU. Converting the 4 byte unsigned numeric value to decimal provides the 8 digit part number assigned by the division. If an Alpha Code is used, then the Alpha Code associated with this part number shall be stored in DID $9D.

Example:

Position	1	2	3	4	---
XMLDFPN	16265965				---
Hex value	00	F8	32	ED	---

4.3.9 MEMR – DID $9D – XML Data File Alpha Code

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$9D	XML Data File Alpha Code	2	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
XML Data File Alpha Code Position 1	0	7	8	ASCII	N/A	N/A
XML Data File Alpha Code Position 2	1	7	8	ASCII	N/A	N/A

This DID contains the 2 character representation of the Alpha Code (or Design Level suffix) (XMLDFAC) associated with the XML Data File Part Number (stored in DID $9C).

Example:

Position	1	2	---
XMLDFAC	RS		---
Hex value	52	53	---

4.3.10 MEMR – DID $9F – Previous Stored Repair Shop Code Or Tester Serial Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$9F	PreviousStoredRepairShopCodeOrTesterSerialNumber (PSRSCOTSN)	20	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Last Stored Repair Shop or Tester Serial # character position 1	0	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 2	1	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 3	2	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 4	3	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 5	4	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 6	5	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 7	6	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 8	7	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 9	8	7	8	ASCII	N/A	N/A
Last Stored Repair Shop or Tester Serial # character position 10	9	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 1	10	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 2	11	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 3	12	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 4	13	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 5	14	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 6	15	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 7	16	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 8	17	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 9	18	7	8	ASCII	N/A	N/A
Second from Last Stored Repair Shop or Tester Serial # character position 10	19	7	8	ASCII	N/A	N/A

This DID contains up to two history values for the contents of DID $98 “Repair Shop Code or Tester Serial Number”. The default value for each byte of DID $9F is $20 (ASCII = blank), if there are no history values to report.

When a new value is programmed into DID $98, the previous value will be automatically copied into bytes 0 through 9 of DID $9F. As additional values are programmed into DID $98, the contents of DID $9F will shift to track up to two historical values.

Example:

1 Position	1	2	3	4	5	6	7	8	9	10	---
LSRSCOSN	S	N	1	2	3	4	5	6	7	8	---
Hex Value	58	31	31	32	33	34	35	36	37	38	---
2 Position	1	2	3	4	5	6	7	8	9	10	---
SLSRSCOSN	S	N	3	3	3	8	5	9	9	7	---
Hex Value	58	33	33	33	33	38	37	39	39	37	---

4.3.11 OSAR – DID $A0 – Manufacturers Enable Counter

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$A0	manufacturers_enable_counter (MEC)	1	PKT	N/A	N/A	R/W_1

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Manufacturers Enable Counter	0	7	8	UNM	0 - $FF	E = N * 1

This DID contains the MEC which is used when determining the current status of ECU security.

W_1 = A node shall not allow the value of the MEC to change once it becomes $00, unless SecurityAccess ($27) is successfully initiated. The ability to allow writing a new value to the MEC for any specific ECU which supports the MEC shall be negotiated by representatives from Service and Assembly Verification, and the responsible DRE, and documented in the CTS.**.**

Example:

Position	1	Description
MEC	254	$FE - $01: ECU's security system not armed
Hex value	FE	$00: ECU's security system armed

4.3.12 MEMR – DID $AB – Compressed Vehicle Partitioning and Product Structure (CVPPS)

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$AB	Compressed Vehicle Partitioning and Product Structure (CVPPS)	14	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
CVPPS 1	0	7	8	ASCII	N/A	N/A
CVPPS 2	1	7	8	ASCII	N/A	N/A
CVPPS 3	2	7	8	ASCII	N/A	N/A
CVPPS 4	3	7	8	ASCII	N/A	N/A
CVPPS 5	4	7	8	ASCII	N/A	N/A

CVPPS 6	5	7	8	ASCII	N/A	N/A
CVPPS 7	6	7	8	ASCII	N/A	N/A
CVPPS 8	7	7	8	ASCII	N/A	N/A
CVPPS 9	8	7	8	ASCII	N/A	N/A
CVPPS 10	9	7	8	ASCII	N/A	N/A
CVPPS 11	10	7	8	ASCII	N/A	N/A
CVPPS 12	11	7	8	ASCII	N/A	N/A
CVPPS 13	12	7	8	ASCII	N/A	N/A
CVPPS 14	13	7	8	ASCII	N/A	N/A

This DID contains the data necessary to meet the component traceability requirements as specified in GMW4710 and GMW15862. This DID contains the compressed VPPS assigned to the component. The first 13 characters are used for the compressed VPPS and the last character is used in the event two components have the same VPPS. This field is populated at the time of component manufacturing by the manufacturer. If less than 13 characters are required to uniquely identify a VPPS, then the trailing characters shall be filled with zeroes ($30). The last character shall be set to a value per GMW15862, Table C1. The default value for the last character shall be "X".

4.3.13 CANR – DID $B0 – ECU Diagnostic Address

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$B0	ECUDiagnosticAddress (ECUADDR)	1	PKT	N/A	N/A	R/W[3]

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
ECUDiagnosticAddress	0	7	8	UNM	0 - $FF	E = N

This DID contains the diagnostic address of an ECU.

W[3] = This DID is usually read only and supported in boot software. An ECU may choose to allow this DID to be written when more than 1 of a given hardware device can exist on a single vehicle.

Examples:

Position	1	Description
ECUADDR	17	ECM1 ECUDiagnosticAddress
Hex value	11	
ECUADDR	18	ECM2 ECUDiagnosticAddress
Hex value	12	
ECUADDR	24	TCM1 ECUDiagnosticAddress
Hex value	18	
ECUADDR	26	TCCM ECUDiagnosticAddress
Hex value	1A	

4.3.14 MEMR – DID $B2 – GM Manufacturing Data

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$B2	GM ManufacturingData (GMMD)	5	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
GM Manufacturing Data Byte 1	0	7	8	UNM	0 - $FF	E = N
GM Manufacturing Data Byte 2	1	7	8	UNM	0 - $FF	E = N

GM Manufacturing Data Byte 3	2	7	8	UNM	0 - $FF	E = N
GM Manufacturing Data Byte 4	3	7	8	UNM	0 - $FF	E = N
GM Manufacturing Data Byte 5	4	7	8	UNM	0 - $FF	E = N

This DID contains 5 bytes of ManufacturingData which is both written and read by tooling in the vehicle assembly plant during the production process. This area in product memory is used as a scratch pad for assembly tooling to pass test results back and forth from the various testers used in the vehicle assembly process. The ECU memory used for this DID shall be permanent memory (e.g. flash or EEPROM).

Example:

Position	1	2	3	4	5	---

4.3.15 MEMR – DID $B3 – Data Universal Numbering System Identification (DUNS)

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$B3	Data Universal Numbering System Identification (DUNS)	9	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
DUNS No. Position 1	0	7	8	ASCII	N/A	N/A
DUNS No. Position 2	1	7	8	ASCII	N/A	N/A
DUNS No. Position 3	2	7	8	ASCII	N/A	N/A
DUNS No. Position 4	3	7	8	ASCII	N/A	N/A
DUNS No. Position 5	4	7	8	ASCII	N/A	N/A
DUNS No. Position 6	5	7	8	ASCII	N/A	N/A
DUNS No. Position 7	6	7	8	ASCII	N/A	N/A
DUNS No. Position 8	7	7	8	ASCII	N/A	N/A
DUNS No. Position 9	8	7	8	ASCII	N/A	N/A

This DID contains the data necessary to meet the component traceability requirements as specified in GMW4710 and GMW15862. This DID contains the DUNS Id assigned to the component supplier's manufacturing site.

Example:

Position	1	2	3	4	5	6	7	8	9	---
DUNS	1	2	3	4	5	6	7	8	9	---
Hex value	31	32	33	34	35	36	37	38	39	---

4.3.16 MEMR – DID $B4 – Manufacturing Traceability Characters

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$B4	ManufacturingTraceabilityCharacters (MTC)	16	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Traceability Character Position 1	0	7	8	ASCII	N/A	N/A

Traceability Character Position 2	1	7	8	ASCII	N/A	N/A
Traceability Character Position 3	2	7	8	ASCII	N/A	N/A
Traceability Character Position 4	3	7	8	ASCII	N/A	N/A
Traceability Character Position 5	4	7	8	ASCII	N/A	N/A
Traceability Character Position 6	5	7	8	ASCII	N/A	N/A
Traceability Character Position 7	6	7	8	ASCII	N/A	N/A
Traceability Number character position 1	7	7	8	ASCII	N/A	N/A
Traceability Number character position 2	8	7	8	ASCII	N/A	N/A
Traceability Number character position 3	9	7	8	ASCII	N/A	N/A
Traceability Number character position 4	10	7	8	ASCII	N/A	N/A
Traceability Number character position 5	11	7	8	ASCII	N/A	N/A
Traceability Number character position 6	12	7	8	ASCII	N/A	N/A
Traceability Number character position 7	13	7	8	ASCII	N/A	N/A
Traceability Number character position 8	14	7	8	ASCII	N/A	N/A
Traceability Number character position 9	15	7	8	ASCII	N/A	N/A

This DID contains the data necessary to meet the component traceability requirements as specified in GMW4710 and GMW15862. The length of this DID is variable but shall not exceed 16 characters. These fields are populated at the time of component manufacturing by the manufacturer. The ECU specific CTS or supplemental diagnostic specification referenced by the CTS or SSTS must document how the fields are to be populated.

Legacy Format (To be phased out per GMW15862) A description of each field follows:

Traceability Character Position 1 & **2** == **Component Identifier (Comp Id)**

The component Identifier is a 2 byte field (starting at byte 1 of the DID) that is used to indicate the component type.

Traceability Character Position 3-6 == Part Number / Broadcast Code (PN/BC)

This is a 4-byte field (starting at byte 3 of the DID) and contains either the ECU Broadcast Code or the last 4 digits of the ECU part number.

Traceability Character Position 7 == Supplier Code (sup)

This is a one byte field occupying the 7th byte of the DID that identifies the supplier and manufacturing location of the ECU.

Traceability Number character position 1-9 == **Traceability Number**

This field starts at byte 8 of the DID and is 9 bytes in length. The content is assigned by the module manufacturer. This field contains information about the ECU at the time of its manufacture. Examples of the type of data included in this field are Julian Date of production, shift build data, production assembly line information and ECU serial number. If less than 9 characters are required to uniquely identify an ECU, then the leading characters shall be filled with zeroes (0x30).

Example:

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	---
MTC	A	S	7	0	8	1	K	1	5	0	6	1	0	0	0	F	
Hex value	41	53	37	30	38	31	4B	31	35	30	36	31	30	30	30	46	---
Field Desc.	Comp Id		PN / BC				sup	Traceability Number									---

GMW15862 Format (To be phased in per GMW15862) A description of each field follows:

Traceability Character Position 1 == **Line Identification (Line Id)**
The Line Identifier is a 1 byte field (starting at byte 1 of the DID) that is used to indicate the component assembly line.
Traceability Character Position 2 == Shift Identification 1, 2 or 3 (Shift Id)
This is a one byte field (starting at byte 2 of the DID) that identifies the shift that assembled the component.
Traceability Character Position 3 & 4 == Last Two Digits of Year (Year)
This is a two byte field (starting at byte 3 of the DID) that contains the last two digits of the year that the component was assembled.
Traceability Character Position 5-7 == Day Of the Year (Day)
This is a three byte field (starting at byte 5 of the DID) that contains the Julian day of the Year that the component was assembled.
Traceability Number character position 1-9 == **Traceability Number**
This value is a supplier assigned serial, lot or batch number. This field starts at byte 8 of the DID and is up to 9 bytes in length. The content is assigned by the module manufacturer. If less than 9 characters are required to uniquely identify an ECU, then the leading characters shall be filled with zeroes (0x30).

Example:

Position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	---
MTC	A	1	0	8	2	8	2	1	5	0	6	1	0	0	0	F	---
Hex value	41	31	30	38	32	38	32	31	35	30	36	31	30	30	30	46	---
Field Desc.	Line	Shift	Year		Day			Traceability Number									---

4.3.17 MEMR – DID $B5 – GM Broadcast Code

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$B5	GM Broadcast Code	4	PKT	N/A	N/A	R/W

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
GM Broadcast Code Position 1	0	7	8	ASCII	N/A	N/A
GM Broadcast Code Position 2	1	7	8	ASCII	N/A	N/A
GM Broadcast Code Position 3	2	7	8	ASCII	N/A	N/A
GM Broadcast Code Position 4	3	7	8	ASCII	N/A	N/A

This DID contains 4 alpha characters which represent the ECU broadcast code. The broadcast code is unique for each combination of hardware, software, and calibrations for an ECU.

Example:

Position	1	2	3	4	---
Broadcast Code	A	B	C	D	---
Hex value	41	42	43	44	---

4.3.18 MEMR/BFPR – DID $C0 – Boot Software Part Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$C0	BootSoftwarePartNumber (BSPN)	4	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Boot Software Part Number Position 1	0	7	8	UNM	$00 - $FF	E = N
Boot Software Part Number Position 2	1	7	8	UNM	$00 - $FF	E = N
Boot Software Part Number Position 3	2	7	8	UNM	$00 - $FF	E = N
Boot Software Part Number Position 4	3	7	8	UNM	$00 - $FF	E = N

This DID contains BootSoftwarePartNumber information which is a 4 byte unsigned numeric representation of the 8 digit GM part number assigned to the boot software of the ECU. If a Boot Software Alpha Code is used, then the Alpha Code associated with this part number shall be stored in dataIdentifier $D0.

Example:

Position	1	2	3	4	---
BootSoftwarePartNumber	16265965				---
Hex value	00	F8	32	ED	---

4.3.19 MEMR – DID $C1 - $C8 – Software Module Identifier

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$C1 - $C8	SoftwareModuleIdentifier (SWMI)	4 / 5-16	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Software Module 1 Identifier character position 1	0	7	8	UNM	N/A	N/A
Software Module 1 Identifier character position 2	1	7	8	UNM	N/A	N/A
Software Module 1 Identifier character position 3	2	7	8	UNM	N/A	N/A
Software Module 1 Identifier character position 4	3	7	8	UNM	N/A	N/A
or						
Software Module 1 ASCII Identifier character position 1	0	7	8	ASCII	N/A	N/A
Software Module 1 ASCII Identifier character position 2	1	7	8	ASCII	N/A	N/A
Software Module 1 ASCII Identifier character position 3	2	7	8	ASCII	N/A	N/A
Software Module 1 ASCII Identifier character position 4	3	7	8	ASCII	N/A	N/A
Software Module 1 ASCII Identifier character position 5	5	7	8	ASCII	N/A	N/A
-	-	-	-	-	-	-

DID's ranging from $C1 to $CA contain Software Module Identifier (SWMI) information used to uniquely identify a given software or calibration module. The SWMI shall be included in each software module that can be separately programmed into the ECU's permanent memory via the SPS. Individual software modules can contain Boot software, Operational software, or Calibration data.
The SWMI information shall be stored using one of 2 allowed formats. The first format consists of a 4 byte UNM representation of the 8 digit GM part number assigned to the software or calibration module. The second allowed format consists of an ASCII representation of either an assigned GM part number or the base name of the file as released by the module supplier. If the ASCII format is used, then the part number/file name shall contain a minimum of 5 characters and a maximum of 16.
The length of the data reported in the response message for a given SWMI is used by the tool to determine which of the 2 formats is being used. A given ECU shall use the same format for all supported SWMI.
If an Alpha Code (or Design Level Suffix) is supported, then this information shall be stored in ASCII format in a dataIdentifier with the upper nibble set to $D and the lower nibble the same as the lower nibble of the dataIdentifier that contains the corresponding SWMI. In other words, if a SWMI is stored in dataIdentifier $C1, then the corresponding Alpha Code (or Design Level Suffix) shall be stored in dataIdentifier $D1
Each SWMI has a corresponding dataIdentifier associated with it. ECU's are only required to support the DID's which correspond to SWMI's valid for that ECU. The relationship between the SWMI and DID number is as follows:

- $C1: SoftwareModule_01_Identifier
- $C2: SoftwareModule_02_Identifier
- $C3: SoftwareModule_03_Identifier
- $C4: SoftwareModule_04_Identifier
- $C5: SoftwareModule_05_Identifier
- $C6: SoftwareModule_06_Identifier
- $C7: SoftwareModule_07_Identifier
- $C8: SoftwareModule_08_Identifier

There are a maximum of ten (10) software modules possible with DID's $C1 - $CA. ECUs that have a combined number of software and calibration modules that exceed 10 shall also support DID $DD. DID $DD provides a way for the test tool to determine which DID's contain the SWMI and SWMIAC for the 11th through the nth software/calibration module(s).

Example:

Position	1	2	3	4	
SWMI	16265965				
Hex value	00	F8	32	ED	

4.3.20 MCPR – DID $C9 - $CA – Software Module Identifier

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$C9 - $CA	SoftwareModuleIdentifier (SWMI)	4 / 5-16	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Software Module 1 Identifier character position 1	0	7	8	UNM	N/A	N/A
Software Module 1 Identifier character position 2	1	7	8	UNM	N/A	N/A
Software Module 1 Identifier character position 3	2	7	8	UNM	N/A	N/A
Software Module 1 Identifier character position 4	3	7	8	UNM	N/A	N/A
or						

Software Module 1 ASCII Identifier character position 1	0	7	8	ASCII	N/A	N/A
Software Module 1 ASCII Identifier character position 2	1	7	8	ASCII	N/A	N/A
Software Module 1 ASCII Identifier character position 3	2	7	8	ASCII	N/A	N/A
Software Module 1 ASCII Identifier character position 4	3	7	8	ASCII	N/A	N/A
Software Module 1 ASCII Identifier character position 5	5	7	8	ASCII	N/A	N/A
-	-	-	-	-	-	-

DID's ranging from $C1 to $CA contain Software Module Identifier (SWMI) information used to uniquely identify a given software or calibration module. The SWMI shall be included in each software module that can be separately programmed into the ECU's permanent memory via the SPS. Individual software modules can contain Boot software, Operational software, or Calibration data.
The SWMI information shall be stored using one of 2 allowed formats. The first format consists of a 4 byte UNM representation of the 8 digit GM part number assigned to the software or calibration module. The second allowed format consists of an ASCII representation of either an assigned GM part number or the base name of the file as released by the module supplier. If the ASCII format is used, then the part number/file name shall contain a minimum of 5 characters and a maximum of 16.
The length of the data reported in the response message for a given SWMI is used by the tool to determine which of the 2 formats is being used. A given ECU shall use the same format for all supported SWMI.
If an Alpha Code (or Design Level Suffix) is supported, then this information shall be stored in ASCII format in a dataIdentifier with the upper nibble set to $D and the lower nibble the same as the lower nibble of the dataIdentifier that contains the corresponding SWMI. In other words, if a SWMI is stored in dataIdentifier $C1, then the corresponding Alpha Code (or Design Level Suffix) shall be stored in dataIdentifier $D1
Each SWMI has a corresponding dataIdentifier associated with it. ECU's are only required to support the DID's which correspond to SWMI's valid for that ECU. The relationship between the SWMI and DID number is as follows:

– $C9: SoftwareModule_09_Identifier
– $CA: SoftwareModule_010_Identifier

There are a maximum of ten (10) software modules possible with DID's $C1 - $CA. ECUs that have a combined number of software and calibration modules that exceed 10 shall also support DID $DD. DID $DD provides a way for the test tool to determine which DID's contain the SWMI and SWMIAC for the 11th through the nth software/calibration module(s).

Example:

Position	1	2	3	4	
SWMI	16265965				
Hex value	00	F8	32	ED	

4.3.21 MEMR – DID $CB – End Model Part Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$CB	EndModelPartNumber (EMPN)	4	PKT	N/A	N/A	R/W_7

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
End Model Part Number position 1	0	7	8	UNM	$00 - $FF	E = N
End Model Part Number position 2	1	7	8	UNM	$00 - $FF	E = N
End Model Part Number position 3	2	7	8	UNM	$00 - $FF	E = N

End Model Part Number position 4	3	7	8	UNM	$00 - $FF	E = N

This DID is used to identify the part number that represents the combination of hardware/software/calibrations present in the ECU as it is received in the vehicle assembly plant. This part number is also used in service to uniquely identify the combination of hardware/software/calibrations programmed into the ECU at the time the part is ordered. Converting the 4 byte USN value to decimal provides the 8 digit part number assigned by the division. If an Alpha Code is used, then the Alpha Code associated with this part number shall be stored in dataIdentifier $DB.

W_7 = This DID may or may not be updated at the conclusion of a programming event based on divisional practices. This DID is not required to be writeable if divisional practices do not require this part number to be updated after module programming.

Example:

Position	1	2	3	4	---
EndModelPartNumber	16265965				---
Hex value	00	F8	32	ED	---

4.3.22 MEMR – DID $CC – Base Model Part Number

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$CC	BaseModelPartNumber (BMPN)	4	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Base Model Part Number position 1	0	7	8	UNM	$00 - $FF	E = N
Base Model Part Number position 2	1	7	8	UNM	$00 - $FF	E = N
Base Model Part Number position 3	2	7	8	UNM	$00 - $FF	E = N
Base Model Part Number position 4	3	7	8	UNM	$00 - $FF	E = N

This DID contains BaseModelPartNumber information which is used during SPS programming to identify a unique combination of ECU hardware and all non-reprogrammable software (e.g. Boot software). The BMPN must be stored in the ECU in a **non-erasable memory area (e.g. boot sector of flash, or EEPROM)**. It shall not be possible to change the BMPN as part of a programming session or through diagnostic services when a programming session is not active. The BMPN of a service replacement ECU shall be identical to that of a “Production ECU” if both ECU’s have the same hardware and non-reprogrammable software. If an Alpha Code is used, then the Alpha Code associated with this part number shall be stored in dataIdentifier $DC.

Example:

Position	1	2	3	4	---
BaseModelPartNumber	16265965				---
Hex value	00	F8	32	ED	---

4.3.23 MEMR/BFPR – DID $D0– Boot Software Part Number Alpha Code

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$D0	BootSoftwarePartNumberAlphaCode	2 - 3	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Boot Software Part Number Alpha Code Position 1	0	7	8	ASCII	N/A	N/A
Boot Software Part Number Alpha Code Position 2	1	7	8	ASCII	N/A	N/A
Boot Software Part Number Alpha Code Position 3	2	7	8	ASCII	N/A	N/A

This DID contains the 2 or 3 character representation of the Alpha Code (or Design Level suffix) associated with the BootSoftwarePartNumber (stored in dataIdentifier $C0)

Example:

Position	1	2	
Alpha Code	RS		
Hex value	52	53	

4.3.24 MEMR – DID $D1 - $D8 – Software Module Identifier Alpha Code

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$D1 - $D8	SoftwareModuleIdentifierAlphaCode (SWMIAC)	2 - 3	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Software Module 1 Alpha Code Identifier character position 1	0	7	8	ASCII	N/A	N/A
Software Module 1 Alpha Code Identifier character position 2	1	7	8	ASCII	N/A	N/A
Software Module 1 Alpha Code Identifier character position 3	2	7	8	ASCII	N/A	N/A

DID's ranging from $D1 to $DA contain a 2 or 3 character representation of the Alpha Code (or Design Level suffix) for the corresponding Software Module Identifier (SWMI). The SWMI's occupy the dataIdentifier range $C1 to $CA. Reference the SWMI DID description for the numerical relationship between the DID's used for the SWMI and the corresponding SWMIAC.

- $D1: SoftwareModule_01_AlphaCode
- $D2: SoftwareModule_02_AlphaCode
- $D3: SoftwareModule_03_AlphaCode
- $D4: SoftwareModule_04_AlphaCode
- $D5: SoftwareModule_05_AlphaCode
- $D6: SoftwareModule_06_AlphaCode
- $D7: SoftwareModule_07_AlphaCode
- $D8: SoftwareModule_08_AlphaCode

The SoftwareModuleIdentifier Alpha Code shall be specified by the vehicle manufacturer.

Example:

Position	1	2	
Alpha Code	RS		
Hex value	52	53	

4.3.25 MCPR – DID $D9 - $DA – Software Module Identifier Alpha Code

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$D9 - $DA	SoftwareModuleIdentifierAlphaCode (SWMIAC)	2 - 3	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Software Module 1 Alpha Code Identifier character position 1	0	7	8	ASCII	N/A	N/A
Software Module 1 Alpha Code Identifier character position 2	1	7	8	ASCII	N/A	N/A
Software Module 1 Alpha Code Identifier character position 3	2	7	8	ASCII	N/A	N/A

DID's ranging from $D1 to $DA contain a 2 or 3 character representation of the Alpha Code (or Design Level suffix) for the corresponding Software Module Identifier (SWMI). The SWMI's occupy the dataIdentifier range $C1 to $CA. Reference the SWMI DID description for the numerical relationship between the DID's used for the SWMI and the corresponding SWMIAC.

– $D9: SoftwareModule_09_AlphaCode
– $DA: SoftwareModule_10_AlphaCode

The SoftwareModuleIdentifier Alpha Code shall be specified by the vehicle manufacturer.

Example:

Position	1	2	
Alpha Code	RS		
Hex value	52	53	

4.3.26 MEMR – DID $DB – End Model Part Number Alpha Code

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$DB	EndModelPartNumberAlphaCode	2 - 3	PKT	N/A	N/A	R/W[8]

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
End Model Part Number Alpha Code character position 1	0	7	8	ASCII	N/A	N/A
End Model Part Number Alpha Code character position 2	1	7	8	ASCII	N/A	N/A
End Model Part Number Alpha Code character position 3	2	7	8	ASCII	N/A	N/A

This DID contains the 2 or 3 character representation of the Alpha Code (or Design Level suffix) associated with the EndModelPartNumber (stored in dataIdentifier $CB)

W_8 = This DID may or may not be updated at the conclusion of a programming event based on divisional practices. This DID is not required to be writeable if divisional practices do not require this part number to be updated after module programming.

Example:

Position	1	2	---
EndModelPartNumberAlphaCode	RS		---
Hex value	52	53	---

4.3.27 MEMR – DID $DC – Base Model Part Number Alpha Code

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$DC	BaseModelPartNumberAlphaCode	2 - 3	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Base Model Part Number Alpha Code character position 1	0	7	8	ASCII	N/A	N/A
Base Model Part Number Alpha Code character position 2	1	7	8	ASCII	N/A	N/A
Base Model Part Number Alpha Code character position 3	2	7	8	ASCII	N/A	N/A

This DID contains the 2 or 3 character representation of the Alpha Code (or Design Level suffix) associated with the BaseModelPartNumber (stored in dataIdentifier $CC)

Example:

Position	1	2	---
BaseModelPartNumberAlphaCode	RS		---
Hex value	52	53	---

4.3.28 CANR – DID $DE – GMLAN Identification Data

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$DE	GMLAN Identification Data	18	PKT	N/A	N/A	R

Signal	Start Byte	Start Bit	Length (Bits)	Data Type	Range	Conversion
Reserved	0	7	1			
Reserved	0	6	1			
Reserved	0	5	1			
Reserved	0	4	1			
Reserved	0	3	1			
Bus Type	0	2	3	ENM	N/A	$0=Low Speed CAN; $1=Infotainment Bus; $2=High Speed CAN; $3=Chassis Expansion Bus; $4=Powertrain Expansion Bus; $5=Front Object Expansion Bus; $6=Rear Object Expansion Bus; $7=High Voltage Energy Management Bus
GMLAN Handler Main Version Number	1	7	8	UNM	0 to 255	E = N * 1
GMLAN Handler Sub Version Number	2	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Model Year)	3	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Planned Release)	4	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Maintenance Release)	5	7	8	UNM	0 to 255	E = N * 1
Reserved	6	7	1			
Reserved	6	6	1			
Reserved	6	5	1			
Reserved	6	4	1			
Reserved	6	3	1			
Bus Type	6	2	3	ENM	N/A	$0=Low Speed CAN; $1=Infotainment Bus; $2=High Speed CAN; $3=Chassis Expansion Bus; $4=Powertrain Expansion Bus; $5=Front Object Expansion Bus; $6=Rear Object Expansion Bus; $7=High Voltage Energy Management Bus
GMLAN Handler Main Version Number	7	7	8	UNM	0 to 255	E = N * 1
GMLAN Handler Sub Version Number	8	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Model Year)	9	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Planned Release)	10	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Maintenance Release)	11	7	8	UNM	0 to 255	E = N * 1
Reserved	12	7	1			
Reserved	12	6	1			
Reserved	12	7	1			
Reserved	12	6	1			
Reserved	12	5	1			
Reserved	12	4	1			
Reserved	12	3	1			

Bus Type	12	2	3	ENM	N/A	$0=Low Speed CAN; $1=Infotainment Bus; $2=High Speed CAN; $3=Chassis Expansion Bus; $4=Powertrain Expansion Bus; $5=Front Object Expansion Bus; $6=Rear Object Expansion Bus; $7=High Voltage Energy Management Bus
GMLAN Handler Main Version Number	13	7	8	UNM	0 to 255	E = N * 1
GMLAN Handler Sub Version Number	14	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Model Year)	15	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Planned Release)	16	7	8	UNM	0 to 255	E = N * 1
GMLAN Data Dictionary Version (Maintenance Release)	17	7	8	UNM	0 to 255	E = N * 1

This DID contains identification data related to the GMLAN network connected to the ECU. One set of data shall be reported for each network that the ECU is connected to. The Powertrain infrastructure implementation of DID $DE will always be 18 bytes long, with the actual populated data dependent on the number of GMLAN networks connected to the ECU. Each populated set of data will contain the following information. If fewer than three GMLAN networks are connected to the ECU, the unused DID locations will be populated with zeros.

Bus Type indicates which GMLAN network the following 5 bytes refer to.

GMLAN Handler Main Version Number and GMLAN Handler Sub Version Number indicate the GMLAN handler version implemented by the ECU on the network specified by the preceding bus type.

GM Database Dictionary Version indicates the GM database revision level implemented by the ECU on the network specified by the preceding bus type. The version is broken up in the following manner.

- GMLAN Data Dictionary Version Model Year
- GMLAN Data Dictionary Version Planned Release
- GMLAN Data Dictionary Version Maintenance Release

4.3.29 VSPR – DID $DF – ECU Odometer Value

DID	Signal	Length (Bytes)	Data Type	Range	Conversion	Access
$DF	ECU Odometer Value	4	UNM	0 - 67,108,863.984375 km	E = N / 64	R/W

This DID contains the ECU odometer value determined from accumulation of Vehicle Speed Sensor (VSS) pulses, converted to kilometers, and maintained for the life of the vehicle. This DID shall allow unrestricted reading but shall allow restricted access for writing, based on the vehicle odometer strategy detailed in the “Global A Electrical Architecture Vehicle Odometer Sense & Display” functional requirements specification.

5. Control Packet Identifiers (CPIDs)

CPIDs are divided into several ranges that determine their usage and the ownership of their requirements, as shown in Table 35: CPID Ranges
Definitions for individual CPIDs and their associated device controls are provided in the following sections.

Range	Usage
$00	Reserved – Cancel Device Controls
$01 to $7F	Application Specific CPIDs
$80 to $FE	Corporate Standard CPIDs
$FF	Reserved

Table 35: CPID Ranges

5.1 Device Control Limit Exceeded Return Codes

The values supported for device control limit exceeded return codes are specified in the Device Limit Exceeded Document found on the Service and Parts Operations web page (http://spo.gm.com/dept/eng/nade/general/index.html) The definitions of the individual device controls in the following sections specify the set of device control limits to be applied for service and dynamic vehicle test (DVT). The service device control limits shall be applied unless they have been bypassed via the SecurityAccess ($27) diagnostic service, in which case the DVT device control limits shall be applied.

5.1.1 Device Control Timer Time-Out

Some device controls support a device control timer time-out as one of their device control limits (return code $0001). The behavior of a timer used to support this device control limit is illustrated in Figure 1. The device control may only be enabled continuously for a limited amount of time (typically 30 seconds), after which it must be disabled for a limited duration (typically 60 seconds) before it may be enabled again. Device control limit exceeded return code $0001 is reported when the device control is first disabled and if requests to enable the device control occur during the limited duration that it is disabled.

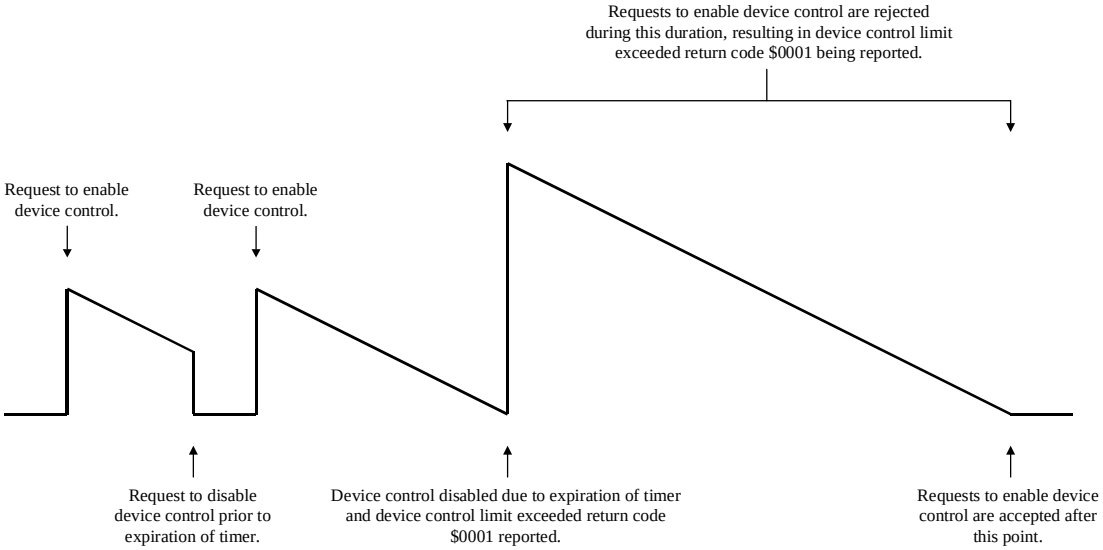

Figure 1: Device Control Timer Example

5.2 CPID $10

CPID $10								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0		Crankshaft Position Variation Learn Trigger	Spark Retard Enable	Disable Idle Spark Enable				
Byte 1	Piston Cooling Oil Control Enable	Piston Cooling Oil Control On					Variable Displacement Oil Pump Enable	Variable Displacement Oil Pump On
Byte 2	Spark Retard Angle							
Byte 3				Displace-ment on Demand Solenoid Enable	Displace-ment on Demand Solenoid A On	Displace-ment on Demand Solenoid B On	Displace-ment on Demand Solenoid C On	Displace-ment on Demand Solenoid D On
Byte 4								

5.2.1 OPCR – Piston Cooling Oil Control Device Control

This device control is used to override the state of the piston cooling oil control output.

Signal	Length (Bits)	Data Type	Range	Conversion
Piston Cooling Oil Control Enable	1	BLN	N/A	$1=True; $0=False
Piston Cooling Oil Control On	1	BLN	N/A	$1=True; $0=False

If (Piston Cooling Oil Control Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The piston cooling oil control output shall be commanded to the value requested in Piston Cooling Oil Control On (on / off).
    EndIf
Else
    Return to normal control.
EndIf

When Engine **is Running**

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$1803	Ignition Voltage Low	Yes	No
$0A5A	Engine Load Out of Range	Yes	No
$0601	Engine Temperature High	Yes	No
$0406	Incorrect Power Mode	Yes	No

When Engine **is Not Running**

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No
$1803	Ignition Voltage Low	Yes	No

5.2 LUBR – Variable Displacement Oil Pump Device Control

This device control is used to override the state of the variable displacement oil pump. When the variable displacement oil pump is turned on, the oil pump will be commanded to a low pressure state.

Signal	Length (Bits)	Data Type	Range	Conversion
Variable Displacement Oil Pump Enable	1	BLN	N/A	$1=True; $0=False
Variable Displacement Oil Pump On	1	BLN	N/A	$1=True; $0=False

If (Variable Displacement Oil Pump Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The variable displacement oil pump shall be commanded to the value requested in Variable Displacement Oil Pump On
    EndIf
Else
    Return to normal control.
EndIf

When Engine **is Running**

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0010	Engine speed out of range.	Yes	No
$0308	Fault Active • Oil Temperature sensor fault • Oil Pressure Sensor fault • PT Relay Fault • Crank Sensor fault • Torque Estimation fault	Yes	No
$030~~C~~D	Engine oil pressure out of range	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0A5A	Engine Load Out of Range	Yes	No

When Engine **is Not Running**

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.2.2 CDAR – Displacement on Demand Solenoid Device Control

This device control is used to override the state of the displacement on demand solenoid outputs. This device control may be used to activate all of the engine's cylinders (all solenoids commanded off), deactivate half of the engine's cylinders (all solenoids commanded on), or deactivate one of the engine's cylinders (exactly one solenoid commanded on). Each of these types of device control has its own set of limit checks, and different limit checks are used when the engine is and is not running for the deactivation of one of the engine's cylinders.

Signal	Length (Bits)	Data Type	Range	Conversion
Displacement on Demand Solenoid Enable	1	BLN	N/A	$1=True; $0=False
Displacement on Demand Solenoid A On	1	BLN	N/A	$1=True; $0=False
Displacement on Demand Solenoid B On	1	BLN	N/A	$1=True; $0=False
Displacement on Demand Solenoid C On	1	BLN	N/A	$1=True; $0=False
Displacement on Demand Solenoid D On	1	BLN	N/A	$1=True; $0=False

```
If (Displacement on Demand Solenoid Enable is True)
    If ((Displacement on Demand Solenoid A On is False) AND
       (Displacement on Demand Solenoid B On is False) AND
       (Displacement on Demand Solenoid C On is False) AND
       (Displacement on Demand Solenoid D On is False))
        All of the engine's cylinders shall be activated by commanding displacement on demand solenoids A,
        B, C, and D off.

    ElseIf ((Displacement on Demand Solenoid A On is True) AND
            (Displacement on Demand Solenoid B On is True) AND
            (Displacement on Demand Solenoid C On is True) AND
            (Displacement on Demand Solenoid D On is True))
        Manage the request to deactivate half of the engine's cylinders as described in Section 5.2.2.1.

    Else
        Clear the indication that the DoD solenoid to command on has been found.
        Clear the indication that too many DoD solenoids are requested to be commanded on.

        If (Displacement on Demand Solenoid A On is True)
            If (the indication that the DoD solenoid to command on has been found is set)
                Set the indication that too many DoD solenoids are requested to be commanded on.
            Else
                Set the DoD solenoid to command on to displacement on demand solenoid A.
                Set the indication that the DoD solenoid to command on has been found.
            EndIf
        EndIf

        If (Displacement on Demand Solenoid B On is True)
            If (the indication that the DoD solenoid to command on has been found is set)
                Set the indication that too many DoD solenoids are requested to be commanded on.
            Else
                Set the DoD solenoid to command on to displacement on demand solenoid B.
                Set the indication that the DoD solenoid to command on has been found.
            EndIf
        EndIf

        If (Displacement on Demand Solenoid C On is True)
            If (the indication that the DoD solenoid to command on has been found is set)
                Set the indication that too many DoD solenoids are requested to be commanded on.
            Else
                Set the DoD solenoid to command on to displacement on demand solenoid C.
                Set the indication that the DoD solenoid to command on has been found.
            EndIf
        EndIf

        If (Displacement on Demand Solenoid D On is True)
            If (the indication that the DoD solenoid to command on has been found is set)
                Set the indication that too many DoD solenoids are requested to be commanded on.
            Else
```

```
                Set the DoD solenoid to command on to displacement on demand solenoid D.
                Set the indication that the DoD solenoid to command on has been found.
            EndIf
        EndIf

        If (the indication that too many DoD solenoids are requested to be commanded on is set)
            Report the appropriate device control limit exceeded return code to the tester.
            Return to normal control.
        Else
            If (the engine is running)
                Manage the request to deactivate one of the engine's cylinders with the engine running as
                described in Section5.2.2.2.
            Else
                Manage the request to deactivate one of the engine's cylinders with the engine not running as
                described in Section5.2.2.3.
            EndIf
        EndIf

    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0312	Invalid displacement on demand device control request.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No

5.2.2.1 Deactivate Half of the Engine's Cylinders

This device control is used to deactivate half of the engine's cylinders by commanding all of the displacement on demand solenoid outputs on.

```
If (any of the device control limits in the table below are exceeded)
    Report the appropriate device control limit exceeded return code to the tester.
    Return to normal control.
Else
    Half of the engine's cylinders shall be deactivated by commanding displacement on demand solenoids A, B, C,
    and D on.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$000C	Maximum deactivation time exceeded.	Yes	Yes
$0010	Engine speed out of range. • Engine speed does not lie between calibratable lower and upper thresholds.	Yes	Yes
$0308	Fault active. • Crank sensor fault. • Cam sensor fault. • Engine oil pressure sensor fault. • Powertrain relay fault. • Intake air temperature sensor fault (service only) (E21240) • Brake vacuum pressure sensor fault (service only). • Displacement on demand solenoid driver fault (service only). • Electronic throttle control reduced engine power active (service only). • Engine coolant temperature sensor fault (service only). • Manifold absolute pressure sensor fault (service only). • Vehicle speed fault (service only). • Variable Displacement Oil Pump Stuck Low	Yes	Yes
$030A	Accelerator pedal position high. • Accelerator pedal effective position exceeds a calibratable threshold.	Yes	No
$030B	Piston protection active.	Yes	No
$030C	Engine oil temperature out of range. • Engine oil temperature does not lie between calibratable lower and upper thresholds.	Yes	No
$030D	Engine oil pressure out of range. • Engine oil pressure does not lie between calibratable lower and upper thresholds.	Yes	Yes
$030F	Engine oil aeration present.	Yes	No
$0310	Engine metal overtemperature protection active.	Yes	No
$0311	Accelerator pedal position increase high. • Change in accelerator pedal effective position exceeds a calibratable threshold.	Yes	No
$0315	Catalyst converter overtemperature protection active.	Yes	No
$0316	Brake vacuum pressure low. • Brake vacuum pressure falls below a calibratable threshold.	Yes	No
$0318	Axle torque limiting active.	Yes	No
$0319	Brake torque management active.	Yes	No
$031D	Engine coolant temperature out of range. • Engine coolant temperature does not lie between calibratable lower and upper thresholds.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0601	Engine coolant temperature too high. • Hot coolant mode active.	Yes	No
$0604	Engine not running.	Yes	No
$0A05	Engine speed too high. • Engine overspeed protection active.	Yes	Yes
$0E02	Vehicle speed not zero.	Yes	No
$2102	Transmission gear incorrect. • Disabled in transmission gears selected via calibration.	Yes	No
$2103	Transmission range incorrect. • Disabled in transmission ranges selected via calibration.	Yes	No
$5003	Ignition voltage out of range. • Powertrain relay voltage does not lie between calibratable lower and upper thresholds.	Yes	No

5.2.2.2 Deactivate One of the Engine's Cylinders with Engine Running

This device control is used to deactivate one of the engine's cylinders with the engine running by commanding the appropriate displacement on demand solenoid output on.

If (any of the device control limits in the table below are exceeded)

Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.

Else

Exactly one of the engine's cylinders shall be deactivated by commanding the appropriate displacement on demand solenoid on (while commanding all remaining displacement on demand solenoids off).

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$000C	Maximum deactivation time exceeded.	Yes	Yes
$0010	Engine speed out of range. • Engine speed does not lie between calibratable lower and upper thresholds.	Yes	Yes
$0308	Fault active. • Crank sensor fault. • Cam sensor fault. • Engine oil pressure sensor fault. • Powertrain relay fault. • Intake air temperature sensor fault (E21240) • Brake vacuum pressure sensor fault. • Displacement on demand solenoid driver fault. • Electronic throttle control reduced engine power active. • Engine coolant temperature sensor fault. • Manifold absolute pressure sensor fault. • Vehicle speed fault.	Yes	Yes
$030A	Accelerator pedal position high. • Accelerator pedal effective position exceeds a calibratable threshold.	Yes	Yes
$030B	Piston protection active.	Yes	Yes
$030C	Engine oil temperature out of range. • Engine oil temperature does not lie between calibratable lower and upper thresholds.	Yes	Yes
$030D	Engine oil pressure out of range. • Engine oil pressure does not lie between calibratable lower and upper thresholds.	Yes	Yes
$030F	Engine oil aeration present.	Yes	Yes
$0310	Engine metal overtemperature protection active.	Yes	Yes
$0311	Accelerator pedal position increase high. • Change in accelerator pedal effective position exceeds a calibratable threshold.	Yes	Yes
$0315	Catalyst converter overtemperature protection active.	Yes	Yes
$0316	Brake vacuum pressure low. • Brake vacuum pressure falls below a calibratable threshold.	Yes	Yes
$0318	Axle torque limiting active.	Yes	Yes
$0319	Brake torque management active.	Yes	Yes
$031D	Engine coolant temperature out of range. • Engine coolant temperature does not lie between calibratable lower and upper thresholds.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No
$0601	Engine coolant temperature too high. • Hot coolant mode active.	Yes	Yes
$0604	Engine not running.	Yes	Yes
$0A05	Engine speed too high. • Engine overspeed protection active.	Yes	Yes
$0E02	Vehicle speed not zero.	Yes	Yes
$2102	Transmission gear incorrect. • Disabled in transmission gears selected via calibration.	Yes	Yes
$2103	Transmission range incorrect. • Disabled in transmission ranges selected via calibration.	Yes	Yes
$5003	Ignition voltage out of range. • Powertrain relay voltage does not lie between calibratable lower and upper thresholds.	Yes	Yes

5.2.2.3 Deactivate One of the Engine's Cylinders with Engine Not Running

This device control is used to deactivate one of the engine's cylinders with the engine not running by commanding the appropriate displacement on demand solenoid output on.

If (any of the device control limits in the table below are exceeded)
    Report the appropriate device control limit exceeded return code to the tester.
    Return to normal control.
Else
    Exactly one of the engine's cylinders shall be deactivated by commanding the appropriate displacement on demand solenoid on (while commanding all remaining displacement on demand solenoids off).
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine speed not zero.	Yes	Yes
$0308	Fault active. • Powertrain relay fault.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No
$0E02	Vehicle speed not zero.	Yes	Yes
$0E03	Transmission not in Park or Neutral.	Yes	Yes
$5003	Ignition voltage out of range. • Powertrain relay voltage does not lie between calibratable lower and upper thresholds.	Yes	Yes

5.2.3 SPKR – Spark Retard Device Control

This device control is used to retard the spark angle from the software calculated spark angle. For example, if the software calculated spark angle was 15 degrees advanced and this device control requested a 3 degree retard, the commanded spark angle that would be delivered would be 12 degrees advanced.

Signal	Length (Bits)	Data Type	Range	Conversion
Spark Retard Enable	1	BLN	N/A	$1=True; $0=False
Spark Retard Angle	8	UNM	0.0 - 89.6484375 deg	E = N * 0.3515625

If (Spark Retard Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The spark angle delivered shall be adjusted by the requested spark retard angle in Spark Retard Angle.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0001	Generic Device Control Timer (30 second enable, 60 second disable) (This limit only applies while the engine is off-idle)	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0701	Requested spark retard is too large	Yes	No

5.2.4 SPKR – Disable Idle Spark Device Control

This device control is used to disable the portion of the spark control algorithm that adjusts spark angle to correct the difference between commanded idle engine speed and actual idle engine speed.

Signal	Length (Bits)	Data Type	Range	Conversion
Disable Idle Spark Enable	1	BLN	N/A	$1=True; $0=False

```
If (Disable Idle Spark Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        Disable Idle Spark
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.2.5 CASR – Crankshaft Position Variation Learn Trigger Device Control

This device control is used to activate the crankshaft position variation learn (CASE) algorithm which calculates the reference pulse correction factors for the misfire diagnostic.

Signal	Length (Bits)	Data Type	Range	Conversion
Crankshaft Position Variation Learn Trigger	1	BLN	N/A	$1=True; $0=False

```
If (Crankshaft Position Variation Learn Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        If  (the crankcase position variation learn algorithm has run and is complete)
        Then
            Return to normal control.
        Else
            The crankcase position variation learn algorithm shall be enabled
        Endif
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description (Non Two-mode hybrid vehicles)	Active for Service	Active for DVT
$0004	Device Control Not Available	Yes	No
$031D	Coolant temperature is not in acceptable range	Yes	Yes
$0401	Vehicle Must Not Be Moving	Yes	Yes
$0406	Incorrect Power Mode	Yes	No
$0903	The vehicle is not in park or neutral	Yes	Yes
$1102	A diagnostic fault is currently active • Crank sensor faults • Cam sensor faults • 5v reference faults • Fan speed faults • Coolant temp sensor fault • Trans engaged fault • Manual trans TOSS fault	Yes	Yes
$1104	Too many failed attempts to learn CASE this power up	Yes	Yes
$1105	CASE already learned this power up	Yes	Yes
$1106	CASE not supported in this controller	Yes	Yes
$1108	PTO is enabled	Yes	Yes
$500A	Propulsion Not Active	Yes	No

Return Code	Device Control Limit Exceeded Description (Two-mode hybrid vehicles)	Active for Service	Active for DVT
$0004	Device Control Not Available	Yes	No
$031D	Coolant temperature is not in acceptable range	Yes	Yes
$0401	Vehicle Must Not Be Moving	Yes	Yes
$0406	Incorrect Power Mode	Yes	No
$0903	The vehicle is not in park or neutral	Yes	Yes
$1102	A diagnostic fault is currently active • Hood switch faults • HCP communication faults • Crank sensor faults • Cam sensor faults • 5v reference faults • Fan speed faults • Coolant temp sensor fault • Trans engaged fault • Manual trans TOSS fault • Brake fault	Yes	Yes
$1104	Too many failed attempts to learn CASE this power up	Yes	Yes
$1105	CASE already learned this power up	Yes	Yes
$1106	CASE not supported in this controller	Yes	Yes
$1107	Brake pedal is not being applied	Yes	No
$1108	PTO is enabled	Yes	Yes
$500A	Propulsion Not Active	Yes	No

5.3 CPID $14

CPID $14								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Throttle Position Enable	Idle Engine Speed Enable	Learned Booster Hardware Adaption Value Reset Trigger	Learned Airflow Adapts / Torque Learn Reset Trigger	~~Electronic Throttle Control Snapshot Trigger~~	Electronic Throttle Control Validation Parameter Reset Trigger	Forced Idle Enable	Wide Open Throttle Self Test Enable
Byte 1	Throttle Position							
Byte 2	Idle Engine Speed							
Byte 3	Super-charger Boost Solenoid Enable		Intake Manifold Runner Control Enable Bank 1	Intake Manifold Runner Control Closed Bank 1	Intake Manifold Tuning Valve Enable	Intake Manifold Tuning Valve Closed	Intercooler Pump Enable	Intercooler Pump On
Byte 4	Supercharger Boost Solenoid Duty Cycle							

5.3.1 TPSR – Forced Idle / Wide Open Throttle Self Test Device Control

The forced idle device control is used to prevent the accelerator pedal contribution to throttle area from being added to the idle contribution to throttle area, effectively forcing idle. This device control allows the accelerator pedal to be moved to its maximum position during DVT, in order to learn the maximum accelerator pedal position, without affecting engine operation.

The wide open throttle (WOT) self test device control is used to limit the duration of time that the throttle spends at WOT while the accelerator pedal is being held at its maximum position. This device control allows the throttle to be moved to WOT during DVT, in order to learn the maximum throttle position, while minimally affecting engine operation. This is accomplished by preventing the accelerator pedal contribution to throttle area from being added to the idle contribution to throttle area, effectively forcing idle, once WOT is achieved (i.e., the throttle position exceeds a WOT threshold) or once the time allowed to achieve WOT has expired.

When terminating these device controls, the learned minimum accelerator pedal position must be reset to 100% in order to prevent a sudden increase in the accelerator pedal contribution to throttle area.

Signal	Length (Bits)	Data Type	Range	Conversion
Forced Idle Enable	1	BLN	N/A	$1=True; $0=False
Wide Open Throttle Self Test Enable	1	BLN	N/A	$1=True; $0=False

```
If ((Forced Idle Enable is True) OR (Wide Open Throttle Self Test Enable is True))
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        If (the indication that this device control is enabled is set)
            Terminate this device control (as described below).
        Else
            Return to normal control.
        EndIf
    Else
        If ((the indication that this device control is enabled is not set) AND
           (engine speed is at idle) AND (accelerator pedal position is at idle))
            Set the indication that this device control is enabled.
            Clear the indication that the WOT self test device control failed.
            Clear the indication that this device control is complete.
        EndIf
        If (the indication that this device control is enabled is set)
            If (Forced Idle Enable is True)
```

```
                Prevent the accelerator pedal contribution to throttle area from being added to the idle contribution
                to throttle area.
            ElseIf (Wide Open Throttle Self Test Enable is True)
                If (throttle position exceeds a calibratable threshold)
                    Terminate this device control (as described below).
                ElseIf (the WOT self test device control has been enabled longer than a calibratable duration)
                    Set the indication that the WOT self test device control failed.
                    Terminate this device control (as described below).
                Else
                    Allow the accelerator pedal contribution to throttle area to be added to the idle contribution to
                    throttle area.
                EndIf
            EndIf
        Else
            Continue normal control.
        EndIf
    EndIf
Else
    If (the indication that this device control is enabled is set)
        Terminate this device control (as described below).
    Else
        Return to normal control.
    EndIf
EndIf
```

The following actions shall be performed upon the termination of this device control:

Prevent the accelerator pedal contribution to throttle area from being added to the idle contribution to throttle area.
Reset the learned minimum accelerator pedal position to 100%.
Set the indication that this device control is complete.
Clear the indication that this device control is enabled.

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0004	Invalid device control selected (used if trying to access a device control in service which is only allowed for assembly line testing).	Yes	No
$0401	Vehicle speed not zero	No	Yes
$0406	Incorrect Power Mode	Yes	No
$0E03	Transmission not in Park or Neutral.	No	Yes
$1603	Failure detected within the throttle control system.	No	Yes

5.3.2 VIMR – Intake Manifold Tuning Valve Device Control

This device control is used to override the state of the intake manifold tuning valve output. A calibration is used to select whether commanding the intake manifold tuning valve output on or off results in the valve being commanded closed or open for a particular application. The intake manifold tuning valve is commanded closed in order to prevent the manifold runners from communicating. (Intake Manifold Tuning (IMT) Valve is the SAE recommended term for any device that controls intake manifold tuning. These devices may be unofficially referred to as long/short runner control or intake manifold communication control.)

Signal	Length (Bits)	Data Type	Range	Conversion
Intake Manifold Tuning Valve Enable	1	BLN	N/A	$1=True; $0=False
Intake Manifold Tuning Valve Closed	1	BLN	N/A	$1=True; $0=False

If (Intake Manifold Tuning Valve Enable is True)

If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
The intake manifold tuning valve output shall be commanded to the value required to result in the position requested in Intake Manifold Tuning Valve Closed.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.3.3 TPSR – Throttle Position Device Control

This device control is used to override the commanded throttle position.

Signal	Length (Bits)	Data Type	Range	Conversion
Throttle Position Enable	1	BLN	N/A	$1=True; $0=False
Throttle Position	8	UNM	0.0 - 99.609375 %	E = N * 0.390625

If (Throttle Position Enable is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
The throttle position shall be commanded to the position requested in Throttle Position.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No
$1601	Engine speed or vehicle speed too high	Yes	Yes

5.3.4 IDLR - Idle Engine Speed Device Control

This device control is used to override the idle engine speed.

Signal	Length (Bits)	Data Type	Range	Conversion
Idle Engine Speed Enable	1	BLN	N/A	$1=True; $0=False
Idle Engine Speed	8	UNM	0.0 - 2040.0 rpm	E = N * 8.0

If (Idle Engine Speed Enable is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else

The idle engine speed shall be commanded to the speed requested in Idle Engine Speed.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0401	Vehicle speed not zero.	Yes	No
$0402	Vehicle speed fault.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0903	Transmission not in park or neutral	Yes	No
$2B00	Transmission gear data is not valid (auto or manual trans)	Yes	No

5.3.5 TPSR/TLNR – Learned Airflow Adapts / Torque Learn Reset Trigger Device Control

This device control is used to reset the learned airflow adaptive values to their initial values.

Signal	Length (Bits)	Data Type	Range	Conversion
Learned Airflow Adapts / Torque Learn Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Learned Airflow Adapts / Torque Learn Reset Trigger is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
ElseIf (the learned airflow adaptive values and torque learn variables are already reset)
Return to normal control.
Else
Reset the learned airflow adaptive values and torque learn variables.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine Speed Not Zero	Yes	No
$0401	Vehicle Speed Not Zero	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.3.6 TPSR - Electronic Throttle Control Validation Parameter Reset Trigger Device Control

This device control is used to trigger a reset of the ETC long term measurement and statistical parameters typically monitored during development and validation.

Signal	Length (Bits)	Data Type	Range	Conversion
Electronic Throttle Control Validation Parameter Reset Trigger	1	BLN	N/A	$1=True; $0=False

```
If (Electronic Throttle Control Validation Parameter Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    ElseIf (the reset is complete)
        Return to normal control.
    Else
        Perform the reset of the ETC validation parameters.
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.3.7 PDAR – Intake Manifold Runner Control Device Control

This device control is used to override the state of the intake manifold runner control output. Commanding the intake manifold runner control output on results in the valve being commanded closed. The intake manifold runner control is commanded closed in order to partially block the intake air traveling to each cylinder. (Intake Manifold Runner Control (IMRC) is the SAE recommended term for any device that controls charge motion. These devices may be unofficially referred to as swirl control, charge motion control, or port deactivation.)

Signal	Length (Bits)	Data Type	Range	Conversion
Intake Manifold Runner Control Enable Bank 1	1	BLN	N/A	$1=True; $0=False
Intake Manifold Runner Control Closed Bank 1	1	BLN	N/A	$1=True; $0=False

```
If (Intake Manifold Runner Control Enable Bank 1 is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Intake Manifold Runner Control output shall be commanded to the value required to result in the
        position requested in Intake Manifold Runner Control Closed Bank 1.
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.4 CPID $17

CPID $17								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Canister Vent Diurnal Control Valve Enable	Canister Vent Diurnal Control Valve Open	ELCP Vacuum Pump Enable	ELCP Vacuum Pump On	ELCP Switching Valve Enable	ELCP Switching Valve On		
Byte 1								
Byte 2								
Byte 3								
Byte 4								

5.4.1 EVPR – Canister Vent Diurnal Control Valve Device Control

This device control is used to override the state of the canister vent diurnal control valve (DCV) output.

Signal	Length (Bits)	Data Type	Range	Conversion
Canister Vent Diurnal Control Valve Enable	1	BLN	N/A	$1=True; $0=False
Canister Vent Diurnal Control Valve Open	1	BLN	N/A	$1=True; $0=False

If (Canister Vent Diurnal Control Valve Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The diurnal control valve shall be commanded to the requested state in Canister Vent Diurnal Control Valve Open.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.4.2 EVPR – ELCP Vacuum Pump Device Control

This device control is used to override the state of the EVAP Leak Check Pump (ELCP) vacuum pump output.

Signal	Length (Bits)	Data Type	Range	Conversion
ELCP Vacuum Pump Enable	1	BLN	N/A	$1=True; $0=False
ELCP Vacuum Pump On	1	BLN	N/A	$1=True; $0=False

If (ELCP Vacuum Pump Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The vacuum pump shall be commanded to the requested state in ELCP Vacuum Pump On.
    EndIf
Else
    Return to normal control.

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0308	DTC Present • Temperature sensor faults (OAT or IAT or other inputs into ambient estimate)	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5009 $1001	Ambient Air Temperature Out of Range Don't operate the pump below 32 °F (0 °C) EVAP System Vacuum Too High	Yes Yes	No Yes

5.4.3 EVPR – ELCP Switching Valve Device Control

This device control is used to override the state of the EVAP Leak Check Pump (ELCP) switching valve output. (OFF = normal state = diverting to atmosphere; ON = diverting to pump)

Signal	Length (Bits)	Data Type	Range	Conversion
ELCP Switching Valve Enable	1	BLN	N/A	$1=True; $0=False
ELCP Switching Valve On	1	BLN	N/A	$1=True; $0=False

If (ELCP Switching Valve Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The switching valve shall be commanded to the requested state in ELCP Switching Valve On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0308	DTC Present • Temperature sensor faults (OAT or IAT or other inputs into ambient estimate)	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5009 $1001	Ambient Air Temperature Out of Range • Don't operate the valve below 32 °F (0 °C) EVAP System Vacuum Too High	Yes Yes	No Yes

5.5 CPID $18

CPID $18								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Fuel Injector Enable	Fuel Injector On	Fuel Closed Loop Enable	Fuel Closed Loop	Fuel Pump On	Fuel Trim Reset Trigger	Fuel Trim Learn Enable	Fuel Trim Learn On
Byte 1	Fuel Injector Identifier							
Byte 2	Canister Purge Enable	Evap Service Test Trigger	Canister Vent Enable	Canister Vent Open	Fuel Pump Enable	Fuel Comp Reset Trigger	Secondary Fuel Pump Enable	Secondary Fuel Pump On
Byte 3	Canister Purge Duty Cycle							
Byte 4					Fuel Pump Prime Suspend			

5.5.1 FULR – Fuel Injector Device Control

This device control is used to override the state of the fuel injector outputs. This device control may be used to trigger a single injector to be commanded on for a short duration in order to test the flow of the injector (port fuel injection engines), to trigger an engine-running pressure drop test (direct fuel injection engines), or to disable injectors in order to cut off fuel to the associated cylinders. Each of these types of device control has its own set of limit checks.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Injector Enable	1	BLN	N/A	$1=True; $0=False
Fuel Injector On	1	BLN	N/A	$1=True; $0=False
Fuel Injector Identifier	8	PKT	N/A	N/A
Cylinder A (most significant bit)	1	BLN	N/A	$1=True; $0=False
Cylinder B	1	BLN	N/A	$1=True; $0=False
Cylinder C	1	BLN	N/A	$1=True; $0=False
Cylinder D	1	BLN	N/A	$1=True; $0=False
Cylinder E	1	BLN	N/A	$1=True; $0=False
Cylinder F	1	BLN	N/A	$1=True; $0=False
Cylinder G	1	BLN	N/A	$1=True; $0=False
Cylinder H (least significant bit)	1	BLN	N/A	$1=True; $0=False

```
If (Fuel Injector Enable is True)
    If (Fuel Injector On is True)
        Manage the fuel injector flow device control as described in Section5.5.1.1 and Section5.5.1.2.
    Else
        Manage the fuel injector disable device control as described in Section5.5.1.3.
    EndIf
Else
    Return to normal control.
EndIf
```

5.5.1.1 Fuel Injector Flow Device Control (Port Fuel Injected Engines)

This device control is used to trigger a single injector to be commanded on for a duration of 500 ms (via approximately 100, 5 ms pulses) in order to test the flow of the injector. Note: The Fuel Injector Flow Device Control applies to Port Fuel Injected (PFI) applications. It does not apply for high pressure Direct Injection (DI) applications.

```
If (any of the device control limits in the table below are exceeded)
    Report the appropriate device control limit exceeded return code to the tester.
    Return to normal control.
Else
    If (fuel injector output specified by the Fuel Injector Identifier has been commanded on for at least 500 ms)
        Return to normal control.
    Else
        The fuel injector output specified by the Fuel Injector Identifier shall be commanded on.
    EndIf
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Output Control not allowed while engine is Running.	Yes	No
$0202	Injector already tested. Unable to restart test during this engine's off cycle.	Yes	No
$0203	Invalid injector number. Check for proper vehicle selection.	Yes	Yes
$0204	Output control not allowed. Injector flow test in progress.	Yes	No
$0205	More than one injector selected.	Yes	No
$0206	Fuel enable not set for theft deterrent system	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No

5.5.1.2 Fuel Injector Flow Device Control (Direct Fuel Injected Engines)

This device control actuates an engine running fuel injector flow test for direct injected (DI) engines. The purpose of the injector flow test is to determine if the requested injector is restricted or plugged. The injector flow test algorithm is completely contained in the engine control module and returns a fuel pressure drop value through DID $6C “Direct Injection Fuel Injector Flow Test Results” when the test is complete. The injector flow test is initiated via request from an offboard tool. The offboard tool sends two pieces of data to the engine control module – the request to enable the injector flow test and the ID of the injector that should be tested.

Once the injector flow test is enabled, the desired fuel rail pressure is commanded to a calibrated fuel pressure. After the fuel rail pressure achieves a stabilized value, the Start Of Inject (SOI) position, DI Scheduling ID event position and commanded pulse width for the injector to be tested are set to calibrated values. The SOI position and DI Scheduling ID event are set to calibrated values in an effort to center the delivered fuel pulse between DI Scheduling ID events so that the captured fuel rail pressure at the next DI ID will correspond to the injector that is being tested. In addition, the high pressure fuel pump is disabled after the fuel rail pressure achieves a stabilized value so that the high pressure fuel pump closed loop control will not attempt to increase the fuel pressure during the test.

The injector flow test algorithm is designed to average the pressure drop over a calibratable number of engine cycles for each injector. A larger pulse width will be commanded on the injector that is requested to be tested. Since a larger pulse width is commanded on the test injector, the observed fuel pressure drop on this injector should be larger than other injectors on the engine. The average pressure drop is reported in DID $6C such that it can be compared with the pressure drops for the remaining engine cylinders during the same key cycle. A determination of injector performance can be made by comparing the DID $6C values from cylinder to cylinder.

```
If (any of the device control limits in the table below are exceeded)
    Report the appropriate device control limit exceeded return code to the tester.
    Return to normal control.
Else
    Trigger the fuel injector flow test for the injector specified by the Fuel Injector Identifier
EndIf
```

Return Code	Device Control Limit Exceeded Description Test Enable Conditions	Active for Service	Active for DVT
$0004	Output control unavailable	Yes	No
$000B	Displacement on Demand Active	Yes	No
$0010	RPM Out of Range	Yes	No
$0203	Invalid injector number. Check for proper vehicle selection.	Yes	No
$0204	Output control not allowed. Injector flow test in progress.	Yes	No
$0205	More than one injector selected	Yes	No
$0206	Fuel enable not set for theft deterrent system	Yes	No
$0308	Output control not allowed, DTC Present. • Fuel injector faults • Fuel pressure sensor faults • High pressure fuel pump fault • Low pressure fuel pump fault • Transmission range / clutch switch fault • Vehicle speed sensor fault	Yes	No
$0401	Vehicle Speed Has Been Detected	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$0903	Transmission must be in Park or Neutral	Yes	No
$0A29	Fuel Pressure Out of Range	Yes	No

5.5.1.3 Fuel Injector Disable Device Control

This device control is used to disable injectors in order to cut off fuel to the associated cylinders. This device control applies to both Port Fuel Injected (PFI) engines and Direct Injected (DI) engines.

If (any of the device control limits in the table below are exceeded)
    Report the appropriate device control limit exceeded return code to the tester.
    Return to normal control.
Else
    The fuel injector output specified by the Fuel Injector Identifier shall be commanded off.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0001	Generic device control timer time-out active (30 second enable, 60 second disable).	Yes	No
$000B	Displacement on demand active.	Yes	Yes
$0301	Injector not valid.	Yes	Yes
$0302	Too many injectors to be disabled simultaneously.	Yes	No
$0303	Injector flow test in progress.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No

5.5.2 FULR – Fuel Closed Loop Device Control

This device control is used to enable or disable closed loop fuel.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Closed Loop Enable	1	BLN	N/A	$1=True; $0=False
Fuel Closed Loop	1	BLN	N/A	$1=True; $0=False

If (Fuel Closed Loop Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else

Closed loop fuel shall be commanded to the requested state in Fuel Closed Loop.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.5.3 FADR – Fuel Trim Learn Device Control

This device control is used to enable or disable fuel trim learning.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Trim Learn Enable	1	BLN	N/A	$1=True; $0=False
Fuel Trim Learn On	1	BLN	N/A	$1=True; $0=False

If (Fuel Trim Learn Enable is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
The fuel trim learn shall be commanded to the requested state in Fuel Trim Learn On
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.5.4 FADR – Fuel Trim Reset Trigger Device Control

This device control is used to reset the learned fuel trim values to their initial values.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Trim Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Fuel Trim Reset Trigger is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
ElseIf (the learned fuel trim values are reset)
Return to normal control.
Else
Reset the learned fuel trim values
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.5.5 EVPR – Canister Purge Device Control

This device control is used to verify functionality of the fuel vapor purge system by allowing the Canister Purge output to be commanded to a specific duty cycle. The software calculated duty cycle should be available to be monitored via a parameter identifier as feedback for the device control override.

Signal	Length (Bits)	Data Type	Range	Conversion
Canister Purge Enable	1	BLN	N/A	$1=True; $0=False
Canister Purge Duty Cycle	8	UNM	0.0 to 99.609375 %	E = N * 100/256

If (Canister Purge Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The canister purge duty cycle shall be commanded to the requested duty cycle in Canister Purge Duty Cycle.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No
$0F01	Vacuum Level too high in purge system	Yes	Yes
$0F02	Failure detected in purge system	Yes	Yes
$0F03	EVAP system test in progress	Yes	No

5.5.6 EVPR – Canister Vent Device Control

This device control is used to either open or close the canister vent.

Signal	Length (Bits)	Data Type	Range	Conversion
Canister Vent Enable	1	BLN	N/A	$1=True; $0=False
Canister Vent Open	1	BLN	N/A	$1=True; $0=False

If (Canister Vent Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The canister vent shall be commanded to the requested state in Canister Vent Open
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0F03	EVAP system test in progress	Yes	No
$1001	Vacuum Level too high in purge system	Yes	Yes
$1002	Failure detected in purge system	Yes	Yes
$0406	Incorrect Power Mode	Yes	No

5.5.7 EVPR – Evap Service Test Trigger Device Control

This device control is used to activate the evaporative emissions service bay test.

Signal	Length (Bits)	Data Type	Range	Conversion
Evap Service Test Trigger	1	BLN	N/A	$1=True; $0=False

```
If (Evap Service Test Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        If (the Evap Service Bay Test is complete)
            Return to normal control.
        Else
            The Evap Service Bay Test shall be enabled
        EndIf
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No
$1301	The service bay test is not available in the controller.	Yes	Yes
$1302	The Evap IM flag is already set.	Yes	Yes
$1303	The engine is running.	Yes	Yes
$1304	The coolant temperature is too high.	Yes	Yes
$1305	Other application specific enable criteria has not been met.	Yes	Yes

5.5.8 FPMR – Fuel Pump Device Control

This device control is used to override the state of the fuel pump output. The device control is a service diagnostic aid that is used to determine if the fuel pump system is functioning properly. This device control is used to operate a relay controlled fuel pump. Different limit checks are used when the engine is and is not running for the operation of the fuel pump.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Pump Enable	1	BLN	N/A	$1=True; $0=False
Fuel Pump On	1	BLN	N/A	$1=True; $0=False
Fuel Pump Prime Suspend	1	BLN	N/A	$1=True; $0=False

5.5.8.1 Fuel Pump Device Control with Engine Not Running

With the engine off, this device control is intended to be used as a discrete control for circuit or pressure diagnosis. The fuel pump cannot be commanded on for more than 3 seconds in service with the engine off.

```
If  (Fuel Pump Enable is True)
    If  (the engine is running)
        Manage the request to control the Fuel Pump as described in "Fuel Pump Device Control with
        Engine Running"
    Else
        If  (the green engine fuel priming routine has not completed)
            Manage the request to control the Fuel Pump as described in "Fuel Pump Device Control with
            Green Engine"
        Else
            If  (any of the device control limits in the table below are exceeded)
                Report the appropriate device control limit exceeded return code to the tester.
                Return to normal control.
            Else
                The fuel pump output shall be commanded to the value requested in Fuel Pump On.
            EndIf
        EndIf
    EndIF
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	The Output Control timer has expired. Please Wait. The fuel pump must not be commanded on (Duty Cycle > 0.0) continuously for more than 3 seconds with engine not running	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range System voltage is above or below a calibrated value	Yes	No

5.5.8.2 Fuel Pump Device Control with Green Engine

When this device control is run on a green engine, this device control will initiate a fuel prime routine which will turn on the fuel pump for a calibratible amount of time. This process will only be activated if the fuel pump has never been primed before and the engine has never been running. There must also be adequate fuel in the tank, or the fuel level data from the tank is failed in which case it is assumed there is adequate fuel in the tank, to allow the prime to occur. The plant EOL green engine fuel pump prime can be suspended to perform additional testing requiring limited current draw. The fuel pump prime can be resumed by initiating the fuel prime routine again.

```
If  (Fuel Pump Enable is True)
    If (Fuel Pump Prime Suspend is True)
        The green engine fuel pump prime routine shall be stopped.
    Else
        The fuel pump output shall be commanded to the value requested in Fuel Pump On.
    EndIf
Else
    Return to normal control.
EndIf
```

5.5.8.3 Fuel Pump Device Control with Engine Not Running

With the engine off, this device control is intended to be used as a discrete control for circuit or pressure diagnosis. With the engine off, the Fuel Pump can not be commanded on for more than 3 seconds.

Note: When this device control is run on a green engine, this device control will initiate a fuel prime routine which will turn on the fuel pump for a calibratible amount of time. This process will only be activated if the fuel pump has never been primed before and the engine has never been running. There must also be adequate fuel in the tank, or the fuel level data from the tank is failed in which case it is assumed there is adequate fuel in the tank, to allow the prime to occur.

```
If   (Fuel Pump Enable is True)
     If   (the engine is running)
          Manage the request to control the Fuel Pump as described in "Fuel Pump Device Control with Engine
     Running"
     Else
          If   (any of the device control limits in the table below are exceeded)
               Report the appropriate device control limit exceeded return code to the tester.
               Return to normal control.
          Else
               The fuel pump output shall be commanded to the value requested in Fuel Pump On.
          EndIf
     EndIf
Else
     Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	The Output Control timer has expired. Please Wait. The fuel pump must not be commanded on (Duty Cycle > 0.0) continuously for more than 3 seconds with engine not running	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range System voltage is above or below a calibrated value	Yes	No

5.5.8.4 Fuel Pump Device Control with Engine Running

This device control is used to operate the fuel pump with the engine running for fuel system pressure and noise diagnosis/troubleshooting. With the engine running, the Fuel Pump cannot be commanded on for more than 30 seconds.

Note: When the low pressure fuel pump is used in conjunction with a separate high pressure fuel pump for Direct Injection (DI) engines, this device control can be used to reduce the high fuel pressure to a low working level. The high fuel pressure control shall monitor the low pressure "Fuel Pump Device Control with Engine Running" such that when the low pressure pump is commanded "off", the high pressure pump will be controlled to a low pressure value. This combination shall be held until either the FPMR device control is released, or the engine stalls.

Note: Plant EOL fuel priming is not allowed while the engine is running in service.

```
If (Fuel Pump Enable is True)
     If (any of the device control limits in the table below are exceeded)
          Report the appropriate device control limit exceeded return code to the tester.
          Return to normal control.
     Else
          The fuel pump output shall be commanded to the value requested in Fuel Pump On.
     EndIf
```

Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	The Output Control timer has expired. Please Wait. • The fuel pump must not be commanded on (Duty Cycle > 0.0) continuously for more than 30 seconds with engine running	Yes	No
$0401	Vehicle speed detected • Vehicle must not be moving	Yes	No
$0402	Vehicle speed DTC present, output control not allowed • DTCs for VSS or TOSS	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • System voltage is above or below a calibrated value	Yes	No

5.5.9 E85R – Fuel Composition Reset Trigger Device Control

This device control is used to reset the learned ethanol content of the fuel.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Comp Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Fuel Comp Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        If (Ethanol Learn is in progress).
            Disable Ethanol Learn.
        EndIf
            If (the reset is complete)
                Return to normal control.
            Else
                Perform the reset of the Fuel Composition parameters.
            EndIf
    EndIf
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0004	Invalid device control selected	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.5.10 FLVR – Secondary Fuel Pump Device Control

This device control is used to override the state of the electric secondary transfer pump output.

Signal	Length (Bits)	Data Type	Range	Conversion
Secondary Fuel Pump Enable	1	BLN	N/A	$1=True; $0=False
Secondary Fuel Pump On	1	BLN	N/A	$1=True; $0=False

If (secondary fuel pump enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Secondary Fuel Pump shall be commanded to the state indicated by Secondary Fuel Pump On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0401	Vehicle speed detected • Vehicle must not be moving	Yes	No
$0403	Fuel pump: time limit exceeded.	Yes	No
$0406	Incorrect power mode.	Yes	No
$0407	Vehicle not configured with an electric secondary transfer fuel pump	Yes	No

5.6 CPID $19

CPID $19								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Cold Start Fuel Reservoir Pump Enable	Cold Start Fuel Reservoir Pump On	Cold Start Fuel Injector Enable	Cold Start Fuel Injector On	Auxiliary Cold Start System Enable	Auxiliary Cold Start System On	SQA Map Reset Trigger	Fuel Setpoint Adaption Learning Adaptive Map Reset Trigger
Byte 1								
Byte 2					Alternative Fuel Mode Indicator Enable	Alternative Fuel Mode Indicator On	Alternative Fuel Mode Control Enable	Alternative Fuel Command FICM Mode
Byte 3			Alternative Fuel Heater 3 Enable	Alternative Fuel Heater 3 On	Alternative Fuel Heater 2 Enable	Alternative Fuel Heater 2 On	Alternative Fuel Heater 1 Enable	Alternative Fuel Heater 1 On
Byte 4	Alternative Fuel Pump Enable	Alternative Fuel Pump On			Alternative Fuel Shutoff Valve 2 Enable	Alternative Fuel Shutoff Valve 1 On	Alternative Fuel Shutoff Valve 1 Enable	Alternative Fuel Shutoff Valve 1 On

5.6.1 FMRR – Alternative Fuel Mode Control

This device control is used to override the state of the alternative fuel mode to either FICM or ECM fuel control. The device control is a service and manufacturing diagnostic aid that is used to determine if the alternative fuel system is functioning properly. This device control may be requested simultaneously with other device control requests to the FICM.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Mode Control Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Command FICM Mode	1	BLN	N/A	$1=True; $0=False

```
If (Alternative Fuel Mode Control Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        If (Alternative Fuel Mode Command FICM Mode is True)
                The alternative fuel mode shall be commanded to FICM Mode.
        Else
                The alternative fuel mode shall be commanded to ECM Mode.
        Endif
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output control timer has expired. • Maximum time allowed calibratable	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • System voltage is above or below a calibrated value	Yes	No
$0308	Fault active. • Vehicle speed sensing fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No

5.6.2 FMRR – Alternative Fuel Mode Indicator Control

This device control is used to override the state of the alternative fuel mode indicator. The device control is a service diagnostic aid that is used to determine if the alternative fuel mode indicator is functioning properly. This device control may be requested simultaneously with other device control requests to the FICM.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Mode Indicator Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Mode Indicator On	1	BLN	N/A	$1=True; $0=False

```
If (Alternative Fuel Mode Indicator Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The alternative fuel mode indicator shall be commanded to the value requested in Alternative Fuel Mode
        Indicator On.
    EndIf
Else
    Return to normal control.
```

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.6.3 FISR – Alternative Fuel Shutoff Valve 1 Control

This device control is used to override the state of the alternative fuel shutoff valve 1. The device control is a service and manufacturing diagnostic aid that is used to determine if the alternative fuel shutoff valve is functioning properly. This device control may be requested simultaneously with other device control requests to the FICM.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Shutoff Valve 1 Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Shutoff Valve 1 On	1	BLN	N/A	$1=True; $0=False

If (Alternative Fuel Shutoff Valve 1 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The alternative fuel shutoff valve 1 output shall be commanded to the value requested in Alternative Fuel Shutoff Valve 1 On.
    EndIf
Else
    Return to normal control.

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output control timer has expired. • Maximum time allowed calibratable	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0308	Fault active. • Vehicle speed sensing fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0A7A	Alternative Fuel Devices Disabled	Yes	No

5.6.4 HPSR – Alternative Fuel Shutoff Valve 2 Control

This device control is used to override the state of the alternative fuel shutoff valve 2. The device control is a service and manufacturing diagnostic aid that is used to determine if the alternative fuel shutoff valve is functioning properly. This device control may be requested simultaneously with other device control requests to the FICM.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Shutoff Valve 2 Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Shutoff Valve 2 On	1	BLN	N/A	$1=True; $0=False

If (Alternative Fuel Shutoff Valve 2 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The alternative fuel shutoff valve 2 output shall be commanded to the value requested in Alternative Fuel Shutoff Valve 2 On.
    EndIf
Else
    Return to normal control.

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output control timer has expired. • Maximum time allowed calibratable	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0308	Fault active. • Vehicle speed sensing fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0A7A	Alternative Fuel Devices Disabled	Yes	No

5.6.5 LPPR – Alternative Fuel Pump Control With Engine Not Running

This device control is used to override the state of the alternative fuel pump output. The device control is a service diagnostic aid that is used to determine if the alternative fuel pump system is functioning properly. This device control is used to operate a relay controlled alternative fuel pump. Different limit checks are used when the engine is and is not running for the operation of the alternative fuel pump.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Pump Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Pump On	1	BLN	N/A	$1=True; $0=False

With the engine off, this device control is intended to be used as a discrete control for circuit or pressure diagnosis. With the engine off, the alternative fuel pump cannot be commanded on for more than 3 seconds.

```
If  (Alternative Fuel Pump Enable is True)
    If  (the engine is running)
        Manage the request to control the Alternative Fuel Pump as described in "Alternative Fuel Pump Device
        Control with Engine Running"
    Else
        If  (any of the device control limits in the table below are exceeded)
            Report the appropriate device control limit exceeded return code to the tester.
            Return to normal control.
        Else
            The alternative fuel pump output shall be commanded to the value requested in Alternative Fuel Pump
            On.
        EndIf
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output control timer has expired. • Maximum time allowed calibratable	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0308	Fault active. Vehicle speed sensing fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$5003	Ignition Voltage Out of Range	Yes	No
$0A7A	Alternative Fuel Devices Disabled	Yes	No

5.6.6 LPPR – Alternative Fuel Pump Control With Engine Running

This device control is used to override the state of the alternative fuel pump output. The device control is a service diagnostic aid that is used to determine if the alternative fuel pump system is functioning properly. This device control is used to operate a relay controlled alternative fuel pump. Different limit checks are used when the engine is and is not running for the operation of the alternative fuel pump. When the alternative fuel system is not active and this device control is requested, the 'engine not running' timer limits are applied.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Pump Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Pump On	1	BLN	N/A	$1=True; $0=False

This device control is used to operate the alternative fuel pump with the engine running for fuel system pressure and noise diagnosis/troubleshooting. With the engine running, the Alternative Fuel Pump cannot be commanded on for more than 30 seconds. When the engine is running on the gasoline/petrol fuel system, the engine off device control limits are applied (i.e. 3 seconds).

```
If (Alternative Fuel Pump Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Alternative Fuel pump output shall be commanded to the value requested in Alternative Fuel Pump On.
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output control timer has expired. • Maximum time allowed calibratable	Yes	No
$0308	Fault active. • Vehicle speed sensing fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Ignition Voltage Out of Range	Yes	No

5.6.7 FVHR – Alternative Fuel Heater Single/1 Control

This device control is used to override the state of the alternative fuel heater output 1. The device control is a service and manufacturing diagnostic aid that is used to determine if the alternative fuel heater output is functioning properly. This device control may be requested simultaneously with other device control requests to the FICM.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Heater 1 Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Heater 1 On	1	BLN	N/A	$1=True; $0=False

If (Alternative Fuel Heater 1 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The alternative fuel heater 1 output shall be commanded to the value requested in Alternative Fuel Heater 1 On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output control timer has expired. • Maximum time allowed calibratable	Yes	No
$5003	Ignition Voltage Out of Range	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0308	Fault active. • Vehicle speed sensing fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0A7A	Alternative Fuel Devices Disabled	Yes	No

5.6.8 FVHR – Alternative Fuel Heater 2 Control

This device control is used to override the state of the alternative fuel heater output 2. The device control is a service and manufacturing diagnostic aid that is used to determine if the alternative fuel heater output is functioning properly. This device control may be requested simultaneously with other device control requests to the FICM.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Heater 2 Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Heater 2 On	1	BLN	N/A	$1=True; $0=False

If (Alternative Fuel Heater 2 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The alternative fuel heater 2 output shall be commanded to the value requested in Alternative Fuel Heater 2 On.
    EndIf
Else
    Return to normal control.

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output control timer has expired. • Maximum time allowed calibratable	Yes	No
$5003	Ignition Voltage Out of Range	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0308	Fault active. • Vehicle speed sensing fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0A7A	Alternative Fuel Devices Disabled	Yes	No

5.6.9 FVHR – Alternative Fuel Heater 3 Control

This device control is used to override the state of the alternative fuel heater output 3. The device control is a service and manufacturing diagnostic aid that is used to determine if the alternative fuel heater relay output is functioning properly. This device control may be requested simultaneously with other device control requests to the FICM.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Heater 3 Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Heater 3 On	1	BLN	N/A	$1=True; $0=False

If (Alternative Fuel Heater 1 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The alternative fuel heater 3 output shall be commanded to the value requested in Alternative Fuel Heater 3 On.
    EndIf
Else
    Return to normal control.

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output control timer has expired. • Maximum time allowed calibratable	Yes	No
$5003	Ignition Voltage Out of Range	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0308	Fault active. • Vehicle speed sensing fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0A7A	Alternative Fuel Devices Disabled	Yes	No

5.6.10 FADR – SQA Map Reset Trigger Device Control

This device control is used to reset the learned values of the small quantity adjustment (SQA) map in case of a fuel rail pressure sensor replacement.

Signal	Length (Bits)	Data Type	Range	Conversion
SQA Map Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (SQA Map Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        Reset the SQA map learned values.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine Is Running	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.6.11 AFIR – Cold Start Fuel Reservoir Pump Device Control

This device control is used to override the state of the fuel pump output for an auxiliary cold start fuel reservoir for circuit or pressure diagnosis. The device control is a service diagnostic aid that is used to determine if the on/off relay controlled fuel pump system is functioning properly.

Signal	Length (Bits)	Data Type	Range	Conversion
Cold Start Fuel Reservoir Pump Enable	1	BLN	N/A	$1=True; $0=False
Cold Start Fuel Reservoir Pump On	1	BLN	N/A	$1=True; $0=False

If (Cold Start Fuel Reservoir Pump Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else

The fuel pump output shall be commanded to the value requested in Cold Start Fuel Reservoir Pump On.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0004	Output control unavailable	Yes	No
$0008	The Output Control Timer has Expired	Yes	No
$0201	Output control not allowed while engine is running	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • System voltage is above or below a calibrated value	Yes	No

5.6.12 AFIR – Cold Start Fuel Injector Device Control

This device control is used to trigger the cold start injector to be commanded on for a calibratable amount of time in order to confirm injector driver operation and circuit integrity or test the flow of the injector. The cold start fuel injector control will appear as discrete to the off-board tool, with the ECU controlling the actual injector duty cycle as defined per calibration.

Signal	Length (Bits)	Data Type	Range	Conversion
Cold Start Fuel Injector Enable	1	BLN	N/A	$1=True; $0=False
Cold Start Fuel Injector On	1	BLN	N/A	$1=True; $0=False

If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
If (the cold start fuel injector output time limit has been exceeded)
Return to normal control.
Else
The cold start fuel injector output shall be commanded to the value requested in Cold Start Fuel Injector On.
EndIf
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0004	Output control unavailable	Yes	No
$0201	Output control not allowed while engine is running	Yes	No
$0202	Injector already tested.	Yes	No
$0206	Fuel disabled by theft deterrent system.	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.6.13 AFIR – Auxiliary Cold Start System Device Control

This device control is used to override the state of the full auxiliary cold start system. This device control will operate the auxiliary cold start reservoir fuel pump and injector in such a way as to mimic in-use control, which will allow service and assembly testing. All cold start system control will appear as discrete to the off-board tool, with the ECU controlling the actual injector duty cycle as defined per calibration.

Signal	Length (Bits)	Data Type	Range	Conversion
Auxiliary Cold Start System Enable	1	BLN	N/A	$1=True; $0=False
Auxiliary Cold Start System On	1	BLN	N/A	$1=True; $0=False

If (Auxiliary Cold Start System Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Auxiliary Cold Start System shall be commanded to the value requested in Auxiliary Cold Start System On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0004	Output control unavailable	Yes	No
$0008	The Output Control Timer has Expired	Yes	No
$0206	Fuel disabled by theft deterrent system.	Yes	No
$0308	Fault Active. • DTCs for VSS or TOS • Cold start fuel injector circuit fault • Cold start fuel reservoir pump circuit fault	Yes	No
$0401	Vehicle speed detected. Vehicle must not be moving	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0604	Engine Must be Running	Yes	No

5.6.14 FADR – Fuel Setpoint Adaptation Learning Adaptive Map Reset Trigger

This device control is used to trigger a reset of the Fuel Setpoint Adaptation (FSA) learning adaptive map.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Setpoint Adaptation Learning Adaptive Map Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Fuel Setpoint Adaptation Learning Adaptive Map Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The reset of the Fuel Setpoint Adaptation Learning Adaptive Map shall be triggered.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No
$0201	Output Control not allowed while engine is Running	Yes	Yes

5.7 CPID $1A

CPID $1A								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Commanded High Fuel Pressure Enabled	Fuel Pressure Regulator 1 Component Test Enable	Fuel Pressure Regulator 1 Component Test On	Fuel Pressure Regulator 2 Component Test Enable	Fuel Pressure Regulator 2 Component Test On	High Pressure Fuel Pump Relief Valve Learn Offset Reset Trigger	High Fuel Pressure Limiter Valve Test Trigger	
Byte 1	High Pressure Fuel Variable Reset Trigger Device Control	Fuel Rail Pressure Sensor First Time Adaptation Reset Trigger	Pressure Sensor Adaptation Suspension Status Enable	Pressure Sensor Adaptation Suspension Status Command	Pressure Sensor Adaptation Service Mode Trigger		Commanded Fuel Pressure Regulator Enable	Commanded Fuel Pressure Regulator Identifier
Byte 2	Reserved							
Byte 3	Commanded High Fuel Pressure (upper byte)							
Byte 4	Commanded High Fuel Pressure (lower byte)							

5.7.1 FHPR – Fuel Pressure Regulator 2 Component Test Device Control

This device control is intended for use on a Direct Injection (DI) high pressure fuel supply system to activate the fuel pressure regulator 2 at ignition on, engine off. Commanding the regulator to “on” will modulate the output driver to a calibratible “maximum” duty cycle. (Note: As this is an electrical circuit test, a regulator solenoid commanded to “on” does not necessarily equate to the regulator state at maximum fuel pressure.)

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Pressure Regulator 2 Component Test Enable	1	BLN	N/A	$1=True; $0=False
Fuel Pressure Regulator 2 Component Test On	1	BLN	N/A	$1=True; $0=False

```
If (Fuel Pressure Regulator 2 Component Test Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        If (Fuel Pressure Regulator 2 Component Test On is True)
            The fuel pressure regulator 2 control output will be commanded on
        Else
            The fuel pressure regulator 2 control output will be commanded to off
        EndIf
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output control limits out of range maximum DC allowed calibratable (ex. 60%)	Yes	No
$0008	The Output Control Timer has Expired	Yes	No
$0201	Output Control not allowed while engine is running	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$5003	Ignition voltage Out of Range	Yes	No

5.7.2 FHPR – Commanded Fuel Pressure Regulator Test Device Control

This device control is intended for use on a Direct Injection (DI) high pressure fuel supply system to govern rail pressure with the specified fuel pressure regulator, when more than one are present. Using this device, both regulator are supplied but only the selected one is modulated by pressure control loop.

Signal	Length (Bits)	Data Type	Range	Conversion
Commanded Fuel Pressure Regulator Enable	1	BLN	N/A	$1=True; $0=False
Commanded Fuel Pressure Regulator Identifier	1	ENM	N/A	$0=Fuel Pres. Regulator 1 $1= Fuel Pres. Regulator 2

```
If (Commanded Fuel Pressure Regulator Enable  is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        If (Commanded Fuel Pressure Regulator Identifier  is $0)
            The fuel pressure regulator 1 is selected to govern rail pressure
    Else
            The fuel pressure regulator 2 is selected to govern rail pressure
        EndIf
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0002	Parameter ID not supported. Check for proper vehicle selection.	Yes	No
$0008	The Output Control Timer has Expired • Pressure Too High for more than allowed time	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$1102	Diagnostic faults currently active or conditions not correct. • Fuel Pressure Regulator 1 or 2 is inhibited	Yes	No
$5003	Voltage Out of Range	Yes	No

5.7.3 FHPR – High Pressure Fuel Variable Reset Trigger Device Control

This device control is used to reset the pump learning status and diagnostic timers in case of high pressure component replacement.

Signal	Length (Bits)	Data Type	Range	Conversion
High Pressure Fuel Variable Reset Trigger	1	BLN	N/A	$1=True; $0=False

```
If (High Pressure Fuel Variable Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control
    Else
        Reset the high pressure fuel variables
    EndIf
Else
```

Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0002	Parameter ID not supported. Check for proper vehicle selection.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No

5.7.4 FHPR – Fuel Rail Pressure Sensor Adaptation Suspension Status Device

This device control is intended for use on a Direct Injection (DI) high pressure fuel supply system to override the suspension state of the fuel rail pressure sensor adaptation while in the assembly plant. The intent of this device control is to allow a user to override and control the ECU's current suspension state of the in plant adaptation test.

Signal	Length (Bits)	Data Type	Range	Conversion
Pressure Sensor Adaptation Suspension Status Enable	1	BLN	N/A	$1=True; $0=False
Pressure Sensor Adaptation Suspension Status Command	1	BLN	N/A	$1=True; $0=False

If (Pressure Sensor AdaptationSuspension Status Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The fuel rail pressure sensor adaptation status shall be commanded to the requested value as indicated by Pressure Sensor Adaptation Suspension Status Command.
        DTCs must be allowed to report under this control.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0002	Device Control Not Supported	none	yes

5.7.5 FHPR – Fuel Rail Pressure Sensor Adaptation Service Mode Trigger Device Control

This device control is intended for use on a Direct Injection (DI) high pressure fuel supply system to modify the fuel rail pressure sensor adaptation enablement while in service. It is intended to be used after the fuel rail pressure sensor has been replaced or the ECU has been replaced.

Signal	Length (Bits)	Data Type	Range	Conversion
Pressure Sensor Adaptation Service Mode Trigger	1	BLN	N/A	$1=True; $0=False

If (Pressure Sensor Adaptation Service Mode Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The ECU shall enter fuel rail pressure sensor adaptation service mode for that key trip.
        DTCs must be allowed to report during this mode.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0002	Device Control Not Supported	yes	none
$0006	Fuel system in open loop operation	yes	none
$0144	Fuel level out of range	yes	none
$0204	Injector flow test in progress	yes	none
$0307	Intake air temperature out of range	yes	none
$0308	Fault Active	yes	none
$0604	Engine not running	yes	none
$0901	Throttle not at idle, control not allowed	yes	none
$0A29	Fuel pressure out of range	yes	none
$0A39	Engine run time is too short	yes	none
$0A4C	Fuel temperature is out of range	yes	none
$0A66	Clutch Pedal Must be Released	yes	none
$1605	Device control not allowed	yes	none
$1803	Ignition voltage too low	yes	none
$2C01	Vehicle speed is too high	yes	none
$3D07	Barometric pressure too low	yes	none
$4004	Accelerator pedal position out of range	yes	none
$8003	Device control requested while other device controls are active	yes	none

5.7.6 FHPR – Fuel Rail Pressure Sensor First Time Adaptation Reset Trigger Device Control

This device control is used to trigger a reset of the adapted offset value for fuel rail pressure sensor. This CPID will reset the adapted offset value and re-enable the first large adaptation. It shall be called after the high fuel rail high pressure sensor is replaced or if ECU is replaced.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Rail Pressure Sensor First Time Adaptation Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Fuel Rail Pressure Sensor First Time Adaptation Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.

Return to normal control.
Else
Perform the reset of the Fuel Rail Pressure Sensor Adaptation value
Perform the reset of Fuel Rail Pressure Sensor First Time Adaptation Complete status bit
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0002	Parameter ID not supported. Check for proper vehicle selection.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No

5.7.7 FHPR - Commanded High Fuel Pressure Device Control

This device control is intended for use on a Direct Injection (DI) high pressure fuel supply system to override the commanded fuel high pressure output from the powertrain controller.

(Note: This device control is also called by the CPID $22 “Fuel System Pressure Test Device Control” to regulate high fuel pressure during the fuel system pressure test. Any future Powertrain software changes to CPID $1A “Commanded High Fuel Pressure Device Control” need to consider this interaction.)

Signal	Length (Bits)	Data Type	Range	Conversion
Commanded High Fuel Pressure Enabled	1	BLN	N/A	$1=True; $0=False
Commanded High Fuel Pressure	16	UNM	0 to 655350 kPa	E = N * 10

If (Commanded High Fuel Pressure Enabled is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
The fuel high pressure output shall be commanded to the value requested in Commanded High Fuel Pressure
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output control limits out of range	Yes	No
$0008	The Output Control Timer has Expired	Yes	No
$0207	No Engine Rotation is Detected	Yes	No
$0401	Vehicle speed has been detected. Vehicle must not be moving.	Yes	No
$0402	Vehicle speed DTC present, output control not allowed	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$0A05	Engine speed too high • Engine not in idle	Yes	No
$1102	Diagnostic faults currently active or conditions not correct. • High pressure Pump circuit DTCs • Low pressure Pump circuit DTCs • High pressure sensor circuit DTCs • (Allow control if pressure-related faults exist)	Yes	No
$5003	Voltage Out of Range	Yes	No

5.7.8 FHPR - Fuel Pressure Regulator 1 Component Test Device Control

This device control is intended for use on a Direct Injection (DI) high pressure fuel supply system to activate the fuel pressure regulator at ignition on, engine off. Both the low side and high side of the regulator are controlled together. Commanding the regulator to "on" will modulate the high side driver to a calibratible "maximum" duty cycle. (Note: As this is an electrical circuit test, a regulator solenoid commanded to "on" does not necessarily equate to the regulator state at maximum fuel pressure.)

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Pressure Regulator 1 Component Test Enable	1	BLN	N/A	$1=True; $0=False
Fuel Pressure Regulator 1 Component Test On	1	BLN	N/A	$1=True; $0=False

```
If (Fuel Pressure Regulator Component Test Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        If (Fuel Pressure Regulator Component Test On is True)
            The fuel pressure regulator control output will be commanded on
        Else
            The fuel pressure regulator control output will be commanded to off
        EndIf
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output control limits out of range maximum DC allowed calibratable (ex. 60%)	Yes	No
$0008	The Output Control Timer has Expired	Yes	No
$0201	Output Control not allowed while engine is running	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$5003	Ignition voltage Out of Range	Yes	No

5.7.9 FHPR – High Pressure Fuel Pump Relief Valve Learn Offset Reset Trigger Device Control

This device control is used to trigger a reset of the learned offset value for the high pressure fuel pump relief valve.

Signal	Length (Bits)	Data Type	Range	Conversion
High Pressure Fuel Pump Relief Valve Learn Offset Reset Trigger	1	BLN	N/A	$1=True; $0=False

```
If (High Pressure Fuel Pump Relief Valve Offset Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        Perform the reset of the High Pressure Fuel Pump Relief Valve Learn Offset value.
    EndIf
Else
    Return to normal control.
```

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0002	Parameter ID not supported. Check for proper vehicle selection.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No

5.7.10 FHPR – High Fuel Pressure Limiter Valve Test Trigger Device Control

This device control is intended for use on a direct injection (DI) high pressure fuel supply system to override the commanded high fuel pressure output from the Powertrain controller. The test is intended to be used after a fuel system overpressure event to verify that the fuel pressure limiter valve was not damaged and will not open at normal operating fuel pressure. This device control requires the "idle engine speed" device control (CPID $14) to command the engine speed above a calibratible value. The result of the high fuel pressure limiter valve test is displayed in PID $2331 and any abort reasons for the high fuel pressure limiter valve test will be displayed in PID $2332.

Signal	Length (Bits)	Data Type	Range	Conversion
High Fuel Pressure Limiter Valve Test Trigger	1	BLN	N/A	$1=True; $0=False

If (High Fuel Pressure Limiter Valve Test Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The high fuel pressure limiter valve test algorithm shall be initiated.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0002	Parameter ID not supported. Check for proper vehicle selection.	Yes	No
$0308	Output control not allowed, DTC Present. • High pressure Pump circuit DTCs • Low pressure Pump circuit DTCs • High pressure sensor circuit DTCs • (Allow control if pressure-related faults exist) • Vehicle speed sensor faults	Yes	No
$0401	Vehicle speed has been detected. Vehicle must not be moving.	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$1605	Device Control Not Allowed • Device control for idle engine speed must be active (CPID $14)	Yes	Yes
$4002	Engine Speed Too Low • Engine speed must be above a calibratible limit	Yes	Yes
$5003	Voltage Out of Range	Yes	No

5.8 CPID $1C

CPID $1C								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Exhaust Gas Recircula-tion Enable	Exhaust Gas Recirculation H-Bridge Duty Cycle Enable	Secondary Air Injection System Enable	Secondary Air Injection Pump On	Secondary Air Injection Valve On	Secondary Air Injection Diagnostic Service Bay Test Trigger	Catalyst Diagnostic Service Bay Test Trigger	
Byte 1	Exhaust Gas Recirculation Pintle Lift							
Byte 2	Exhaust Cam Phase Solenoid Bank 1 Enable	Exhaust Cam Phase Solenoid Bank 1 On	Exhaust Cam Phase Solenoid Bank 2 Enable	Exhaust Cam Phase Solenoid Bank 2 On	Intake Cam Phase Solenoid Bank 1 Enable	Intake Cam Phase Solenoid Bank 1 On	Intake Cam Phase Solenoid Bank 2 Enable	Intake Cam Phase Solenoid Bank 2 On
Byte 3	Exhaust Gas Recirculation H-Bridge Duty Cycle							
Byte 4							EGR Cooler Coolant Pump Enable	EGR Cooler Coolant Pump On

5.8.1 PHSR – Exhaust and Intake Cam Phase Solenoid Device Controls

These device controls are used to override the states of the exhaust and intake cam phase solenoid outputs for banks 1 and 2.

Signal	Length (Bits)	Data Type	Range	Conversion
Exhaust Cam Phase Solenoid Bank 1 Enable	1	BLN	N/A	$1=True; $0=False
Exhaust Cam Phase Solenoid Bank 1 On	1	BLN	N/A	$1=True; $0=False
Exhaust Cam Phase Solenoid Bank 2 Enable	1	BLN	N/A	$1=True; $0=False
Exhaust Cam Phase Solenoid Bank 2 On	1	BLN	N/A	$1=True; $0=False
Intake Cam Phase Solenoid Bank 1 Enable	1	BLN	N/A	$1=True; $0=False
Intake Cam Phase Solenoid Bank 1 On	1	BLN	N/A	$1=True; $0=False
Intake Cam Phase Solenoid Bank 2 Enable	1	BLN	N/A	$1=True; $0=False
Intake Cam Phase Solenoid Bank 2 On	1	BLN	N/A	$1=True; $0=False

If (Exhaust/Intake Cam Phase Solenoid Bank *N* Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The exhaust/intake cam phase solenoid output for bank *N* shall be commanded to the value requested in Exhaust/Intake Cam Phase Solenoid Bank *N* On (i.e., 100% duty cycle or 0% duty cycle).
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0001	Device control timer time-out active (5 second enable, 60 second disable).	Yes	No
$0004	Invalid device control selected (vehicle not equipped with camshaft position actuators).	Yes	No
$0201	Engine speed not zero.	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.8.2 AIRR – Secondary Air Injection Diagnostic Service Bay Test Trigger Device Control

This device control is used to trigger the secondary air injection (SAI) diagnostic service bay test, which causes the on-board SAI diagnostic to execute until it is completed or aborted. The SAI diagnostic consists of three phases, SAI on test, SAI valve shut test, and SAI pump off test, all of which are triggered by this device control.

Signal	Length (Bits)	Data Type	Range	Conversion
Secondary Air Injection Diagnostic Service Bay Test Trigger	1	BLN	N/A	$1=True; $0=False

If (Secondary Air Injection Diagnostic Service Bay Test Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The secondary air injection diagnostic service bay test shall be triggered.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0001	Device control timer time-out active	Yes	Yes
$0002	Device control not supported.	Yes	Yes
$000A	DTC present. • Intake Air Temperature (IAT) Sensor Circuit Low • Intake Air Temperature (IAT) Sensor Circuit High • Intake Air Temperature (IAT) Sensor Performance • Secondary Air Injection (AIR) Solenoid Control Circuit Bank 1 • Secondary Air Injection (AIR) Pump Control Circuit Bank 1 • ECM/PCM Internal Engine Off Timer Performance	Yes	Yes
$0201	Engine speed not zero.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No
$0601	Coolant temperature too high	Yes	Yes
$0E02	Vehicle speed not zero.	Yes	Yes
$0E03	Transmission not in Park or Neutral.	Yes	Yes
$5003	Ignition voltage out of range.	Yes	Yes

5.8.3 AIRR – Secondary Air Injection Reaction System Device Control

This device control is used to either turn the AIR Pump and/or AIR Valve on or off.

Signal	Length (Bits)	Data Type	Range	Conversion
Secondary Air Injection Reaction System Enable	1	BLN	N/A	$1=True; $0=False
Secondary Air Injection Reaction Pump On	1	BLN	N/A	$1=True; $0=False
Secondary Air Injection Reaction Valve On	1	BLN	N/A	$1=True; $0=False

If (Secondary Air Injection Reaction System Enable is True)
    If (Secondary Air Injection Reaction Pump On is True OR Secondary Air Injection Reaction Valve On is True)
        If (any of the device control limits in the table below are exceeded)
            Report the appropriate device control limit exceeded return code to the tester.
            Return to normal control.
        Else

The AIR Pump and Valve shall be commanded to the states requested by Secondary Air Injection Reaction Pump On and Secondary Air Injection Reaction Valve On
Endif
Else
Secondary Air Injection Reaction Pump On is False AND Secondary Air Injection Reaction Valve On is False (The AIR Pump and Valve shall be commanded off)
Endif
Else
Return to normal control
Endif

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0001	Generic Device Control Timer (30 second enable, 60 second disable)	Yes	No
$0406	Incorrect Power Mode	Yes	No
$1103	Engine coolant temperature low	Yes	No

5.8.4 EGRR – Exhaust Gas Recirculation Device Control

This device control is used to verify functionality of the Exhaust Gas Recirculation (EGR) system by allowing the Exhaust Gas Recirculation output to be commanded to a specific lift. The software calculated pintle lift should be available to be monitored via a parameter identifier as feedback for the device control override.

Signal	Length (Bits)	Data Type	Range	Conversion
Exhaust Gas Recirculation Enable	1	BLN	N/A	$1=True; $0=False
Exhaust Gas Recirculation Pintle Lift	8	UNM	0.0 to 99.609375 %	E = N * 0.390625

If (Exhaust Gas Recirculation Enable is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
The position shall be commanded to the requested position in Exhaust Gas Recirculation Position
Endif
Else
Return to normal control
Endif

Return Code	Device Control Limit Exceeded Description (Compression Ignition Engine)	Active for Service	Active for DVT
$0003	Device Control out of range • Calibratable minimum or maximum position limit exceeded	Yes	No
$0008	The Output Control Timer has Expired, please wait • The device control shall not be active for more than a calibrateable time • The device control must be disabled for a calibrateable time before it is reactivated	Yes	No
$0308	Output control not allowed, DTC Present. • EGR valve position sensor	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • Ignition Voltage too High or too Low	Yes	No

Return Code	Device Control Limit Exceeded Description (Spark Ignition Engine - Not Running)	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

Return Code	Device Control Limit Exceeded Description (Spark Ignition Engine - Running)	Active for Service	Active for DVT
$0001	Generic Device Control Timer (30 second enable, 60 second disable)	Yes	No
$0003	Device Control out of range • Calibratable minimum or maximum Position limit exceeded	Yes	No
$0008	The Output Control Timer has Expired, please wait	Yes	No
$0308	Output control not allowed, DTC Present. • Electrical fault on position feedback.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • Ignition Voltage too High or too Low	Yes	No

5.8.5 CATR – Catalyst Diagnostic Service Bay Test Trigger Device Control

This device control is used to activate the catalyst diagnostic service bay test.

Signal	Length (Bits)	Data Type	Range	Conversion
Catalyst Diagnostic Service Bay Test Trigger	1	BLN	N/A	$1=True; $0=False

If (Catalyst diagnostic service bay test trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    ElseIf (Catalyst diagnostic service bay test is complete)
        Return to normal control.
    Else
        Catalyst diagnostic service bay test shall be triggered.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

1.1.1 EGRR - Exhaust Gas Recirculation H-Bridge Duty Cycle Control

This device control is used to verify functionality of the Exhaust Gas Recirculation Valve by allowing the Exhaust Gas Recirculation Valve output to be commanded to a specific duty cycle. The software calculated duty cycle should be available to be monitored via a parameter identifier as feedback for the device control override.

Signal	Length (Bits)	Data Type	Range	Conversion
Exhaust Gas Recirculation H-Bridge Duty Cycle Enable	1	BLN	N/A	$1=True; $0=False
Exhaust Gas Recirculation H-Bridge Duty Cycle	8	SNM	-128 to +127 %	E = N * 1

If (Exhaust Gas Recirculation Valve Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.

Else
The Exhaust Gas Recirculation H-Bridge duty cycle shall be commanded to the requested duty cycle in Exhaust Gas Recirculation H-Bridge Duty Cycle.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Device Control out of range • Calibratable minimum or maximum position limit exceeded	Yes	No
$0008	The Output Control Timer has Expired, please wait • The Device control shall not be active for more then a calibratable time. Device Control shall be disabled for a calibratable time before its reactivated.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$1805	Ignition Voltage too High	Yes	No

5.8.6 EGRR – EGR Cooler Coolant Pump Device Control

This device control is used to override the state of the EGR cooler coolant pump.

Signal	Length (Bits)	Data Type	Range	Conversion
EGR Cooler Coolant Pump Enable	1	BLN	N/A	$1=True; $0=False
EGR Cooler Coolant Pump On	1	BLN	N/A	$1=True; $0=False

If (EGR Cooler Coolant Pump Enable is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
The EGR cooler coolant pump output shall be commanded to the state requested in EGR Cooler Coolant Pump On.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	The Output Control Timer Has Expired, Please Wait	Yes	No
$0307	Intake Air Temperature Out of Range • Intake Manifold Air Temperature higher than a calibrate-able threshold	Yes	No
$0406	Incorrect Power Mode	Yes	Yes
$5003	Voltage Out of Range	Yes	No

5.9 CPID $1E

CPID $1E								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Oxygen Sensor Heater Bank 1 Sensor 1 Enable	Oxygen Sensor Heater Bank 2 Sensor 1 Enable	Oxygen Sensor Heater Bank 1 Sensor 2 Enable	Oxygen Sensor Heater Bank 2 Sensor 2 Enable	Oxygen Sensor Heater Resistance Reset Trigger	~~Oxygen Sensor Drift Compensation Correction Value Reset Trigger~~	Oxygen Sensor Service Bay Heater Resistance Learn Trigger	Oxygen Sensor Service Bay Response Test Trigger
Byte 1	Oxygen Sensor Heater Bank 1 Sensor 1 Duty Cycle							
Byte 2	Oxygen Sensor Heater Bank 2 Sensor 1 Duty Cycle							
Byte 3	Oxygen Sensor Heater Bank 1 Sensor 2 Duty Cycle							
Byte 4	Oxygen Sensor Heater Bank 2 Sensor 2 Duty Cycle							

5.9.1 OXYR – Oxygen Sensor Heater Resistance Reset Trigger Device Control

This device control is used to trigger the reset of the learned resistances for the oxygen sensor heaters following a replacement in service. The learned resistances are reset to a calibration used to prevent overheating of the sensors until more accurate resistances may be learned following an extended engine off time (typically 12 or more hours).

Signal	Length (Bits)	Data Type	Range	Conversion
Oxygen Sensor Heater Resistance Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Oxygen Sensor Heater Resistance Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The reset of the learned resistances for all oxygen sensor heaters shall be triggered.
    EndIf
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.9.2 OXYR – Oxygen Sensor Heaters Device Control

This device control is used to override the duty cycle of the oxygen sensor heaters.

Signal	Length (Bits)	Data Type	Range	Conversion
Oxygen Sensor Heater Bank 1 Sensor 1 Enable	1	BLN	N/A	$1=True; $0=False
Oxygen Sensor Heater Bank 1 Sensor 1 Duty Cycle	8	UNM	0.0 - 99.609375 %	E = N * 0.390625
Oxygen Sensor Heater Bank 2 Sensor 1 Enable	1	BLN	N/A	$1=True; $0=False
Oxygen Sensor Heater Bank 2 Sensor 1 Duty Cycle	8	UNM	0.0 - 99.609375 %	E = N * 0.390625
Oxygen Sensor Heater Bank 1 Sensor 2 Enable	1	BLN	N/A	$1=True; $0=False
Oxygen Sensor Heater Bank 1 Sensor 2 Duty Cycle	8	UNM	0.0 - 99.609375 %	E = N * 0.390625
Oxygen Sensor Heater Bank 2 Sensor 2 Enable	1	BLN	N/A	$1=True; $0=False
Oxygen Sensor Heater Bank 2 Sensor 2 Duty Cycle	8	UNM	0.0 - 99.609375 %	E = N * 0.390625

5.9.2.1 Oxygen Sensor Heater Device Control with Engine Not Running

With the engine off, this device control is intended to be used as a discrete control for circuit diagnosis. With the engine off, none of the oxygen sensor heaters can be commanded on for more than 3.0 seconds, after which the selected oxygen sensor heater must be disabled for a limited duration (typically 60 seconds) before it may be enabled again. Device control limit exceeded return code $0008 is reported when the device control is first disabled and if requests to enable the device control occur during the limited duration that it is disabled.

For all applicable oxygen sensor heaters:

If (Oxygen Sensor Heater Bank *N* Sensor *Y* Enable is True)
    If (the engine is running)
        Manage the request to control the Oxygen Sensor Heaters as described in Oxygen Sensor Heater Device Control with Engine Running Below
    Else
        If (any of the device control limits in the table below are exceeded)
            Report the appropriate device control limit exceeded return code to the tester.
            Return to normal control.
        Else
            The oxygen sensor heater output for bank *N* sensor *Y* shall be commanded to the value requested in Oxygen Sensor Heater Bank *N* Sensor *Y* Duty Cycle. (Typical values will be 100% for on and 0% for off)
        EndIf
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	The Output Control timer has expired. Please Wait. - The selected oxygen sensor heater must not be commanded on (Duty Cycle > 0.0) continuously for more than 3.0 seconds with engine not running.	Yes	Yes
$0307	Intake Air Temperature out of range	Yes	No
$0406	Incorrect Power Mode	Yes	No
$1805	Ignition Voltage too high during O2 Heater device control.	Yes	No

5.9.2.2 Oxygen Sensor Heater Device Control with Engine Running

With the engine running, this device control is used to operate the oxygen sensor heaters at the various duty cycles available to the engine controller.

For all applicable oxygen sensor heaters:

If (Oxygen Sensor Heater Bank *N* Sensor *Y* Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The oxygen sensor heater output for bank *N* sensor *Y* shall be commanded to the value requested in Oxygen Sensor Heater Bank *N* Sensor *Y* Duty Cycle.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0305	Exhaust Temperature Too Low	Yes	No
$0406	Incorrect Power Mode	Yes	No
$1802	O2 Heater: O2 Heaters are over temperature	Yes	No
$1805	Ignition Voltage too high during O2 Heater device control.	Yes	No
$2E04	Requested Duty Cycle is Out of Range	Yes	No

5.9.3 OXYR – Oxygen Sensor Service Bay Heater Resistance Learn Trigger

This device control is used to trigger a service bay procedure to learn the oxygen sensor heater resistance following a sensor replacement in service. This procedure is required before executing the Service Bay Oxygen Sensor Response Test if PID $231C indicates the oxygen sensor heater resistance learn is not complete.

Signal	Length (Bits)	Data Type	Range	Conversion
Oxygen Sensor Service Bay Heater Resistance Learn Trigger	1	BLN	N/A	$1=True; $0=False

If (Oxygen Sensor Service Bay Heater Resistance Learn Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The service bay procedure to learn the oxygen sensor heater resistance shall be triggered.
    EndIf
Else
    Return to normal control
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0006	Fuel system in Open Loop operation	Yes	No
$0144	Fuel level out of range	Yes	No
$0308	Output control not allowed, DTC Present. • Engine coolant temperature sensor faults • Vehicle speed sensor faults • Park/neutral switch faults • O2 heater faults	Yes	No
$031D	Engine coolant temperature out of range	Yes	No
$0401	Vehicle speed has been detected, Vehicle must not be moving	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch	Yes	No
$0604	Engine must be running	Yes	No
$0903	Transmission must be in Park or Neutral	Yes	No
$0A36	A/C Must be Off	Yes	No
$1805	Voltage too high	Yes	No

5.9.4 OXYR – Oxygen Sensor Service Bay Response Test Trigger

This device control is used to trigger a service bay procedure to execute the oxygen sensor response test following a sensor replacement in service. This procedure is required to set the oxygen sensor IM ready flag. The PID $231C must indicate the oxygen sensor heater resistance learn is complete before executing this procedure. If the learn is not complete, use the Oxygen Sensor Service Bay Heater Resistance Learn Trigger device control to perform the learn.

Signal	Length (Bits)	Data Type	Range	Conversion
Oxygen Sensor Service Bay Response Test Trigger	1	BLN	N/A	$1=True; $0=False

```
If (Oxygen Sensor Service Bay Response Test Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The service bay procedure to run the oxygen sensor response test shall be triggered.
    EndIf
Else
    Return to normal control
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Limits Out of Range	Yes	Yes
$0006	Fuel system in Open Loop operation	Yes	Yes
$0010	RPM Out of Range	Yes	Yes
$0144	Fuel level out of range	Yes	Yes
$0307	Intake Air Temperature Out of Range	Yes	Yes
$0308	Output control not allowed, DTC Present. • Engine coolant temperature sensor faults • Vehicle speed sensor faults • Park/neutral switch faults • O2 sensor faults • Misfire faults • Intake air temperature sensor faults • Mass Airflow Sensor faults • MAP sensor faults • EVAP system faults • Air Injection system faults • Ethanol sensor fault • Electronic Throttle faults • Fuel injector faults	Yes	Yes
$031D	Engine Coolant Temperature Out of Range	Yes	Yes
$0401	Vehicle speed has been detected, Vehicle must not be moving	Yes	Yes
$0406	Incorrect Power Mode. Check Ignition Switch	Yes	Yes
$0604	Engine must be running	Yes	Yes
$0903	Transmission must be in Park or Neutral	Yes	Yes
$0A17	O2 Heater in Water Shock Protection Mode	Yes	Yes
$0A36	A/C Must be Off	Yes	Yes
$0A37	HO2S Heater Learn not Complete	Yes	Yes
$0A38	Fuel Alcohol Content too High	Yes	Yes
$0A39	Engine Run Time too Short	Yes	Yes
$3D07	Baro Pressure Too Low	Yes	Yes
$5003	Voltage Out of Range	Yes	Yes

5.10 CPID $20

CPID $20								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Malfunction Indicator Lamp Enable	Malfunction Indicator Lamp On	Air Condi-tioning Compres-sor Enable	Air Condi-tioning Compres-sor On	Generator L-Terminal Enable	Generator L-Terminal On	Powertrain Relay Enable	Powertrain Relay On
Byte 1	Cooling Fan Speed Enable		Reverse Inhibit Enable	Reverse Inhibit On	Skip Shift Enable	Skip Shift On		
Byte 2	Cooling Fan Speed							
Byte 3	Cooling Fan Relay Enable	Cooling Fan Relay 1 On	Cooling Fan Relay 2 On	Cooling Fan Relay 3 On	Starter Relay Enable	Starter Relay On	Neutral Gear Sensor Learning Trigger	Remote Vehicle Start Disable History Reset Trigger
Byte 4	Shared High Side Driver 1 Enable	Shared High Side Driver 1 On	Shared High Side Driver 2 Enable	Shared High Side Driver 2 On	Clutch Slip Detection Reset Trigger	Manual Transmissio n Gear Position Sensor Learning Trigger	Brake Pedal Fully Released Learn Position Reset Trigger	Clutch Pedal Fully Applied Position Learn Trigger

5.10.1 MCTR – Clutch Slip Detection Reset Trigger Device Control

This device control is used to reset all clutch slip detection values. The reset should be triggered when the clutch system was exchanged or when the clutch system was checked but showed no clutch wear.

Signal	Length (Bits)	Data Type	Range	Conversion
Clutch Slip Detection Reset Trigger	1	BLN	N/A	$1=True; $0=False

```
If (Clutch Slip Detection Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The reset of the clutch slip detection values shall be triggered.
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.10.2 SPMR – Manual Transmission Gear Position Sensor Learning Trigger Device Control

This device control is used to force the ECU to perform a gear position sensor learning after a sensor or ECU replacement.

Signal	Length (Bits)	Data Type	Range	Conversion
Manual Transmission Gear Position Sensor Learning Trigger	1	BLN	N/A	$1=True; $0=False

If (Manual Transmission Gear Position Sensor Learning Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        Reset the manual transmission gear position sensor learning values to default values and perform a manual transmission gear position sensor learn.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode • Run/crank must be active	Yes	No
$0A52	Transmission not in Neutral 3-4 gate.	Yes	No

5.10.3 TRGR – Neutral Gear Sensor Learning Trigger Device Control

This device control is used to force the ECU to perform a Neutral Gear Sensor (NGS) learning after a NGS component exchange.

Signal	Length (Bits)	Data Type	Range	Conversion
Neutral Gear Sensor Learning Trigger	1	BLN	N/A	$1=True; $0=False

If (Neutral Gear Sensor Learning Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        Reset the neutral gear sensor learning values and perform a neutral gear sensor learn next time all the preconditions are fulfilled.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode • Run/crank must be active	Yes	No

5.10.4 BRKR – Brake Pedal Fully Released Learn Position Reset Trigger Device Control

This device control is used to reset the initialized values for the brake pedal fully released learned position.

Signal	Length (Bits)	Data Type	Range	Conversion
Brake Pedal Fully Released Learn Position Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Brake Pedal Fully Released Learn Position Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The brake pedal fully released position value shall be reset to the initialized value.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0308	Fault active. • Brake pedal position sensor fault active. • Buffered 5V supply fault active. • Vehicle speed sensor faults	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.10.5 MTCR – Clutch Pedal Fully Applied Position Learn Trigger Device Control

This device control is used to trigger the procedure to learn the clutch pedal fully applied position.

Signal	Length (Bits)	Data Type	Range	Conversion
Clutch Pedal Fully Applied Position Learn Trigger	1	BLN	N/A	$1=True; $0=False

If (Clutch Pedal Fully Applied Position Learn Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The procedure to learn the clutch pedal fully applied position shall be triggered.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine speed not zero.	Yes	No
$0308	Fault active. • Clutch pedal position sensor fault active.	Yes	Yes
$0309	Learn procedure already performed this key cycle.	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.10.6 ACCR – Air Conditioning Compressor Device Control

This device control is used to override the state of the air conditioning compressor output.

Signal	Length (Bits)	Data Type	Range	Conversion
Air Conditioning Compressor Enable	1	BLN	N/A	$1=True; $0=False
Air Conditioning Compressor On	1	BLN	N/A	$1=True; $0=False

```
If (Air Conditioning Compressor Enable is True)
    If (Air Conditioning Compressor On is False)
        The air conditioning compressor output shall be commanded off.
    Else
        If (the engine is not running)
            The air conditioning compressor output shall be commanded on.
        Else
            If (the air conditioning compressor output has been commanded on for less than 5 seconds)
                If (the device control limits associated with return codes of $0A05 or $0A06 are exceeded)
                    Report the appropriate device control limit exceeded return code to the tester.
                    Return to normal control.
                Else
                    The air conditioning compressor output shall be commanded on.
                EndIf
            Else
                If (any of the device control limits in the table below are exceeded)
                    Report the appropriate device control limit exceeded return code to the tester.
                    Return to normal control.
                Else
                    The air conditioning compressor output shall be commanded on.
                EndIf
            EndIf
        EndIf
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No
$0A03	Air conditioning refrigerant high side fluid pressure out of range.	Yes	No
$0A04	Air conditioning compressor commanded off by platform.	Yes	No
$0A05	Engine speed out of range.	Yes	No
$0A06	Ignition voltage out of range.	Yes	No
$0A07	Air conditioning refrigerant high side fluid pressure fault.	Yes	No

5.10.7 FANR – Cooling Fan Speed Device Control

This device control is used to override the state of the cooling fan control output.

Signal	Length (Bits)	Data Type	Range	Conversion
Cooling Fan Speed Enable	1	BLN	N/A	$1=True; $0=False
Cooling Fan Speed	8	UNM	0.0 - 99.609375 %	E = N * 0.390625

If (Cooling Fan Speed Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The cooling fan speed shall be commanded to the value requested in Cooling Fan Speed.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output control limits out of range	Yes	No
$0004	Output control unavailable • Relay fan Electronic PWM fan	Yes	Yes
$0308	DTC Present. • Engine coolant temperature sensing fault • Air Conditioning pressure sensing fault	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0601	Engine coolant temperature too high. • Engine coolant temperature exceeds a calibrateable threshold when reducing fan speed with engine running. Limit does not apply when increasing fan speed.	Yes	No
$0604	Engine not running.	Yes	No
$0A03	Air conditioning refrigerant high side fluid pressure out of range.	Yes	No
$0A1E	Diesel Particulate Filter Regeneration is Active	Yes	No

5.10.8 GENR – Generator L-Terminal Device Control

This device control is used to override the state of the generator L-terminal output.

Signal	Length (Bits)	Data Type	Range	Conversion
Generator L-Terminal Enable	1	BLN	N/A	$1=True; $0=False
Generator L-Terminal On	1	BLN	N/A	$1=True; $0=False

If (Generator L-Terminal Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The generator L-terminal output shall be commanded to the state requested in Generator L-Terminal On.
    EndIf
Else
    Return to normal control.

EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	The Output Control Timer Has Expired.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0E01	Engine is not running - must be running to turn off L-terminal.	Yes	No
$0E02	Vehicle speed is not zero - vehicle must not be moving.	Yes	No
$0E03	Transmission range is not in Park or Neutral.	Yes	No

5.10.9 RVSR – Remote Vehicle Start Disable History Reset Trigger Device Control

This device control is used to trigger the reset of the last eight reasons that powertrain has disabled remote vehicle start (RVS).

Signal	Length (Bits)	Data Type	Range	Conversion
Remote Vehicle Start Disable History Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Remote Vehicle Start Disable History Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The reset of the last eight reasons that powertrain has disabled RVS shall be triggered.
    EndIf
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.10.10PMDR – Powertrain Relay Device Control

This device control is used to override the state of the Powertrain Relay.

Signal	Length (Bits)	Data Type	Range	Conversion
Powertrain Relay Enable	1	BLN	N/A	$1=True; $0=False
Powertrain Relay On	1	BLN	N/A	$1=True; $0=False

If (Powertrain Relay Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Powertrain Relay shall be commanded to the state requested in Powertrain Relay On
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine is rotating	Yes	No
$1600	Engine rotating data is not valid	Yes	No
$0401	Vehicle speed is not equal to zero	Yes	No
$0402	Vehicle speed data is not valid	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.10.11RINR – Reverse Inhibit Device Control

This device control is used to override the state of the Reverse Inhibit output.

Signal	Length (Bits)	Data Type	Range	Conversion
Reverse Inhibit Enable	1	BLN	N/A	$1=True; $0=False
Reverse Inhibit On	1	BLN	N/A	$1=True; $0=False

If (the vehicle is equipped with a reverse enable solenoid (reverse is inhibited with solenoid commanded off))
    Manage the request to inhibit reverse as described in Section 5.8.7.1.
Else (for reverse inhibit applications (reverse is inhibited with solenoid commanded on))
    Manage the request to inhibit reverse as described in Section 5.8.7.2.
EndIf

5.10.11.1 RINR – Reverse Inhibit for Reverse Enable Solenoid Applications

If (Reverse Inhibit Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The reverse inhibit output shall be commanded to the state requested in Reverse Inhibit On
            If Reverse Inhibit On = True
                De-energize Reverse Inhibit Output (command solenoid off to inhibit reverse).
            Else
                Energize Reverse Inhibit Output (command solenoid on to allow reverse)
            EndIf
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0004	Invalid device control selected.	Yes	Yes
$0401	Vehicle speed is not equal to zero	Yes	No
$0402	Vehicle speed data is not valid	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.10.11.2 RINR – Reverse Inhibit for Reverse Inhibit Solenoid Applications

If (Reverse Inhibit Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.

```
        Return to normal control.
    Else
        The reverse inhibit output shall be commanded to the state requested in Reverse Inhibit On
            If Reverse Inhibit On = True
                Energize Reverse Inhibit Output (command solenoid on to inhibit reverse).
            Else
                De-energize Reverse Inhibit Output (command solenoid off to allow reverse)
            EndIf
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0004	Invalid device control selected.	Yes	Yes
$0401	Vehicle speed is not equal to zero	Yes	No
$0402	Vehicle speed data is not valid	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.10.12SKSR – Skip Shift Device Control

This device control is used to override the state of the Skip Shift output.

Signal	Length (Bits)	Data Type	Range	Conversion
Skip Shift Enable	1	BLN	N/A	$1=True; $0=False
Skip Shift On	1	BLN	N/A	$1=True; $0=False

```
If (Skip Shift Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The skip shift output shall be commanded to the state requested in Skip Shift On
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0001	Generic Device Control Timer (30 second enable, 60 second disable)	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.10.13STRR – Starter Relay Device Control

This device control is used to override the state of the Starter Relay output. On a tandem start system, the starter motor and starter pinion outputs will be commanded on in the same manner as a conventional key start. Fuel and spark are disabled to prevent the engine from starting if the starter relay output is commanded on.

Signal	Length (Bits)	Data Type	Range	Conversion
Starter Relay Enable	1	BLN	N/A	$1=True; $0=False
Starter Relay On	1	BLN	N/A	$1=True; $0=False

```
If (Starter Relay Enable is True)
    If (starter relay output(s) is currently commanded off)
        If (Starter Relay On is True)
            If (Limits $1600, $0201, $0206, $2B00, $0903, $2C07, $0402, $0401, $0405, $1603, and $0008 in the
            following table are exceeded)
                Report the appropriate device control limit exceeded return code to the tester.
                Return to normal control.
            Else
                The starter relay output(s) shall be commanded on
            EndIf
        Else
            The starter relay output(s) shall be commanded off
        EndIf
    Else
        If (Starter Relay On is True)
            If (Limits $2B00, $0903, $2C07, $0207, $0402, $0401, $0405, $1603, and $0008 in the following
            table are exceeded)
                Report the appropriate device control limit exceeded return code to the tester.
                Return to normal control.
            Else
                The starter relay output(s) shall be commanded on
                    (fuel and spark control are also disabled to prevent the engine from starting)
            EndIf
        Else
            The starter relay output(s) shall be commanded off
        EndIf
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Not enough time has elapsed since the last time the starter has been enabled via device control	Yes	No
$0201	Engine is rotating	Yes	No
$0206	Theft system inhibiting starter enable	Yes	No
$0207	No engine rotation has been detected after starter has been enabled	Yes	No
$0401	Vehicle speed is not equal to zero	Yes	No
$0402	Vehicle speed data is not valid	Yes	No
$0405	Run/Crank voltage is too low	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0903	Transmission is in gear (automatic transmission)	Yes	No
$1600	Engine rotating data is not valid	Yes	No
$1603	ETC system inhibiting starter enable	Yes	No
$2B00	Transmission gear data is not valid (automatic or manual transmission)	Yes	No
$2C07	Clutch Pedal Must Be Applied	Yes	No

5.10.14DFIR – Malfunction Indicator Lamp Device Control

This device control is used to either turn the Malfunction Indicator Lamp (MIL) on or off.

Signal	Length (Bits)	Data Type	Range	Conversion
Malfunction Indicator Lamp Enable	1	BLN	N/A	$1=True; $0=False
Malfunction Indicator Lamp On	1	BLN	N/A	$1=True; $0=False

If (Malfunction Indicator Lamp Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    ElseIf (Malfunction Indicator Lamp On is False)
        The Malfunction Indicator Lamp shall be commanded off
    Else
        The Malfunction Indicator Lamp shall be commanded on
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.10.15FADR – Cooling Fan Relay Driver Device Control

This device control is used to override the state of the Discrete Cooling Fan Relay outputs.

Signal	Length (Bits)	Data Type	Range	Conversion
Cooling Fan Relay Enable	1	BLN	N/A	$1=True; $0=False
Cooling Fan Relay 1 On	1	BLN	N/A	$1=True; $0=False
Cooling Fan Relay 2 On	1	BLN	N/A	$1=True; $0=False
Cooling Fan Relay 3 On	1	BLN	N/A	$1=True; $0=False

If (Cooling Fan Relay Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The cooling fan relay 1 output shall be commanded to the state requested in Cooling Fan Relay 1 On
        The cooling fan relay 2 output shall be commanded to the state requested in Cooling Fan Relay 2 On
        The cooling fan relay 3 output shall be commanded to the state requested in Cooling Fan Relay 3 On
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	**`Output control limits out of range**	**Yes**	**No**
$0004	Device Control Not Available • Mechanical fan • Electronic PWM fan	Yes	Yes
$0201	Output Control not allowed while engine is Running	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0601	Engine coolant temperature too high.	Yes	No

5.10.16SHDR – Shared High Side Driver 1 Device Control

This device control is used to override the state of the Shared High Side Driver 1 output. (A shared high side driver serves as a controlled actuator supply voltage for devices with individual low side control.)

Signal	Length (Bits)	Data Type	Range	Conversion
Shared High Side Driver 1 Enable	1	BLN	N/A	$1=True; $0=False
Shared High Side Driver 1 On	1	BLN	N/A	$1=True; $0=False

If (Shared High Side Driver 1 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Shared high side driver 1 output shall be commanded to the state requested in Shared High Side Driver 1 On
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No

5.10.17SHDR – Shared High Side Driver 2 Device Control

This device control is used to override the state of the Shared High Side Driver 2 output. (A shared high side driver serves as a controlled actuator supply voltage for devices with individual low side control.)

Signal	Length (Bits)	Data Type	Range	Conversion
Shared High Side Driver 2 Enable	1	BLN	N/A	$1=True; $0=False
Shared High Side Driver 2 On	1	BLN	N/A	$1=True; $0=False

If (Shared High Side Driver 2 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Shared high side driver 2 output shall be commanded to the state requested in Shared High Side Driver 2 On
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No

5.11 CPID $22

CPID $22								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0			Compression Test Enable	Compression Test Engine Crank		Fuel Injector Flow Test Trigger	Fuel System Pressure Test Enable	Fuel System Pressure Test On
Byte 1								
Byte 2	Compression Test Engine Cranking Speed (Hybrid)							
Byte 3					Starter Pinion Relay Enable	Starter Pinion Relay On	Starter Motor Relay Enable	Starter Motor Relay On
Byte 4								

5.11.1 STRR –Starter Motor Relay Device Control

This device control is used to override the state of the starter motor relay output.

Signal	Length (Bits)	Data Type	Range	Conversion
Starter Motor Relay Enable	1	BLN	N/A	$1=True; $0=False
Starter Motor Relay On	1	BLN	N/A	$1=True; $0=False

If (Starter Motor Relay Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The starter motor relay shall be commanded to the state indicated by Starter Motor Relay On
            (fuel and spark control are also disabled to prevent the engine from starting)
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Not enough time has elapsed since the last time the starter has been enabled via device control	Yes	No
$0201	Engine is rotating	Yes	No
$0405	Run/Crank voltage is too low	Yes	No
$0406	Incorrect Power Mode	Yes	No
$1600	Engine rotating data is not valid	Yes	No

5.11.2 STRR –Starter Pinion Relay Device Control

This device control is used to override the state of the starter pinion relay output.

Signal	Length (Bits)	Data Type	Range	Conversion
Starter Pinion Relay Enable	1	BLN	N/A	$1=True; $0=False
Starter Pinion Relay On	1	BLN	N/A	$1=True; $0=False

If (Starter Pinion Relay Enable is True)
    If (any of the device control limits in the table below are exceeded)

Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
The starter pinion relay shall be commanded to the state indicated by Starter Pinion Relay On.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine is rotating	Yes	No
$0405	Run/Crank voltage is too low	Yes	No
$0406	Incorrect Power Mode	Yes	No
$1600	Engine rotating data is not valid	Yes	No

5.11.3 ASSR/STRR – Compression Test Device Control

This device control is used to set up the internal combustion engine for a compression test. The fuel pump, fuel injectors, and spark are all disabled. The throttle is opened to a Wide Open Throttle (WOT) position. For conventional engine vehicles, the extended crank feature is not active, and for strong hybrid vehicles, the Hybrid Powertrain Control Module is configured to crank at the speed specified in the device control command. Engine cranking can be controlled by the technician via the Compression Test Engine Crank bit, or through the ignition key, or through a remote starter switch as applicable. (A remote starter switch is a specific tool that a technician connects in place of the starter relay. This is not to be confused with a customer-activated remote start feature.)

The device control limits and rejects will apply normally upon entry to the device control. Once the device control is active, the limits will be monitored (to prevent unwanted starter operation, for example), but in the case of a limit exceeded or a tester disconnect, device control of the fuel and spark will not be released until the Power Mode switches to “accessory” or “off”. This will prevent unexpected fuel and spark operation in the case of a device control abort.

Signal	Length (Bits)	Data Type	Range	Conversion
Compression Test Enable	1	BLN	N/A	$1=True; $0=False
Compression Test Engine Crank	1	BLN	N/A	$1=True; $0=False
Compression Test Engine Cranking Speed (Hybrid)	8	UNM	0 - 2040 rpm	E = N * 8

If (Compression Test Enable is True)

If (Limits $0010, $0201, $0308, $0401, $0402, $0405, $0406 or $0903 are exceeded) and (test is not active)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
Disable electric fuel pump
Disable fuel injectors
Disable spark
Open throttle to WOT
(Continue to monitor device control limits, but do not release control of the above items until the Power Mode switches to “accessory” or “off”.)

If (vehicle is a strong hybrid)
Send message to HPCM for compression test cranking behavior using speed specified in Compression Test Engine Cranking Speed (Hybrid).

```
        EndIf
    EndIf


        If (starter relay output(s) is currently commanded off)
                If (Compression Test Engine Crank is True)
                        If (Limits $0008, $0406, $0903, or $2C07 are exceeded)
                                Report the appropriate device control limit exceeded return code to the tester.
                                Return to normal control.
                        Else
                                The starter relay output(s) shall be commanded on.
                        EndIf
                Else
                        The starter relay output(s) shall be commanded off.
                EndIf
    EndIf


        If (starter relay output(s) is currently commanded on)
                If (Compression Test Engine Crank is True)
                        If (Limits $0008, $0207, $0308, $0401, $0402, $0405, $0406, $0903, $0A14 or $2C07
                        are exceeded)
                                Report the appropriate device control limit exceeded return code to the tester.
                                Return to normal control.
                        Else
                                The starter relay output(s) shall continue to be commanded on.
                        EndIf
                Else
                        The starter relay output(s) shall be commanded off.
                EndIf
    EndIf

Else
    Return to normal control
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Timer expired. Please wait. • Not enough time has elapsed since the last time the compression test has been enabled via device control	Yes	No
$0010	Engine Speed Out of Range	Yes	No
$0201	Engine is running	Yes	No
$0207	No Engine Rotation is Detected	Yes	No
$0308	DTC Present • Crank sensor fault • Electronic throttle control fault • Fuel pump relay fault • Communication fault with HPCM (hybrid applications) • Park / Neutral Position Fault	Yes	No
$0401	Vehicle Speed Detected	Yes	No
$0402	Vehicle Speed DTC Present	Yes	No
$0405	Battery Voltage Exceeds Limits • Run/Crank voltage is too low	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$0903	Transmission must be in Park or Neutral.	Yes	No
$0A14	Hybrid System Disabled Engine Crank	Yes	No
$2C07	Clutch Pedal Must be Applied	Yes	No

	•		

5.11.4 FULR – Fuel Injector Flow Test Trigger

This device control actuates an engine running fuel injector flow test for direct injected (DI) engines. The purpose of the fuel injector flow test is to determine if there is a restricted or plugged fuel injector. The fuel injector flow test algorithm is completely contained in the engine control module and returns a fuel pressure drop value through DID $6C "Direct Injection Fuel Injector Flow Test Results" when the test is complete. The fuel injector flow test is initiated via request from an off-board tool and will test all fuel injectors in sequence.

Once the fuel injector flow test is enabled, the desired fuel rail pressure is commanded to a calibrated fuel pressure. After the fuel rail pressure achieves a stabilized value, the Start of Inject (SOI) position, and the commanded pulse width for the injector to be tested are set to calibrated values. The SOI position is set to calibrated values so that the captured fuel rail pressure at the next DI ID will correspond to the injector that is being tested. In addition, the high pressure fuel pump closed loop control is disabled after the fuel rail pressure achieves a stabilized value so that the high pressure fuel pump closed loop control will not attempt to increase the fuel pressure during the test.

The fuel injector flow test algorithm is designed to average the pressure drop over a calibratable number of engine cycles for each injector. A larger pulse width will be commanded on the injector that is requested to be tested. Since a larger pulse width is commanded on the test injector, the observed fuel pressure drop on this injector should be larger than other injectors on the engine. The average pressure drop is reported in DID $6C such that it can be compared with the pressure drops for the remaining engine cylinders during the same key cycle. A determination of injector performance can be made by comparing the DID $6C values from cylinder to cylinder. The progress of the test is displayed in PIDs $2334 and 2335.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Injector Flow Test Trigger	1	BLN	N/A	$1=True; $0=False

```
If (Fuel Injector Flow Test Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        Trigger the fuel injector flow test and present the result of the test in DID $6C.
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description Test Enable Conditions	Active for Service	Active for DVT
$0004	Output control unavailable.	Yes	No
$0010	RPM Out of Range	Yes	No
$0204	Injector flow test in progress.	Yes	No
$0308	Output control not allowed, DTC Present. • Fuel injector faults • Fuel pressure sensor faults • High pressure fuel pump fault • High pressure fuel pump not learned • Low pressure fuel pump fault • Transmission range / clutch switch fault (plausibility for clutch pedal) • Vehicle speed sensor fault • Acceleration pedal Fault • Engine Coolant temperature sensor Fault	Yes	No
$031D	Engine coolant temperature out of range	Yes	No
$0401	Vehicle Speed Has Been Detected	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch	Yes	No
$0903	Transmission must be in Park or Neutral	Yes	No
$0A1E	DPF regeneration active (normal running or service procedure)	Yes	No
$0A36	A/C Must be Off	Yes	No
$1108	PTO is Enabled	Yes	No
$2C07	Clutch Pedal Must be Depressed	Yes	No
$4004	Accelerator Pedal Position	Yes	No
$8003	Device Control Requested While Other Device Controls Active	Yes	No

5.11.5 STRR – Fuel System Pressure Test Device Control

This device control is used to charge the fuel rail on a high pressure direct injection system before completing the assembly of the engine in order to check for fuel leaks.

This feature will allow the technician to inspect the fuel system for leaks after servicing the fuel system or after any other service procedure that required disconnection or removal of fuel system components.

The starter relay or starter motor relay and pinion relay on a tandem start system, is commanded on to crank the engine. Spark (SIDI) and fuel injection are inhibited to prevent the engine from starting. The high pressure fuel pump/ regulator(s) are controlled to ensure maximum fuel system pressure is achieved while the engine is cranking and fuel pressure maintained after the engine stops rotating.

(Note: This device control calls CPID $1A “Commanded High Fuel Pressure Device Control” to regulate high fuel pressure during the fuel system pressure test. Any future Powertrain software changes to CPID $22 “Fuel System Pressure Test Device Control” need to consider the CPID $1A “Commanded High Fuel Pressure Device Control” interaction.)

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel System Pressure Test Enable	1	BLN	N/A	$1=True; $0=False
Fuel System Pressure Test On	1	BLN	N/A	$1=True; $0=False

If (Fuel System Pressure Test Enable is True)

If (starter relay output(s) is currently commanded off) and (Fuel System Pressure Test On is True)
If (Limits $0004, $0008, $0144, $0201, $0308, $0401, $0405, $0406, $0903, $2B00 or $2C07 are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control
Else
Reset all timers.
Command starter relay output(s) on.

```
            Start crank timer.
            Disable fuel injectors.
            Disable spark plugs.
            Command low pressure pump (if equipped) on or to maximum pressure.
            Control high pressure fuel pump/regulator for maximum fuel pressure.
        EndIf
    EndIf

    If (starter relay output(s) is currently commanded on)
        If (Limits $0141, $0207, $0308, $0401, $0405, $0406, $0903, $0A05, $0A14, $2B00 or $2C07 are
        exceeded)
            Report the appropriate device control limit exceeded return code to the tester.
                Return to normal control.
        Else
            If (fuel pressure has reached calibrated value
                Start activation timer.
                Start delay timer.
                Reset crank timer.
                Command starter relay output(s) off.
            EndIf
        EndIf
    EndIf

    If (starter relay output(s) is currently commanded off)
        If (Limits $0406 or $0C02 are exceeded)
            Report the appropriate device control limit exceeded return code to the tester.
                        Return to normal control.
        Endif
    Endif

EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0004	Device Control Not Available • This device control is only available for SIDI applications. Device control will be rejected for PFI or CIDI engines.	Yes	No
$0008	Timer expired. Please wait.	Yes	No
$0141	Maximum Crank Time Exceeded. Fuel System Operating Pressure Cannot be Achieved.	Yes	No
$0144	Fuel Level Out of Range	Yes	No
$0201	Engine is Running.	Yes	No
$0207	No Engine Rotation is Detected	Yes	No
$0308	DTC Present. • Crankshaft position sensing fault • Clutch pedal position sensor/switch faults • High pressure fuel pump/regulator fault • Fuel pressure sensor faults • Low pressure fuel pump fault • Starter (motor) relay fault • Starter pinion relay fault • Vehicle speed sensing fault • Park/Neutral position switch faults • Communication fault with fuel pump control module • Communication fault with HCP • Fuel pump control module requesting MIL illumination • Engine position sensing faults • Fuel level sensor faults	Yes	No
$0401	Vehicle Speed Detected	Yes	No
$0405	Battery Voltage Exceeds Limits	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$0903	Transmission must be in Park or Neutral.	Yes	No
$0A05	Engine speed too high.	Yes	No
$0A14	Hybrid System Disabled Engine Crank	Yes	No
$0C02	Activation Timer Has Expired	Yes	No
$2B00	Transmission Range Malfunction	Yes	No
$2C07	Clutch Pedal Must be Applied.	Yes	No

5.12 CPID $23

CPID $23								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Engine Coolant Thermostat Heater Enable			Engine Coolant Pump Clutch Enable	Engine Coolant Pump Clutch On	~~Active Grill Air Shutter 1 Enable~~	~~Active Grill Air Shutter 2 Enable~~	EGR Cooler Bypass Valve Enable
Byte 1	Engine Coolant Thermostat Heater Duty Cycle							
Byte 2	EGR Cooler Bypass Valve Duty Cycle							
Byte 3	~~Active Grill Air Shutter 1 Position~~							
Byte 4	~~Active Grill Air Shutter 2 Position~~							

5.12.1 THMR – Engine Coolant Thermostat Heater Device Control

This device control is used to override the state of the Engine Coolant Thermostat Heater.

Signal	Length (Bits)	Data Type	Range	Conversion
Engine Coolant Thermostat Heater Enable	1	BLN	N/A	$1=True; $0=False
Engine Coolant Thermostat Heater Duty Cycle	8	UNM	0 - 100 %	E = N * 100 / 255

If (Engine Coolant Thermostat Heater Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The engine coolant thermostat Heater output shall be commanded to the value requested in Engine Coolant Thermostat Heater Duty Cycle.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output Control timer has expired (Only applicable with engine off.)	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.12.2 CEBR – Exhaust Gas Recirculation Cooler Bypass Valve Duty Cycle Device Control

This device control is used to verify functionality of the exhaust gas recirculation cooler bypass valve system by allowing the exhaust gas recirculation cooler bypass valve output to be commanded to a specific duty cycle.

Signal	Length (Bits)	Data Type	Range	Conversion
EGR Cooler Bypass Valve Enable	1	BLN	N/A	$1=True; $0=False
EGR Cooler Bypass Valve Duty Cycle	8	UNM	0 to 100 %	E = N * 100 / 255

If (Exhaust Gas Recirculation Cooler Bypass Valve Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Exhaust Gas Recirculation Cooler Bypass Valve shall be commanded to the requested position in Exhaust Gas Recirculation Cooler Bypass Valve Duty Cycle.
    Endif
Else
    Return to normal control
Endif

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Device Control out of range • Calibratable minimum or maximum duty cycle limit exceeded	Yes	No
$0008	The Output Control Timer has Expired, please wait • Limited to a calibrated time, Device Control shall be disabled for a calibratable time before its reactivated	Yes	No
$0308	Fault Detected • Intake Air Temperature sensor fault (IAT or EGR temp depending on system)	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0A28	EGR Cooler Outlet Temperature out of Range • Air Temperature Too High	Yes	No
$1805	Ignition Voltage too High	Yes	No

5.12.3 THMR – Engine Coolant Pump Clutch Device Control

This device control is used to override the state of the engine coolant pump clutch output. When the clutch is commanded on, the coolant pump is not spinning, so coolant flow is stopped.

Signal	Length (Bits)	Data Type	Range	Conversion
Engine Coolant Pump Clutch Enable	1	BLN	N/A	$1=True; $0=False
Engine Coolant Pump Clutch On	1	BLN	N/A	$1=True; $0=False

If (Engine Coolant Pump Clutch Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The engine coolant pump clutch shall be commanded to the state indicated by Engine Coolant Pump Clutch On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Timer expired. Please wait. • This timer only applies when the clutch is commanded on, meaning the pump is not spinning and coolant flow is stopped.	Yes	No
$0308	DTC Present • Engine metal temperature sensor faults • Engine coolant temperature sensor faults	Yes	No
$031D	Engine Coolant Temperature Out Of Range	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0A57	Engine Block/Cylinder Head Temperature Out of Range	Yes	No

5.13 CPID $25

CPID $25								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Mass Air Flow Sensor Supply Voltage Enable	Mass Air Flow Sensor Supply Voltage On	Cooling Fan Duty Cycle Enable	Transmission Cooling Fan Duty Cycle Enable	Exhaust Gas Recircula-tion 2 Enable	Crankcase Ventilation Heater Enable	Crankcase Ventilation Heater On	Exhaust Gas Recirculation 2 H-Bridge Duty Cycle Enable
Byte 1	Cooling Fan Duty Cycle							
Byte 2	Transmission Cooling Fan Duty Cycle							
Byte 3	Exhaust Gas Recirculation 2 Position							
Byte 4	Exhaust Gas Recirculation 2 H-Bridge Duty Cycle							

5.13.1 FANR – Transmission Cooling Fan Duty Cycle Device Control

This device control is used to override the state of the transmission cooling fan control output.

Signal	Length (Bits)	Data Type	Range	Conversion
Transmission Cooling Fan Duty Cycle Enable	1	BLN	N/A	$1=True; $0=False
Transmission Cooling Fan Duty Cycle	8	UNM	0.0 to 100 %	E = N * 100 / 255

If (Transmisson Cooling Fan Duty Cycle Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The transmission cooling fan duty cycle shall be commanded to the value requested in Transmission Cooling Fan Duty Cycle.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$2E04	Requested duty cycle out of range.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$2E06	Transmission oil temp out of range.	Yes	No
$0308	DTC Present • Transmission fluid temperature sensing fault • Transmission control module communication fault • Engine control module communication fault	Yes	No

5.13.2 LPER – Exhaust Gas Recirculation 2 Device Control

This device control is used to verify functionality of the Exhaust Gas Recirculation 2 system by allowing the Exhaust Gas Recirculation 2 output to be commanded to a specific position. The software calculated position should be available to be monitored via a parameter identifier as feedback for the device control override.

Signal	Length (Bits)	Data Type	Range	Conversion
Exhaust Gas Recirculation 2 Enable	1	BLN	N/A	$1=True; $0=False
Exhaust Gas Recirculation 2 Position	8	UNM	0 to 100%	E = N * 100 / 255

If (Exhaust Gas Recirculation 2 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The position shall be commanded to the requested position in Exhaust Gas Recirculation 2 Position
    Endif
Else
    Return to normal control
Endif

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Device Control out of range • Calibratable minimum or maximum Position limit exceeded	Yes	No
$0008	The Output Control Timer has Expired, please wait • The Device control shall not be active for more than a calibratable time. The Device Control shall be disabled for a calibratable time before its reactivated.	Yes	No
$0308	Output control not allowed, DTC Present. • Electrical fault on position feedback.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • System voltage is above or below a calibrated value	Yes	No

5.13.3 LPER – Exhaust Gas Recirculation 2 H-Bridge Duty Cycle Control

This device control is used to verify functionality of the Exhaust Gas Recirculation 2 valve by allowing the Exhaust Gas Recirculation 2 valve output to be commanded to a specific duty cycle. The software calculated duty cycle should be available to be monitored via a parameter identifier as feedback for the device control override.

Signal	Length (Bits)	Data Type	Range	Conversion
Exhaust Gas Recirculation 2 H-Bridge Duty Cycle Enable	1	BLN	N/A	$1=True; $0=False
Exhaust Gas Recirculation 2 H-Bridge Duty Cycle	8	SNM	-128 to +127 %	E = N * 1

If (Exhaust Gas Recirculation 2 H-Bridge Duty Cycle Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Exhaust Gas Recirculation 2 H-Bridge duty cycle shall be commanded to the requested duty cycle in Exhaust Gas Recirculation 2 H-Bridge Duty Cycle.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output Control Limits Out of Range • Calibratable minimum or maximum position limit exceeded	Yes	No
$0008	The Output Control Timer has Expired, please wait • The Device control shall not be active for more than a calibratable time. The Device Control shall be disabled for a calibratable time before its reactivated.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$1805	Ignition Voltage too High	Yes	No

5.13.4 CVHR – Crankcase Ventilation Heater Device Control

This device control is used to turn the crankcase ventilation heater output on or off.

Signal	Length (Bits)	Data Type	Range	Conversion
Crankcase Ventilation Heater Enable	1	BLN	N/A	$1=True; $0=False
Crankcase Ventilation Heater On	1	BLN	N/A	$1=True; $0=False

If (Crankcase Ventilation Heater Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The crankcase ventilation heater output shall be commanded to the requested state in Crankcase Ventilation Heater On.
    Endif
Else
    Return to normal control
Endif

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0405	Battery Voltage Exceeds Limits	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No

5.13.5 FANR – Cooling Fan Duty Cycle Device Control

This device control is used to override the state of cooling fan duty cycle control output.

Signal	Length (Bits)	Data Type	Range	Conversion
Cooling Fan Duty Cycle Enable	1	BLN	N/A	$1=True; $0=False
Cooling Fan Duty Cycle	8	UNM	0 to 100%	E = N * 100 / 255

If (Cooling Fan Duty Cycle Enable is True)
  If (any of the device control limits in the table below are exceeded)
    Report the appropriate device control limit exceeded return code to the tester.
    Return to normal control.
  Else
    The cooling fan duty cycle shall be commanded to the value requested in Cooling Fan Duty Cycle.
  EndIf
Else
  Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output control limits out of range	Yes	No
$0004	Output control unavailable • Relay fan	Yes	Yes
$0201	Output Control not allowed while engine is Running • Mechanical fan	Yes	No
$0308	Output control not allowed, DTC Present. • Engine coolant temperature sensing fault • Air Conditioning pressure sensing fault	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0601	Engine Coolant Temperature Too High • Engine Coolant Temperature exceeds a calibrateable threshold when reducing the duty cycle with the engine running. Limit does not apply when increasing the duty cycle or when the engine is not running. Value shall be continuously monitored during the test.	Yes	No
$0A03	Air conditioning refrigerant high side fluid pressure out of range.	Yes	No

5.13.6 MAFR – Mass Air Flow Sensor Supply Voltage Device Control

This device control is used to override the state of the mass air flow sensor supply voltage output.

Signal	Length (Bits)	Data Type	Range	Conversion
Mass Air Flow Sensor Supply Voltage Enable	1	BLN	N/A	$1=True; $0=False
Mass Air Flow Sensor Supply Voltage On	1	BLN	N/A	$1=True; $0=False

If (Mass Air Flow Sensor Supply Voltage Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The mass air flow sensor supply voltage shall be commanded to the state indicated by Mass Air Flow Sensor Supply Voltage On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine is running	Yes	No
$0406	Incorrect Power Mode	Yes	No

5.14 CPID $26

CPID $26								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Fuel Pump Enable	Desired Fuel Supply Pressure Control Enable	Fuel Pump Closed-Loop Correction Reset Trigger	Fuel Pump Soft Start Trigger				
Byte 1	Fuel Pump Duty Cycle							
Byte 2	Desired Fuel Supply Pressure							
Byte 3	Reserved							
Byte 4	Reserved							

5.14.1 FPMR – Fuel Pump UpIntegrated Device Control

This device control is used to override the normal control of the fuel pump output. The device control is a service diagnostic aid that is used to determine if the fuel pump system is functioning properly. Different limit checks are used when the engine is and is not running for the operation of the fuel pump. This device control can also be used to initiate a fuel pump soft start. This function is primarily used by fuel pump suppliers to perform durability testing.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Pump Enable	1	BLN	N/A	$1=True; $0=False
Fuel Pump Soft Start Trigger	1	BLN	N/A	$1=True; $0=False
Fuel Pump Duty Cycle	8	UNM	0 to 100 %	E = N * 100 / 255

5.14.1.1 Fuel Pump Device Control with Engine Not Running UpIntegrated

With the engine off, this device control is intended to be used as a discrete control for circuit or pressure diagnosis. With the engine off, the Fuel Pump cannot be commanded on for more than 3 seconds.

Note: When this device control is run on a green engine, this device control will initiate a fuel prime routine which will turn on the fuel pump for a calibratible amount of time. This process will only be activated if the fuel pump has never been primed before and the engine has never been running. There must also be adequate fuel in the tank, or the fuel level data from the tank is failed in which case it is assumed there is adequate fuel in the tank, to allow the prime to occur.

```
If   (Fuel Pump Enable is True)
     If   (the engine is running)
          Manage the request to control the Fuel Pump as described in "Fuel Pump Device Control with Engine
     Running UpIntegrated"
     Else
          If   (any of the device control limits in the table below are exceeded)
               Report the appropriate device control limit exceeded return code to the tester.
               Return to normal control.
          Else
               If (Fuel Pump Soft Start Trigger is True)
                    The fuel pump output shall be ramped to the value requested in Fuel Pump Duty Cycle.
               Else
                    The fuel pump output shall be immediately commanded to the value requested in Fuel Pump Duty
                    Cycle.
               EndIf
          EndIf
     EndIf
Else
     Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output Control Limits Out of Range	Yes	No
$0008	The Output Control timer has expired. Please Wait. • The fuel pump must not be commanded on (Duty Cycle > 0.0) continuously for more than 3 seconds with engine not running	Yes	No
$000A	Device Control Lost Due to DTC • Engine speed fault	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • System voltage is above or below a calibrated value	Yes	No

5.14.1.2 Fuel Pump Device Control with Engine Running UpIntegrated

This device control is used to operate the fuel pump with the engine running for fuel system pressure and noise diagnosis/troubleshooting. With the engine running, the Fuel Pump cannot be commanded on for more than 30 seconds.

Note: When the low pressure fuel pump is used in conjunction with a separate high pressure fuel pump for Direct Injection (DI) engines, this device control can be used to reduce the high fuel pressure to a low working level. The high fuel pressure control shall monitor the low pressure "Fuel Pump Device Control with Engine Running" such that when the low pressure pump is commanded "off", the high pressure pump will be controlled to a low pressure value. This combination shall be held until either the FPMR device control is released, or the engine stalls.

If (Fuel Pump Enable is True)

```
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        If (Fuel Pump Soft Start Trigger is True)
            The fuel pump output shall be ramped to the value requested in Fuel Pump Duty Cycle.
        Else
            The fuel pump output shall be immediately commanded to the value requested in Fuel Pump Duty
            Cycle.
        EndIf

    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output Control Limits Out of Range	Yes	No
$0008	The Output Control timer has expired. Please Wait. • The fuel pump must not be commanded on (Duty Cycle > 0.0) continuously for more than 30 seconds with engine running	Yes	No
$000A	Device Control Lost Due to DTC • Engine speed fault	Yes	No
$0401	Vehicle speed detected • Vehicle must not be moving	Yes	No
$0402	Vehicle speed DTC present, output control not allowed • DTCs for VSS or TOSS	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • System voltage is above or below a calibrated value	Yes	No

5.14.2 FRPR – Desired Fuel Supply Pressure Device Control

This device control is used to override the pressure of the fuel supply.

Signal	Length (Bits)	Data Type	Range	Conversion
Desired Fuel Supply Pressure Control Enable	1	BLN	N/A	$1=True; $0=False
Desired Fuel Supply Pressure	8	UNM	0 to 765 kPa	E = N * 3

```
If (Desired Fuel Supply Pressure Control Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The fuel supply pressure shall be commanded to the value requested in Desired Fuel Supply Pressure.
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0003	Output Control Limits Out of Range	Yes	No
$0008	The Output Control timer has expired. Please Wait. • The fuel pump must not be commanded on (Duty Cycle > 0.0) continuously for more than 30 seconds with engine running	Yes	No
$000A	Device Control Lost Due to DTC • Engine speed fault • Vehicle speed fault • Pressure sensor circuit fault • Fuel pump circuit fault	Yes	No
$0401	Vehicle speed detected	Yes	No
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • System voltage is above or below a calibrated value	Yes	No
$0604	Engine Must Be Running	Yes	No

5.14.3 FRPR – Fuel Pump Closed-Loop Correction Reset Trigger Device Control

This device control is used to reset the fuel pump closed-loop correction value.

Signal	Length (Bits)	Data Type	Range	Conversion
Fuel Pump Closed-Loop Correction Reset Trigger	1	BLN	N/A	$1=True; $0=False

If (Fuel Pump Closed-Loop Correction Reset Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        Reset the Fuel Pump Closed-Loop Correction value.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0401	Vehicle Speed Detected	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0402	Vehicle speed DTC present, output control not allowed • DTCs for VSS or TOSS	Yes	No

5.15 CPID $28

CPID $28								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Ignition Off Controller Shutdown Trigger	Immediate Controller Shutdown Trigger						
Byte 1								
Byte 2								
Byte 3								
Byte 4								

5.15.1 RBSR – Ignition Off Controller Shutdown Trigger Device Control

This device control is used to trigger a controller shutdown once the Run/Crank input is low in order to force a write of the RAM copy of NVM parameters to flash memory. When the Run/Crank input is low, the ignition key has transitioned out of the Run position and into either the Accessory or Off position. This device control will be sent at the successful completion of a DVT test in the assembly plant. If power is being supplied to the controller (i.e., if the Accessory input is high), the controller will reset immediately following the shutdown sequence. In the case of a Hybrid Control Processor (HCP), the shutdown sequence will include powering down the Motor Control Processors (MCPs) and opening the high voltage contactors.

Signal	Length (Bits)	Data Type	Range	Conversion
Ignition Off Controller Shutdown Trigger	1	BLN	N/A	$1=True; $0=False

```
If (Ignition Off Controller Shutdown Trigger is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        Set an indication that a controller shutdown should be performed (once the Run/Crank input is low).
    EndIf
EndIf

If (an indication that a controller shutdown should be performed has been set) AND (the Run/Crank input is low)
    Trigger controller shutdown.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.15.2 RBSR – Immediate Controller Shutdown Trigger Device Control

This device control is used to trigger a controller shutdown immediately in order to force a write of the RAM copy of NVM parameters to flash memory. This device control will be sent following a reprogramming event in the assembly plant. If the controller is an Engine Control Module (ECM), the immediate controller shutdown will not be allowed if the engine is rotating. If the controller is a Hybrid Control Processor (HCP), the immediate controller shutdown will not be allowed if the propulsion system is active. If power is being supplied to the controller (i.e., if the Run/Crank or Accessory inputs are high), the controller will reset immediately following the shutdown sequence. In the case of an HCP, the shutdown sequence will include powering down the Motor Control Processors (MCPs) and opening the high voltage contactors.

Signal	Length (Bits)	Data Type	Range	Conversion
Immediate Controller Shutdown Trigger	1	BLN	N/A	$1=True; $0=False

```
If (Immediate Controller Shutdown Trigger is True)
    If (the controller is an ECM) AND (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
    Else
        Trigger controller shutdown.
    EndIf
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine speed not zero (ECM only)	Yes	Yes
$0406	Incorrect Power Mode	Yes	No
$0A13	Propulsion System Active (HCP only)	Yes	Yes

5.16 CPID $2F

CPID $2F								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Alternative Fuel Injector Enable	Alternative Fuel Injector On			Alternative Fuel System Purge Enable	Alternative Fuel System Purge On	Alternative Fuel Disable Requested	Alternative Fuel Disable Active
Byte 1	Alternative Fuel Injector Identifier							
Byte 2								
Byte 3								
Byte 4								

5.16.1 FULR - Alternative Fuel System Purge Device Control

This device control is used to trigger all injector to be commanded on for a calibrated number of pulses (i.e. 250). The injector pulse width (i.e. 450 ms) and related off time (i.e. 50 ms) shall be set to fixed values, and the frequency of injector pulses should be set to a fixed value (i.e. 2 Hz). The sole purpose of this device control is to facilitate de-pressurization of the fuel system following the alternative fuel system assembly leak test, and it will not be used as part of field service procedures. This device control is used to purge the fuel system (i.e. fuel tank and fuel lines) of the compressible fluid used as part of the assembly leak check (i.e. Nitrogen or compressed air). This device control is expected to be active for up to 10 minutes, and several 'test cycles' are allowed to accommodate the need for some vehicles that may need to be repaired and re-tested. This device control is permanently disallowed following detection of the first fuel tank fill (i.e. > 10% fuel level), if the vehicle has been driven more than a calibrated distance (i.e. 1.0 km), if a calibrated amount of alternative fuel has been consumed with the engine running, or when the maximum number if injection cycles have completed.

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel System Purge Enable	1	BLN	N/A	$1=True; $0=False
Alternative Fuel System Purge On	1	BLN	N/A	$1=True; $0=False

```
If   (Alternative Fuel System Purge is True)
     If (any of the device control limits in the table below are exceeded)
          Report the appropriate device control limit exceeded return code to the tester.
          Return to normal control.
     Else
          The alternative fuel system purge routine shall be commanded to the value requested in Alternative Fuel
          System Purge On.
     EndIf
Else
     Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Output Control not allowed while engine is Running.	Yes	Yes
$0A7B	Alternative Fuel System Purge Already Finished	Yes	Yes
$0308	Fault active. • Vehicle speed sensing fault • Park / Neutral position fault • Alternative Fuel Rail Pressure sensor fault • Alternative Fuel High Pressure Shutoff valve fault • Alternative Fuel Low Pressure Shutoff valve fault • Alternative Fuel Injector fault • Alternative Fuel Level Sensor fault	Yes	Yes
$0401	Vehicle speed not zero.	Yes	Yes
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	Yes
$0A7A	Alternative Fuel Devices Disabled	Yes	Yes
$1803	Ignition voltage too low.	Yes	Yes
$0903	Transmission must be in Park or Neutral	Yes	Yes

5.16.1.1 Fuel Injector Disable Device Control

This device control is used to disable injectors in order to cut off fuel to the associated cylinders. This device control applies to both Port Fuel Injected (PFI) engines and Direct Injected (DI) engines. For gaseous fuel applications using a fuel injector control module (FICM), the gaseous fuel injectors can be disabled using this device control (i.e. by the ECM) when the gaseous fuel system is active. For gaseous fuel applications using a fuel injector control module (FICM), the gaseous fuel injectors can be disabled using this device control (i.e. by the ECM) when the gaseous fuel system is active.

If (any of the device control limits in the table below are exceeded)
    Report the appropriate device control limit exceeded return code to the tester.
    Return to normal control.
Else
    The fuel injector output specified by the Fuel Injector Identifier shall be commanded off.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0001	Generic device control timer time-out active (30 second enable, 60 second disable).	Yes	No
$000B	Displacement on demand active.	Yes	Yes
$0301	Injector not valid.	Yes	Yes
$0302	Too many injectors to be disabled simultaneously.	Yes	No
$0303	Injector flow test in progress.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No

5.16.1.2 Alternative Fuel Injector Flow Device Control (CNG/LPG Port Injection)

This device control actuates an engine not running fuel injector flow test for gaseous fuel injected engines (i.e. CNG/LPG). The purpose of the injector flow test is to determine if the requested injector is restricted or plugged. The injector flow test algorithm is completely contained in the fuel injector control module (FICM) and returns a fuel pressure drop value through DID $6C “Direct Injection Fuel Injector Flow Test Results” when the test is complete. The injector flow test is initiated via request from an off-board tool. The off-board tool sends two pieces of data to the fuel injector control module – the request to enable the injector flow test and the ID of the injector that should be tested.

Once the injector flow test is enabled, the desired fuel rail pressure is commanded to a calibrated fuel pressure and fuel Shut-Off valves are opened. After the fuel rail pressure achieves a stabilized value, the fuel Shut-Off valves are closed to prevent the fuel pressure regulator from being replenished with fuel that would increase the fuel pressure during the test.

The injector flow test algorithm is designed to average the pressure drop over a calibrate-able number of injections for each cylinder. The calibrated pulse width and number of pulses will be delivered only for the cylinder requested by the off-board tool. The average pressure drop is reported in DID $6C such that it can be compared with the pressure drops for the remaining engine cylinders during the same key cycle. A determination of injector performance can be made by comparing the DID $6C values from cylinder to cylinder.

If (any of the device control limits in the table below are exceeded)
    Report the appropriate device control limit exceeded return code to the tester.
    Return to normal control.
Else
    Trigger the fuel injector flow test for the injector specified by the Fuel Injector Identifier
EndIf

Return Code	Device Control Limit Exceeded Description Test Enable Conditions	Active for Service	Active for DVT
$0002	Device control not supported. • Alternative Fuel Injector Disable not supported in a fuel injector control module. Fuel Injector Disable must be done from the ECM while running on alternative fuel.	Yes	No
$0201	Output Control not allowed while engine is Running.	Yes	Yes
$0202	Injector already tested. Unable to restart test during this engine's off cycle.	Yes	No
$0203	Invalid injector number. Check for proper vehicle selection.	Yes	No
$0204	Output control not allowed. Injector flow test in progress.	Yes	No
$0205	More than one injector selected	Yes	No
$0308	Fault active. • Vehicle speed sensing fault • Park / Neutral position fault • Alternative Fuel Rail Pressure sensor fault • Alternative Fuel High Pressure Shutoff valve fault • Alternative Fuel Low Pressure Shutoff valve fault • Alternative Fuel Injector fault	Yes	Yes
$0401	Vehicle Speed Has Been Detected	Yes	No
$0406	Incorrect Power Mode. Check Ignition Switch.	Yes	No
$0903	Transmission must be in Park or Neutral	Yes	No
$0A29	Fuel Pressure Out of Range	Yes	No
$0A7A	Alternative Fuel Devices Disabled	Yes	Yes
$1803	Ignition voltage too low.	Yes	Yes

5.16.2 FMCR – Alternative Fuel Disable Device Control

This device control is used to override the state of the alternative fuel disable output. The device control is a service diagnostic aid that is used to determine if the alternative fuel disable system is functioning properly. This device control is used to exercise the ECM output used to disable FICM controlled alternative fuel system devices. This device control (to the ECM) may be requested simultaneously with other device control requests to the FICM. The Alternative Fuel Disable (output from ECM) must be 'off' in order for FICM output devices to be powered

Signal	Length (Bits)	Data Type	Range	Conversion
Alternative Fuel Disable Requested	1	BLN	N/A	$1=True; $0=False
Alternative Fuel Disable Active	1	BLN	N/A	$1=True; $0=False

```
If  (Alternative Fuel Disable Requested is True)
    If  (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The alternative fuel disable output shall be commanded to the value requested in Alternative Fuel
        Disable Active.
    EndIF
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No
$5003	Voltage Out of Range • System voltage is above or below a calibrated value	Yes	No
$0308	Fault active. • Vehicle speed sensing fault • Park / Neutral position fault	Yes	No
$0401	Vehicle speed not zero.	Yes	No
$0903	Transmission must be in Park or Neutral	Yes	No

5.17 CPID $40

CPID $40								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Heater Core Pump Relay Enable	Heater Core Pump Relay On	Motor A Coolant Pump Relay/Discrete Enable	Motor A Coolant Pump Relay/Discrete On				
Byte 1							Auxiliary Transmission Pump Relay Enable	Auxiliary Transmission Pump Relay On
Byte 2								
Byte 3								
Byte 4								

5.17.1 PCOR – Motor A Coolant Pump Relay/Discrete Device Control

This device control is used to override the state of the electric motor A coolant pump relay output.

Signal	Length (Bits)	Data Type	Range	Conversion
Motor A Coolant Pump Relay/Discrete Enable	1	BLN	N/A	$1=True; $0=False
Motor A Coolant Pump Relay/Discrete On	1	BLN	N/A	$1=True; $0=False

If (Motor A Coolant Pump Relay/Discrete Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The outputs for the motor A coolant pump relay/discrete shall be commanded to the requested state in Motor A Coolant Pump Relay/Discrete On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0308	Output control not allowed, DTC Present. • Motor A stator temperature sensor faults (Relay Pump only)	Yes	No
$0406	Incorrect Power Mode	Yes	No
$2E01	Over Temperature Condition Detected (Discrete Pump only)	Yes	No
$1803	System Voltage Too Low (Discrete Pump only)	Yes	Yes

5.17.2 AXPR – Auxiliary Transmission Pump Relay Device Control

This device control is used to override the state of the auxiliary transmission pump relay output.

Signal	Length (Bits)	Data Type	Range	Conversion
Auxiliary Transmission Pump Relay Enable	1	BLN	N/A	$1=True; $0=False
Auxiliary Transmission Pump Relay On	1	BLN	N/A	$1=True; $0=False

If (Auxiliary Transmission Pump Relay Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The auxiliary transmission pump relay shall be commanded to the state indicated by Auxiliary Transmission Pump Relay On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0008	Output Control Timer has expired	Yes	No
$0401	Vehicle speed is not zero (or validity bit not set)	Yes	No
$0406	Incorrect Power Mode	Yes	No
$1102	Fault Active – Limit checking to vehicle speed, engine running , or power mode	Yes	No
$1303	Engine running (or validity bit not set)	Yes	No

5.17.3 THMR – Heater Core Pump Relay Device Control

This device control is used to override the state of the heater core pump relay output.

Signal	Length (Bits)	Data Type	Range	Conversion
Heater Core Pump Relay Enable	1	BLN	N/A	$1=True; $0=False
Heater Core Pump Relay On	1	BLN	N/A	$1=True; $0=False

```
If (Heater Core Pump Relay Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The heater core pump relay shall be commanded to the state indicated by Heater Core Pump Relay On.
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0406	Incorrect Power Mode	Yes	No

5.18 CPID $58

CPID 58								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Sensor Bus Relay Enable	Sensor Bus Relay On						
Byte 1								
Byte 2								
Byte 3								
Byte 4								

5.18.1 SBRR – Sensor Bus Relay Device Control

This device control is used to control the sensor bus supply relay

Signal	Length (Bits)	Data Type	Range	Conversion
Sensor Bus Relay Enable	1	BLN	N/A	$1=True; $0=False
Sensor Bus Relay On	1	BLN	N/A	$1=True; $0=False

If (Sensor Bus Relay Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The sensor bus relay shall be commanded to the value requested in Sensor Bus Relay On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT*
$0406	Incorrect Power Mode • Run/crank must be active (ODAR)	Yes	No
$0201	Engine Rotating	Yes	No
$1600	Engine Rotating Data Fault	Yes	No
$0401	Vehicle Speed not Zero	Yes	No
$$0402	Vehicle Speed Error	Yes	No

5.19 CPID $59

CPID $59								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	A Rocker Arm Actuator Control Circuit Bank 1 Enable	A Rocker Arm Actuator Control Circuit Bank 1 On	A Rocker Arm Actuator 2 Control Circuit Bank 1 On					Intake Rocker Arm Lift Position Diagnostic Trigger
Byte 1								
Byte 2								
Byte 3								
Byte 4								

5.19.1 VVLR – A Rocker Arm Actuator Bank 1 Device Control

This device control is used to override the state of the A Rocker Arm Actuator Control Circuit Bank 1outputs. The behavior and limits of this device control will be different based on the state of the internal combustion engine. When the engine is running, this device control may be used to command low lift mode either as a system, or separately as controlled by the algorithm. When the engine is not running, this device control may be used for direct electrical control of the A rocker arm solenoid valves for Bank 1.

Signal	Length (Bits)	Data Type	Range	Conversion
A Rocker Arm Actuator Control Circuit Bank 1 Enable	1	BLN	N/A	$1=True; $0=False
A Rocker Arm Actuator Control Circuit Bank 1 On	1	BLN	N/A	$1=True; $0=False
A Rocker Arm Actuator 2 Control Circuit Bank 1 On	1	BLN	N/A	$1=True; $0=False

5.19.1.1 A Rocker Arm Actuator Bank 1 Device Control with Engine Running

When the engine is running, this device control may be used to command intake valve opening to low or high (high being default when the solenoid is not energized) valve lift mode either in cylinder pairs or as a system. Since engine timing is critical to the operation of the intake cam variable valve lift system, commanding the solenoids "on" or "off" with the engine running will cause the algorithm to transition between low and high valve lift mode (or vice versa) as it is safe to do so.

```
If (A Rocker Arm Actuator Control Circuit Bank 1 Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The A Rocker Arm Actuator Control Circuit Bank 1outputs shall be commanded to the states requested in
        A Rocker Arm Actuator (or Actuator 2) Control Circuit Bank 1 On.
    EndIf
Else
    Return to normal control.
EndIf
```

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0010	Engine speed out of range. • Engine speed does not lie between calibratable lower and upper thresholds.	Yes	Yes
$0308	Fault active. • Crank sensor fault. • Cam sensor fault. • Engine oil pressure sensor fault. • Powertrain relay fault. • Intake air temperature sensor fault (E21240) • Brake vacuum pressure sensor fault. • Electronic throttle control reduced engine power active. • Engine coolant temperature sensor fault. • Manifold absolute pressure sensor fault. • Vehicle speed fault.	Yes	Yes
$030B	Piston protection active.	Yes	Yes
$030C	Engine oil temperature out of range. • Engine oil temperature does not lie between calibratable lower and upper thresholds.	Yes	Yes
$030D	Engine oil pressure out of range. • Engine oil pressure does not lie between calibratable lower and upper thresholds.	Yes	Yes
$030F	Engine oil aeration present.	Yes	Yes
$0310	Engine metal overtemperature protection active.	Yes	Yes
$0315	Catalyst converter overtemperature protection active.	Yes	Yes
$0316	Brake vacuum pressure low. • Brake vacuum pressure falls below a calibratable threshold.	Yes	Yes
$0318	Axle torque limiting active.	Yes	Yes
$0319	Brake torque management active.	Yes	Yes
$031D	Engine coolant temperature out of range. • Engine coolant temperature does not lie between calibratable lower and upper thresholds.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No
$0601	Engine coolant temperature too high. • Hot coolant mode active.	Yes	Yes
$0604	Engine not running.	Yes	Yes
$0A05	Engine speed too high. • Engine overspeed protection active.	Yes	Yes
$0E02	Vehicle speed not zero.	Yes	Yes
$2102	Transmission gear incorrect. • Disabled in transmission gears selected via calibration.	Yes	Yes
$2103	Transmission range incorrect. • Disabled in transmission ranges selected via calibration.	Yes	Yes
$5003	Ignition voltage out of range. • Powertrain relay voltage does not lie between calibratable lower and upper thresholds.	Yes	Yes

5.19.1.2 A Rocker Arm Actuator Bank 1 Device Control with Engine Not Running

When the engine is not running, this device control may be used for direct electrical control of the A rocker arm solenoid valves (A Rocker Arm Actuator Control Circuit Bank 1).

If (A Rocker Arm Actuator Control Circuit Bank 1Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else

The A rocker arm actuator control circuit Bank 1outputs shall be commanded to the states requested in A Rocker Arm Actuator (or Actuator 2) Control Circuit Bank 1 On.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0201	Engine speed not zero.	Yes	Yes
$0308	Fault active. • Powertrain relay fault.	Yes	Yes
$0406	Incorrect Power Mode	Yes	No
$0E02	Vehicle speed not zero.	Yes	Yes
$0E03	Transmission not in Park or Neutral.	Yes	Yes
$5003	Ignition voltage out of range. • Powertrain relay voltage does not lie between calibratable lower and upper thresholds.	Yes	Yes

5.19.2 VVLR – Intake Rocker Arm Bank 1 Lift Position Diagnostic Trigger Device Control

This device control is used to initiate the on-board intake cam variable valve lift diagnostic test sequence. Under normal vehicle operation this diagnostic will run once per key cycle, transparent to the operator.

Signal	Length (Bits)	Data Type	Range	Conversion
Intake Rocker Arm Lift Position Diagnostic Trigger	1	BLN	N/A	$1=True; $0=False

If (Intake Rocker Arm Lift Position Diagnostic Trigger is True)
If (any of the device control limits in the table below are exceeded)
Report the appropriate device control limit exceeded return code to the tester.
Return to normal control.
Else
The Intake Rocker Arm Bank 1 lift position diagnostic testing sequence shall be triggered.
EndIf
Else
Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT
$0010	Engine speed out of range.	Yes	No
$0308	Fault active. • Crank sensor fault. • Cam sensor fault. • Engine oil pressure sensor fault. • Engine Oil Pressure Sensor B FaultPowertrain relay fault. • Intake/Exhaust Cam actuator fault • Intake rocker arm solenoid valve driver fault.	Yes	Yes
$030C	Engine oil temperature out of range.	Yes	No
$030D	Engine oil pressure out of range.	Yes	No
$0406	Incorrect Power Mode	Yes	No
$0604 $0A5A	Engine not running. Engine Torque out of range / not steady state	Yes Yes	Yes Yes
$0004	Device Control Not Available: • Oil Pump Intrusive request rejected • Variable Valve Lift Intrusive request rejected	Yes	No

5.20 CPID $5A

CPID $5A								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0	Cruise Control DVT Mass Trigger		Cruise Control Integral Scalar Trigger					
Byte 1	Cruise Control DVT Mass							
Byte 2	Cruise Control Integral Scalar							
Byte 3	Reserved							
Byte 4	Reserved							

5.20.1 CRZR – Cruise Control DVT Mass Device Control

This device control is used to trigger the use of a “DVT Mass” sent by the DVT control software to the test booth to allow modification of the mass value used to calculate the cruise control axle torque request. This modification is only allowed while the DVT test is being run in the vehicle assembly plant.

Signal	Length (Bits)	Data Type	Range	Conversion
Cruise Control DVT Mass Trigger	1	BLN	N/A	$1=True; $0=False
Cruise Control DVT Mass	8	UNM	0 to 2550 kg	E = N * 10

If (Cruise Control DVT Mass Trigger is True)
    The Cruise Control DVT Mass will be used in the cruise control axle torque calculation.
Else
    Return to normal control.
EndIf

5.20.2 CRZR – Cruise Control DVT Integral Scalar Device Control

This device control is used to trigger the use of an “Integral Scalar” sent by the DVT control software to the test booth to allow modification of the closed loop integrator used to calculate the cruise control axle torque request. This modification is only allowed while the DVT test is being run in the vehicle assembly plant.

Signal	Length (Bits)	Data Type	Range	Conversion
Cruise Control Integral Scalar Trigger	1	BLN	N/A	$1=True; $0=False
Cruise Control Integral Scalar	8	UNM	0 to 1	E = N / 255

If (Cruise Control DVT Integral Scalar Trigger is True)
    The Cruise Control Integral Scalar will be used to modify the closed loop integrator in the cruise control axle torque calculation.
Else
    Return to normal control.
EndIf

5.21 CPID $74

CPID $74								
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Byte 0							Transmission Surge Accumulator Solenoid Enable	Transmission Surge Accumulator Solenoid On
Byte 1								
Byte 2								
Byte 3								
Byte 4								

5.21.1 SACR – Transmission Surge Accumulator Solenoid Device Control

This device control is used to override the state of the transmission surge accumulator solenoid output.

Signal	Length (Bits)	Data Type	Range	Conversion
Transmission Surge Accumulator Solenoid Enable	1	BLN	N/A	$1=True; $0=False
Transmission Surge Accumulator Solenoid On	1	BLN	N/A	$1=True; $0=False

If (Transmission Surge Accumulator Solenoid Enable is True)
    If (any of the device control limits in the table below are exceeded)
        Report the appropriate device control limit exceeded return code to the tester.
        Return to normal control.
    Else
        The Transmission Surge Accumulator Solenoid output shall be commanded to the value requested in Transmission Surge Accumulator Solenoid On.
    EndIf
Else
    Return to normal control.
EndIf

Return Code	Device Control Limit Exceeded Description	Active for Service	Active for DVT*
$0201	Engine speed not zero.	Yes	No
$0401	Vehicle speed is not zero (or validity bit not set)	Yes	No
$0406	Incorrect Power Mode • Run/crank must be active	Yes	No
$0903	Transmission not in park or neutral	Yes	No
$1102	Fault Active – • Transmission output speed sensor fault • Engine speed fault • Transmission range sensor fault	Yes	No

6. Diagnostic Trouble Codes (DTCs)

All of the indicators within the DTC status byte (i.e., the value provided for the DTCStatusByte parameter of a positive response to sub-function readStatusOfDTCByNumber ($80) or sub-function readStatusOfDTCByStatusMask ($81) of the ReadDiagnosticInformation ($A9) service) shall be implemented according to Appendix E of [1]. Additional details of the implementation of these indicators are specified below.

The behavior of the indicators within the DTC status byte is controlled by the light enable and fault type calibrations associated with each DTC. The recommended values for these calibrations for individual DTCs are specified in [6].

The boolean light enable calibration is used to enable the setting of the warningIndicatorRequestedState indicator. If the light enable calibration for a DTC is true, the associated warningIndicatorRequestedState indicator shall be set upon the reporting of a failure that results in the associated historyDTC indicator being set. If the light enable calibration for a DTC is false, the associated warningIndicatorRequestedState indicator shall always be cleared.

The fault type calibration may assume the values described in Table 36: Descriptions of DTC Fault Types . It is used to enable (type A, B, and C) or disable (type X) DTCs, to differentiate emission related (type A and B) and non emission related (type C) DTCs, and to determine whether to set the historyDTC indicator for a DTC on the first trip (type A and C) or the second consecutive trip (type B) during which a failure is reported.

Fault Type	Description
A	Emission related DTC The historyDTC and (if enabled) warningIndicatorRequestedState indicators shall be set on the first trip during which a failure is reported.
B	Emission related DTC The historyDTC and (if enabled) warningIndicatorRequestedState indicators shall be set on the second consecutive trip during which a failure is reported.
C	Non emission related DTC The historyDTC and (if enabled) warningIndicatorRequestedState indicators shall be set on the first trip during which a failure is reported.
X	DTC not supported

Table 36: Descriptions of DTC Fault Types

For emission related (type A and B) DTCs, the associated warningIndicatorRequestedState indicator shall be cleared after the third consecutive trip during which a pass condition is reported and a failure is not reported. For non emission related (type C) DTCs, the associated warningIndicatorRequestedState indicator shall be cleared after the first trip during which a pass condition is reported and a failure is not reported.

The behavior of the set of emission related DTCs shown in Table 37: One Trip MIL Off DTCs , one trip MIL off DTCs, differs slightly from that of other emission related DTCs. For these DTCs, the associated warningIndicatorRequestedState indicator shall be cleared after the first trip during which a pass condition is reported and a failure is not reported.

DTC	Description
P0455	Evaporative Emission System Large Leak Detected

Table 37: One Trip MIL Off DTCs

The behavior of the set of type B emission related DTCs shown in Table 38: Special Operating Condition DTCs , special operating condition DTCs, differs slightly from that of other type B emission related DTCs. For these DTCs, the associated historyDTC and (if enabled) warningIndicatorRequestedState indicators shall be set on the second, but not necessarily consecutive, trip during which a failure is reported if the operating conditions (engine speed, engine load, and warm up status) under which the failure was reported on the first trip and second trip match. In addition, the associated warningIndicatorRequestedState indicator shall be cleared after the third consecutive trip during which the operating conditions under which the stored failure was reported are matched, a pass condition is reported, and a failure is not reported.

DTC	Description
P0171	Fuel Trim System Lean Bank 1
P0172	Fuel Trim System Rich Bank 1
P0174	Fuel Trim System Lean Bank 2
P0175	Fuel Trim System Rich Bank 2
P0300	Engine Misfire Detected

Table 38: Special Operating Condition DTCs

In addition to the use of the ClearDiagnosticInformation ($04) diagnostic service to reset the indicators within the DTC status byte as described in [1], an automatic code clear function (based upon the completion of a number of warm up cycles without a failure being reported) shall be supported. An automatic code clear for a DTC shall affect any failure records associated with the DTC and all indicators within the associated DTC status byte in the same manner as the ClearDiagnosticInformation ($04) diagnostic service with the exception of the testNotPassedSinceDTCCleared indicator, which shall be unaffected by an automatic code clear.

The automatic code clear function shall be implemented as three independent functions: one for all emission related (type A and B) DTCs that illuminate the MIL, one for all emission related (type A and B) DTCs that do not illuminate the MIL and one for all non emission related (type C) DTCs.

- An automatic code clear shall be performed for all emission related DTCs upon the completion of 40 warm up cycles without an emission related failure being reported if the code has illuminated the MIL.
- An automatic code clear shall be performed for all emission related DTCs upon the completion of 5 warm up cycles without an emission related failure being reported if the code has not illuminated the MIL.
- An automatic code clear shall be performed for all non emission related DTCs upon the completion of 40 warm up cycles without a non emission related failure being reported.

7. Failure Records / Freeze Frame

- Mode 02 is used to obtain OBD required (legislated) freeze frame data
  - OBD required freeze frame data is a subset of the FFRR freeze frame
  - Only PIDs $0000 to $00FF can be requested
  - Only the ECM need support Mode 02
- Mode 12 is used to obtain both corporate standard freeze frame data and failure record data
  - All PIDs ($0000 to $FFFF) can be delivered
  - ECM, TCM, and TCCM can support Mode 12

7.1 Corporate Standard Freeze Frame

Refer to [7] and [8] for the rules for PID selection to create an application specific freeze frame PID List.

8. Compliance with Standards

8.1 Compliance with GMPT Product Engineering Standards

This document is in compliance with the following standards.

Document Number	Version	Document Title
TL.07.0043	R08	Document Format Standard

8.2 Compliance with Industry Standards

No applicable industry standards were identified for this document.

Appendix A - Revision History

Revision	Date	Author	Description	EACR / CR Number(s)

Appendix B - List of Acronyms

The following acronyms are used throughout this document.

ABS	Antilock Brake System
A/C	Air Conditioning
ASC	ASCII Encoded
AWD	All Wheel Drive
BASS	Brake Apply Sensing System
BCD	Binary Coded Decimal (Numeric Signal)
BLN	Boolean
C2C	Clutch to Clutch
CAN	Controller Area Network
CARB	California Air Resource Board
CASS	Clutch Apply Sensing System
CASE	Crank Angle Sensing Error
CPID	Control Packet Identifier
CTS	Component Technical Specification
DID	Data Identifier
DPID	Data Packet Identifier
DRI	Drive Reverse Indicator
DTC	Diagnostic Trouble Code
E	Engineering Units
EACR	Enhancement and Anomaly Change Record
ECCC	Electronically Controlled Converter Clutch
ECM	Engine Control Module
ECU	Electronic Control Unit
EGR	Exhaust Gas Recirculation
ENM	Enumerated
EONV	Engine Off Natural Vacuum
EPA	Environmental Protection Agency
ERS	Energy Reduction Software
ETEI	Engine Control Module – Transmission Control Module Electrical Interface
EV	Electro Viscous
EWMA	Exponentially Weighted Moving Average
GM	General Motors
GMLAN	General Motors Local Area Network
GMPT	General Motors Powertrain
ICM	Immobilizer Control Module
IMS	Internal Mode Switch
ISO	The International Organization for Standardization
LORES	Low Resolution
MAF	Mass Airflow
MAP	Manifold Absolute Pressure

MBT	Mean Best Torque
MEC	Manufacturers Enable Counter
MEDRES	Medium Resolution
MRF	Magneto Rheological Fan
MIL	Malfunction Indicator Lamp
N	Numerical Units
NSBU	Neutral Safety Backup
OBD	On Board Diagnostic
OSC	Oxygen Storage Capacity
PCS	Pressure Control Solenoid
PDS	Periodic DPID Scheduler
PID	Parameter Identifier
PKT	Packeted
PPEI	Platform-Powertrain Electrical Interface
PPM	Pulse Per Mile
PRNDL	Park Reverse Neutral Drive Low
PSM	Pressure Switch Manifold
PTO	Power Take Off
PWM	Pulse Width Modulation
RDM	Rear Drive Module
RVC	Regulated Voltage Control
RVS	Remote Vehicle Start
SAI	Secondary Air Injection
SCIAP	Supercharger Inlet Absolute Pressure
SNM	2´s Complement Signed Numeric
SPS	Service Programming System
STP	Standard Temperature and Pressure
SVS	Service Vehicle Soon
TCC	Torque Converter Clutch
TCCM	Transfer Case Control Module
TCM	Transmission Control Module
TIS	Transmission Input Speed
TLS	Temperature Limiting Software
TOS	Transmission Output Speed
UNM	Unsigned Numeric
USDT	Unacknowledged Segmented Data Transfer
UUDT	Unacknowledged Unsegmented Data Transfer
VTD	Vehicle Theft Deterrent
WOT	Wide Open Throttle

Appendix C - Related Documents

The following documents are either referenced by this document or contain related information. These documents may provide useful information to the reader and should be reviewed for compatibility in the event that this document (or any of the related documents) undergoes revision.

	Document Number	Version	Document Title
[1]	GMW 3110	1.4	GMLAN Enhanced Diagnostic Test Mode Specification http://www.ived.gm.com/electrical/warren/ssltexpt/gmlan/specs/index.html
[2]	GMW 3203	0.2	GMLAN Signal and Frame Definition Guidelines Section 3, Standard Data Types http://www.ived.gm.com/electrical/warren/ssltexpt/gmlan/specs/index.html
[3]	ISO 15031-5		Road Vehicles - Communication Between Vehicle and External Equipment - Part 5: Emission-Related Diagnostic Services
[4]	ISO 15031-6		Road Vehicles - Communication Between Vehicle and External Equipment - Part 6: Diagnostic Trouble Code Definitions
[5]	corp_pid.xls		Parameter Identification (PID) List (All) http://so.gm.com/product/tech2/tech2a.htm
[6]	GM_Master_DTC_List.xls	PCC.08.02.00	DTC Master List http://powertrain.gm.com/engineer/controls/eis/algo_private/inf/dfir/index.htm
[7]	ffmaster.txt	PT1#08.02.00	Freeze Frame / Failure Record Master List http://powertrain.gm.com/engineer/controls/eis/algo_private/inf/ffrr/index.htm
[8]	FreezeFrameFailureRecordList.xls		Freeze Frame Failure Record List.xls
[9]	ISO 15031-4		Road Vehicles - Communication Between Vehicle and External Equipment - Part 5: External Test Equipment
[10]	ovirkmode06tmp.doc	08.02.00	OBD Mode 6 Test Value and Limit Template
[11]	ovirkgmptJ1979.doc	08.02.00	Implementation Specification for SAE J1979 Test Modes http://powertrain.gm.com/engineer/controls/eis/algo_private/inf/ovir/index.htm